



**Shiva Testing & Research Laboratory
Private Limited**

Geo-Technical Investigation Report

Name of the Work

**GEOTECHNICAL INVESTIGATION FOR THE
CONSTRUCTION OF BQ HOUSING COMPLEX AT
DULIAJAN, ASSAM**

Location

Duliajan, Assam

Client

MITES LIMITED

Report Number

SR-2507170001

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Test Conducted By:

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1.0 INTRODUCTION

There is a proposal for **Geotechnical Investigation for the construction of BQ Housing Complex at Duliajan, Assam**. Geotechnical investigation is required to know the type of substrata and its safe bearing capacity at the proposed site of structures for design of foundation system.

The job for boring bore holes, in situ testing, soil sampling, determination of safe bearing capacity (SBC) and report preparation was awarded by RITES LIMITED.

2.0 SCOPE OF WORK

1. Drilling of boreholes of dia 150mm as directed and specified- 11 no.
2. Conducting Standard Penetration Test as per IS 2131 at every 1.5 M depth or any change of strata.
3. Collection disturbed and undisturbed samples of soil.
4. Conducting 6 trial pit (TP-1 to TP-6) of size (1.0 mtr x 1.0 mtr x 1.2 mtr) including collection and testing of samples.
5. Recording of Ground Water Table up to the depth of investigation.
6. Conducting various laboratory tests on soil samples collected from borehole.
7. Preparation and submission of this technical report stating safe bearing capacity of soil sub strata.

3.0 OBJECTIVE OF SITE INVESTIGATION

The Object of the current geotechnical investigation was to explore and determine the existing sub-strata conditions such as stratification, Denseness or hardness of the strata, position of ground water table at Bore Holes etc. and to evaluate approximate range of safe bearing capacity for the proposed structure using empirical formulas provided in the relevant IS codes. The goal of our investigation was to identify the key geotechnical issues that could

potentially impact the proposed project and to develop geotechnical recommendations for design and construction of the project.

4.0 INVESTIGATIONS CONDUCTED

This report contains details of field investigations, Laboratory tests and the recommendations based upon them.

The various results have been presented under two heads, i.e.

- (i) The field tests
- (ii) The laboratory tests

4.1 The field tests include:

- (a) Boring, borehole of 150 mm diameter at the site for the purpose of knowing the type of soil and thickness of every stratum encountered by the boring tools.
- (b) Standard Penetration tests (SPT) is meant for measuring the penetration resistance of the soil, which is measure of its bearing capacity insitu. This test is particularly best suited to cohesionless soils which are difficult and expensive to be sampled in the undisturbed state.
- (c) Sampling, depending upon the type of soil, both disturbed & undisturbed soil samples are collected. They are needed for various tests to be conducted in the laboratory for confirmation of soil properties.

4.2 The laboratory tests include:

- (a) Bulk density, Dry Density and Moisture Content, of the soil samples as per IS: 1892-2021 at various depths for the estimation of overburden pressure required for overburden correction in the observed penetration resistance values of the soil and for bearing capacity computation from shear failure consideration.
- (b) Grain Size analysis (sieve analysis and hydrometer analysis) of the soil samples as per IS 2720 Part 4: 1985 RA: 2020 for the purpose of soil classification and preparation of log of the borehole.
- (c) Atterberg limits, i.e. the liquid limit, plastic limit tests and plasticity index of fine-grained soils passing through 425 micron IS sieve as per IS 2720 Part 5: 1985 RA: 2020. They give an idea about the type of soil and its consistency, i.e., the degree of firmness of a soil to bear external loads imposed on it.

- (d) Shear strength parameters: i.e the soil cohesion (c) and its angle of internal friction (f). These are needed for the estimation of bearing capacity (B.C.) from shear consideration. They are determined by Tri-axial shear Tests as per IS 2720: Part 11: 1993 RA 2021/ IS 2720: Part 12: 1981 RA: 2021 and Direct shear Tests as per IS 2720: Part 13:1986 RA: 2021.
- (e) Consolidation Properties: The consolidation parameters are required for knowing the ultimate settlements of clayey strata. The test is performed as per the procedure laid down in IS: 2720 Part 15: 1965 RA: 2021.
- (f) Determination of specific gravity test as per the procedure laid down in IS: 2720 Part 3 Sec 1 &2 : 1980 RA: 2021.
- (g) Determination of free swell index of soils as per IS: 2720 Part 40: 1977 RA 2021.

5.0 FIELD TESTS

The purpose of various field tests required during sub-surface explorations for the foundation of a structure has already been discussed; a brief discussion of these tests is given below:

5.1 BORING AND SAMPLING

Boring was done for Eleven boreholes for the **Geotechnical Investigation for the construction of BQ Housing Complex at Duliajan, Assam**. These boreholes have been designated as BH 1 - BH 11.

Representative undisturbed soil samples were collected in a Shelby tube (a thin tube sampler) of 100 mm dia and 450 mm long by inserting it in the soil with hydraulic pressure to minimize disturbance. These samples were collected in accordance with the procedure laid down in IS: 1892-2021 “Code of Practice for Subsurface Investigation for Foundations”. Also, disturbed soil samples were collected from these boreholes. Disturbed soil samples were collected from SPT sampler and may be used for classification of soil where undisturbed soil could not be obtained. Depending upon the type of soil strata that exists in the borehole, a number of undisturbed & disturbed soil samples were collected, numbered and packed. The undisturbed soil samples were sealed by wax on both sides to avoid any moisture loss. All the soil samples collected were transported to the laboratory for further testing.

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5.2 STANDARD PENETRATION TEST

The boreholes were supplemented by standard penetration tests (S.P.T.) at 1.5 m vertical depth interval from top of existing ground to drilled depth in both boreholes. These tests were conducted in accordance with the procedure laid down in IS:2131-1981 “Method of Standard Penetration Tests for soils” and N-Values (Soil Penetration resistance) were obtained by driving a split spoon sampler 30cm into the ground after giving 15 cm of initial seating drive at respective intervals or at change of strata. N value is the number of blows required to penetrate the SPT sampler 30 cm in the sub strata when it is subjected to an impact load given by a 63.5 kg hammer when it falls freely from 75 cm height.

6.0 LABORATORY TESTS

To know about the type of soil and to confirm & supplement the information obtained from field tests about the nature of soils strata, laboratory tests as given below were conducted on the soil samples collected from the boreholes at the time of drilling.

6.1 BULK DENSITY TEST

Bulk density determinations were carried out on undisturbed soil samples collected from the boreholes. The results of dry density are calculated from bulk density.

The results of average bulk density, insitu moisture content and dry density tests of the undisturbed soil sample collected from these boreholes is tabulated in Log of Boreholes of each borehole.

6.2 MOISTURE CONTENT

In situ Moisture Content of undisturbed soil samples was determined by oven drying method. It is used to find out In situ Dry Density of soil and also helpful in knowing present condition of wetness of strata.

6.3 PARTICLE SIZE ANALYSIS

This test was done as per Indian Standard IS: 2720 (Part-IV) – 1985. Samples collected from the boreholes were subjected to mechanical analysis for particle size distribution and classification of the soil at proposed site and depth. The analysis was done by wet sieving for particles bigger than 75 micron and by hydrometer analysis for particles finer than 75 micron.

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The results of Particle Size analysis of the soil samples collected from the boreholes are tabulated in Log of Boreholes of each borehole. Grain size distribution curves of all the soil samples were plotted and are given in figures next to tables. The description of various soil strata encountered in these boreholes is shown in Log of boreholes of each borehole. The I.S. Classification for various soils is also shown in this table.

6.4 ATTERBERG LIMITS

For fine-grained cohesive soils, the soil consistency was determined to check their firmness. This was done by determining the liquid limit, plastic limit and plasticity index of such soil samples according to the procedure laid down in IS :2720 Part-V – 1985, “Determination of Liquid and Plastic Limits”.

The results of liquid Limit (L.L.), Plastic Limit (P.L.) tests and Plasticity Index (P.I.) for various fine-grained soil samples collected from these boreholes is tabulated in log of boreholes.

6.5 SHEAR STRENGTH PARAMETERS

The Shear strength Parameters are determined by Direct Shear tests and Triaxial tests. Triaxial tests are conducted on cohesive (Clayey soil) while Direct Shear tests are conducted on cohesionless soils (Sandy/silty) by packing them at insitu density in a remoulded state and at saturated conditions. These shear parameters are important from the point of view to check the bearing capacity of the soil of proposed site against shear failure and for the design of foundation.

For this project direct shear tests and triaxial shear tests were performed on some selected soil samples in saturated conditions from each borehole. The samples having some clays were tested by triaxial test and others by direct shear test. The results of these shear tests on soil samples collected from various depths are given in Log of Boreholes of each borehole. The results of specific gravity tests are also given in this table.

6.6 SPECIFIC GRAVITY TEST

Specific Gravity of all the undisturbed soil samples was determined by specific gravity bottles.

7.0 TESTS RESULTS

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From the field and laboratory tests description of (BH-1 to BH-11) at the site of **Geotechnical Investigation for the construction of BQ Housing Complex at Duliajan, Assam** refer Summary of Findings clause 9.0

The exact soil strata, results of field tests while doing sub-surface explorations by means of borehole at the site of the proposed site are given in log of borehole. The results obtained from laboratory investigations/tests carried out on soil samples collected from the borehole at the proposed site are given in Log of Borehole

7.1 WATER TABLE

The water table is given below in table that was met at respectively depth in the boreholes at the site **Geotechnical Investigation for the construction of BQ Housing Complex at Duliajan**, during investigation period in the month of June-July 2025. Actual Water table may rise if investigation conducted during monsoon.

Bore Hole Number	Drilling Depth in Meter	Ground Water Table in Meter
1	30	1.20
2	30	1.00
3	30	1.50
4	10	1.00
5	30	0.65
6	30	0.80
7	30	0.70
8	20	1.00
9	30	1.10
10	30	1.00
11	10	1.00

7.2 STANDARD PENETRATION TEST

The result of these tests for both boreholes are calculated penetration resistance curves for observed penetration resistance (N-Values) as well as corrected penetration resistance (N'-Values). These results are given in Log of Boreholes and plots are given in Fig.2 of each borehole.

The observed values of 'N' have been corrected for overburden pressure by the following formula.

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$$N' = (0.77 \log_{10} 20/p) N \quad \dots\dots\dots (1)$$

where N = observed value for penetration resistance in no. of blows per 30cm penetration of sampler.

N' = Value of penetration resistance corrected for overburden pressure

p = Overburden pressure in Kg/cm^2

The observed N values have also been corrected for dilatancy wherever applicable.

8. COMPUTATION OF BEARING CAPACITY

A properly designed foundation must satisfy the following basic requirements:

- (i) Foundation must be safe against the shear failure of the supporting soil.
- (ii) The settlement of foundation must be within permissible limit.

The bearing capacities have been determined separately to satisfy both the above-mentioned requirements and the smaller of the two values has been recommended as the allowable bearing capacity.

8.1 BEARING CAPACITY FROM SHEAR CONSIDERATIONS

In shear, a foundation can fail in any of the three different ways viz.

- (i) Local shear failure
- (ii) General shear failure
- (iii) In between the above two

The local Shear Failure is assumed to occur for ϕ less than 28.5° & general shear failure for $\phi > 36^\circ$ & between these limiting value of ϕ , interpolated values for local & general shear failure can be determined. In this case bearing capacity for shear failure consideration has been determined.

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Adopting a continuous foundation of width 2.0m and taking the depths from 1.50 m to 4.50m below O.G.L. and using Eq.2 below as per code IS 6403 – 1981: “Code of practice for determination of bearing capacity of shallow foundations” and also using codes IS 8009 (Part I) - 1978: Code of practice for calculation of settlements of foundations and IS: 1904-1978 “Structural Safety for Buildings: Shallow Foundation” SBC has been determined.

$$q_a = \frac{1}{F} \left[\frac{2}{3} C N_c s_c d_c i_c + \gamma D_f (N_q - 1) s_q d_q i_q + 0.5 \gamma B N_\gamma s_\gamma d_\gamma i_\gamma W' \right] + \gamma D_f \quad \dots\dots\dots (2)$$

where q_a = Allowable Safe Bearing Capacity

F = Factor of safety taken as 2.5

C_m = Modified cohesive strength

γ = Unit Weight of Soil

D_f = Depth of foundation

B = Width of foundation

N_c, N_q, N_γ = Modified bearing capacity factors

s_c, s_q, s_γ = Shape factors

d_c, d_q, d_γ = Depth factors

i_c, i_q, i_γ = Inclination factor

8.2 BEARING CAPACITY FROM SETTLEMENT CRITERIA

The net allowable Bearing Capacity based on the maximum settlement of foundation to 50 mm has been computed as per IS: 8009.

The foundation should not settle or deflect to an extent causing damage to structure or impair its usefulness. The bearing capacity calculation for foundation shall be governed by IS 1904-1986, IS 6403-1981, and I.S. 8009-1981 (part-I & II) - 1976 on the basis of available information regarding the proposed design have been done.

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9.0 SUMMARY OF FINDINGS

The safe bearing capacities have been calculated from above two criteria i.e. from settlement criteria and shear failure criteria. The lower of these two values have been recommended for adoption as SBC.

- a.** Stratum of borehole, BH-1, consists of clay of intermediate compressibility (of CI Group) from GL to 4.5 m depth, consists of Silty sand (of SM Group) from depth 4.50 m to 30.40 m. The water Table was met at the depth of 1.20 m.
- b.** Stratum of borehole, BH-2, consists of clay of low compressibility (of CL Group) from GL to 1.5 m depth, consists of Clayey soil with intermediate compressibility (of CI Group) from depth 1.50 m to 4.50 m, consists of silty sand (of SM Group) from depth 4.50 m to 8.25 m, consists of Poorly graded sand with silty fines (of SP-SM Group) from depth 8.25 m to 30.45m. The water Table was met at the depth of 1.00 m from ground level.
- c.** Stratum of borehole, BH-3, consists of clay of low compressibility (of CL Group) from GL to 4.5 m depth, consists of Poorly graded sand with silty fines (of SP-SM Group) from depth 4.50 m to 8.25m, consists of poorly graded sand (of SP Group) from depth 8.25 m to 30.45 m. The water table was met at the depth of 1.50 m from ground level.
- d.** Stratum of borehole, BH-4, consists of clay of low compressibility (of CL Group) from GL to 1.5 m depth, consists of Clayey soil with intermediate compressibility (of CI Group) from depth 1.50 m to 4.50 m, consists of silty sand (of SM Group) from depth 4.50 m to 10.95 m . The water table was met at the depth of 1.00 m from ground level.
- e.** Stratum of borehole, BH-5, consists of clay of intermediate compressibility (of CI Group) from GL to 3.0 m depth, consists of Silty sand (of SM Group) from depth 3.00 m to 30.45 m. The water Table was met at the depth of 0.65 m.
- f.** Stratum of borehole, BH-6, consists of clay of low compressibility (of CL Group) from GL to 1.5 m depth, consists of Clayey soil with intermediate compressibility (of CI Group) from depth 1.50 m to 4.50 m, consists of silty sand (of SM Group) from depth 4.50 m to 30.45 m . The water table was met at the depth of 0.80 m from ground level.
- g.** Stratum of borehole, BH-7, consists of Clay of Low compressibility (of CL Group) from GL to 5.25 m depth, consists of Silty sand (of SM Group) from depth 5.25 m to 30.45 m. The water Table was met at the depth of 0.70 m.

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h. Stratum of borehole, BH-8, consists silty sand (of SM Group) from GL to 1.5 m depth, consists of Clayey soil with intermediate compressibility (of CI Group) from depth 1.50 m to 5.25 m, consists of Clay of Low compressibility (of CL Group) from depth 5.25 m to 10.50 m, consists of silty sand (of SM Group) from depth 10.50 m to 20.45 m. The water table was met at the depth of 1.00 m from ground level.

i. Stratum of borehole, BH-9, consists of clay of intermediate compressibility (of CI Group) from GL to 4.5 m depth, consists of Silty sand (of SM Group) from depth 4.50 m to 30.45 m. The water Table was met at 1.10 from Ground level.

j. Stratum of borehole, BH-10, consists of clay of intermediate compressibility (of CI Group) from GL to 4.5 m depth, consists of Silty sand (of SM Group) from depth 4.50 m to 30.45 m. The water Table was met at the depth of 1.00 m from ground level.

k. Stratum of borehole, BH-11, consists of clay of low compressibility (of CL Group) from GL to 1.5 m depth, consists of Clayey soil with intermediate compressibility (of CI Group) from depth 1.50 m to 4.50 m, consists of silty sand (of SM Group) from depth 4.50 m to 10.95 m . The water table was met at the depth of 1.00 m from ground level.

10.0 RECOMMENDATIONS

Soil Bearing Capacity for the shallow as well as deep foundation are enlisted below in tabulated form: -

FOR SHALLOW FOUNDATION

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Type of Structure	BH	Foundation Type and Size	Soil Bearing Capacity T/m ²		
			For Embedment Depth of Footing (D _f)		
			D _f =1.5m	D _f =3m	D _f =4.5m
Residential Towers	1	10m x 10m Raft	11.07	21.31	23.44
	2		11.44	11.77	27.57
	3		11.99	12.34	36.44
	5		10.88	24.35	24.35
	6		11.62	11.96	30.49
	7		11.25	11.58	11.91
	9		12.17	12.53	25
	10		11.99	12.34	25.34
	1	20m x 20m Raft	10.91	22.87	23.44
	2		11.27	11.44	27.57
	3		11.81	11.99	39.47
	5		10.72	25.68	25.51
	6		11.45	11.62	30.74
	7		11.09	11.25	11.42
	9		12	12.17	25
	10		11.81	11.99	25.34
	1	30m x 30m Raft	10.85	22.87	23.44
	2		11.21	11.32	27.57
	3		11.76	11.87	39.47
	5		10.67	25.68	25.51
	6		11.39	11.51	30.74
	7		11.03	11.14	11.25
	9		11.94	12.06	25
	10		11.76	11.87	25.34

Type of Structure	BH	Foundation Type and Size	Soil Bearing Capacity T/m ²		
			For Embedment Depth of Footing (D _f)		
			D _f =1.5m	D _f =3m	D _f =4.5m
Community Centre	4	2m x 2m Square Footing	3.75	7.5	14.5
		3m x 3m Square Footing	2.56	5.89	13.89
		4m x 4m Square Footing	2.06	4.88	13.13
Guest House	8	2m x 2m Square Footing	3.5	6.25	10
		3m x 3m Square Footing	2.556	3.778	6.000
		4m x 4m Square Footing	2.1875	3.25	4.0625
		5mx5m Raft	3.4	4.6	6.2
		10m x 10m Raft	1.9	2.5	3.4
		15m x 15m Raft	1.51	2	2.67
Service Infrastructure	11	2m x 2m Square Footing	3.75	8.5	24.5
		3m x 3m Square Footing	2.67	6.67	20.56
		4m x 4m Square Footing	2.25	5.63	17.5

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FOR PILE FOUNDATION:

Guest House

GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

Type of Structure	BH	Pile Diameter and Length	Pile Capacity in Tons		
			Net Vertical Capacity	Uplift Capacity	Lateral Capacity
Guest House	8	0.5m Dia & 10m Length	15.517	14.947	2.033
		0.6m Dia & 10m Length	18.042	18.512	3.046
		0.75m Dia & 10m Length	21.471	24.223	4.904
		0.5m Dia & 15m Length	22.080	22.095	2.342
		0.6m Dia & 15m Length	27.580	28.866	3.532
		0.75m Dia & 15m Length	35.340	39.676	5.728

Residential Towers

Length of Pile- 15m

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Type of Structure	BH	Pile Diameter and Length	Pile Capacity in Tons		
			Net Vertical Capacity	Uplift Capacity	Lateral Capacity
Residential Towers	1	0.5m Dia and 15m Length	4.335	9.674	1.709
	2		20.781	13.845	1.546
	3		24.221	14.156	1.944
	5		10.110	13.716	2.156
	6		18.987	19.930	2.096
	7		13.135	15.834	1.630
	9		8.871	12.849	1.501
	10		10.992	14.334	1.773
	1	0.6m Dia and 15m Length	5.629	13.501	2.541
	2		33.200	19.703	2.289
	3		39.664	20.302	2.907
	5		13.577	19.065	3.240
	6		25.638	27.507	3.145
	7		18.739	22.678	2.419
	9		13.060	18.702	2.221
	10		15.747	20.583	2.640
	1	0.75m Dia and 15m Length	6.163	19.253	4.058
	2		54.745	28.645	3.639
	3		67.127	29.663	4.670
	5		17.292	27.042	5.231
	6		34.155	38.847	5.071
	7		25.735	32.953	3.855
	9		18.130	27.629	3.526
	10		21.565	30.033	4.223

Length of Pile- 20m

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Type of Structure	BH	Pile Diameter and Length	Pile Capacity in Tons		
			Net Vertical Capacity	Uplift Capacity	Lateral Capacity
Residential Towers	1	0.5m Dia and 20m Length	10.513	16.552	2.171
	2		26.473	20.383	1.950
	3		30.275	20.948	2.229
	5		17.649	21.547	2.363
	6		26.198	27.531	2.337
	7		20.225	23.350	2.065
	9		14.569	19.391	1.630
	10		16.968	21.070	2.017
	1	0.6m Dia and 20m Length	14.959	23.709	3.263
	2		41.748	29.364	2.916
	3		48.883	30.433	3.353
	5		24.912	30.676	3.566
	6		36.459	38.759	3.525
	7		29.497	33.885	3.096
	9		21.772	28.478	2.419
	10		24.887	30.659	3.020
	1	0.75m Dia and 20m Length	20.093	34.749	5.271
	2		67.441	43.278	4.685
	3		80.982	45.106	5.424
	5		34.245	44.655	5.786
	6		50.225	55.841	5.716
	7		41.896	50.011	4.988
	9		31.311	42.601	3.855
	10		35.345	45.425	4.861

Length of Pile- 25 M

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Type of Structure	BH	Pile Diameter and Length	Pile Capacity in Tons		
			Net Vertical Capacity	Uplift Capacity	Lateral Capacity
Residential Towers	1	0.5m Dia and 25m Length	16.691	23.430	2.303
	2		32.164	26.921	2.144
	3		36.329	27.739	2.345
	5		25.567	29.643	2.446
	6		43.838	27.531	2.427
	7		44.848	30.866	2.226
	9		20.226	25.932	1.953
	10		37.170	27.807	2.191
	1	0.6m Dia and 25m Length	24.289	33.917	3.470
	2		50.295	39.024	3.221
	3		58.102	40.564	3.537
	5		36.815	42.685	3.697
	6		67.961	38.759	3.666
	7		71.843	45.093	3.349
	9		30.483	38.253	2.921
	10		59.781	40.734	3.295
	1	0.75m Dia and 25m Length	34.023	50.245	5.623
	2		80.138	57.911	5.200
	3		94.837	60.550	5.737
	5		52.056	62.868	6.011
	6		109.310	55.841	5.957
	7		117.840	67.069	5.417
	9		44.492	57.573	4.694
	10		98.057	60.816	5.325

- It is advised that any significant deviations in the soil profile or ground conditions observed during construction activities be promptly communicated so that necessary reassessments can be undertaken to ensure foundation stability.
- The groundwater table has been conservatively assumed at ground surface level to account for potential flooding conditions, which may occur during monsoon or other high-inflow scenarios. This assumption ensures a safe and robust foundation design under adverse conditions.

- Necessary measures—such as improved site drainage, use of durable construction materials, protective coatings, and concrete mix modifications—should be considered to enhance the long-term durability and structural integrity of the substructure.
- Due to practical limitations and difficulties in obtaining undisturbed soil samples at deeper depths, the SBC values were conservatively estimated by correlating with similar strata encountered at shallower depths, and their tested engineering properties were adopted for analysis.
- Additionally, consolidation settlement has been duly considered, for the boreholes indicates the presence of significant clay layers at depth. The zone of influence for settlement calculations has been conservatively taken as twice the width of the foundation, in accordance with established geotechnical practice, ensuring that deeper compressible strata contributing to settlement are adequately captured in the analysis.
- For shallow foundations in cohesionless strata, the average N-value over the influence depth is first determined. Individual N-values greater than this average are disregarded, and the average of the remaining lesser values is used for SBC calculations based on the settlement criterion.
- In certain locations for raft foundation, clay layers are encountered up to a depth of approximately 3.0 m; however, within the calculated zone of influence, the strata are predominantly cohesionless. Accordingly, settlement analysis has been carried out based on the procedures for cohesionless soils as per IS 8009 (Part 1).
- For the boreholes containing predominant clay layers in the influence zone column loads are initially assumed for consolidation settlement checks, the column loads were adjusted up to a corresponding to a permissible foundation settlement of 50 mm. This load, when divided by the plan area of the foundation, yields the net safe bearing capacity under the settlement criterion for the clay stratum.
- In many sites, shallow layers have also been found to be liquefiable up to depths of approximately 3.0 m to 4.5 m. Although SBC values are provided for foundation depths of 1.5 m and 3.0 m, without appropriate liquefaction mitigation measures, the adoption of such depths is not advisable. In such cases, it is recommended that the foundation depth be increased to 4.5 m or deeper to ensure safety and serviceability under adverse conditions.
- As per IS 6403, an SPT N-value greater than 50 indicates the presence of very dense cohesionless deposits. Where the pile founding stratum and the underlying strata are entirely composed of such very dense deposits, the end bearing capacity has been adopted for pile capacity calculations as discussed with the client. In cases where one or more underlying layers exhibit N-values less than

50, only skin friction resistance is considered in the pile capacity evaluation. To confirm the calculated pile capacities, it is recommended to carry out a pile load test at the site.

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Akshya Sharma Director

Authorized Signatory
STR Labs



Geotech. Expert
STR Labs

11.0 ABBREVIATION USED

UDS = Undisturbed Soil Sample

DS= Disturbed Sample

DIR = Direct Shear Test

Tx = Triaxial Shear Test

SPT= Standard Penetration Test

NP= Non-Plastic (Soil has no plasticity; plasticity index = 0)

NL= Non-liquefiable

C = Cohesion

ϕ = Angle of Shearing Resistance

UU= Unconsolidated Undrained (triaxial compression test type)

RA= Reaffirmed Year

12.0 CLOSURE

We appreciate the opportunity given to M/s Shiva Testing and Research Laboratory Pvt. Ltd. Baddi (Himachal Pradesh) to conduct the soil investigation (lab and analysis) and to submit this technical report. This presented report is based on observations made in field and tests conducted on samples collected from Eleven boreholes and as briefed by driller-in-charge. In case of any deviation in soil strata from reported one or abnormal/abrupt change in soil sub strata is noticed during actual excavation, kindly contact us before proceeding with further construction to take remedial measure, if any, required for safety of structure. This Geotechnical Report is specifically applicable to the locations where the site investigation was conducted. The findings and recommendations presented herein may not be representative or valid for areas

beyond the explored locations. Any changes to the area of exploration or site conditions may necessitate additional investigation and reevaluation.



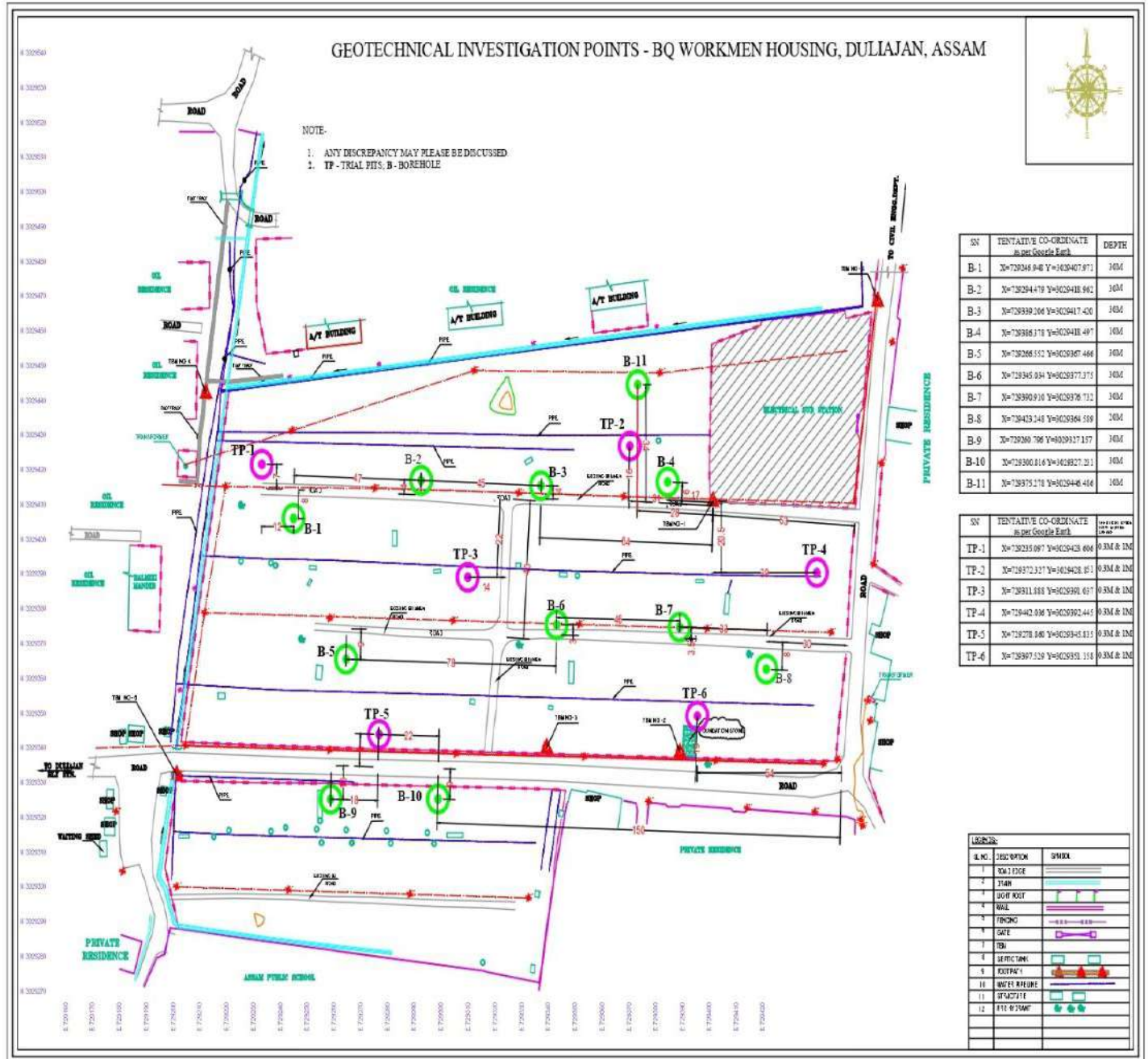
For, Shiva Testing & Research Laboratory Pvt. Ltd.
Akshya Sharma Director

For M/s STR Labs,
Baddi (Himachal Pradesh)

13.0 REFERENCES

1. IS 6403 – 1981: Code of practice for determination of bearing capacity of shallow foundations.
 2. IS 8009 (Part I) - 1978: Code of practice for calculation of settlements of foundations.
 3. IS: 1904-1978 “Structural Safety for Buildings: Shallow Foundation”.
 4. Foundation design manual, N. V. Nayak, 4th edition, 1996.
 5. Foundation analysis and design, Joseph E. bowels.
 6. IS-1498, IS-2131, IS-1892 and various parts of IS-2720
-

14.0 SITE LAYOUT PLAN



For Shiva Testing & Research Laboratory Pvt. Ltd.

ASW
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COORDONATES & REDUCED LEVELS OF BOREHOLES AND TRIAL PITS

Sr. no.	Co-ordinates	Natural Ground Level (NGL) (m)
BH-1	X=729246.948; Y =3029407.971	120.68
BH-2	X=729294.479; Y=3029418.962	120.81
BH-3	X=729339.206; Y=3029417.420	120.86
BH-4	X=729386.378; Y=3029418.497	120.68
BH-5	X=729266.552; Y=3029367.466	120.59
BH-6	X=729345.034; Y=3029377.375	120.78
BH-7	X=729390.910; Y=3029376.732	120.76
BH-8	X=729423.248; Y=3029364.589	120.62
BH-9	X=729260.796; Y=3029327.157	120.71
BH-10	X=729300.816; Y=3029327.231	120.66
BH-11	X=729375.278; Y=3029446.486	120.77
	TRIAL-PITS	
TP-1	X=729235.097 Y=3029423.606	120.55
TP-2	X=729372.327 Y=3029428.851	120.74
TP-3	X=729311.888 Y=3029391.037	120.46
TP-4	X=729442.036 Y=3029392.445	120.68
TP-5	X=729278.860 Y=3029345.835	120.53
TP-6	X=729397.529 Y=3029351.158	120.54

For Shiva Testing & Research Laboratory Pvt. Ltd.



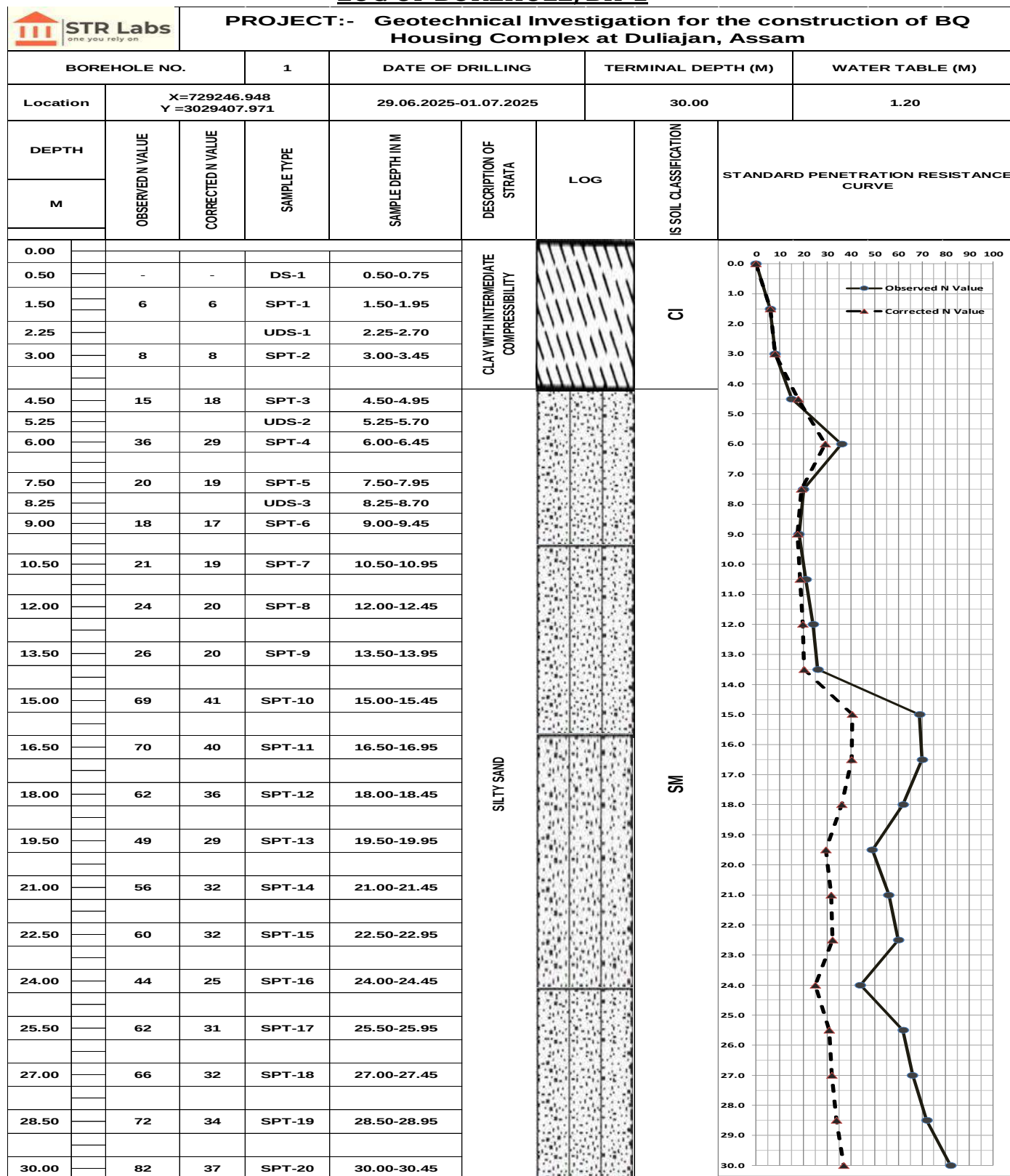
Authorized Signatory

15.0 LOG OF BOREHOLES

For Shiva Testing & Research Laboratory Pvt. Ltd.

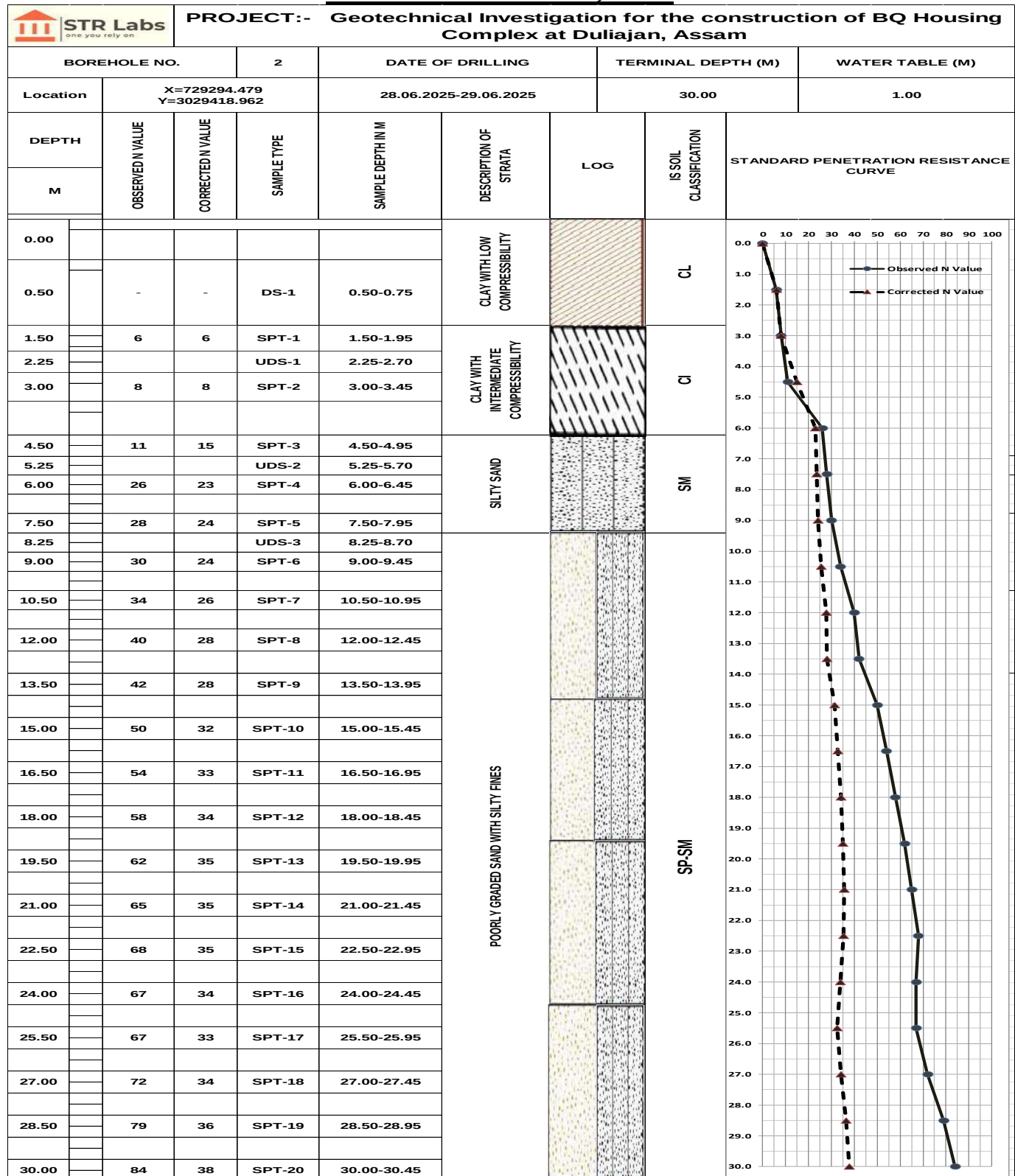
Authorized Signatory

LOG OF BOREHOLE, BH-1



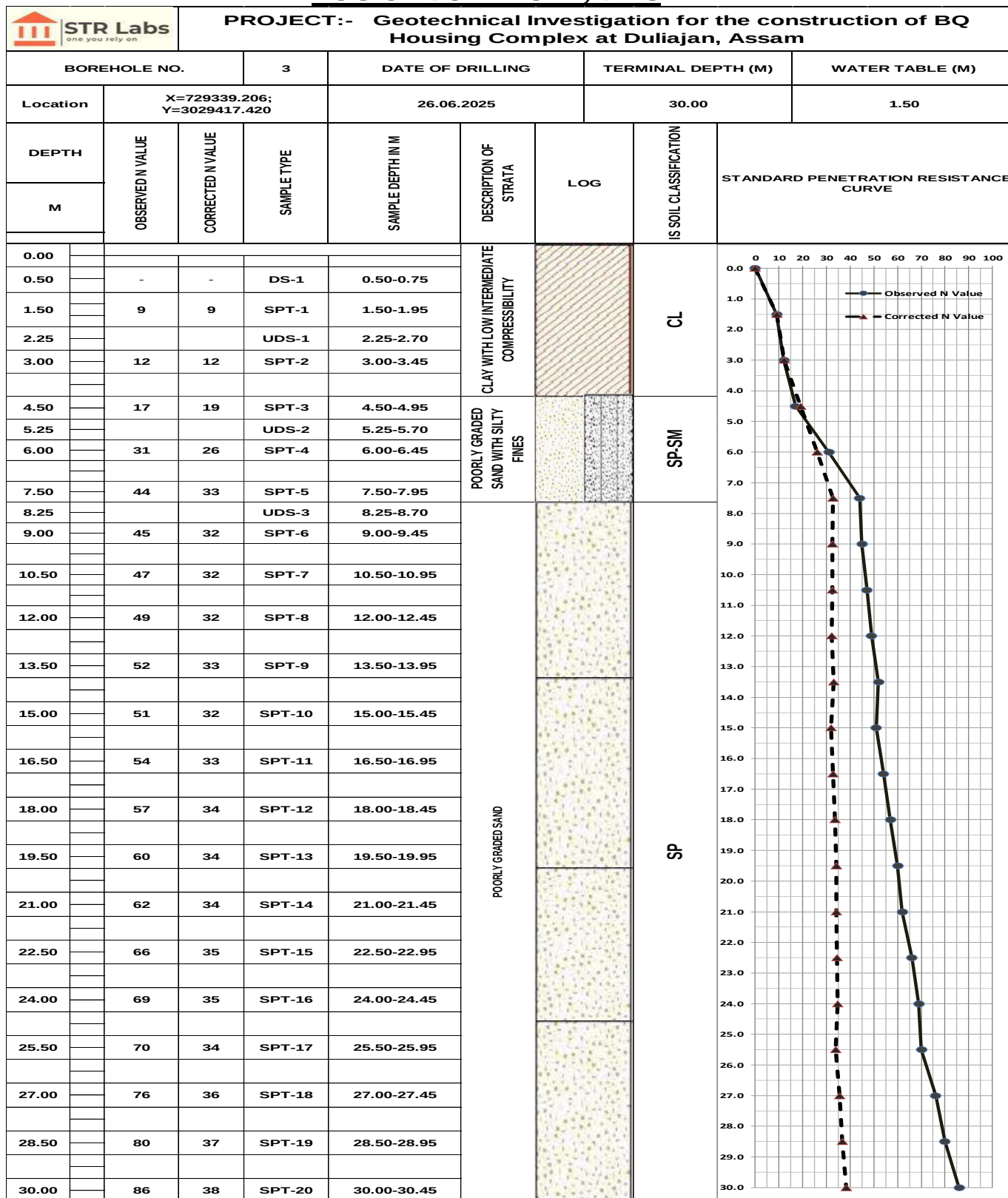
GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-2



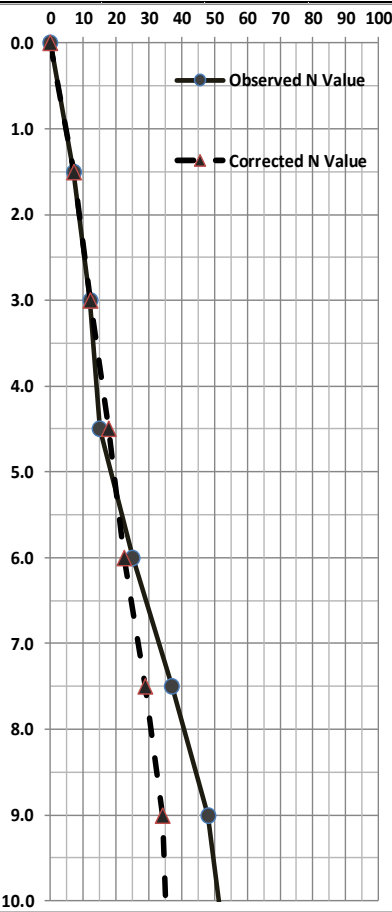
GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING
COMPLEX AT DULIAJAN, ASSAM

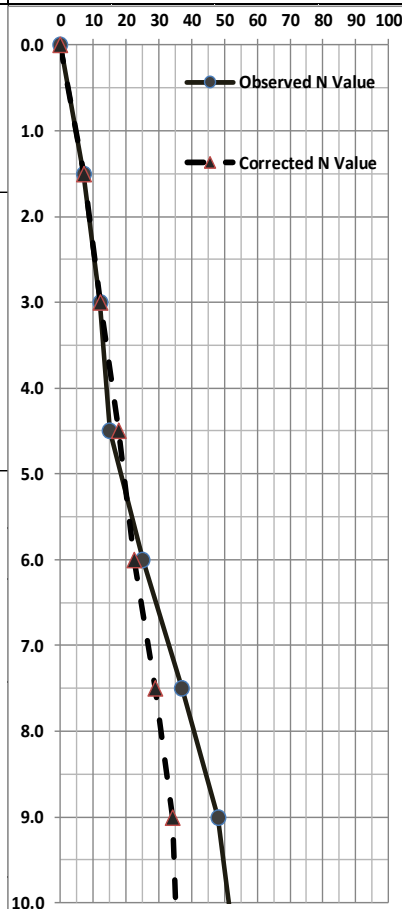
LOG OF BOREHOLE, BH-3



GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-4

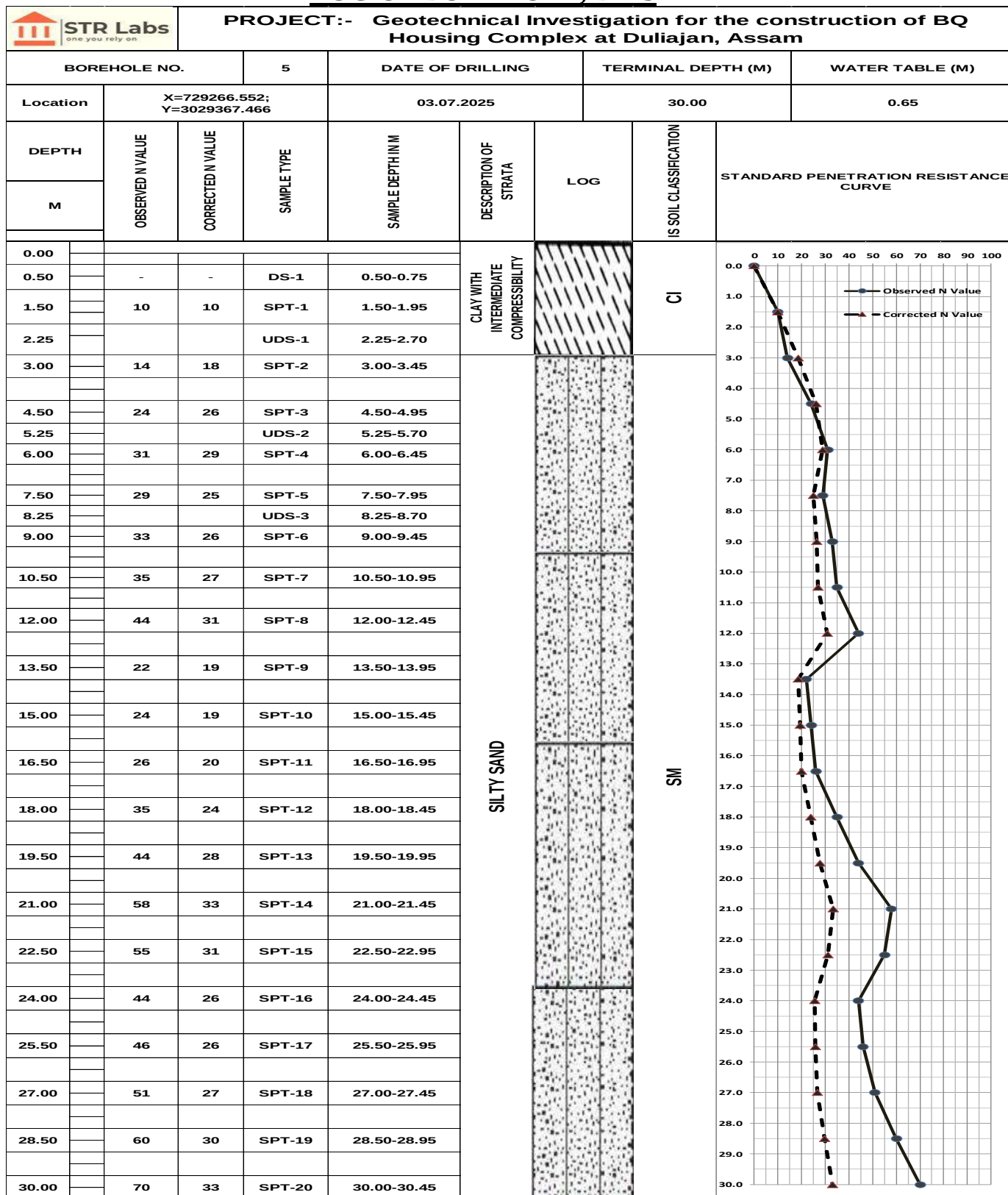
BOREHOLE NO.		4		DATE OF DRILLING		TERMINAL DEPTH (M)		WATER TABLE (M)	
Location		X=729386.378; Y=3029418.497		24.06.2025		10.00		1.00	
DEPTH		OBSERVED N VALUE	CORRECTED N VALUE	SAMPLE TYPE	SAMPLE DEPTH IN M	DESCRIPTION OF STRATA	LOG	IS SOIL CLASSIFICATION	STANDARD PENETRATION RESISTANCE CURVE
M									
0.00						CLAY WITH LOW COMPRESSIBILITY		CL	
0.50		-	-	DS-1	0.30-0.50				
1.50		7	7	SPT-1	1.50-1.95	CLAY WITH INTERMEDIATE COMPRESSIBILITY		CI	
2.25				UDS-1	2.25-2.70				
3.00		12	12	SPT-2	3.00-3.45				
						SILTY SAND		SM	
4.50		15	18	SPT-3	4.50-4.95				
5.25				UDS-2	5.25-5.70				
6.00		25	23	SPT-4	6.00-6.45				
7.50		37	29	SPT-5	7.50-7.95				
8.25				UDS-3	8.25-8.70				
9.00		48	34	SPT-6	9.00-9.45				
10.50		53	36	SPT-7	10.50-10.95				



For Shiva Testing & Research Laboratory Pvt. Ltd.

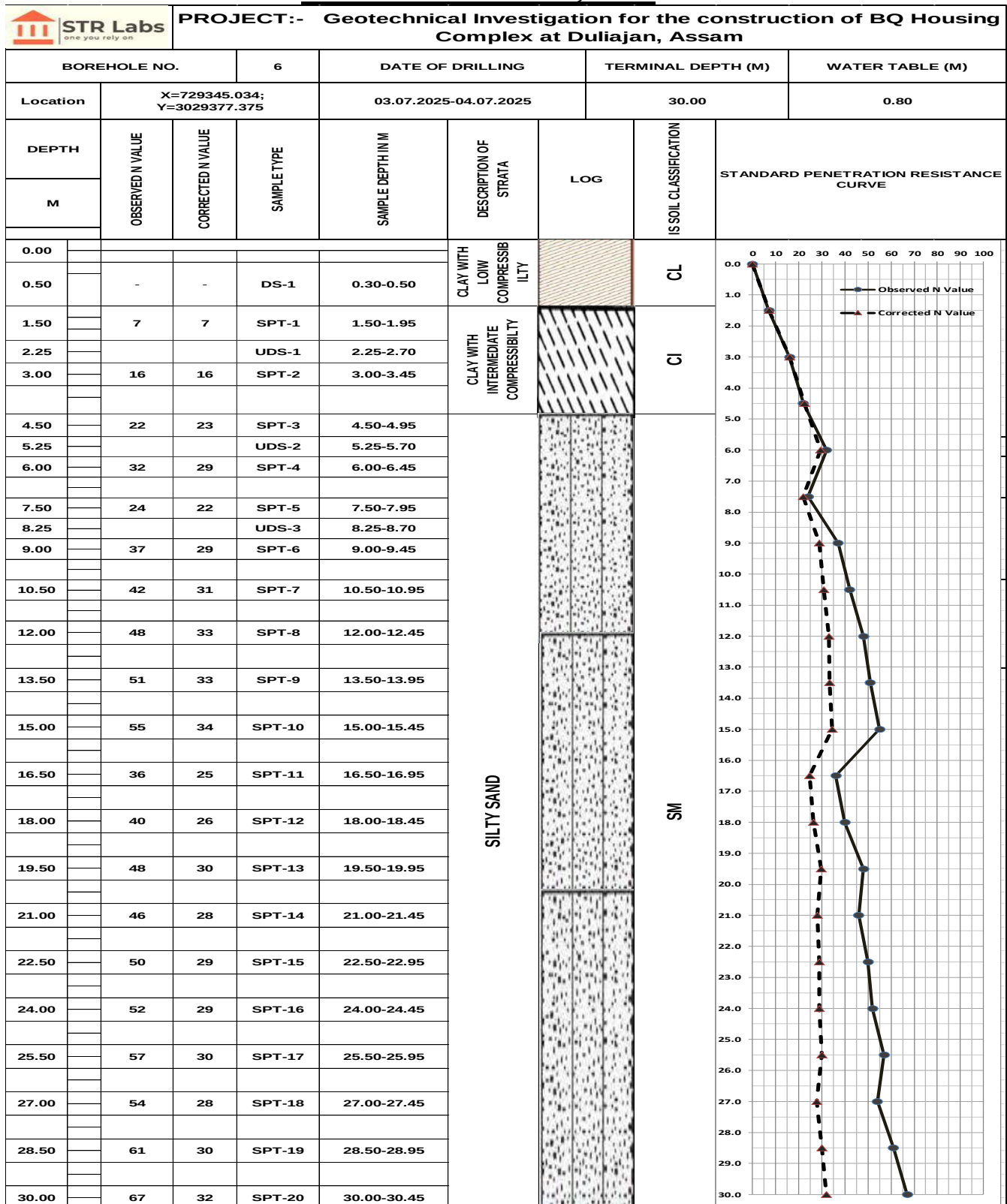
Authorized Signatory

LOG OF BOREHOLE, BH-5



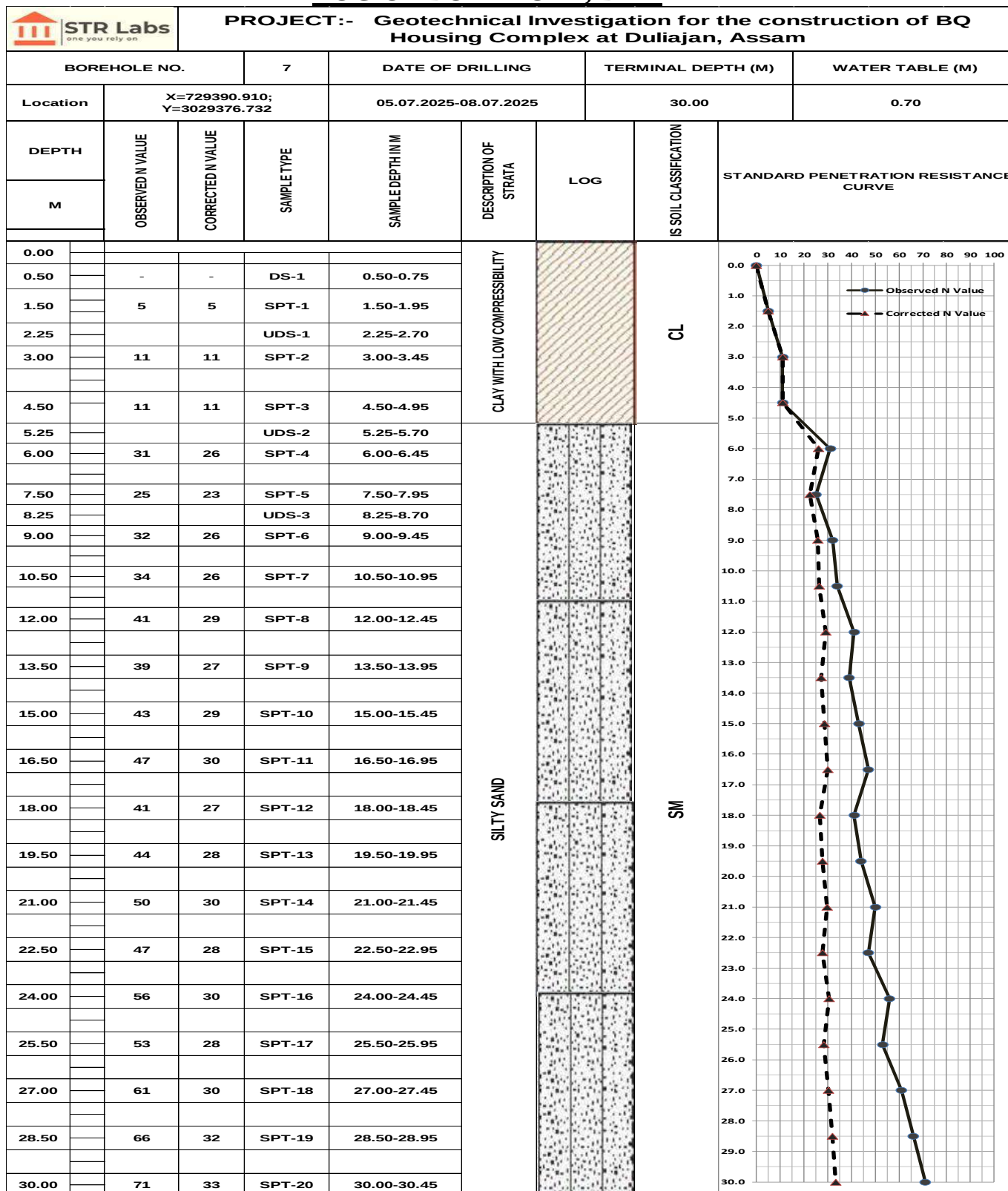
GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-6



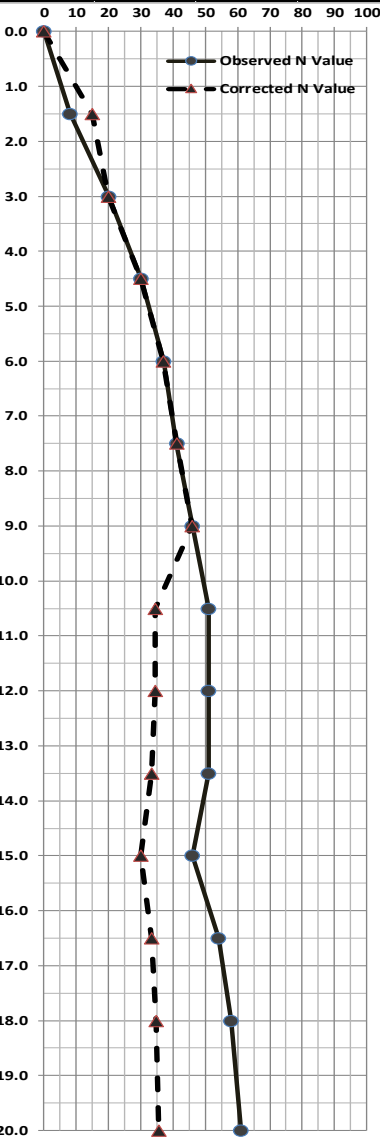
GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-7

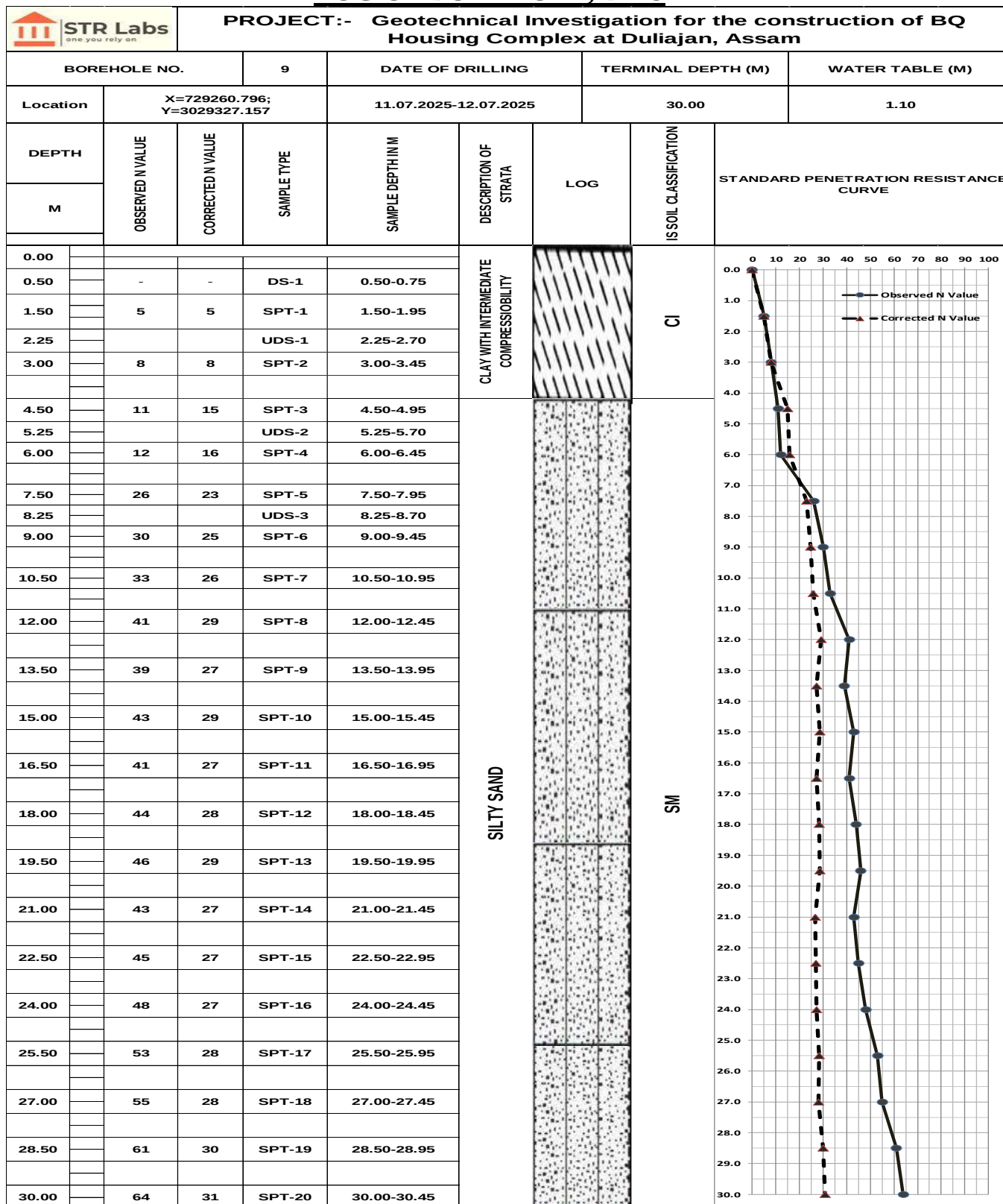


GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-8

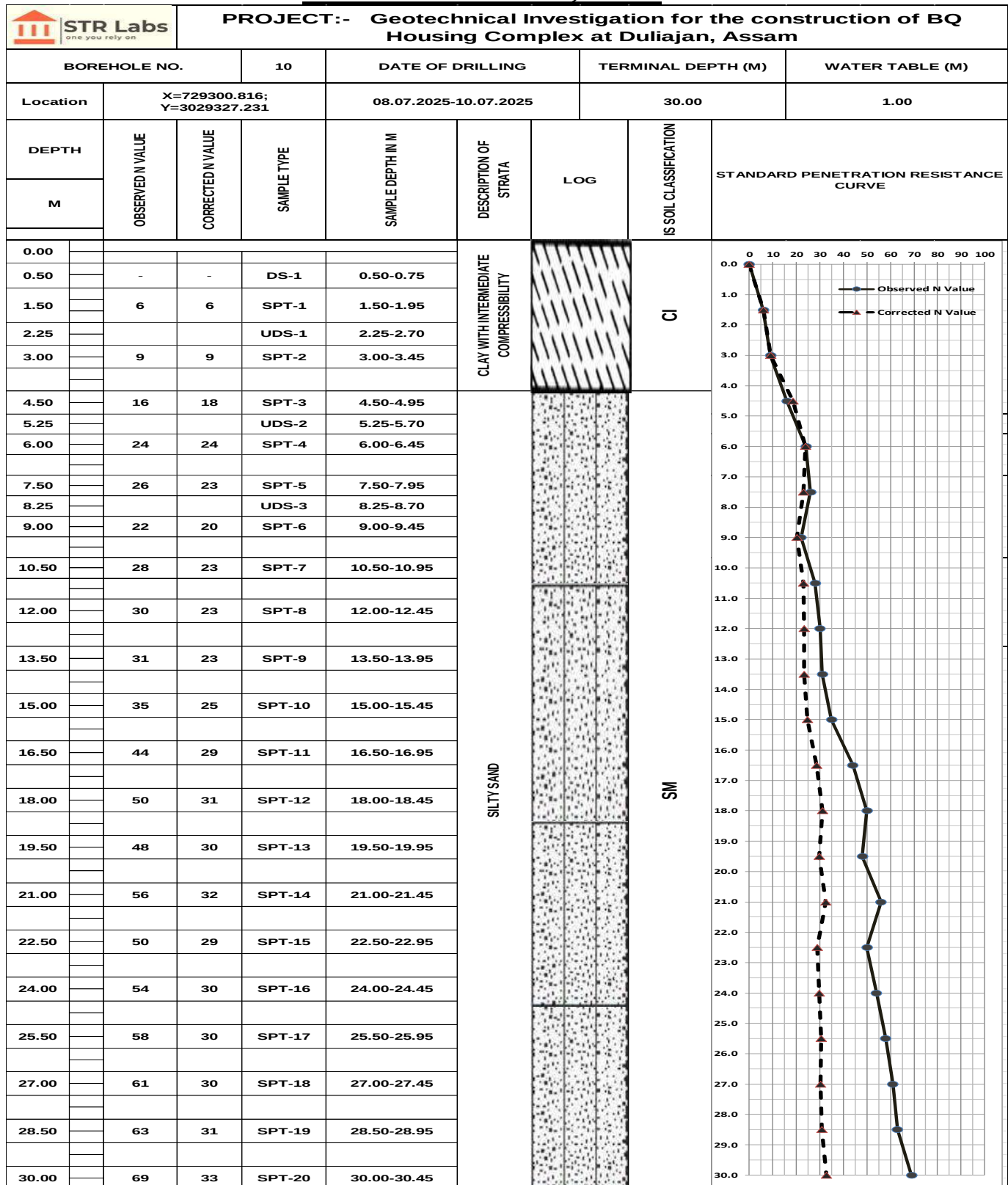
BOREHOLE NO.		8		DATE OF DRILLING		TERMINAL DEPTH (M)		WATER TABLE (M)	
Location		X=729423.248; Y=3029364.589		22.06.2025		20.00		1.00	
DEPTH		OBSERVED N VALUE	CORRECTED N VALUE	SAMPLE TYPE	SAMPLE DEPTH IN M	DESCRIPTION OF STRATA	LOG	IS SOIL CLASSIFICATION	STANDARD PENETRATION RESISTANCE CURVE
M									
0.00						SILTY SAND		SM	
0.50		-	-	DS-1	0.30-0.50				
1.50		8	15	SPT-1	1.50-1.95	CLAY WITH INTERMEDIATE COMPRESSIBILITY		CI	
2.25				UDS-1	2.25-2.70				
3.00		20	20	SPT-2	3.00-3.45				
4.50		30	30	SPT-3	4.50-4.95	CLAY WITH LOW COMPRESSIBILITY		CL	
5.25				UDS-2	5.25-5.70				
6.00		37	37	SPT-4	6.00-6.45				
7.50		41	41	SPT-5	7.50-7.95	SILTY SAND		SM	
8.25				UDS-3	8.25-8.70				
9.00		46	46	SPT-6	9.00-9.45				
10.50		51	35	SPT-7	10.50-10.95				
12.00		51	35	SPT-8	12.00-12.45				
13.50		51	33	SPT-9	13.50-13.95				
15.00		46	30	SPT-10	15.00-15.45				
16.50		54	33	SPT-11	16.50-16.95				
18.00		58	35	SPT-12	18.00-18.45				
20.00		61	36	SPT-13	20.00-20.45				

LOG OF BOREHOLE, BH-9



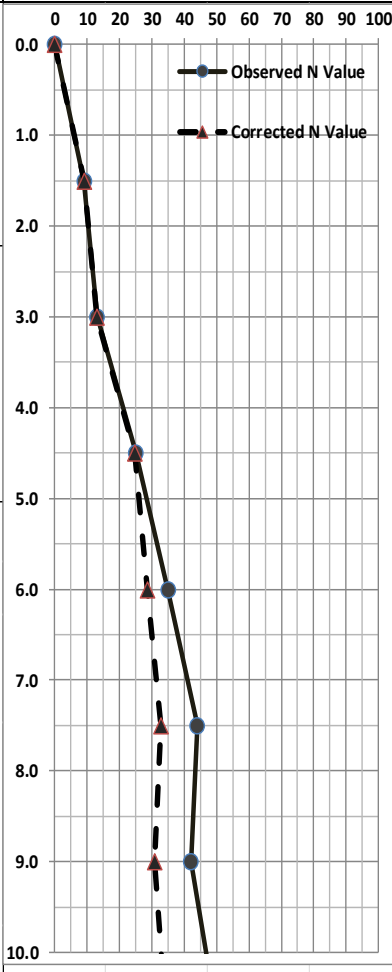
GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-10



GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM

LOG OF BOREHOLE, BH-11

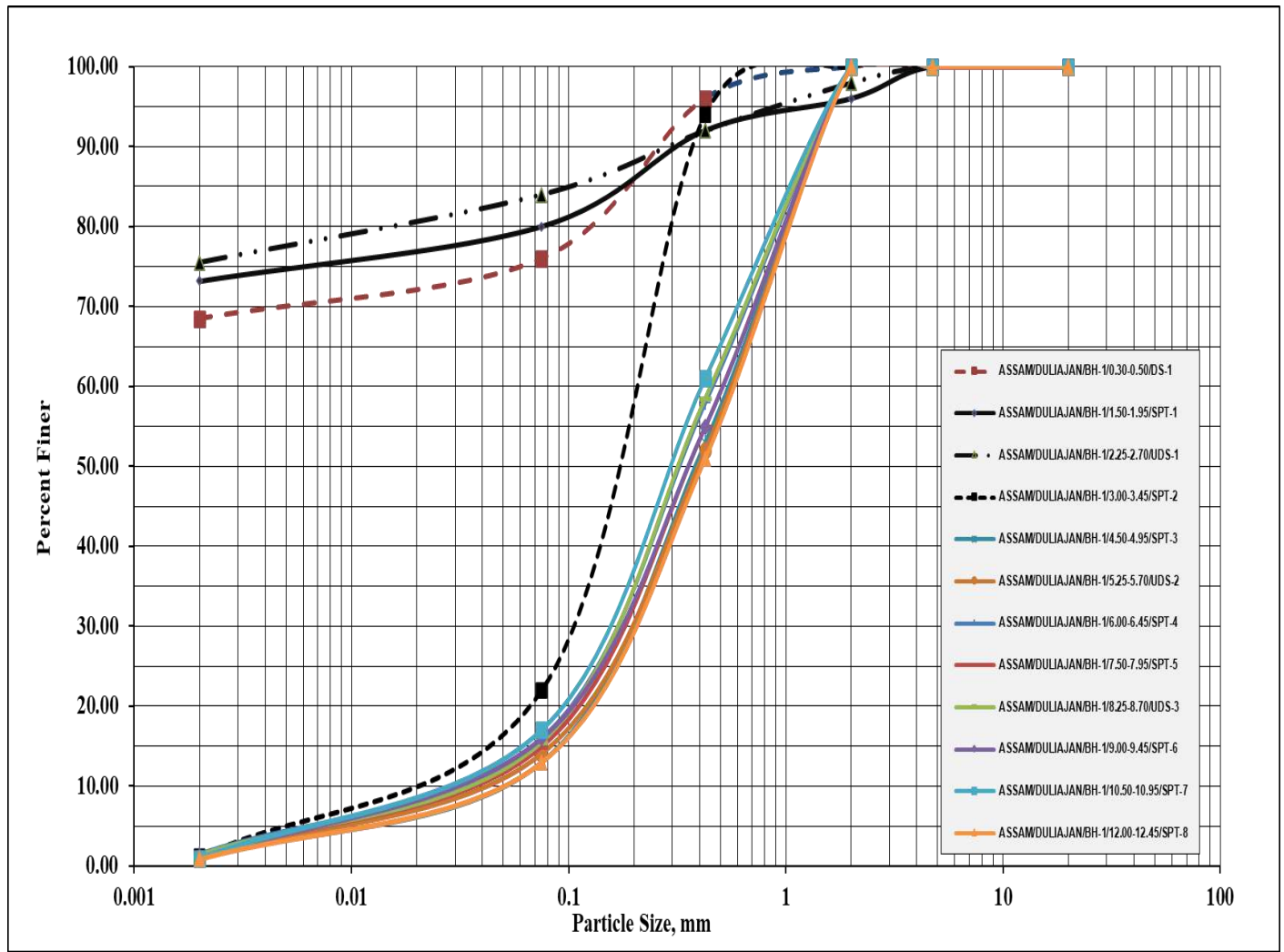
 STR Labs one you rely on		PROJECT:- Geotechnical Investigation for the construction of BQ Housing Complex at Duliajan, Assam										
BOREHOLE NO.			11		DATE OF DRILLING			TERMINAL DEPTH (M)		WATER TABLE (M)		
Location		X=729375.278; Y=3029446.486			25.06.2025			10.00		1.00		
DEPTH		OBSERVED N VALUE	CORRECTED N VALUE	SAMPLE TYPE	SAMPLE DEPTH IN M	DESCRIPTION OF STRATA	LOG	IS SOIL CLASSIFICATION	STANDARD PENETRATION RESISTANCE CURVE			
M												
0.00						CLAY WITH LOW COMPRESSIBILITY		CL	 Observed N Value (Blue circles) Corrected N Value (Black triangles)			
0.50		-	-	DS-1	0.30-0.50							
1.50		9	9	SPT-1	1.50-1.95	CLAY WITH INTERMEDIATE COMPRESSIBILITY		CI				
2.25				UDS-1	2.25-2.70							
3.00		13	13	SPT-2	3.00-3.45							
4.50		25	25	SPT-3	4.50-4.95	SILTY SAND		SM				
5.25				UDS-2	5.25-5.70							
6.00		35	29	SPT-4	6.00-6.45							
7.50		44	33	SPT-5	7.50-7.95							
8.25				UDS-3	8.25-8.70							
9.00		42	31	SPT-6	9.00-9.45							
10.50		49	34	SPT-7	10.50-10.95							

16.0 LABORATORY TEST RESULTS

For Shiva Testing & Research Laboratory Pvt. Ltd.


Authorized Signatory

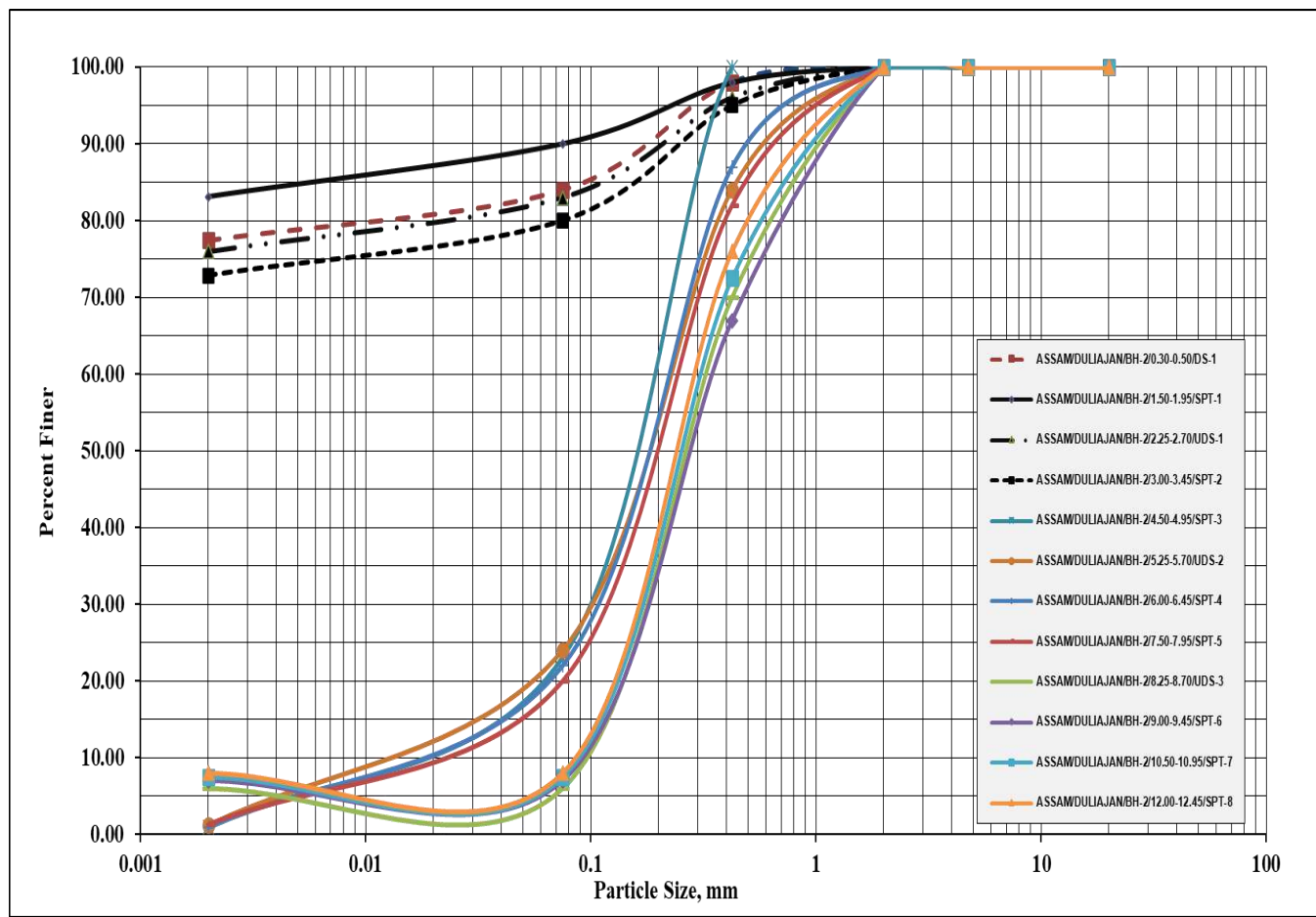
TEST ON SOIL SAMPLES																			
BH-1			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS		
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	FINE SAND	MED SAND	COARSE SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm											
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ
0.30-0.50	DS-1	CI	68.50	7.50	20.00	4.00	0.00	0.00	42.21	24.36	17.85	-	25	-	2.59	-	-	-	13
1.50-1.95	SPT-1	CI	73.15	6.85	12.00	4.00	4.00	0.00	45.25	25.45	19.80	-	29	-	2.60	-	-	-	-
2.25-2.70	UDS-1	CI	75.50	8.50	8.00	6.00	2.00	0.00	47.50	25.17	22.33	1.60	28	1.25	2.60	1.08	TX-UU	0.60	21.31
3.00-3.45	SPT-2	SM	1.15	20.85	72.00	6.00	0.00	0.00	NL	NP	NIL	-	20	-	2.61	-	-	-	-
4.50-4.95	SPT-3	SM	1.00	12.00	40.00	47.00	0.00	0.00	NL	NP	NIL	-	22	-	2.63	-	-	-	-
5.25-5.70	UDS-2	SM	1.05	12.95	38.00	48.00	0.00	0.00	NL	NP	NIL	1.71	21	1.41	2.64	0.87	DIR-DRAINED	0.05	31.29
6.00-6.45	SPT-4	SM	1.55	14.45	42.00	42.00	0.00	0.00	NL	NP	NIL	-	22	-	2.60	-	-	-	-
7.50-7.95	SPT-5	SM	1.40	13.60	40.00	45.00	0.00	0.00	NL	NP	NIL	-	18	-	2.65	-	-	-	-
8.25-8.70	UDS-3	SM	1.30	14.20	43.00	41.50	0.00	0.00	NL	NP	NIL	1.76	16	1.52	2.62	0.73	DIR-DRAINED	0.01	34.05
9.00-9.45	SPT-6	SM	1.00	15.00	39.00	45.00	0.00	0.00	NL	NP	NIL	-	18	-	2.63	-	-	-	-
10.50-10.95	SPT-7	SM	0.95	16.05	44.00	39.00	0.00	0.00	NL	NP	NIL	-	19	-	2.64	-	-	-	-
12.00-12.45	SPT-8	SM	0.90	12.10	38.00	49.00	0.00	0.00	NL	NP	NIL	-	20	-	2.60	-	-	-	-
13.50-13.95	SPT-9	SM	0.80	14.20	32.00	53.00	0.00	0.00	NL	NP	NIL	-	17	-	2.63	-	-	-	-
15.00-15.45	SPT-10	SM	0.45	12.55	34.00	53.00	0.00	0.00	NL	NP	NIL	-	15	-	2.60	-	-	-	-
16.50-16.95	SPT-11	SM	0.55	13.45	30.00	56.00	0.00	0.00	NL	NP	NIL	-	19	-	2.65	-	-	-	-
18.00-18.45	SPT-12	SM	0.60	11.90	36.00	51.50	0.00	0.00	NL	NP	NIL	-	20	-	2.64	-	-	-	-
19.50-19.95	SPT-13	SM	0.60	13.40	40.00	46.00	0.00	0.00	NL	NP	NIL	-	17	-	2.63	-	-	-	-
21.00-21.45	SPT-14	SM	0.55	12.45	45.00	42.00	0.00	0.00	NL	NP	NIL	-	16	-	2.64	-	-	-	-
22.50-22.95	SPT-15	SM	0.75	12.25	46.00	41.00	0.00	0.00	NL	NP	NIL	-	15	-	2.63	-	-	-	-
24.00-24.45	SPT-16	SM	0.60	14.40	45.00	40.00	0.00	0.00	NL	NP	NIL	-	14	-	2.64	-	-	-	-
25.50-25.95	SPT-17	SM	0.90	13.10	50.00	36.00	0.00	0.00	NL	NP	NIL	-	20	-	2.63	-	-	-	-
27.00-27.45	SPT-18	SM	0.95	11.55	44.00	43.50	0.00	0.00	NL	NP	NIL	-	19	-	2.64	-	-	-	-
28.50-28.95	SPT-19	SM	0.80	12.20	41.00	46.00	0.00	0.00	NL	NP	NIL	-	17	-	2.60	-	-	-	-
30.00-30.45	SPT-20	SM	0.75	14.25	43.00	42.00	0.00	0.00	NL	NP	NIL	-	16	-	2.63	-	-	-	-



GRAIN-SIZE DISTRIBUTION CURVE BH-1

For Shiva Testing & Research Laboratory Pvt. Ltd.
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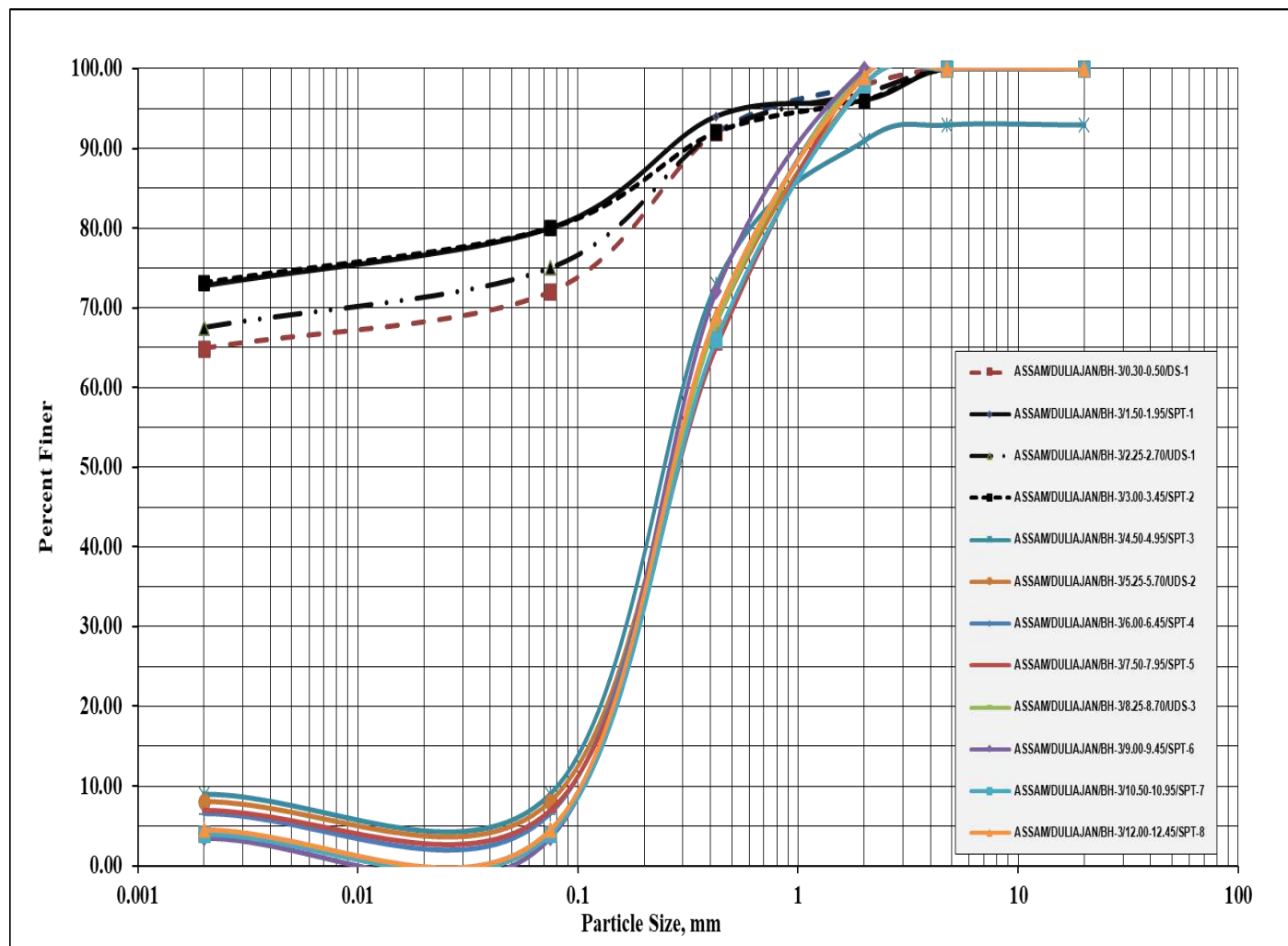
TEST ON SOIL SAMPLES																					
BH-2			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS				FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degrees		
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm													
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%	
0.30-0.50	DS-1	CL	77.45	6.55	14.00	2.00	0.00	0.00	34.50	21.57	12.93	-	24	-	2.58	-	-	-	-	12	
1.50-1.95	SPT-1	CI	83.10	6.90	8.00	2.00	0.00	0.00	46.00	26.04	19.96	-	28	-	2.57	-	-	-	-	-	
2.25-2.70	UDS-1	CI	75.95	7.05	13.00	4.00	0.00	0.00	47.52	26.22	21.30	1.62	25	1.30	2.56	0.98	TX-UU	0.62	21.01	14	
3.00-3.45	SPT-2	CI	72.85	7.15	15.00	5.00	0.00	0.00	49.12	26.17	22.95	-	27	-	2.59	-	-	-	-	-	
4.50-4.95	SPT-3	SM	1.00	22.00	77.00	0.00	0.00	0.00	NL	NP	NIL	-	17	-	2.60	-	-	-	-	-	
5.25-5.70	UDS-2	SM	1.20	22.80	60.00	16.00	0.00	0.00	NL	NP	NIL	1.70	18	1.44	2.61	0.81	DIR-DRAINED	0.01	31.05	0	
6.00-6.45	SPT-4	SM	0.95	21.05	65.00	13.00	0.00	0.00	NL	NP	NIL	-	20	-	2.62	-	-	-	-	-	
7.50-7.95	SPT-5	SM	1.35	18.65	62.00	18.00	0.00	0.00	NL	NP	NIL	-	8	-	2.60	-	-	-	-	-	
8.25-8.70	UDS-3	SP-SM	6.00	0.00	64.00	30.00	0.00	0.00	NL	NP	NIL	1.75	11	1.58	2.63	0.67	DIR-DRAINED	0.00	33.18	0	
9.00-9.45	SPT-6	SP-SM	7.00	0.00	60.00	33.00	0.00	0.00	NL	NP	NIL	-	8	-	2.60	-	-	-	-	-	
10.50-10.95	SPT-7	SP-SM	7.50	0.00	65.00	27.50	0.00	0.00	NL	NP	NIL	-	18	-	2.64	-	-	-	-	-	
12.00-12.45	SPT-8	SP-SM	8.00	0.00	68.00	24.00	0.00	0.00	NL	NP	NIL	-	17	-	2.62	-	-	-	-	-	
13.50-13.95	SPT-9	SP-SM	8.50	0.00	60.00	31.50	0.00	0.00	NL	NP	NIL	-	16	-	2.63	-	-	-	-	-	
15.00-15.45	SPT-10	SP-SM	5.00	0.00	67.00	28.00	0.00	0.00	NL	NP	NIL	-	15	-	2.62	-	-	-	-	-	
16.50-16.95	SPT-11	SP-SM	8.00	0.00	56.00	36.00	0.00	0.00	NL	NP	NIL	-	17	-	2.61	-	-	-	-	-	
18.00-18.45	SPT-12	SP-SM	5.50	0.00	52.00	42.50	0.00	0.00	NL	NP	NIL	-	18	-	2.60	-	-	-	-	-	
19.50-19.95	SPT-13	SP-SM	6.00	0.00	50.00	44.00	0.00	0.00	NL	NP	NIL	-	18	-	2.63	-	-	-	-	-	
21.00-21.45	SPT-14	SP-SM	6.50	0.00	48.00	45.50	0.00	0.00	NL	NP	NIL	-	20	-	2.62	-	-	-	-	-	
22.50-22.95	SPT-15	SP-SM	7.00	0.00	54.00	39.00	0.00	0.00	NL	NP	NIL	-	17	-	2.61	-	-	-	-	-	
24.00-24.45	SPT-16	SP-SM	7.50	0.00	52.00	40.50	0.00	0.00	NL	NP	NIL	-	16	-	2.60	-	-	-	-	-	
25.50-25.95	SPT-17	SP-SM	5.00	0.00	40.00	55.00	0.00	0.00	NL	NP	NIL	-	15	-	2.62	-	-	-	-	-	
27.00-27.45	SPT-18	SP-SM	6.00	0.00	46.00	48.00	0.00	0.00	NL	NP	NIL	-	12	-	2.63	-	-	-	-	-	
28.50-28.95	SPT-19	SP-SM	8.00	0.00	44.00	48.00	0.00	0.00	NL	NP	NIL	-	18	-	2.64	-	-	-	-	-	
30.00-30.45	SPT-20	SP-SM	6.50	0.00	47.00	46.50	0.00	0.00	NL	NP	NIL	-	15	-	2.61	-	-	-	-	-	



GRAIN-SIZE DISTRIBUTION CURVE BH-2

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TEST ON SOIL SAMPLES																			
BH-3			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS		
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm											
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ
0.30-0.50	DS-1	CL	64.90	7.10	20.00	6.00	2.00	0.00	34.47	20.17	14.30	-	13	-	2.58	-	-	-	11
1.50-1.95	SPT-1	CL	72.75	7.25	14.00	2.00	4.00	0.00	33.80	21.55	12.25	-	29	-	2.59	-	-	-	-
2.25-2.70	UDS-1	CL	67.50	7.50	17.00	5.00	3.00	0.00	34.58	20.80	13.78	1.59	27	1.25	2.57	1.05	TX-UU	0.65	20.95
3.00-3.45	SPT-2	CL	73.15	6.85	12.00	4.00	4.00	0.00	33.17	20.69	12.48	-	24	-	2.55	-	-	-	-
4.50-4.95	SPT-3	SP-SM	9.00	0.00	64.00	18.00	2.00	0.00	NL	NP	NIL	-	22	-	2.60	-	-	-	-
5.25-5.70	UDS-2	SP-SM	8.00	0.00	60.00	30.00	2.00	0.00	NL	NP	NIL	1.73	19	1.45	2.61	0.80	DIR-DRAINED	0.03	32.01
6.00-6.45	SPT-4	SP-SM	6.50	0.00	62.00	31.50	0.00	0.00	NL	NP	NIL	-	21	-	2.63	-	-	-	-
7.50-7.95	SPT-5	SP-SM	7.00	0.00	58.00	35.00	0.00	0.00	NL	NP	NIL	-	23	-	2.60	-	-	-	-
8.25-8.70	UDS-3	SP	4.00	0.00	64.00	32.00	0.00	0.00	NL	NP	NIL	1.79	21	1.48	2.63	0.78	DIR-DRAINED	0.00	34.66
9.00-9.45	SPT-6	SP	3.50	0.00	68.50	28.00	0.00	0.00	NL	NP	NIL	-	23	-	2.64	-	-	-	-
10.50-10.95	SPT-7	SP	4.00	0.00	62.00	32.00	2.00	0.00	NL	NP	NIL	-	20	-	2.62	-	-	-	-
12.00-12.45	SPT-8	SP	4.50	0.00	64.50	30.00	1.00	0.00	NL	NP	NIL	-	22	-	2.62	-	-	-	-
13.50-13.95	SPT-9	SP	3.00	0.00	66.00	31.00	0.00	0.00	NL	NP	NIL	-	22	-	2.63	-	-	-	-
15.00-15.45	SPT-10	SP	2.00	0.00	68.00	30.00	0.00	0.00	NL	NP	NIL	-	22	-	2.60	-	-	-	-
16.50-16.95	SPT-11	SP	2.50	0.00	62.50	35.00	0.00	0.00	NL	NP	NIL	-	12	-	2.61	-	-	-	-
18.00-18.45	SPT-12	SP	4.00	0.00	66.00	30.00	0.00	0.00	NL	NP	NIL	-	21	-	2.62	-	-	-	-
19.50-19.95	SPT-13	SP	3.00	0.00	60.00	37.00	0.00	0.00	NL	NP	NIL	-	21	-	2.61	-	-	-	-
21.00-21.45	SPT-14	SP	2.00	0.00	74.00	24.00	0.00	0.00	NL	NP	NIL	-	19	-	2.62	-	-	-	-
22.50-22.95	SPT-15	SP	3.00	0.00	77.00	20.00	0.00	0.00	NL	NP	NIL	-	15	-	2.64	-	-	-	-
24.00-24.45	SPT-16	SP	2.00	0.00	75.00	23.00	0.00	0.00	NL	NP	NIL	-	16	-	2.60	-	-	-	-
25.50-25.95	SPT-17	SP	1.00	0.00	74.00	25.00	0.00	0.00	NL	NP	NIL	-	16	-	2.61	-	-	-	-
27.00-27.45	SPT-18	SP	2.00	0.00	74.00	24.00	0.00	0.00	NL	NP	NIL	-	13	-	2.62	-	-	-	-
28.50-28.95	SPT-19	SP	4.00	0.00	52.00	44.00	0.00	0.00	NL	NP	NIL	-	14	-	2.63	-	-	-	-
30.00-30.45	SPT-20	SP	3.00	0.00	57.00	40.00	0.00	0.00	NL	NP	NIL	-	21	-	2.61	-	-	-	-



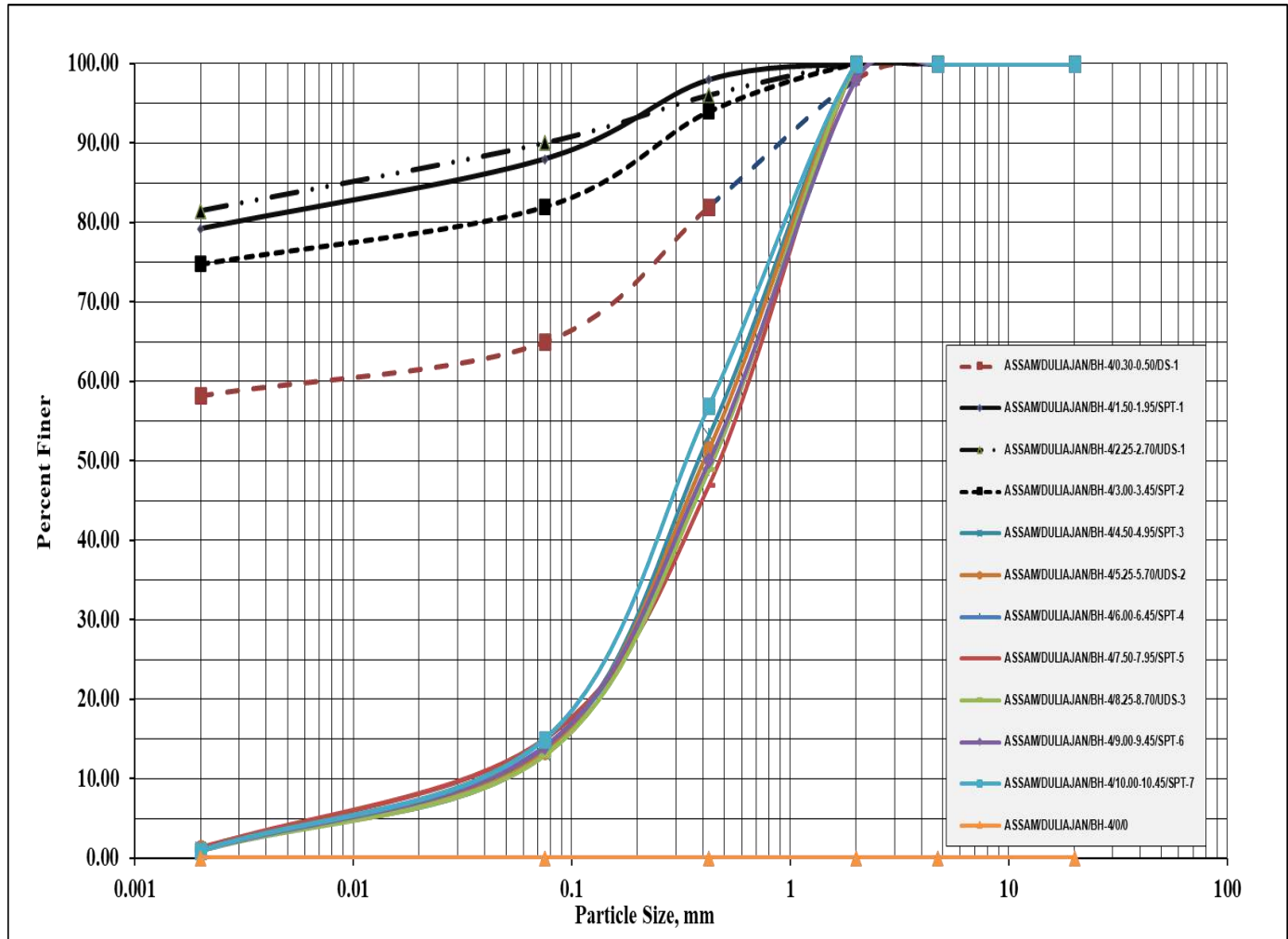
GRAIN-SIZE DISTRIBUTION CURVE BH-3

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TEST ON SOIL SAMPLES																				
BH-4			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.30-0.50	DS-1	CL	58.25	6.75	17.00	16.00	2.00	0.00	29.70	17.53	12.17	-	20	-	2.56	-	-	-	-	13
1.50-1.95	SPT-1	CI	79.20	8.80	10.00	2.00	0.00	0.00	38.48	23.98	14.50	-	26	-	2.57	-	-	-	-	-
2.25-2.70	UDS-1	CI	81.50	8.50	6.00	4.00	0.00	0.00	39.05	24.11	14.94	1.63	23	1.33	2.59	0.95	TX-UU	0.64	21.36	12
3.00-3.45	SPT-2	CI	74.80	7.20	12.00	6.00	0.00	0.00	40.18	24.63	15.55	-	27	-	2.59	-	-	-	-	-
4.50-4.95	SPT-3	SM	1.20	12.00	40.00	46.80	0.00	0.00	NL	NP	NIL	-	20	-	2.62	-	-	-	-	-
5.25-5.70	UDS-2	SM	1.35	12.15	38.00	48.50	0.00	0.00	NL	NP	NIL	1.74	21	1.44	2.63	0.83	DIR-DRAINED	0.02	30.48	0
6.00-6.45	SPT-4	SM	1.05	12.95	35.00	51.00	0.00	0.00	NL	NP	NIL	-	22	-	2.62	-	-	-	-	-
7.50-7.95	SPT-5	SM	1.25	13.75	32.00	53.00	0.00	0.00	NL	NP	NIL	-	18	-	2.60	-	-	-	-	-
8.25-8.70	UDS-3	SM	1.00	12.00	36.00	51.00	0.00	0.00	NL	NP	NIL	1.80	22	1.48	2.63	0.78	DIR-DRAINED	0.00	33.75	0
9.00-9.45	SPT-6	SM	1.00	13.00	36.00	48.00	2.00	0.00	NL	NP	NIL	-	18	-	2.60	-	-	-	-	-
10.00-10.45	SPT-7	SM	0.95	14.05	42.00	43.00	0.00	0.00	NL	NP	NIL	-	20	-	2.61	-	-	-	-	-

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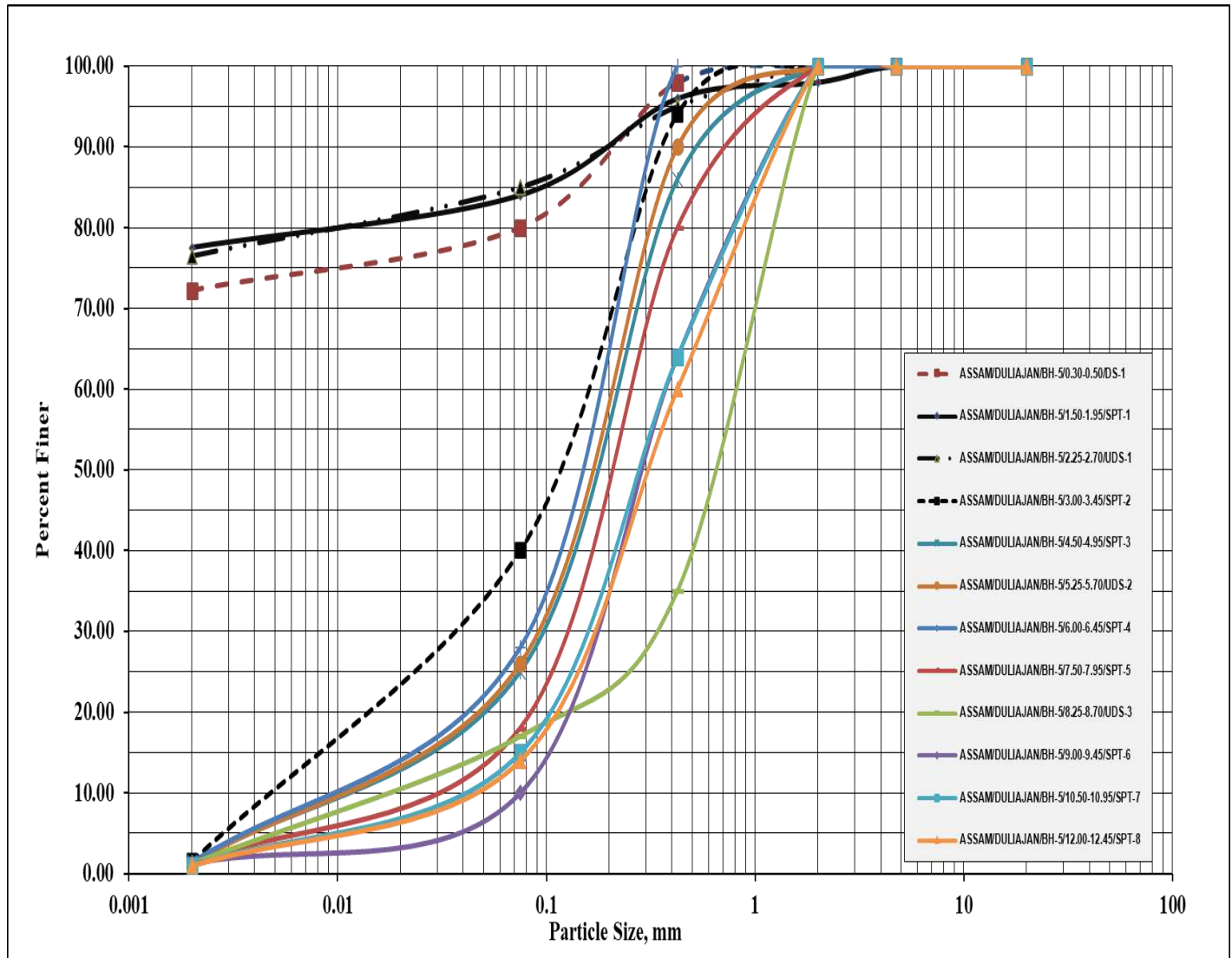

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GRAIN-SIZE DISTRIBUTION CURVE BH-4

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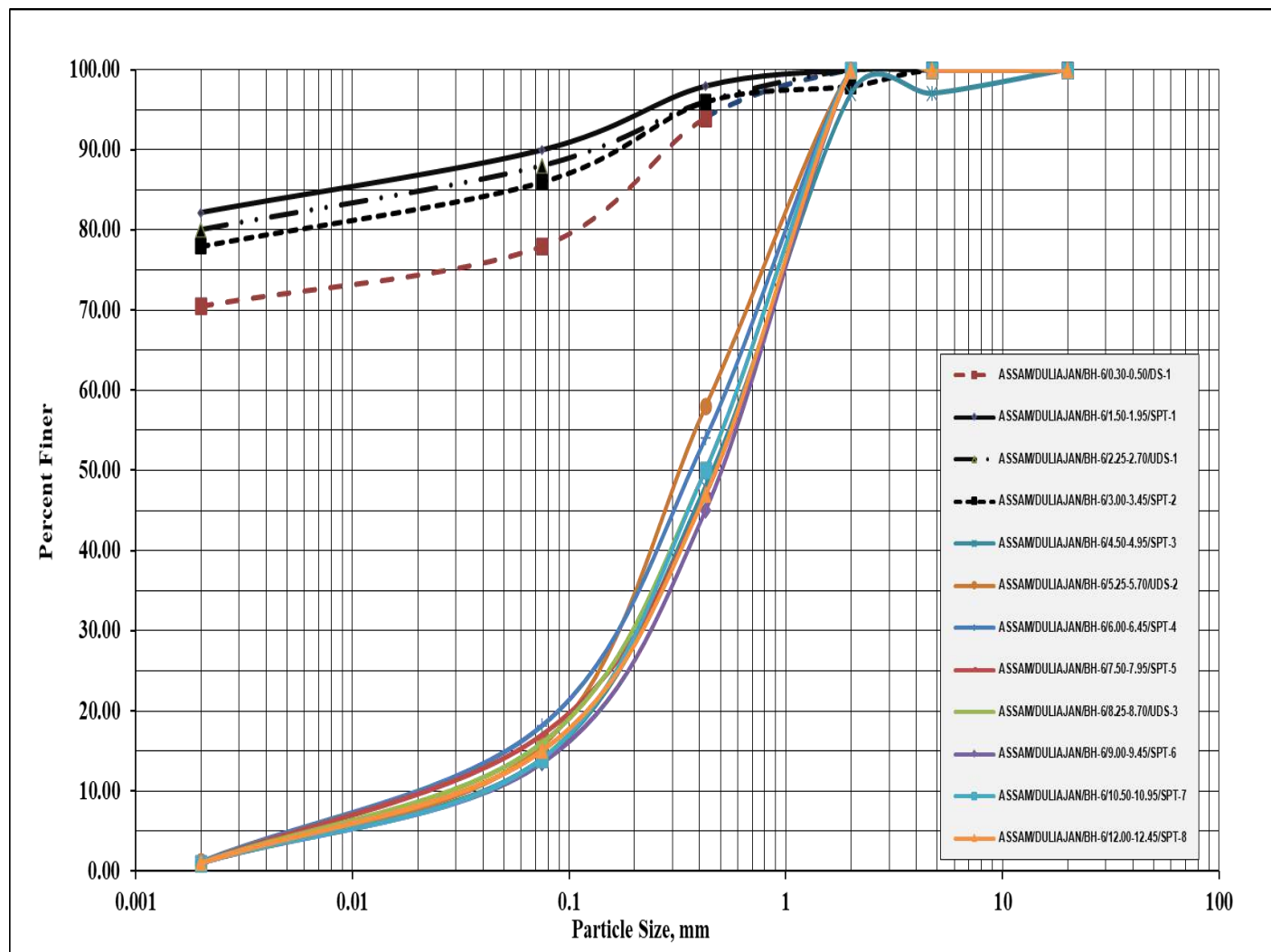
TEST ON SOIL SAMPLES																			
BH-5			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS		
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm											
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ
0.30-0.50	DS-1	CI	72.20	7.80	18.00	2.00	0.00	0.00	37.51	20.12	17.39	-	27	-	2.59	-	-	-	13
1.50-1.95	SPT-1	CI	77.50	6.50	12.00	2.00	2.00	0.00	38.65	20.87	17.78	-	28	-	2.58	-	-	-	-
2.25-2.70	UDS-1	CI	76.45	8.55	10.00	5.00	0.00	0.00	40.20	21.03	19.17	1.67	29	1.29	2.59	1.00	TX-UU	0.59	23.74
3.00-3.45	SPT-2	SM	1.50	38.50	54.00	6.00	0.00	0.00	25.67	NP	NIL	-	25	-	2.60	-	-	-	-
4.50-4.95	SPT-3	SM	1.45	23.55	61.00	14.00	0.00	0.00	20.65	NP	NIL	-	20	-	2.61	-	-	-	-
5.25-5.70	UDS-2	SM	1.20	24.80	64.00	10.00	0.00	0.00	21.18	NP	NIL	1.75	17	1.50	2.63	0.76	DIR-DRAINED	0.03	33.48
6.00-6.45	SPT-4	SM	1.35	26.65	72.00	0.00	0.00	0.00	20.19	NP	NIL	-	18	-	2.60	-	-	-	-
7.50-7.95	SPT-5	SM	1.15	16.85	62.00	20.00	0.00	0.00	NL	NP	NIL	-	13	-	2.63	-	-	-	-
8.25-8.70	UDS-3	SM	1.20	15.80	18.00	65.00	0.00	0.00	NL	NP	NIL	1.82	14	1.60	2.64	0.65	DIR-DRAINED	0.01	36.15
9.00-9.45	SPT-6	SM	1.25	8.75	54.00	36.00	0.00	0.00	NL	NP	NIL	-	13	-	2.65	-	-	-	-
10.50-10.95	SPT-7	SM	1.00	14.00	49.00	36.00	0.00	0.00	NL	NP	NIL	-	10	-	2.62	-	-	-	-
12.00-12.45	SPT-8	SM	1.00	13.00	46.00	40.00	0.00	0.00	NL	NP	NIL	-	19	-	2.63	-	-	-	-
13.50-13.95	SPT-9	SM	1.15	11.85	42.00	45.00	0.00	0.00	NL	NP	NIL	-	18	-	2.64	-	-	-	-
15.00-15.45	SPT-10	SM	1.00	13.00	24.00	62.00	0.00	0.00	NL	NP	NIL	-	17	-	2.63	-	-	-	-
16.50-16.95	SPT-11	SM	1.05	17.95	33.00	48.00	0.00	0.00	NL	NP	NIL	-	13	-	2.60	-	-	-	-
18.00-18.45	SPT-12	SM	1.00	17.00	38.00	44.00	0.00	0.00	NL	NP	NIL	-	17	-	2.61	-	-	-	-
19.50-19.95	SPT-13	SM	0.95	12.05	47.00	40.00	0.00	0.00	NL	NP	NIL	-	12	-	2.62	-	-	-	-
21.00-21.45	SPT-14	SM	1.00	16.00	43.00	40.00	0.00	0.00	NL	NP	NIL	-	10	-	2.62	-	-	-	-
22.50-22.95	SPT-15	SM	1.05	14.95	48.00	36.00	0.00	0.00	NL	NP	NIL	-	10	-	2.61	-	-	-	-
24.00-24.45	SPT-16	SM	1.00	14.00	36.00	49.00	0.00	0.00	NL	NP	NIL	-	7	-	2.63	-	-	-	-
25.50-25.95	SPT-17	SM	0.95	12.55	31.50	55.00	0.00	0.00	NL	NP	NIL	-	8	-	2.64	-	-	-	-
27.00-27.45	SPT-18	SM	0.90	13.10	28.00	58.00	0.00	0.00	NL	NP	NIL	-	12	-	2.63	-	-	-	-
28.50-28.95	SPT-19	SM	0.85	12.15	24.00	63.00	0.00	0.00	NL	NP	NIL	-	12	-	2.60	-	-	-	-
30.00-30.45	SPT-20	SM	0.80	14.20	20.00	65.00	0.00	0.00	NL	NP	NIL	-	11	-	2.61	-	-	-	-



GRAIN-SIZE DISTRIBUTION CURVE BH-5

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TEST ON SOIL SAMPLES																					
BH-6			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS				FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree		
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm													
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%	
0.30-0.50	DS-1	CL	70.50	7.50	16.00	6.00	0.00	0.00	33.01	18.17	14.84	-	30	-	2.56	-	-	-	-	12	
1.50-1.95	SPT-1	CI	82.15	7.85	8.00	2.00	0.00	0.00	35.66	19.91	15.75	-	29	-	2.57	-	-	-	-	-	
2.25-2.70	UDS-1	CI	80.00	8.00	8.00	4.00	0.00	0.00	36.00	21.22	14.78	1.68	25	1.34	2.59	0.93	TX-UU	0.63	22.27	14	
3.00-3.45	SPT-2	CI	77.95	8.05	10.00	2.00	2.00	0.00	36.71	21.35	15.36	-	29	-	2.58	-	-	-	-	-	
4.50-4.95	SPT-3	SM	1.00	13.00	34.00	49.00	0.00	3.00	NL	NP	NIL	-	19	-	2.61	-	-	-	-	-	
5.25-5.70	UDS-2	SM	1.20	14.50	42.30	42.00	0.00	0.00	NL	NP	NIL	1.75	17	1.50	2.62	0.75	DIR-DRAINED	0.01	33.74	0	
6.00-6.45	SPT-4	SM	1.15	17.00	35.85	46.00	0.00	0.00	NL	NP	NIL	-	18	-	2.63	-	-	-	-	-	
7.50-7.95	SPT-5	SM	1.10	16.00	29.90	53.00	0.00	0.00	NL	NP	NIL	-	13	-	2.61	-	-	-	-	-	
8.25-8.70	UDS-3	SM	1.00	15.00	34.00	50.00	0.00	0.00	NL	NP	NIL	1.78	15	1.55	2.60	0.68	DIR-DRAINED	0.00	35.62	0	
9.00-9.45	SPT-6	SM	1.05	12.50	31.45	55.00	0.00	0.00	NL	NP	NIL	-	13	-	2.64	-	-	-	-	-	
10.50-10.95	SPT-7	SM	1.00	13.00	36.00	50.00	0.00	0.00	NL	NP	NIL	-	10	-	2.63	-	-	-	-	-	
12.00-12.45	SPT-8	SM	1.05	14.00	31.95	53.00	0.00	0.00	NL	NP	NIL	-	17	-	2.65	-	-	-	-	-	
13.50-13.95	SPT-9	SM	0.90	16.00	23.10	60.00	0.00	0.00	NL	NP	NIL	-	13	-	2.63	-	-	-	-	-	
15.00-15.45	SPT-10	SM	0.90	14.00	21.10	64.00	0.00	0.00	NL	NP	NIL	-	12	-	2.62	-	-	-	-	-	
16.50-16.95	SPT-11	SM	0.75	17.00	22.25	60.00	0.00	0.00	NL	NP	NIL	-	11	-	2.60	-	-	-	-	-	
18.00-18.45	SPT-12	SM	0.85	18.00	34.15	44.00	0.00	-0.10	NL	NP	NIL	-	18	-	2.61	-	-	-	-	-	
19.50-19.95	SPT-13	SM	0.70	19.00	38.30	42.00	0.00	0.00	NL	NP	NIL	-	17	-	2.63	-	-	-	-	-	
21.00-21.45	SPT-14	SM	0.80	13.00	39.20	47.00	0.00	0.00	NL	NP	NIL	-	13	-	2.65	-	-	-	-	-	
22.50-22.95	SPT-15	SM	0.95	18.00	45.05	36.00	0.00	0.00	NL	NP	NIL	-	10	-	2.64	-	-	-	-	-	
24.00-24.45	SPT-16	SM	0.65	18.50	50.85	30.00	0.00	0.00	NL	NP	NIL	-	11	-	2.63	-	-	-	-	-	
25.50-25.95	SPT-17	SM	0.70	17.00	47.30	35.00	0.00	0.00	NL	NP	NIL	-	17	-	2.62	-	-	-	-	-	
27.00-27.45	SPT-18	SM	0.70	16.00	55.30	28.00	0.00	0.00	NL	NP	NIL	-	19	-	2.61	-	-	-	-	-	
28.50-28.95	SPT-19	SM	0.65	14.00	59.45	26.00	0.00	0.00	NL	NP	NIL	-	18	-	2.60	-	-	-	-	-	
30.00-30.45	SPT-20	SM	0.65	15.00	62.45	22.00	0.00	0.00	NL	NP	NIL	-	8	-	2.63	-	-	-	-	-	

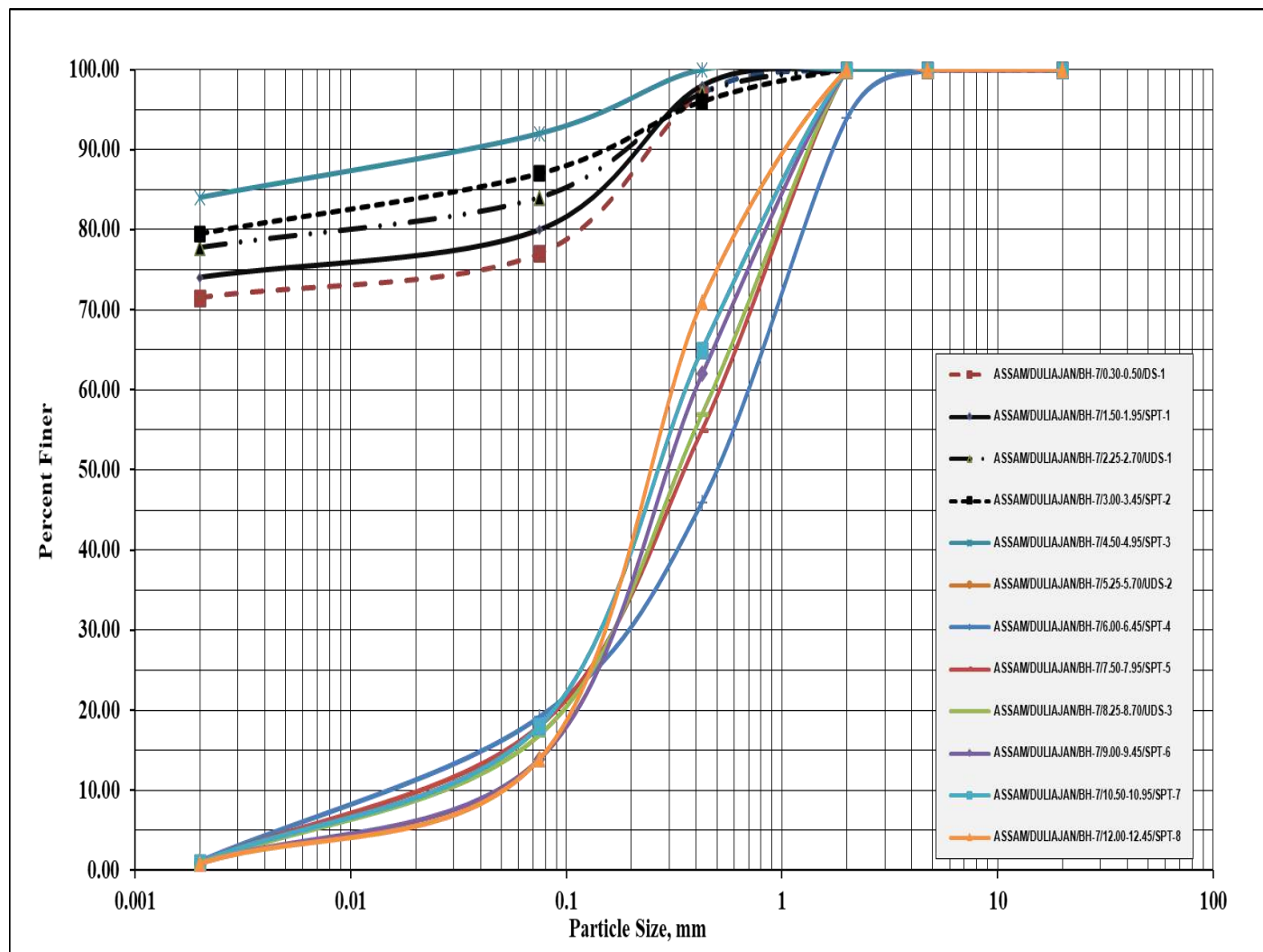


GRAIN-SIZE DISRIBUTION CURVE BH-6

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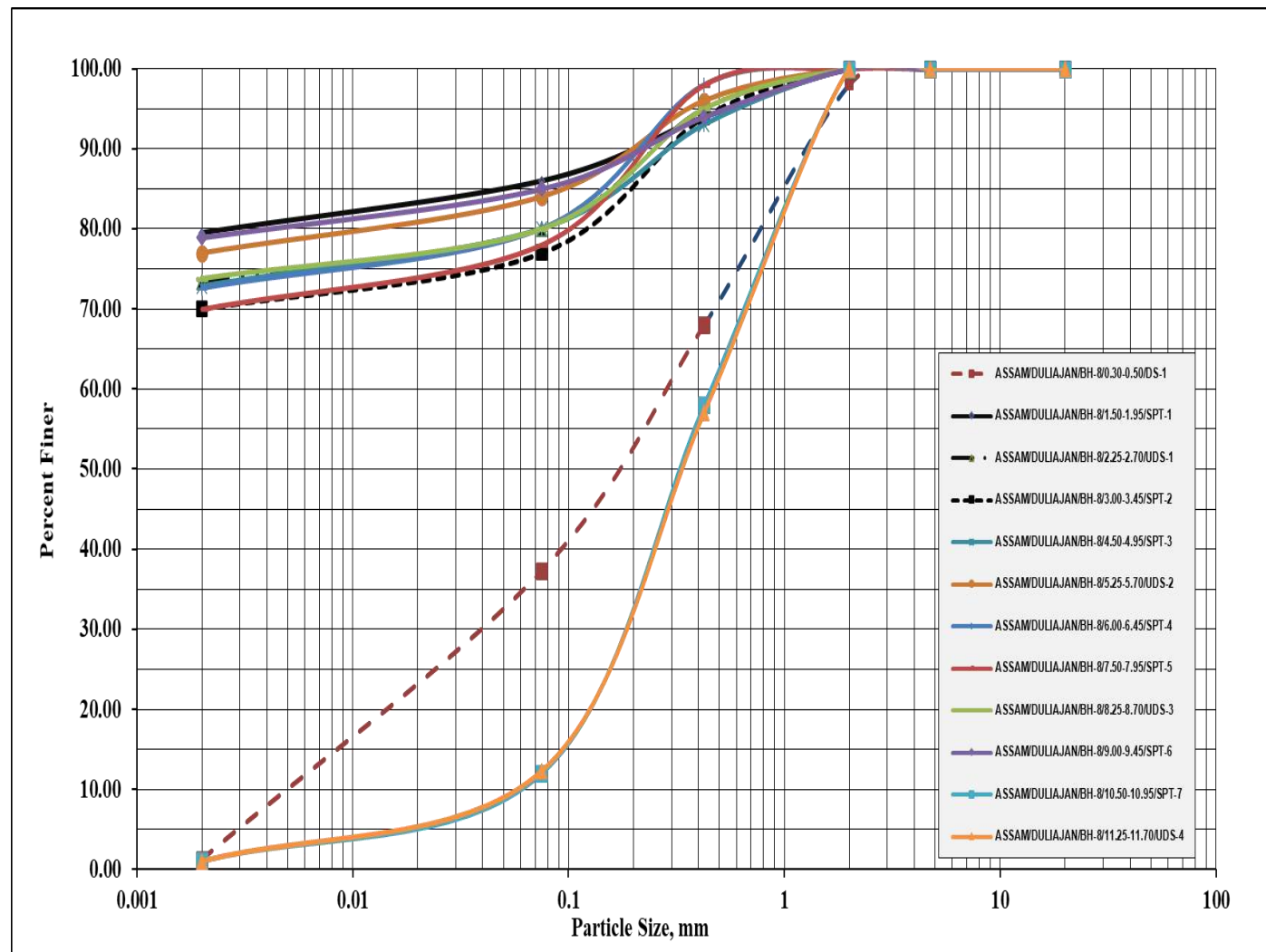
TEST ON SOIL SAMPLES																				
BH-7			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degrees	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.30-0.50	DS-1	CL	71.50	5.50	20.00	3.00	0.00	0.00	30.18	21.24	8.94	-	23	-	2.58	-	-	-	-	11
1.50-1.95	SPT-1	CL	74.05	5.95	18.00	2.00	0.00	0.00	31.50	21.63	9.87	-	26	-	2.56	-	-	-	-	-
2.25-2.70	UDS-1	CL	77.75	6.25	13.00	3.00	0.00	0.00	31.39	21.36	10.03	1.66	22	1.36	2.59	0.90	TX-UU	0.61	23.64	15
3.00-3.45	SPT-2	CL	79.50	7.50	9.00	4.00	0.00	0.00	32.15	21.47	10.68	-	25	-	2.57	-	-	-	-	-
4.50-4.95	SPT-3	CL	84.00	8.00	8.00	0.00	0.00	0.00	33.72	21.76	11.96	-	27	-	2.55	-	-	-	-	-
5.25-5.70	UDS-2	SM	1.00	17.00	47.00	35.00	0.00	0.00	NL	NP	NIL	1.76	18	1.49	2.60	0.74	DIR-DRAINED	0.02	33.56	0
6.00-6.45	SPT-4	SM	1.20	18.00	26.80	48.00	6.00	0.00	NL	NP	NIL	-	19	-	2.61	-	-	-	-	-
7.50-7.95	SPT-5	SM	1.10	17.00	36.90	45.00	0.00	0.00	NL	NP	NIL	-	15	-	2.63	-	-	-	-	-
8.25-8.70	UDS-3	SM	0.95	16.00	40.05	43.00	0.00	0.00	NL	NP	NIL	1.82	13	1.61	2.60	0.61	DIR-DRAINED	0.01	35.65	0
9.00-9.45	SPT-6	SM	0.95	13.00	48.05	38.00	0.00	0.00	NL	NP	NIL	-	15	-	2.64	-	-	-	-	-
10.50-10.95	SPT-7	SM	1.00	17.00	47.00	35.00	0.00	0.00	NL	NP	NIL	-	12	-	2.63	-	-	-	-	-
12.00-12.45	SPT-8	SM	0.90	13.00	57.10	29.00	0.00	0.00	NL	NP	NIL	-	11	-	2.61	-	-	-	-	-
13.50-13.95	SPT-9	SM	0.85	18.00	39.15	42.00	0.00	0.00	NL	NP	NIL	-	14	-	2.62	-	-	-	-	-
15.00-15.45	SPT-10	SM	0.85	19.00	35.15	45.00	0.00	0.00	NL	NP	NIL	-	14	-	2.63	-	-	-	-	-
16.50-16.95	SPT-11	SM	0.95	17.00	39.05	43.00	0.00	0.00	NL	NP	NIL	-	8	-	2.64	-	-	-	-	-
18.00-18.45	SPT-12	SM	0.80	16.00	35.20	48.00	0.00	0.00	NL	NP	NIL	-	8	-	2.62	-	-	-	-	-
19.50-19.95	SPT-13	SM	0.85	15.00	31.15	53.00	0.00	0.00	NL	NP	NIL	-	12	-	2.65	-	-	-	-	-
21.00-21.45	SPT-14	SM	0.80	13.00	35.20	51.00	0.00	0.00	NL	NP	NIL	-	10	-	2.63	-	-	-	-	-
22.50-22.95	SPT-15	SM	0.95	14.00	31.05	54.00	0.00	0.00	NL	NP	NIL	-	17	-	2.64	-	-	-	-	-
24.00-24.45	SPT-16	SM	0.90	15.00	34.10	50.00	0.00	0.00	NL	NP	NIL	-	17	-	2.63	-	-	-	-	-
25.50-25.95	SPT-17	SM	0.80	15.65	48.55	35.00	0.00	0.00	NL	NP	NIL	-	11	-	2.61	-	-	-	-	-
27.00-27.45	SPT-18	SM	0.80	16.75	40.45	42.00	0.00	0.00	NL	NP	NIL	-	11	-	2.63	-	-	-	-	-
28.50-28.95	SPT-19	SM	0.75	13.75	47.50	38.00	0.00	0.00	NL	NP	NIL	-	11	-	2.62	-	-	-	-	-
30.00-30.45	SPT-20	SM	0.65	13.00	44.35	42.00	0.00	0.00	NL	NP	NIL	-	10	-	2.60	-	-	-	-	-



GRAIN-SIZE DISTRIBUTION CURVE BH-7

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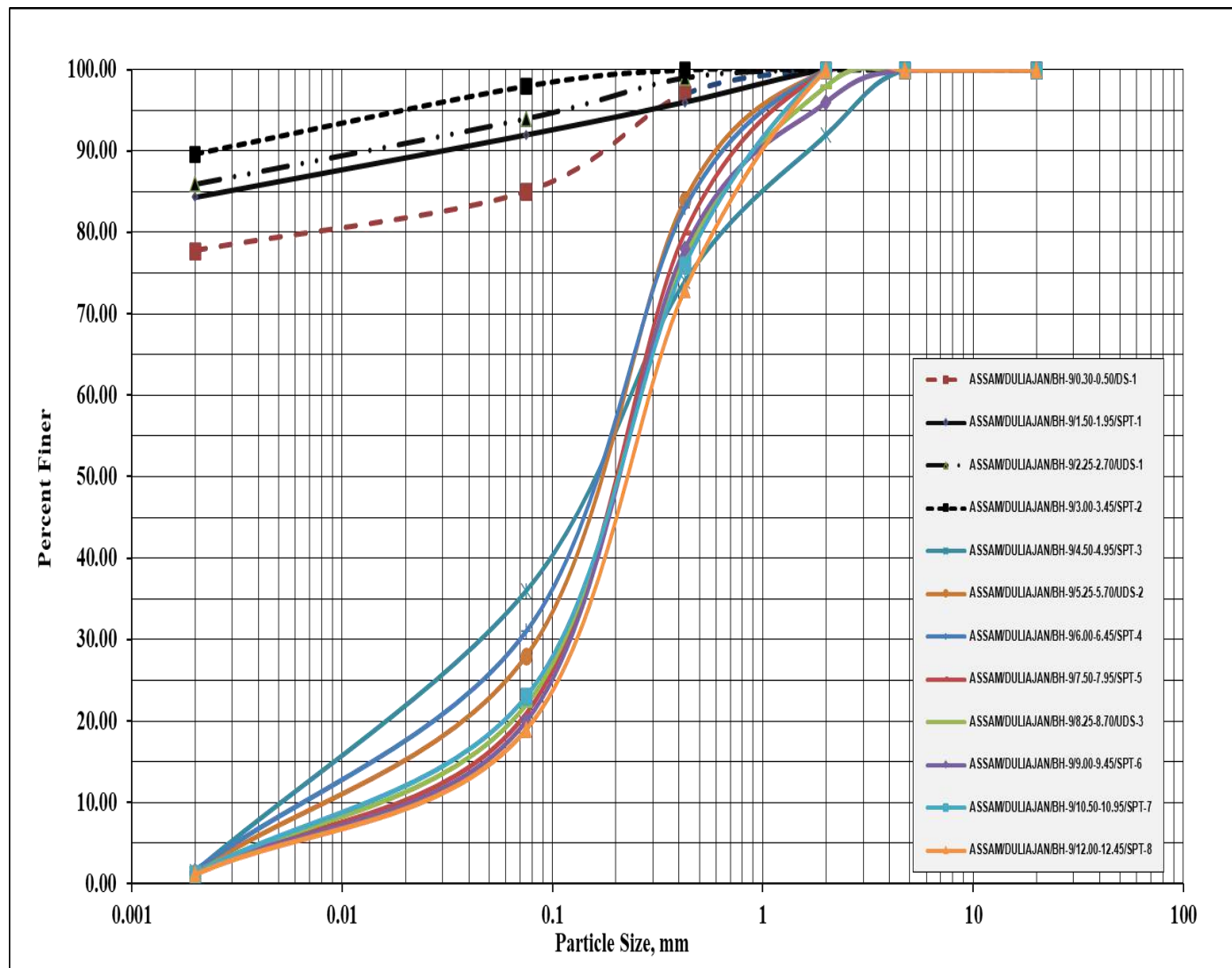
TEST ON SOIL SAMPLES																				
BH-8			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.30-0.50	DS-1	SM	1.25	36.00	30.75	30.00	2.00	0.00	30.82	NP	NIL	-	26	-	2.62	-	-	-	-	0
1.50-1.95	SPT-1	CI	79.50	6.50	8.00	6.00	0.00	0.00	43.55	24.41	19.14	-	26	-	2.59	-	-	-	-	-
2.25-2.70	UDS-1	CI	73.35	6.65	15.00	5.00	0.00	0.00	40.00	23.58	16.42	1.62	24	1.31	2.55	0.95	TX-UU	0.61	23.32	13
3.00-3.45	SPT-2	CI	70.00	7.00	17.00	6.00	0.00	0.00	41.36	24.60	16.76	-	31	-	2.57	-	-	-	-	-
4.50-4.95	SPT-3	CI	72.85	7.15	13.00	7.00	0.00	0.00	42.84	24.68	18.16	-	31	-	2.56	-	-	-	-	-
5.25-5.70	UDS-2	CL	76.95	7.05	12.00	4.00	0.00	0.00	34.51	18.00	16.51	1.67	28	1.30	2.50	0.92	TX-UU	0.65	22.95	15
6.00-6.45	SPT-4	CL	72.55	7.45	18.00	2.00	0.00	0.00	30.75	18.09	12.66	-	27	-	2.56	-	-	-	-	-
7.50-7.95	SPT-5	CL	69.95	8.05	20.00	2.00	0.00	0.00	30.22	18.23	11.99	-	25	-	2.53	-	0.630	-	-	-
8.25-8.70	UDS-3	CL	73.80	6.20	15.00	5.00	0.00	0.00	31.42	18.42	13.00	1.72	24	1.39	2.57	0.85	TX-UU	0.63	23.00	18
9.00-9.45	SPT-6	CL	78.90	6.10	9.00	6.00	0.00	0.00	32.63	18.14	14.49	-	25	-	2.58	-	-	-	-	-
10.50-10.95	SPT-7	SM	1.05	10.95	46.00	42.00	0.00	0.00	NL	NP	NIL	-	20	-	2.61	-	-	-	-	-
11.25-11.70	UDS-4	SM	1.00	11.25	44.75	43.00	0.00	0.00	NL	NP	NIL	1.85	21	1.53	2.64	0.73	DIR- DRAINED	0.00	35.08	0
12.00-12.45	SPT-8	SM	1.10	11.90	40.00	47.00	0.00	0.00	NL	NP	NIL	-	22	-	2.63	-	-	-	-	-
13.50-13.95	SPT-9	SM	1.25	13.75	42.00	43.00	0.00	0.00	NL	NP	NIL	-	23	-	2.62	-	-	-	-	-
15.00-15.45	SPT-10	SM	1.10	12.90	44.00	42.00	0.00	0.00	NL	NP	NIL	-	20	-	2.64	-	-	-	-	-
16.50-16.95	SPT-11	SM	1.00	16.00	40.00	43.00	0.00	0.00	NL	NP	NIL	-	21	-	2.62	-	-	-	-	-
18.00-18.45	SPT-12	SM	0.95	17.05	45.00	37.00	0.00	0.00	NL	NP	NIL	-	21	-	2.63	-	-	-	-	-
19.50-19.95	SPT-13	SM	0.90	12.10	40.00	47.00	0.00	0.00	NL	NP	NIL	-	22	-	2.60	-	-	-	-	-
20.00-20.45	SPT-14	SM	0.90	13.60	42.00	43.50	0.00	0.00	NL	NP	NIL	-	21	-	2.61	-	-	-	-	-



GRAIN-SIZE DISTRIBUTION CURVE, BH-8

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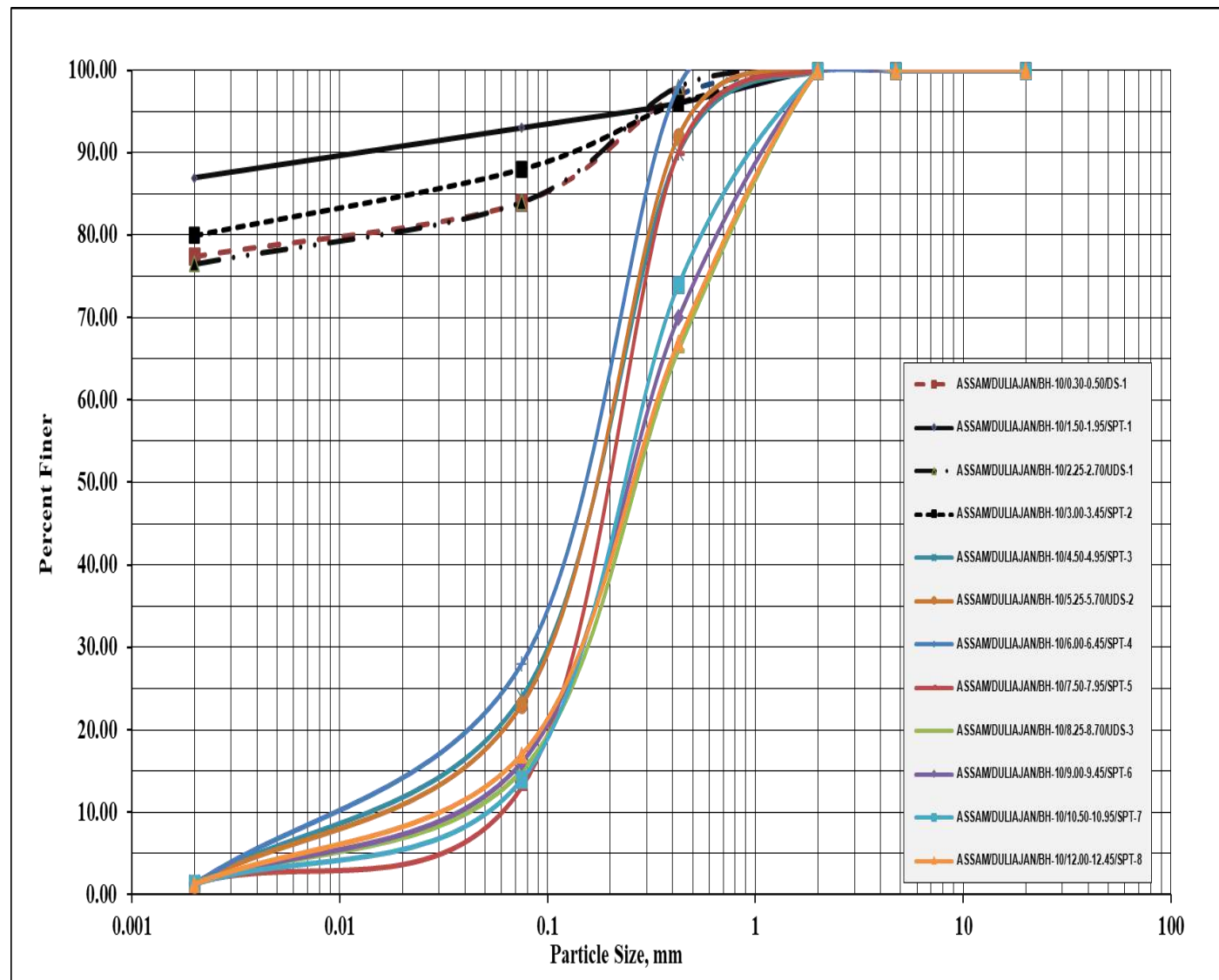
TEST ON SOIL SAMPLES																			
BH-9			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS		
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm											
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ
0.30-0.50	DS-1	CI	77.75	7.25	12.00	3.00	0.00	0.00	44.64	21.74	22.90	-	30	-	2.59	-	-	-	14
1.50-1.95	SPT-1	CI	84.35	7.65	4.00	4.00	0.00	0.00	47.46	22.96	24.50	-	31	-	2.57	-	-	-	-
2.25-2.70	UDS-1	CI	85.95	8.05	5.00	1.00	0.00	0.00	45.47	23.00	22.47	1.58	29	1.22	2.58	1.11	TX-UU	0.66	21.08
3.00-3.45	SPT-2	CI	89.60	8.40	2.00	0.00	0.00	0.00	44.05	23.21	20.84	-	33	-	2.59	-	-	-	-
4.50-4.95	SPT-3	SM	1.55	34.45	38.00	18.00	8.00	0.00	NL	NP	NIL	-	14	-	2.63	-	-	-	-
5.25-5.70	UDS-2	SM	1.40	26.60	56.00	16.00	0.00	0.00	NL	NP	NIL	1.66	15	1.44	2.64	0.83	DIR- DRAINED	0.03	32.39
6.00-6.45	SPT-4	SM	1.65	29.35	52.00	17.00	0.00	0.00	NL	NP	NIL	-	16	-	2.62	-	-	-	-
7.50-7.95	SPT-5	SM	1.50	19.50	59.00	20.00	0.00	0.00	NL	NP	NIL	-	19	-	2.63	-	-	-	-
8.25-8.70	UDS-3	SM	1.40	20.60	55.00	21.00	2.00	0.00	NL	NP	NIL	1.75	12	1.56	2.64	0.69	DIR- DRAINED	0.01	34.68
9.00-9.45	SPT-6	SM	1.45	18.55	58.00	18.00	4.00	0.00	NL	NP	NIL	-	19	-	2.65	-	-	-	-
10.50-10.95	SPT-7	SM	1.25	21.75	53.00	24.00	0.00	0.00	NL	NP	NIL	-	18	-	2.63	-	-	-	-
12.00-12.45	SPT-8	SM	1.20	17.80	54.00	27.00	0.00	0.00	NL	NP	NIL	-	19	-	2.62	-	-	-	-
13.50-13.95	SPT-9	SM	1.30	15.70	58.00	25.00	0.00	0.00	NL	NP	NIL	-	17	-	2.61	-	-	-	-
15.00-15.45	SPT-10	SM	1.10	11.90	60.00	27.00	0.00	0.00	NL	NP	NIL	-	19	-	2.60	-	-	-	-
16.50-16.95	SPT-11	SM	1.15	12.85	54.00	32.00	0.00	0.00	NL	NP	NIL	-	15	-	2.63	-	-	-	-
18.00-18.45	SPT-12	SM	1.00	14.00	52.00	33.00	0.00	0.00	NL	NP	NIL	-	17	-	2.62	-	-	-	-
19.50-19.95	SPT-13	SM	1.00	19.00	45.00	35.00	0.00	0.00	NL	NP	NIL	-	19	-	2.61	-	-	-	-
21.00-21.45	SPT-14	SM	0.95	15.55	53.50	30.00	0.00	0.00	NL	NP	NIL	-	18	-	2.63	-	-	-	-
22.50-22.95	SPT-15	SM	0.90	14.10	51.00	34.00	0.00	0.00	NL	NP	NIL	-	17	-	2.61	-	-	-	-
24.00-24.45	SPT-16	SM	0.90	12.10	57.00	30.00	0.00	0.00	NL	NP	NIL	-	12	-	2.62	-	-	-	-
25.50-25.95	SPT-17	SM	0.80	13.20	57.00	29.00	0.00	0.00	NL	NP	NIL	-	15	-	2.63	-	-	-	-
27.00-27.45	SPT-18	SM	0.75	14.25	50.00	35.00	0.00	0.00	NL	NP	NIL	-	11	-	2.64	-	-	-	-
28.50-28.95	SPT-19	SM	0.85	13.15	48.00	38.00	0.00	0.00	NL	NP	NIL	-	10	-	2.60	-	-	-	-
30.00-30.45	SPT-20	SM	0.90	16.10	43.00	40.00	0.00	0.00	NL	NP	NIL	-	11	-	2.63	-	-	-	-



GRAIN-SIZE DISTRIBUTION CURVE BH-9

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TEST ON SOIL SAMPLES																				
BH-10			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.30-0.50	DS-1	CI	77.45	6.55	13.00	3.00	0.00	0.00	45.80	23.14	22.66	-	30	-	2.59	-	-	-	-	14
1.50-1.95	SPT-1	CI	86.95	6.05	3.00	4.00	0.00	0.00	48.05	24.60	23.45	-	32	-	2.58	-	-	-	-	-
2.25-2.70	UDS-1	CI	76.50	7.50	14.00	2.00	0.00	0.00	47.51	24.33	23.18	1.58	31	1.21	2.58	1.14	TX-UU	0.66	21.01	15
3.00-3.45	SPT-2	CI	79.95	8.05	8.00	4.00	0.00	0.00	46.87	24.02	22.85	-	23	-	2.59	-	-	-	-	-
4.50-4.95	SPT-3	SM	1.45	22.55	66.00	10.00	0.00	0.00	NL	NP	NIL	-	14	-	2.62	-	-	-	-	-
5.25-5.70	UDS-2	SM	1.25	21.75	69.00	8.00	0.00	0.00	NL	NP	NIL	1.69	15	1.47	2.63	0.79	DIR- DRAINED	0.01	33.67	0
6.00-6.45	SPT-4	SM	1.30	26.70	70.00	2.00	0.00	0.00	NL	NP	NIL	-	16	-	2.60	-	-	-	-	-
7.50-7.95	SPT-5	SM	1.35	11.65	77.00	10.00	0.00	0.00	NL	NP	NIL	-	12	-	2.61	-	-	-	-	-
8.25-8.70	UDS-3	SM	1.40	13.60	51.00	34.00	0.00	0.00	NL	NP	NIL	1.75	18	1.48	2.62	0.77	DIR- DRAINED	0.00	35.14	0
9.00-9.45	SPT-6	SM	1.25	14.75	54.00	30.00	0.00	0.00	NL	NP	NIL	-	12	-	2.63	-	-	-	-	-
10.50-10.95	SPT-7	SM	1.35	12.65	60.00	26.00	0.00	0.00	NL	NP	NIL	-	11	-	2.64	-	-	-	-	-
12.00-12.45	SPT-8	SM	1.30	15.70	50.00	33.00	0.00	0.00	NL	NP	NIL	-	17	-	2.62	-	-	-	-	-
13.50-13.95	SPT-9	SM	1.20	18.30	49.50	31.00	0.00	0.00	NL	NP	NIL	-	13	-	2.63	-	-	-	-	-
15.00-15.45	SPT-10	SM	1.25	19.75	51.00	28.00	0.00	0.00	NL	NP	NIL	-	14	-	2.60	-	-	-	-	-
16.50-16.95	SPT-11	SM	1.10	11.90	63.00	24.00	0.00	0.00	NL	NP	NIL	-	13	-	2.61	-	-	-	-	-
18.00-18.45	SPT-12	SM	1.15	20.85	49.00	29.00	0.00	0.00	NL	NP	NIL	-	10	-	2.62	-	-	-	-	-
19.50-19.95	SPT-13	SM	1.20	21.80	56.00	21.00	0.00	0.00	NL	NP	NIL	-	14	-	2.63	-	-	-	-	-
21.00-21.45	SPT-14	SM	1.00	12.50	60.50	24.00	2.00	0.00	NL	NP	NIL	-	10	-	2.64	-	-	-	-	-
22.50-22.95	SPT-15	SM	0.95	14.05	57.00	28.00	0.00	0.00	NL	NP	NIL	-	11	-	2.63	-	-	-	-	-
24.00-24.45	SPT-16	SM	0.90	13.10	60.00	26.00	0.00	0.00	NL	NP	NIL	-	10	-	2.60	-	-	-	-	-
25.50-25.95	SPT-17	SM	0.90	12.10	67.00	20.00	0.00	0.00	NL	NP	NIL	-	11	-	2.61	-	-	-	-	-
27.00-27.45	SPT-18	SM	0.95	17.05	60.00	22.00	0.00	0.00	NL	NP	NIL	-	14	-	2.63	-	-	-	-	-
28.50-28.95	SPT-19	SM	0.85	13.15	66.00	20.00	0.00	0.00	NL	NP	NIL	-	13	-	2.60	-	-	-	-	-
30.00-30.45	SPT-20	SM	0.80	16.20	57.00	26.00	0.00	0.00	NL	NP	NIL	-	13	-	2.61	-	-	-	-	-



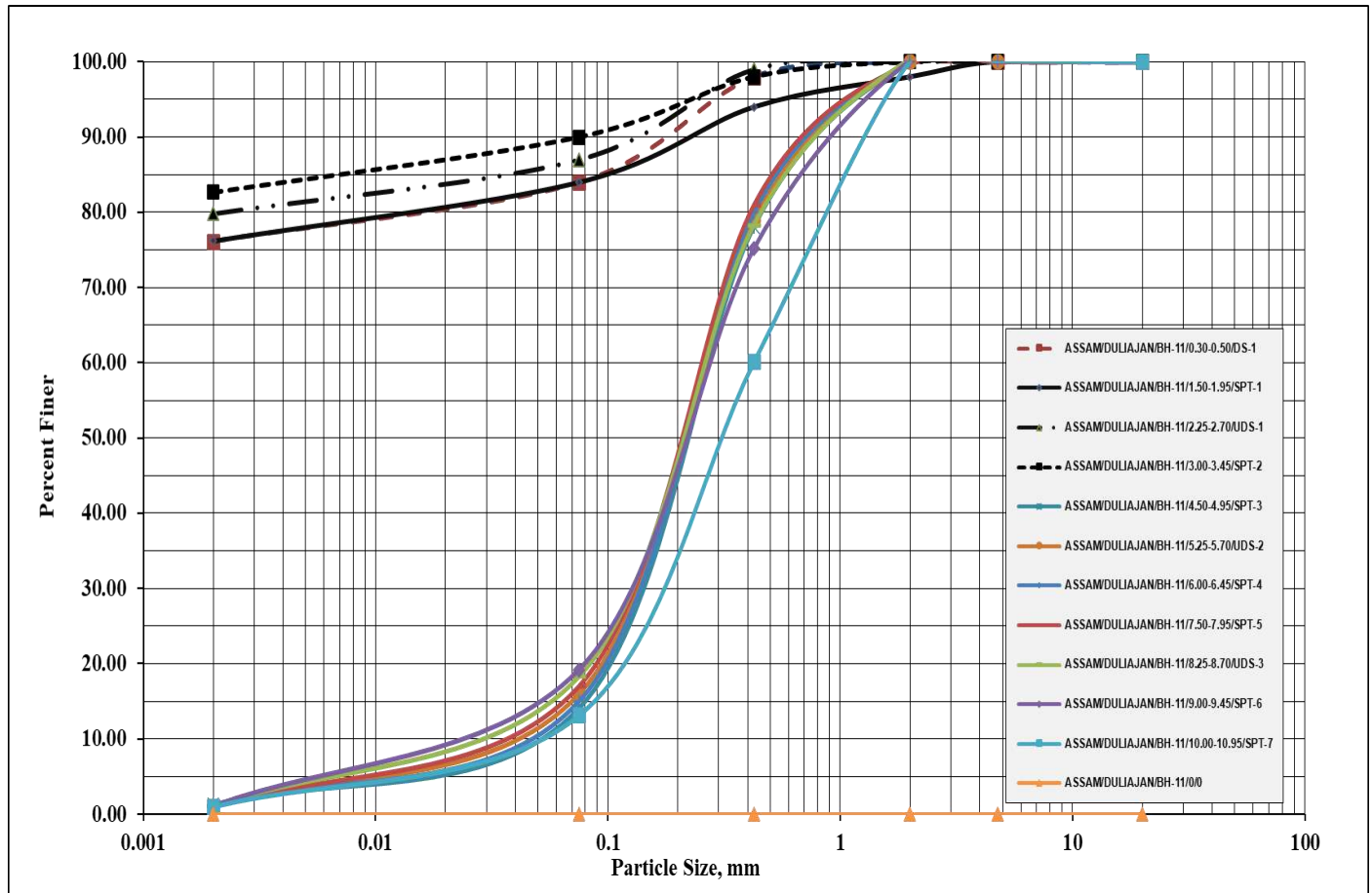
GRAIN-SIZE DISTRIBUTION CURVE BH-10

For Shiva Testing & Research Laboratory Pvt. Ltd.
(Signature)
Authorized Signatory

TEST ON SOIL SAMPLES																				
BH-11			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.30-0.50	DS-1	CL	76.15	7.85	14.00	2.00	0.00	0.00	32.80	22.36	10.44	-	24	-	2.59	-	-	-	-	17
1.50-1.95	SPT-1	CI	76.20	7.80	10.00	4.00	2.00	0.00	48.77	25.21	23.56	-	32	-	2.57	-	-	-	-	-
2.25-2.70	UDS-1	CI	79.80	7.20	12.00	1.00	0.00	0.00	46.05	26.47	19.58	1.60	30	1.23	2.58	1.10	TX-UU	0.62	21.32	14
3.00-3.45	SPT-2	CI	82.65	7.35	8.00	2.00	0.00	0.00	45.21	22.18	23.03	-	28	-	2.56	-	-	-	-	-
4.50-4.95	SPT-3	SM	1.25	12.85	64.00	22.00	0.00	-0.10	NL	NP	NIL	-	20	-	2.62	-	-	-	-	-
5.25-5.70	UDS-2	SM	1.10	14.90	63.00	21.00	0.00	0.00	NL	NP	NIL	1.74	21	1.44	2.61	0.82	DIR- DRAINED	0.04	32.84	0
6.00-6.45	SPT-4	SM	1.20	13.80	65.00	20.00	0.00	0.00	NL	NP	NIL	-	22	-	2.60	-	-	-	-	-
7.50-7.95	SPT-5	SM	1.00	16.05	64.00	19.00	0.00	-0.05	NL	NP	NIL	-	22	-	2.63	-	-	-	-	-
8.25-8.70	UDS-3	SM	1.15	17.20	60.00	22.00	0.00	-0.35	NL	NP	NIL	1.80	19	1.51	2.62	0.73	DIR- DRAINED	0.00	35.23	0
9.00-9.45	SPT-6	SM	1.05	18.15	56.00	25.00	0.00	-0.20	NL	NP	NIL	-	22	-	2.61	-	-	-	-	-
10.00-10.95	SPT-7	SM	1.00	12.15	47.00	40.00	0.00	-0.15	NL	NP	NIL	-	20	-	2.63	-	-	-	-	-

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory



GRAIN-SIZE DSITRIBUTION CURVE BH-11

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

17.0 TRIAL-PIT

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

Trail-Pit

A trial pit (or test pit) is an excavation of ground to study or sample the composition and structure of the subsurface, usually dug during a site investigation, a soil survey or a geological survey. Trial pits are dug before the construction. They are dug to determine the geology and the water table of that site.

Trial pits are usually between 1 and 4 metres deep and are dug either by hand or using a mechanical digger. Building and construction regulations clearly state that any trial pits that concedes deeper than 1.2 metres should be secured against structural collapse.

As per the client's requirement field investigation as well as the laboratory test are conducted to complies with the client requirement.

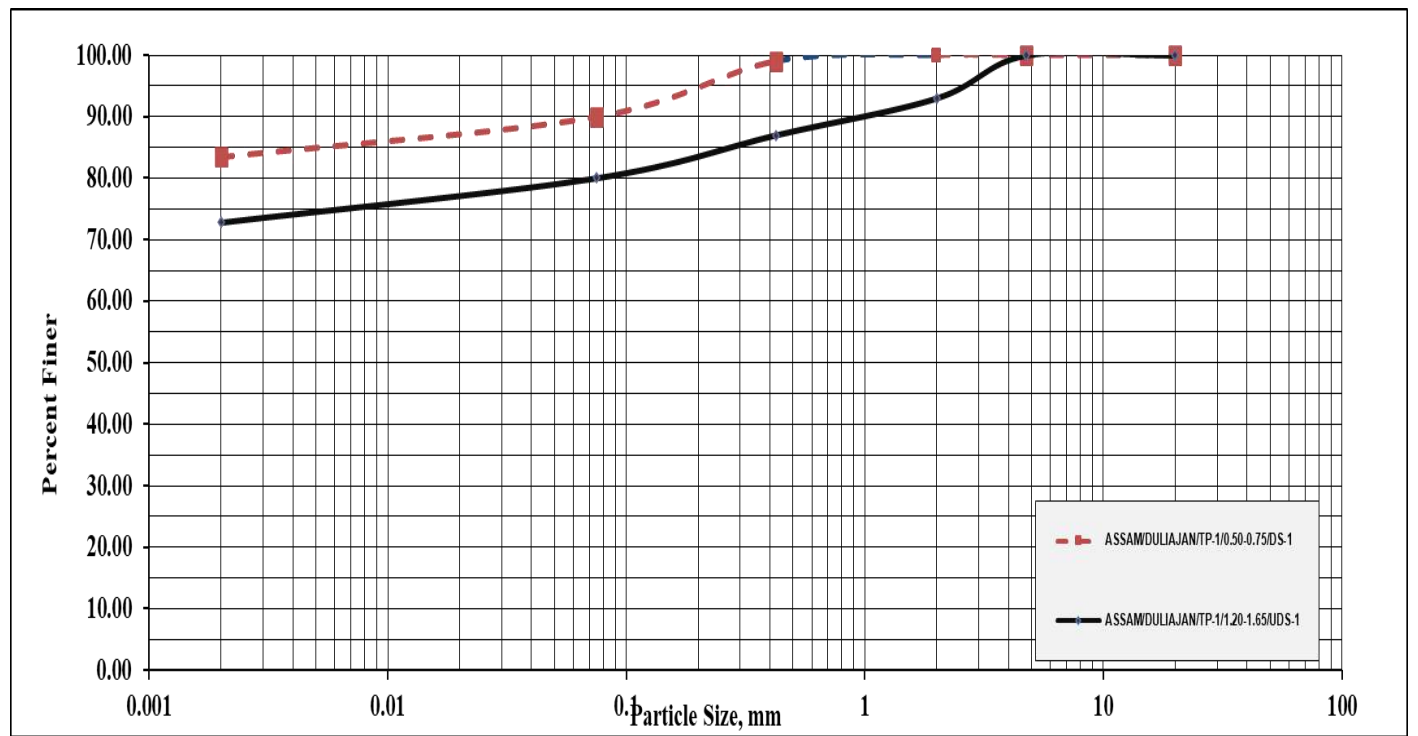
Field Observation: -

- 06 no. of the Trial Pit (TP-1 to TP-6) of size 1.0 mtr x 1.0 mtr x 1.2 mtr have been executed at the location identifies the site engineer and in the presence of the concern engineer.
- During excavation it was found that Clayey soil of low to intermediate compressibility up to the depth of the 2.00 meter (of CI & CL Group) and whole layers are well compacted.

For Shiva Testing & Research Laboratory Pvt. Ltd.

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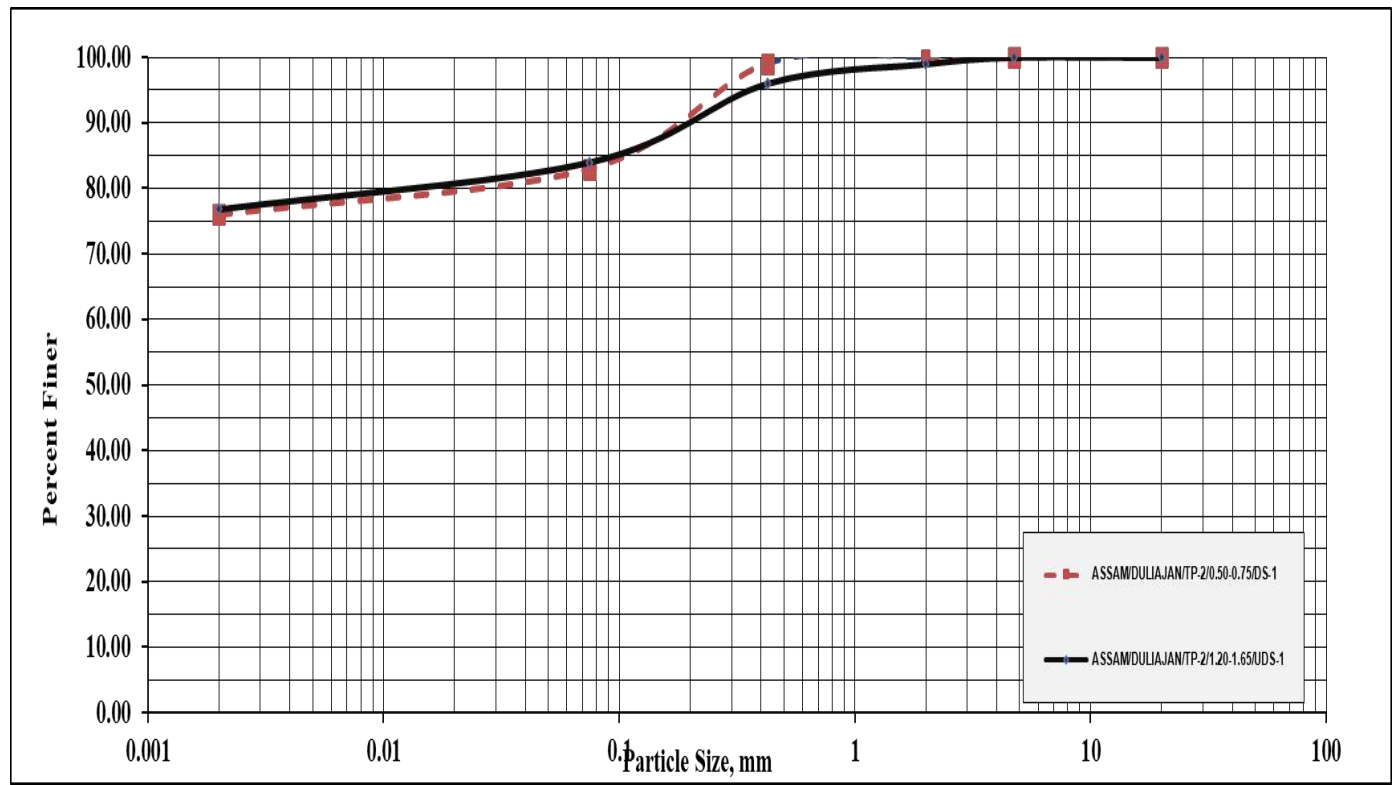
TEST ON SOIL SAMPLES																				
TP-1			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729235.097 Y=3029423.606																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	FREE SWELL INDEX
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	%
0.50-0.75	DS-1	CI	83.50	6.50	9.00	1.00	0.00	0.00	36.39	19.17	17.22	-	26	-	2.51	-	-	-	-	10
1.20-1.65	UDS-1	CI	72.75	7.25	7.00	6.00	7.00	0.00	37.51	20.18	17.33	1.78	37	1.30	2.57	0.98	TX	0.65	20.7	12



GRAIN SIZE DISTRIBUTION CURVE, TP-1

For Shiva Testing & Research Laboratory Pvt. Ltd.
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TEST ON SOIL SAMPLES																				
TP-2			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729372.327 Y=3029428.851																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	
0.50-0.75	DS-1	CL	76.00	7.00	16.00	1.00	0.00	0.00	33.42	20.60	12.82	-	22	-	2.52	-	-	-	-	
1.20-1.65	UDS-1	CL	76.80	7.20	12.00	3.00	1.00	0.00	34.12	20.57	13.55	1.76	31	1.34	2.54	0.89	TX	0.66	20.9	

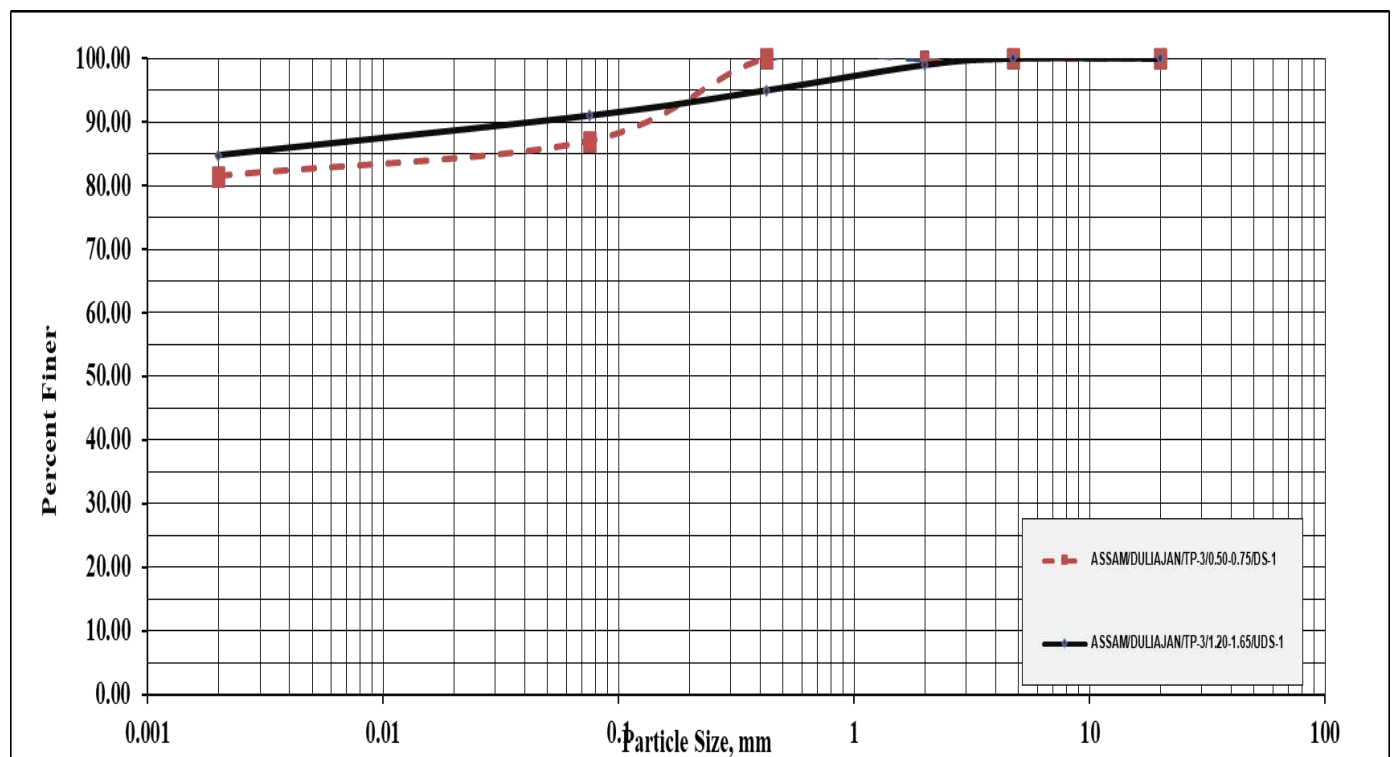


GRAIN SIZE DISTRIBUTION CURVE, TP-2

For Shiva Testing & Research Laboratory Pvt. Ltd.

(Signature)
 Authorized Signatory

TEST ON SOIL SAMPLES																				
TP-3			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729311.888 Y=3029391.037																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	FREE SWELL INDEX
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	
0.50-0.75	DS-1	CI	81.50	5.50	13.00	0.00	0.00	0.00	49.77	17.85	31.92	-	23	-	2.54	-	-	-	-	12
1.20-1.65	UDS-1	CI	84.75	6.25	4.00	4.00	1.00	0.00	48.52	17.23	31.29	1.77	31	1.35	2.56	0.89	TX	0.62	22.8	14

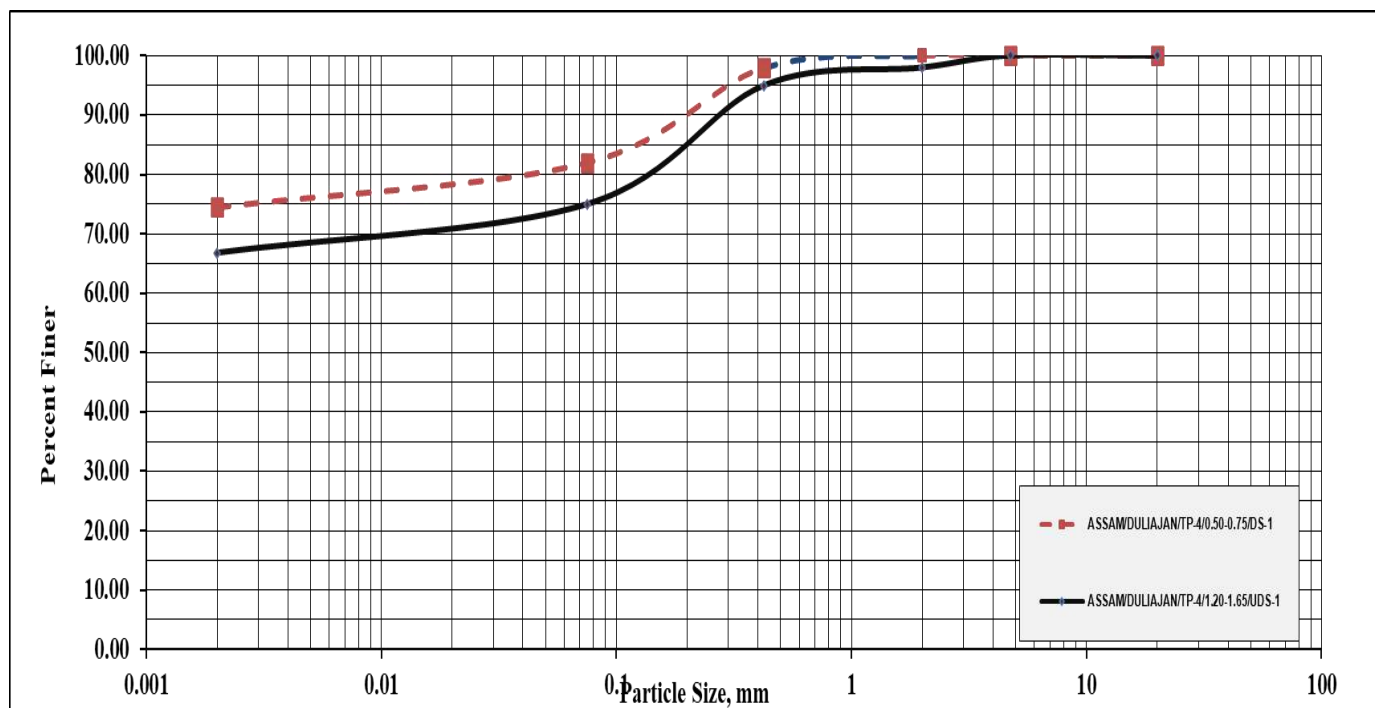


GRAIN SIZE DISTRIBUTION CURVE, TP-3

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

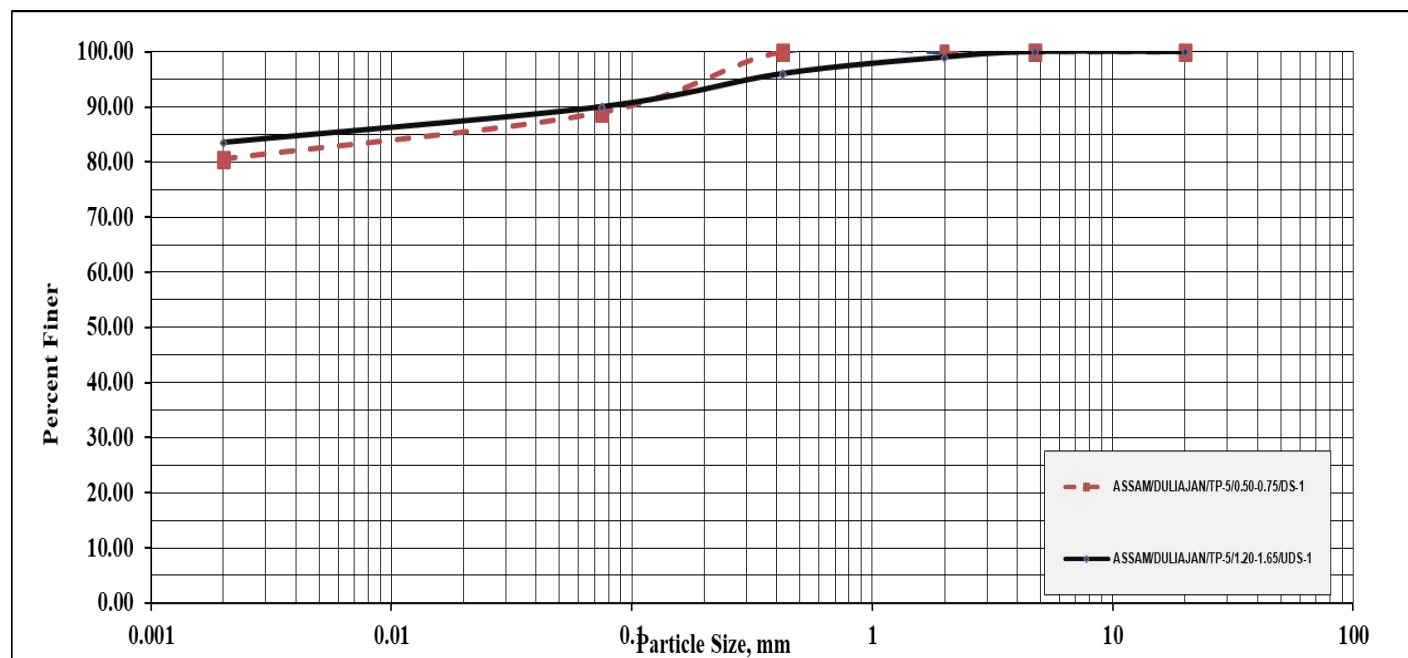
TEST ON SOIL SAMPLES																				
TP-4			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729442.036 Y=3029392.445																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	FREE SWELL INDEX
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	ϕ	%
0.50-0.75	DS-1	CL	74.50	7.50	16.00	2.00	0.00	0.00	32.83	18.05	14.78	-	28	-	2.55	-	-	-	-	11
1.20-1.65	UDS-1	CL	66.80	8.20	20.00	3.00	2.00	0.00	33.28	18.14	15.14	1.75	30	1.35	2.57	0.91	TX	0.59	20.6	13



GRAIN SIZE DISTRIBUTION CURVE, TP-4

For Shiva Testing & Research Laboratory Pvt. Ltd.
 Authorized Signatory

TEST ON SOIL SAMPLES																				
TP-5			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729278.860 Y=3029345.835																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	FREE SWELL INDEX
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm	%	%	%	gm/cc	%	gm/cc	G	e		c	ϕ	%
0.50-0.75	DS-1	CL	80.50	8.50	11.00	0.00	0.00	0.00	33.68	20.02	13.66	-	20	-	2.54	-	-	-	-	12
1.20-1.65	UDS-1	CL	83.50	6.50	6.00	3.00	1.00	0.00	34.40	20.52	13.88	1.76	35	1.30	2.58	0.98	TX	0.61	23.2	15

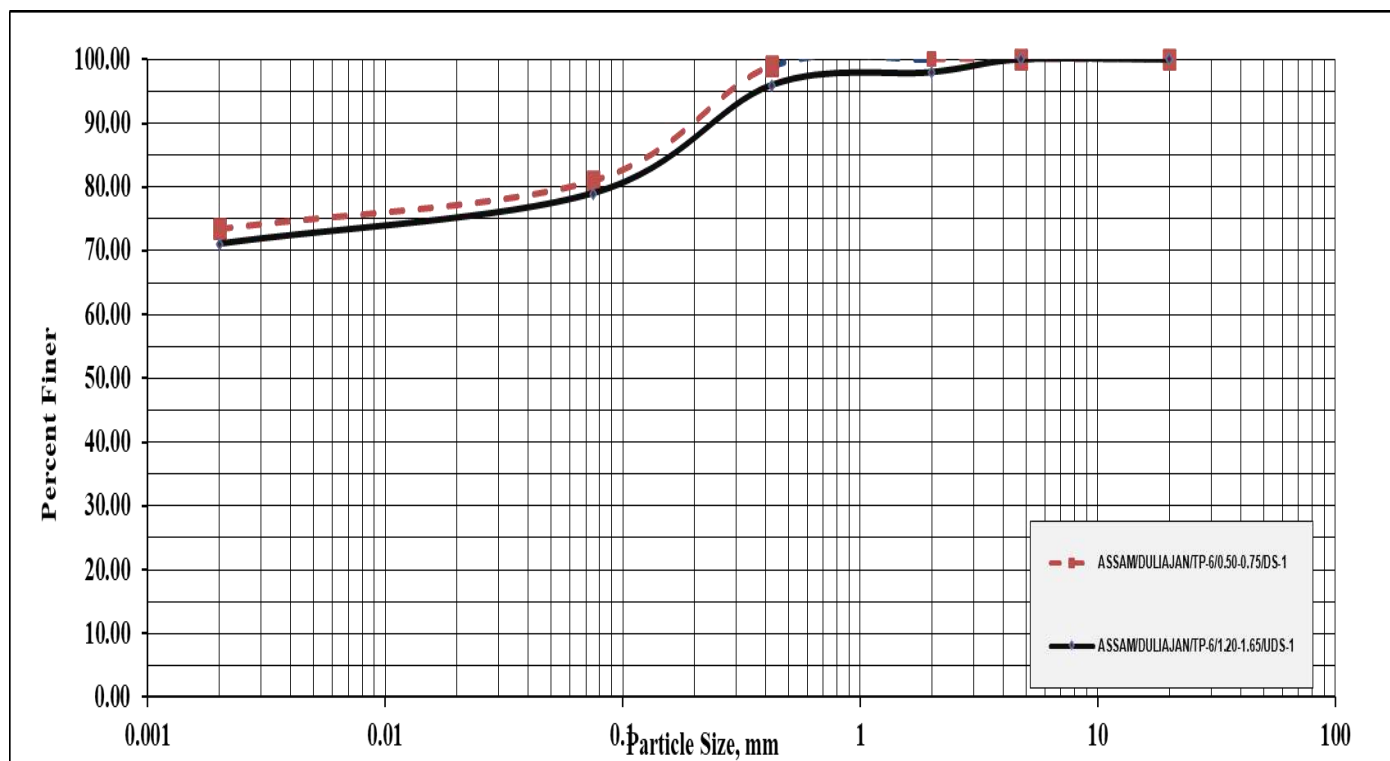


GRAIN SIZE DISTRIBUTION CURVE, TP-5

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

TEST ON SOIL SAMPLES																				
TP-6			GRAIN SIZE ANALYSIS						ATTERBERG LIMITS			DENSITY & MOISTURE CONTENT					SHEAR PARAMETERS			FREE SWELL INDEX
Location	X=729397.529 Y=3029351.158																			
SAMPLE DEPTH IN M	SAMPLE NO.	IS CLASSIFICATION OF SOIL	CLAY	SILT	fine SAND	med SAND	coarse SAND	GRAVEL	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	BULK DENSITY	MOISTURE CONTENT	DRY DENSITY	SPECIFIC GRAVITY	VOID RATIO	TEST TYPE	COHESION, in kg/cm ²	ANGLE OF INTERNAL FRICTION, in Degree	
			<0.002 mm	0.002-0.075	0.075-0.425	0.425-2.00	2.00-4.75	>4.75 mm												
			%	%	%	%	%	%	%	%	%	gm/cc	%	gm/cc	G	e		c	φ	
0.50-0.75	DS-1	CL	73.45	7.55	18.00	1.00	0.00	0.00	33.32	19.27	14.05	-	22	-	2.57	-	-	-	-	
1.20-1.65	UDS-1	CL	71.05	7.95	17.00	2.00	2.00	0.00	32.21	19.05	13.16	1.77	31	1.35	2.58	0.91	TX	0.64	24.1	



GRAIN SIZE DISTRIBUTION CURVE, TP-6

For Shiva Testing & Research Laboratory Pvt. Ltd.
 Authorized Signatory

18.0 CBR TESTS FOR TRIAL-PITS

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

For Trail-Pit-1

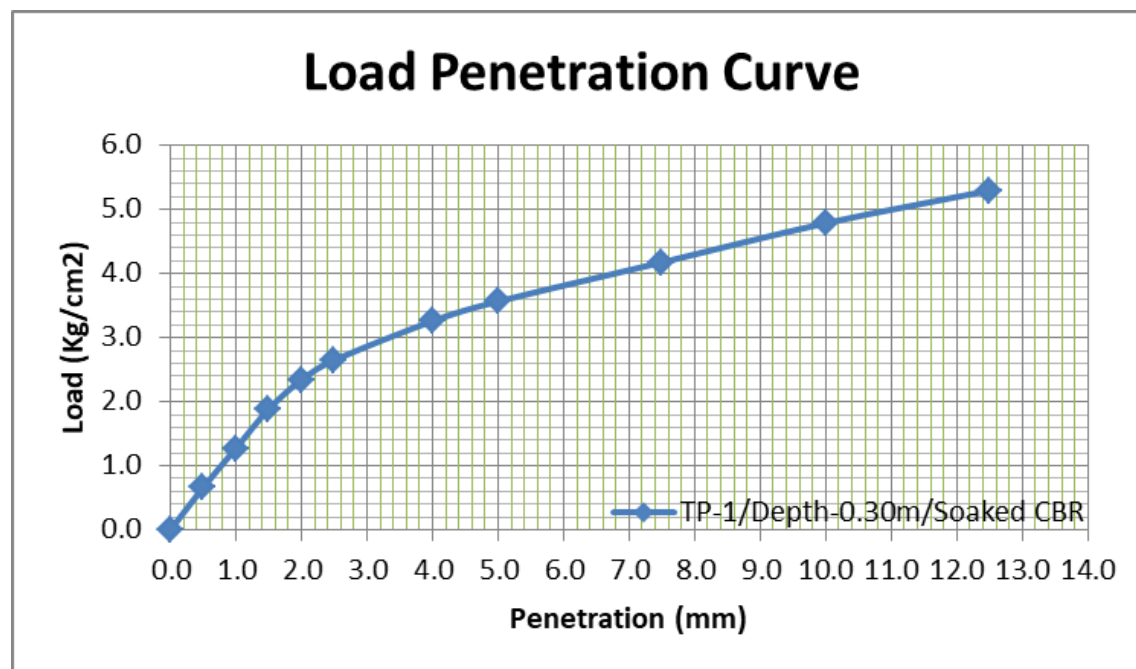
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.13	13	0.66	0.5
0.25	25	1.27	1.0
0.37	37	1.88	1.5
0.46	46	2.34	2.0
0.52	52	2.65	2.5
0.64	64	3.26	4.0
0.70	70	3.57	5.0
0.82	82	4.18	7.5
0.94	94	4.79	10.0
1.04	104	5.30	12.5

CBR at 2.5mm= 3.78

CBR at 5.0mm= 3.40


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

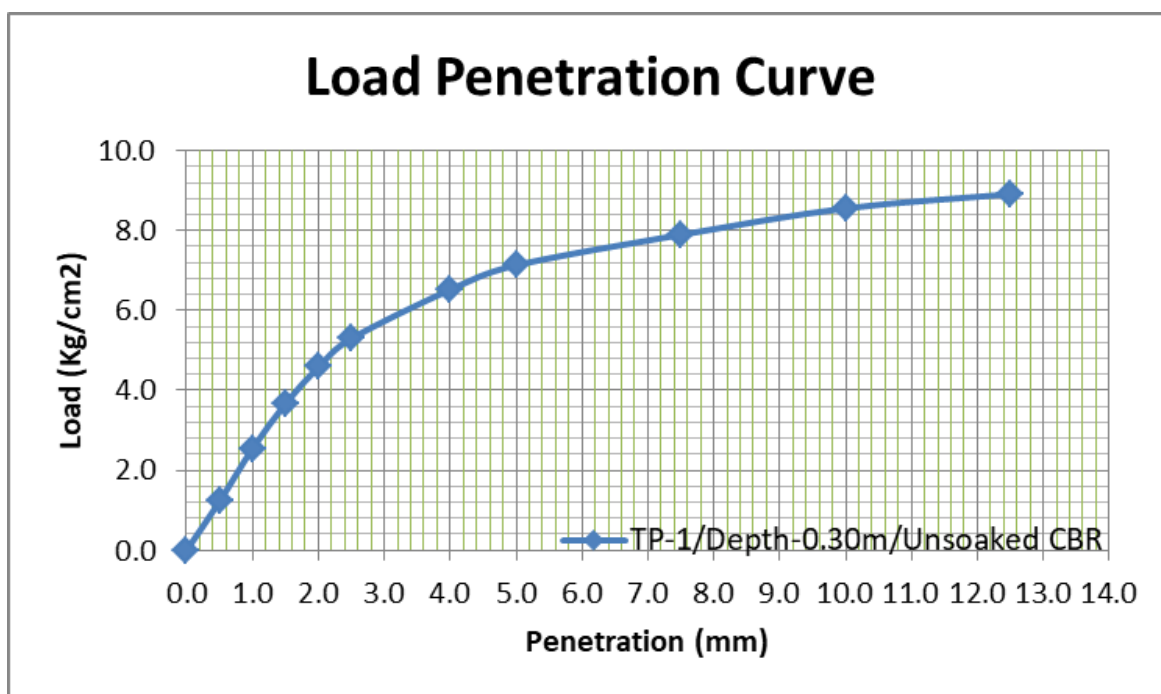
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.24	24	1.22	0.5
0.50	50	2.55	1.0
0.72	72	3.67	1.5
0.90	90	4.58	2.0
1.04	104	5.30	2.5
1.28	128	6.52	4.0
1.40	140	7.13	5.0
1.55	155	7.90	7.5
1.68	168	8.56	10.0
1.75	175	8.91	12.5

CBR at 2.5mm= 7.57

CBR at 5.0mm= 6.79


 For Shiva Testing & Research Laboratory Pvt. Ltd.

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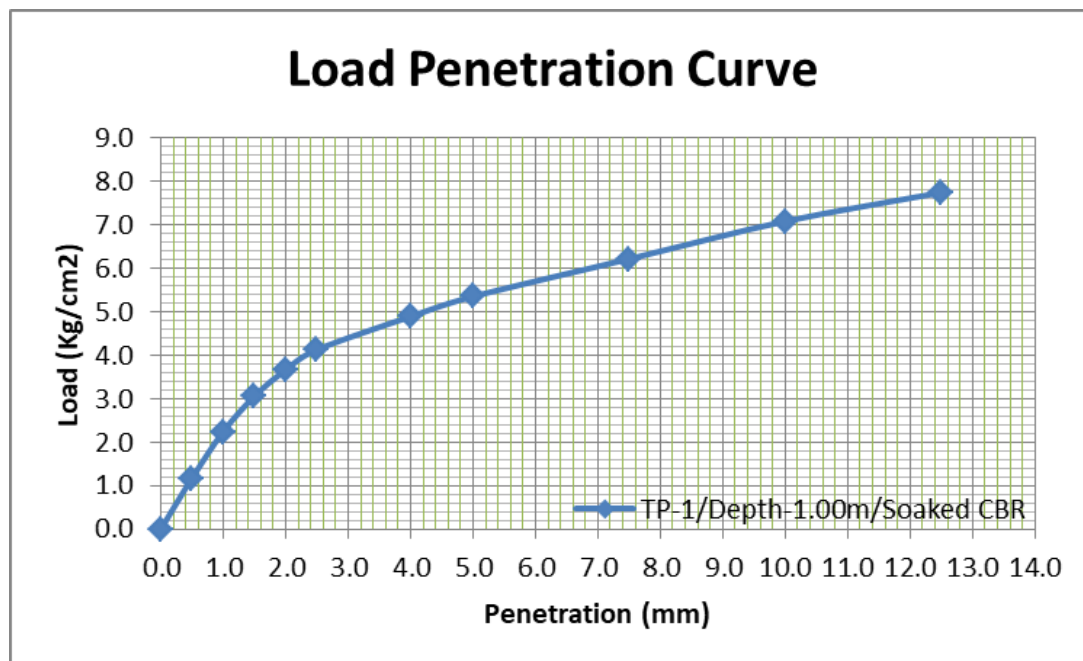
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.23	23	1.17	0.5
0.44	44	2.24	1.0
0.60	60	3.06	1.5
0.72	72	3.67	2.0
0.81	81	4.13	2.5
0.96	96	4.89	4.0
1.05	105	5.35	5.0
1.22	122	6.21	7.5
1.39	139	7.08	10.0
1.52	152	7.74	12.5

CBR at 2.5mm= 5.89

CBR at 5.0mm= 5.09


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Secretary

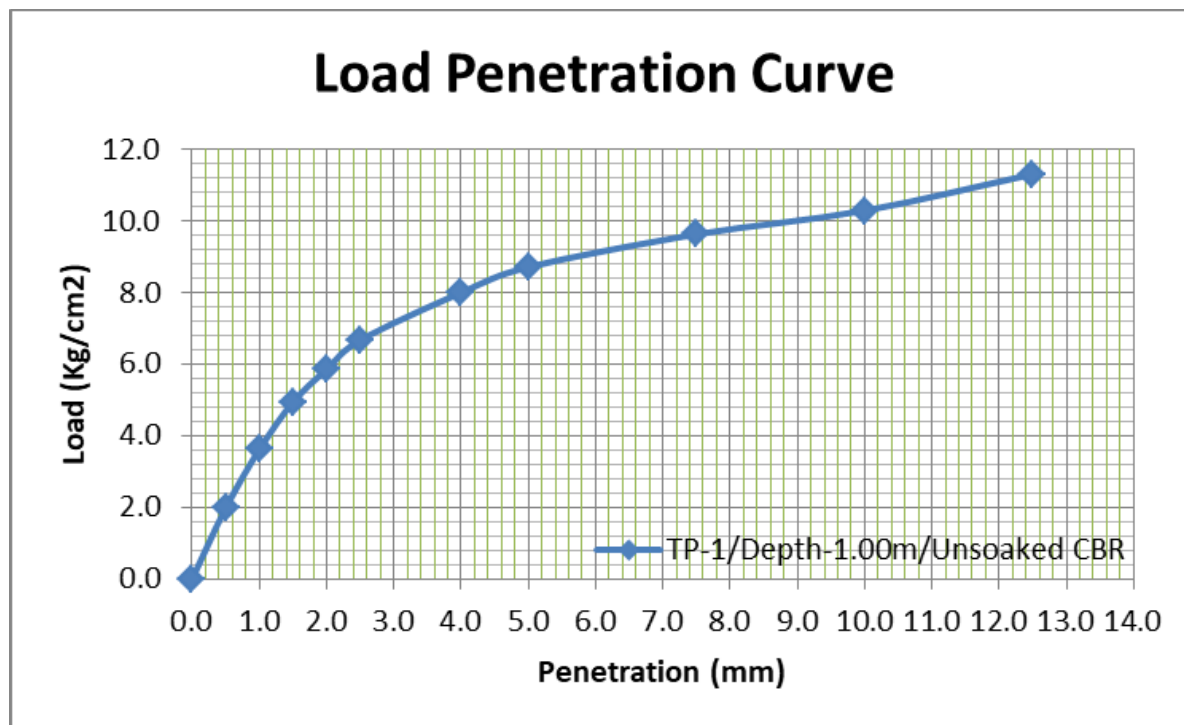
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.39	39	1.99	0.5
0.71	71	3.62	1.0
0.97	97	4.94	1.5
1.15	115	5.86	2.0
1.31	131	6.67	2.5
1.57	157	8.00	4.0
1.71	171	8.71	5.0
1.89	189	9.63	7.5
2.02	202	10.29	10.0
2.22	222	11.31	12.5

CBR at 2.5mm= 9.53

CBR at 5.0mm= 8.30



FOR TRIAL PIT-2

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

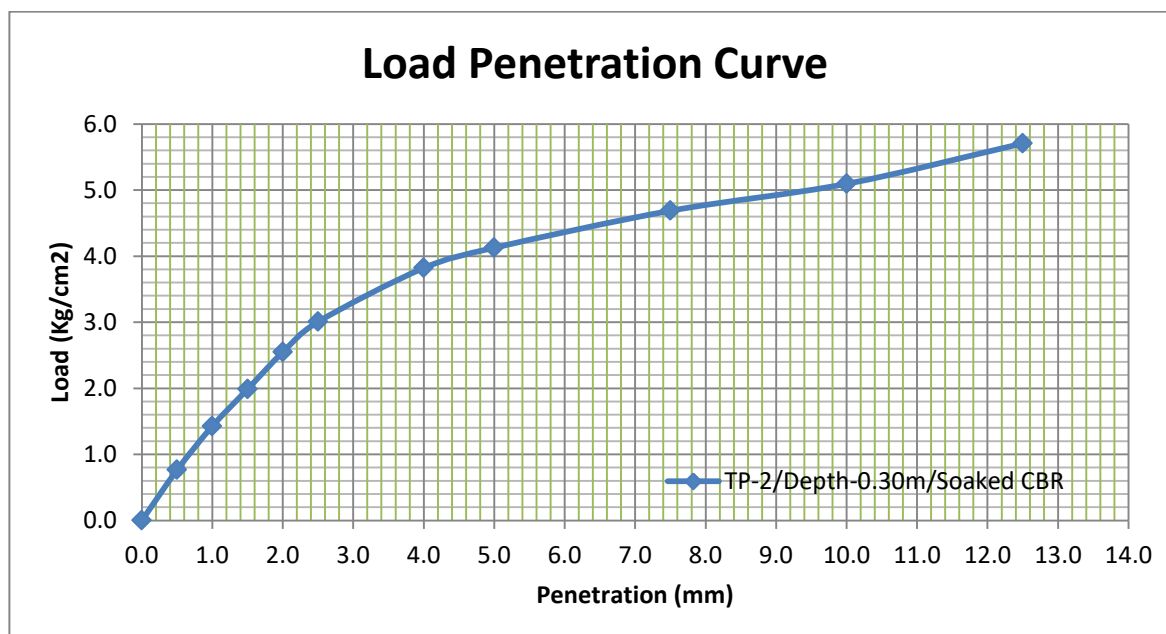
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.15	15	0.76	0.5
0.28	28	1.43	1.0
0.39	39	1.99	1.5
0.50	50	2.55	2.0
0.59	59	3.01	2.5
0.75	75	3.82	4.0
0.81	81	4.13	5.0
0.92	92	4.69	7.5
1.00	100	5.09	10.0
1.12	112	5.71	12.5

CBR at 2.5mm= 4.29

CBR at 5.0mm= 3.93


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

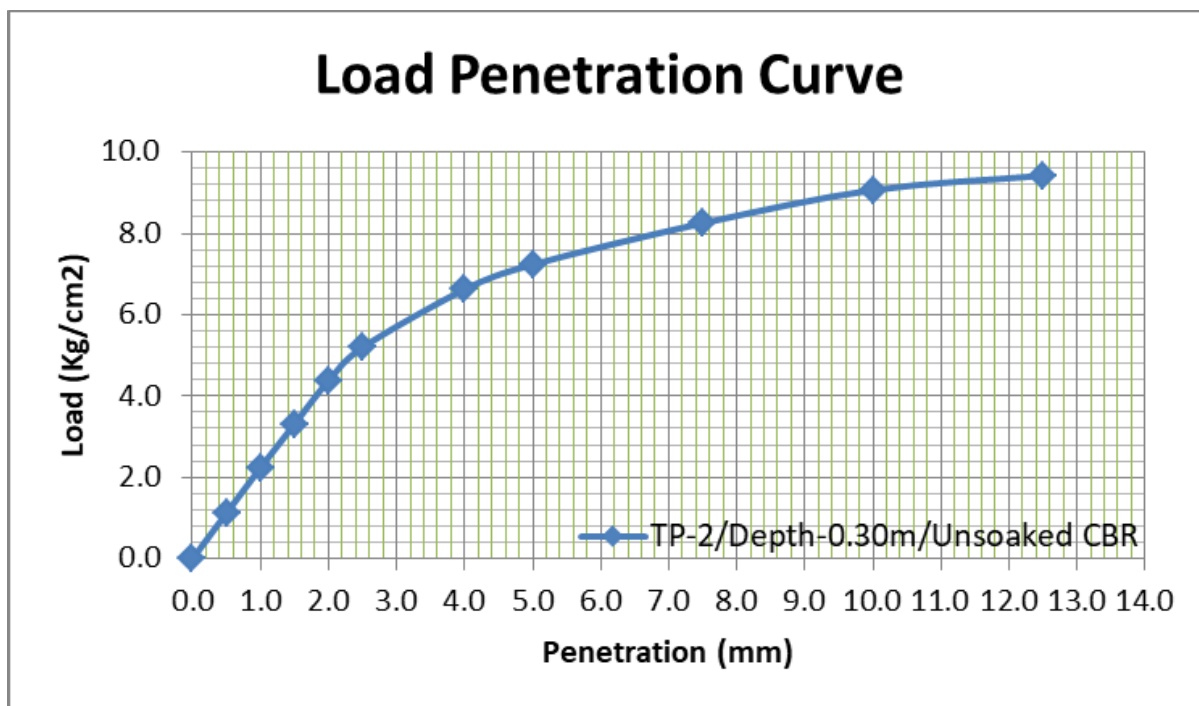
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.22	22	1.12	0.5
0.44	44	2.24	1.0
0.65	65	3.31	1.5
0.86	86	4.38	2.0
1.02	102	5.20	2.5
1.30	130	6.62	4.0
1.42	142	7.23	5.0
1.62	162	8.25	7.5
1.78	178	9.07	10.0
1.85	185	9.42	12.5

CBR at 2.5mm= 7.42

CBR at 5.0mm= 6.89


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

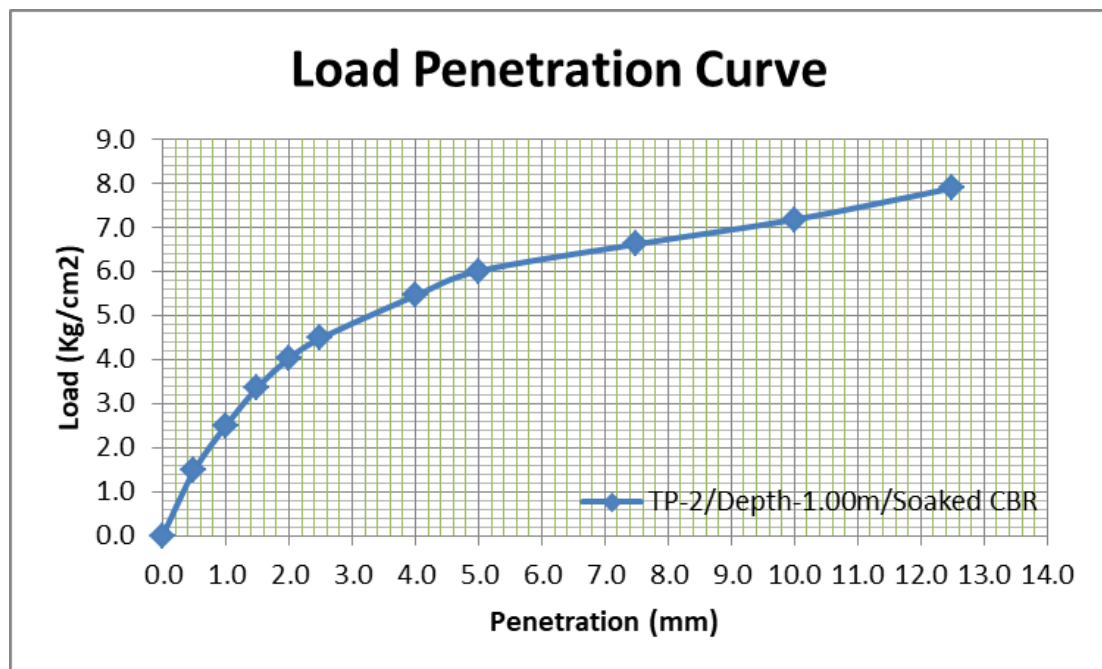
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.29	29	1.48	0.5
0.49	49	2.50	1.0
0.66	66	3.36	1.5
0.79	79	4.02	2.0
0.88	88	4.48	2.5
1.07	107	5.45	4.0
1.18	118	6.01	5.0
1.30	130	6.62	7.5
1.41	141	7.18	10.0
1.55	155	7.90	12.5

CBR at 2.5mm= 6.40

CBR at 5.0mm= 5.72


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

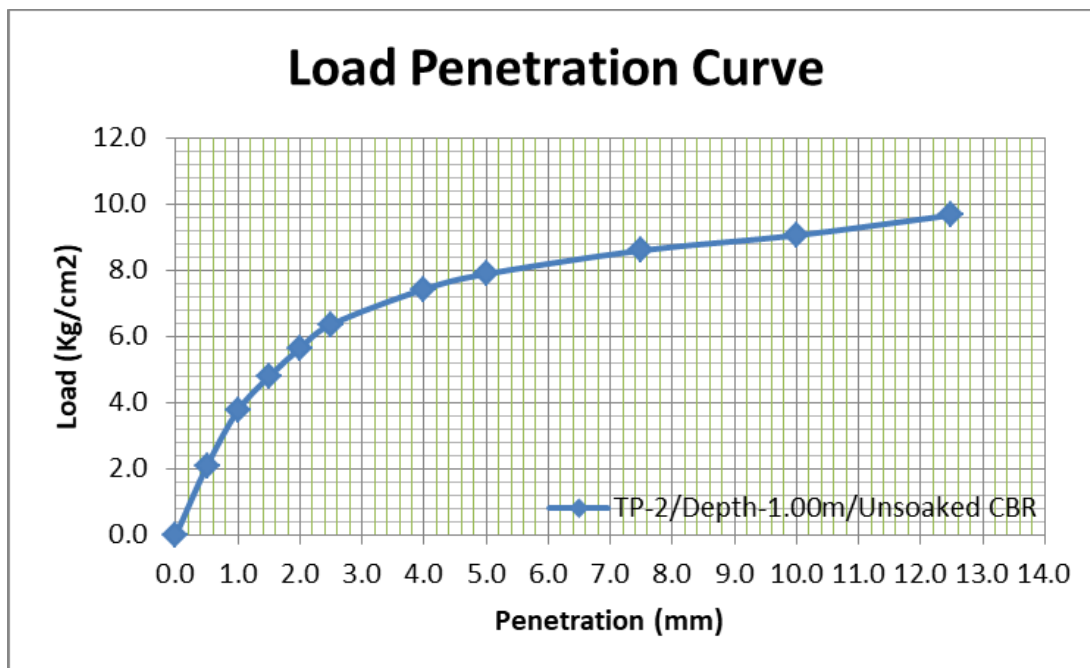
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.41	41	2.09	0.5
0.74	74	3.77	1.0
0.94	94	4.79	1.5
1.11	111	5.65	2.0
1.25	125	6.37	2.5
1.46	146	7.44	4.0
1.55	155	7.90	5.0
1.69	169	8.61	7.5
1.78	178	9.07	10.0
1.90	190	9.68	12.5

CBR at 2.5mm= 9.10

CBR at 5.0mm= 7.52



For Shiva Testing & Research Laboratory Pvt. Ltd.


 Authorized Signatory

FOR TRIAL PIT-3

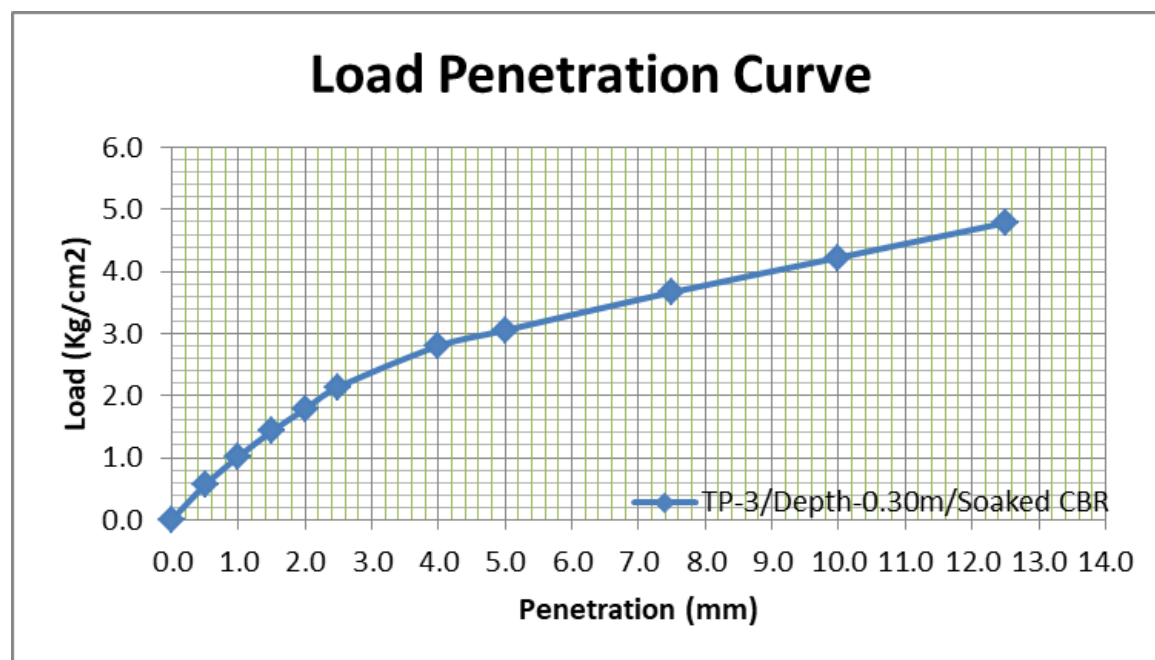
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.11	11	0.56	0.5
0.20	20	1.02	1.0
0.28	28	1.43	1.5
0.35	35	1.78	2.0
0.42	42	2.14	2.5
0.55	55	2.80	4.0
0.60	60	3.06	5.0
0.72	72	3.67	7.5
0.83	83	4.23	10.0
0.94	94	4.79	12.5

CBR at 2.5mm= 3.06

CBR at 5.0mm= 2.91



For Shiva Testing & Research Laboratory Pvt. Ltd.



Authorized Signatory

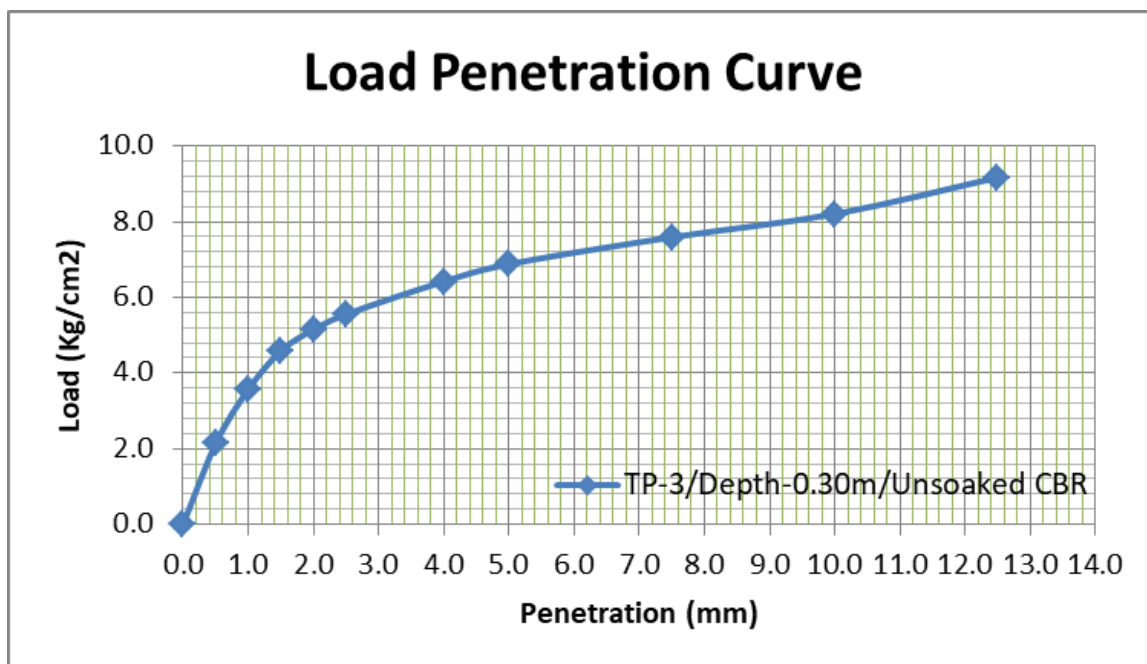
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.42	42	2.14	0.5
0.70	70	3.57	1.0
0.90	90	4.58	1.5
1.01	101	5.15	2.0
1.09	109	5.55	2.5
1.26	126	6.42	4.0
1.35	135	6.88	5.0
1.49	149	7.59	7.5
1.61	161	8.20	10.0
1.80	180	9.17	12.5

CBR at 2.5mm= 7.93

CBR at 5.0mm= 6.55


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

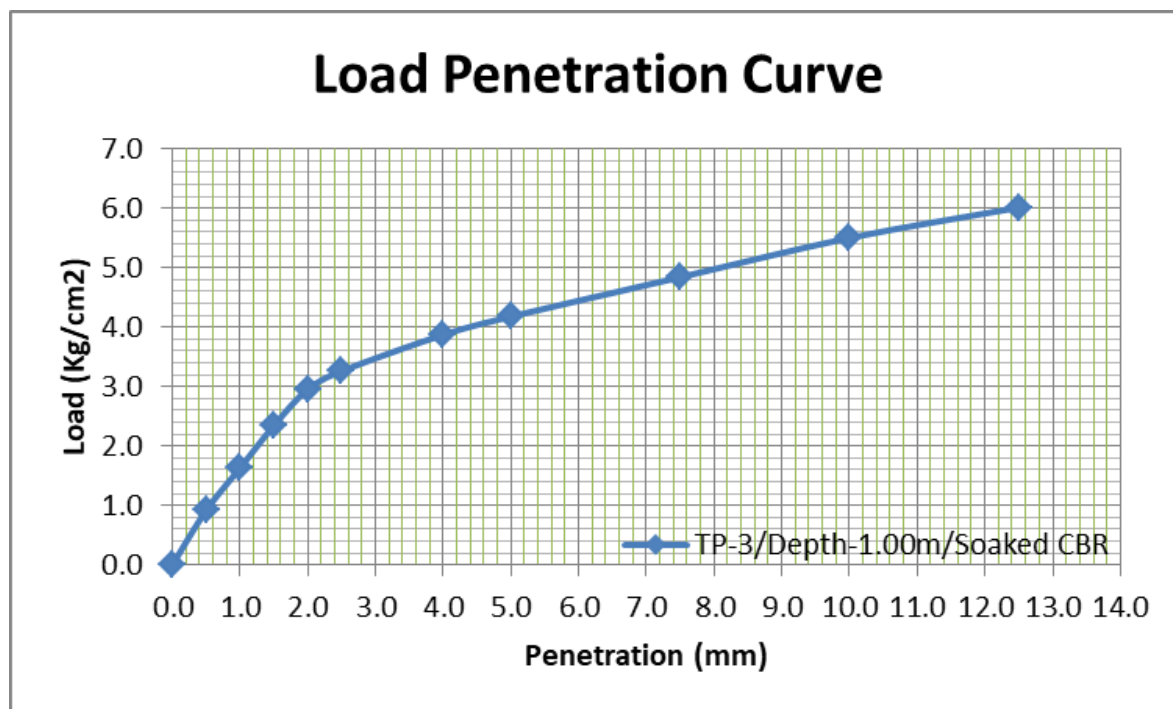
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.18	18	0.92	0.5
0.32	32	1.63	1.0
0.46	46	2.34	1.5
0.58	58	2.95	2.0
0.64	64	3.26	2.5
0.76	76	3.87	4.0
0.82	82	4.18	5.0
0.95	95	4.84	7.5
1.08	108	5.50	10.0
1.18	118	6.01	12.5

CBR at 2.5mm= 4.66

CBR at 5.0mm= 3.98


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

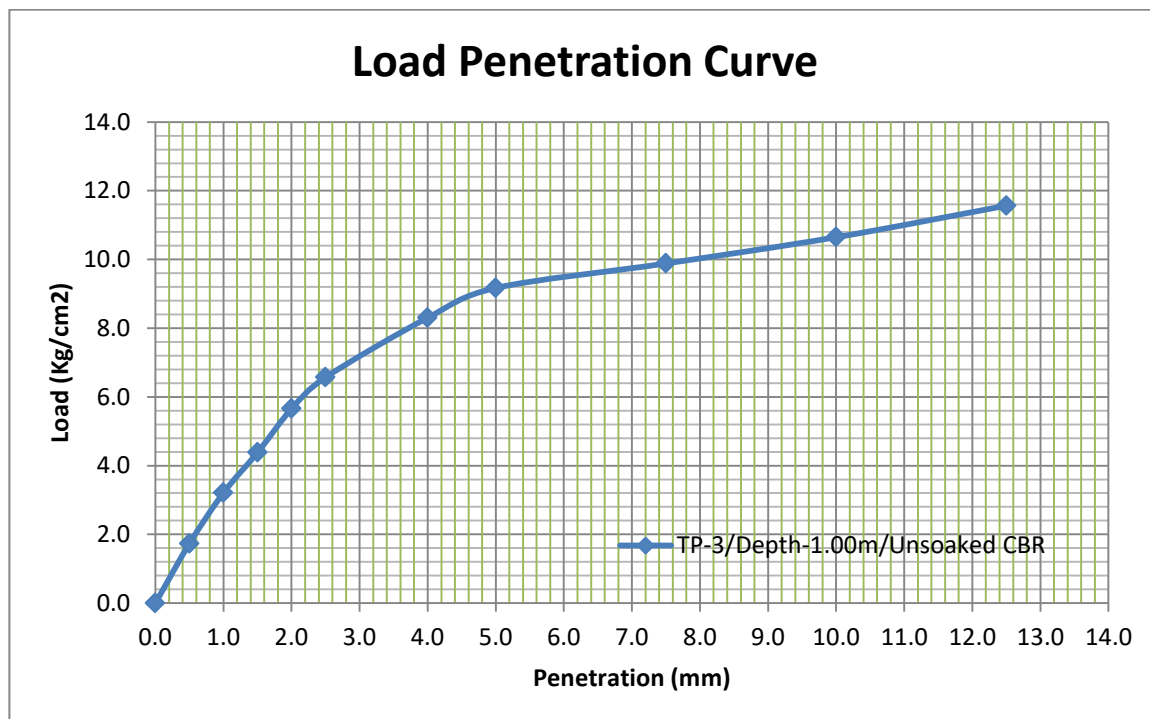
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.34	34	1.73	0.5
0.63	63	3.21	1.0
0.86	86	4.38	1.5
1.11	111	5.65	2.0
1.29	129	6.57	2.5
1.63	163	8.30	4.0
1.80	180	9.17	5.0
1.94	194	9.88	7.5
2.09	209	10.65	10.0
2.27	227	11.56	12.5

CBR at 2.5mm= 9.39

CBR at 5.0mm= 8.73


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

FOR TRIAL PIT-4

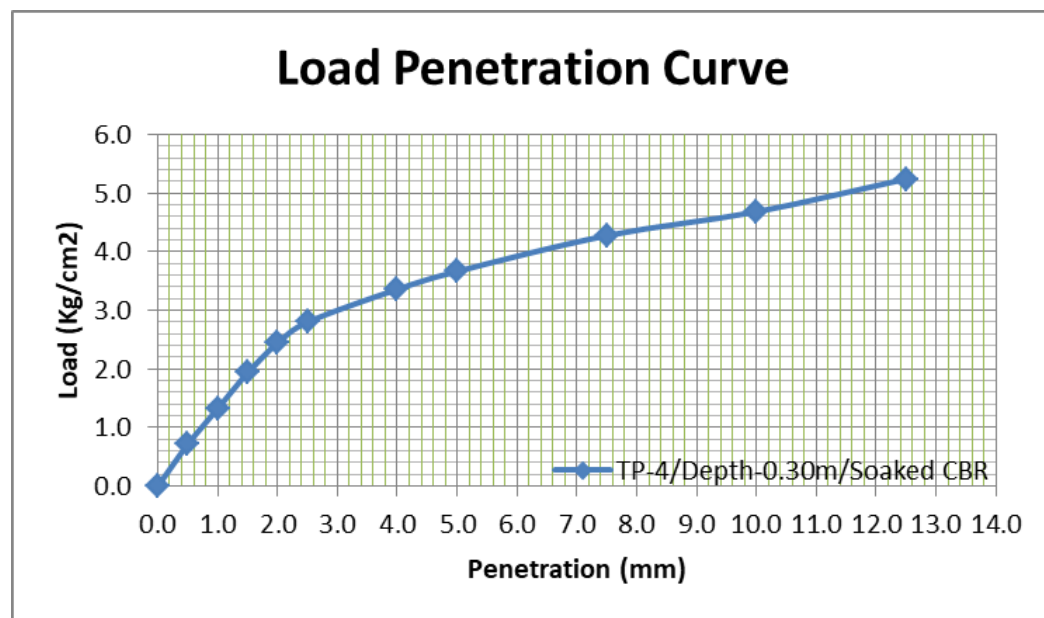
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.14	14	0.71	0.5
0.26	26	1.32	1.0
0.38	38	1.94	1.5
0.48	48	2.45	2.0
0.55	55	2.80	2.5
0.66	66	3.36	4.0
0.72	72	3.67	5.0
0.84	84	4.28	7.5
0.92	92	4.69	10.0
1.03	103	5.25	12.5

CBR at 2.5mm= 4.00

CBR at 5.0mm= 3.49


 For: Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

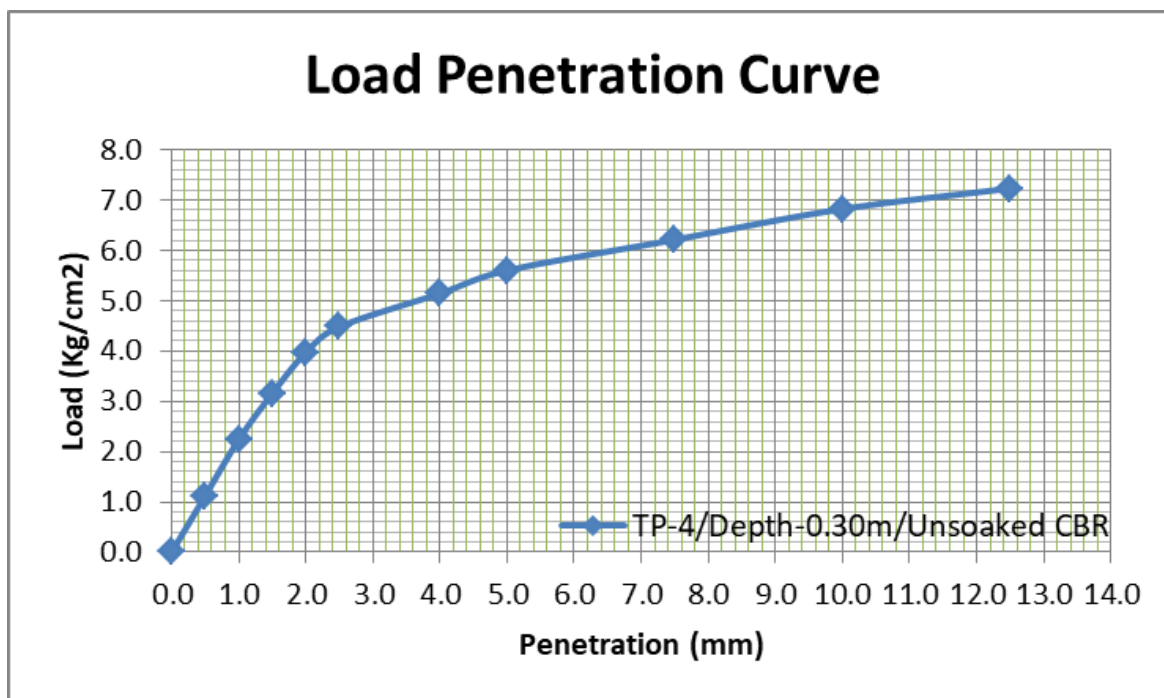
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.22	22	1.12	0.5
0.44	44	2.24	1.0
0.62	62	3.16	1.5
0.78	78	3.97	2.0
0.88	88	4.48	2.5
1.01	101	5.15	4.0
1.10	110	5.60	5.0
1.22	122	6.21	7.5
1.34	134	6.83	10.0
1.42	142	7.23	12.5

CBR at 2.5mm= 6.40

CBR at 5.0mm= 5.34


 For Shiva Testing & Research Laboratory Pvt. Ltd.
 Authorized Signatory

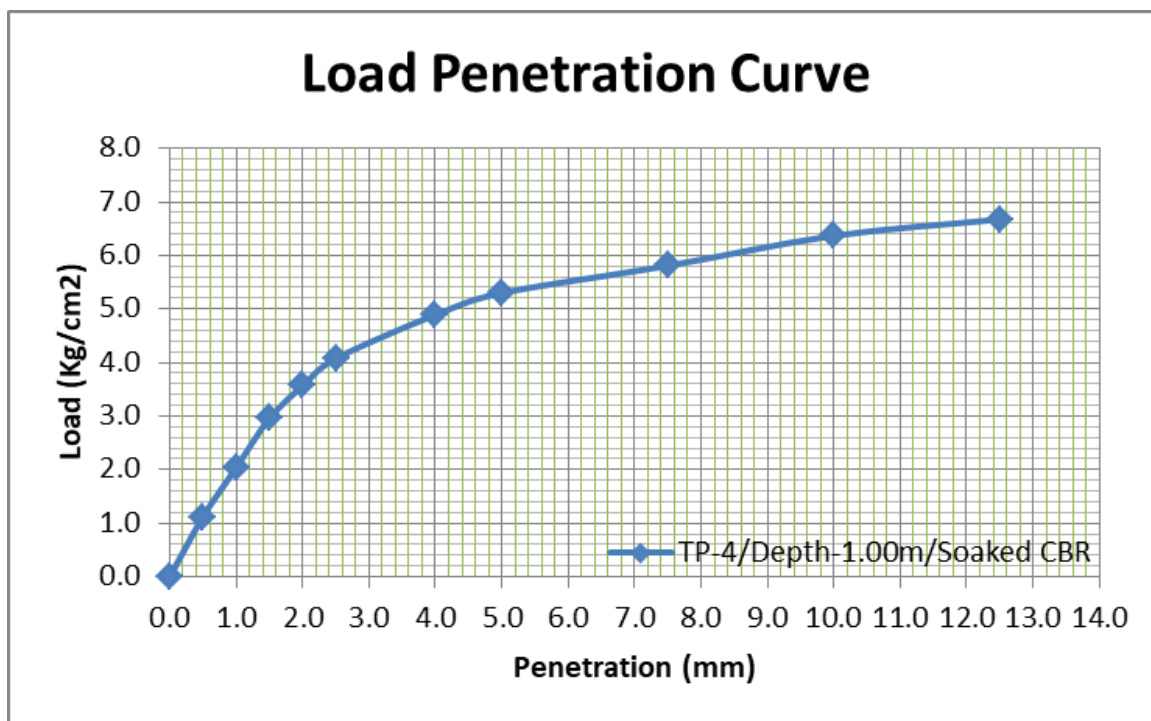
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.22	22	1.12	0.5
0.40	40	2.04	1.0
0.58	58	2.95	1.5
0.70	70	3.57	2.0
0.80	80	4.08	2.5
0.96	96	4.89	4.0
1.04	104	5.30	5.0
1.14	114	5.81	7.5
1.25	125	6.37	10.0
1.31	131	6.67	12.5

CBR at 2.5mm= 5.82

CBR at 5.0mm= 5.05


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

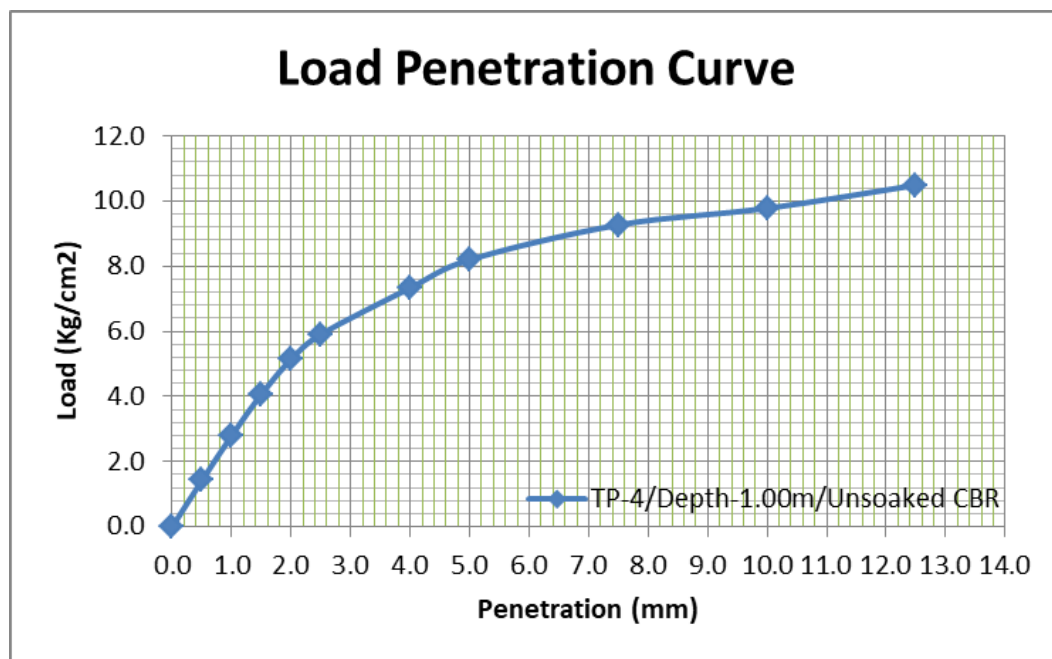
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.28	28	1.43	0.5
0.55	55	2.80	1.0
0.80	80	4.08	1.5
1.01	101	5.15	2.0
1.16	116	5.91	2.5
1.44	144	7.34	4.0
1.61	161	8.20	5.0
1.82	182	9.27	7.5
1.92	192	9.78	10.0
2.06	206	10.49	12.5

CBR at 2.5mm= 8.44

CBR at 5.0mm= 7.81


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

FOR TRIAL PIT-5

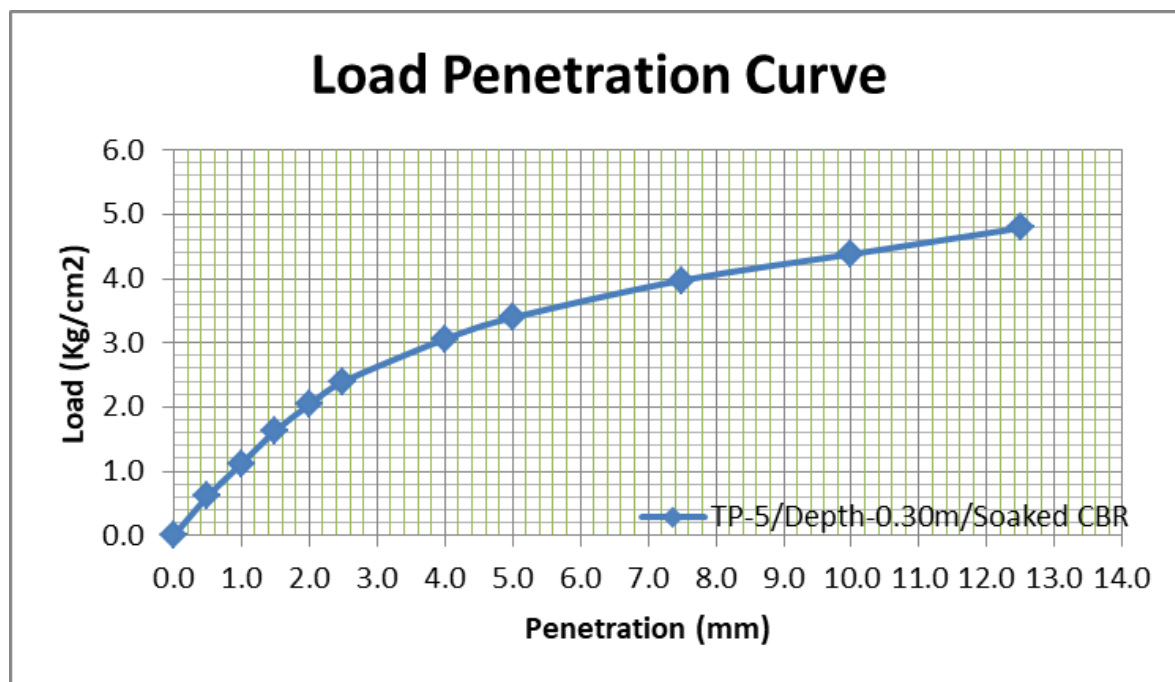
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.12	12	0.61	0.5
0.22	22	1.12	1.0
0.32	32	1.63	1.5
0.40	40	2.04	2.0
0.47	47	2.39	2.5
0.60	60	3.06	4.0
0.67	66.6	3.39	5.0
0.78	78	3.97	7.5
0.86	86	4.38	10.0
0.94	94	4.79	12.5

CBR at 2.5mm= 3.42

CBR at 5.0mm= 3.23


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signature

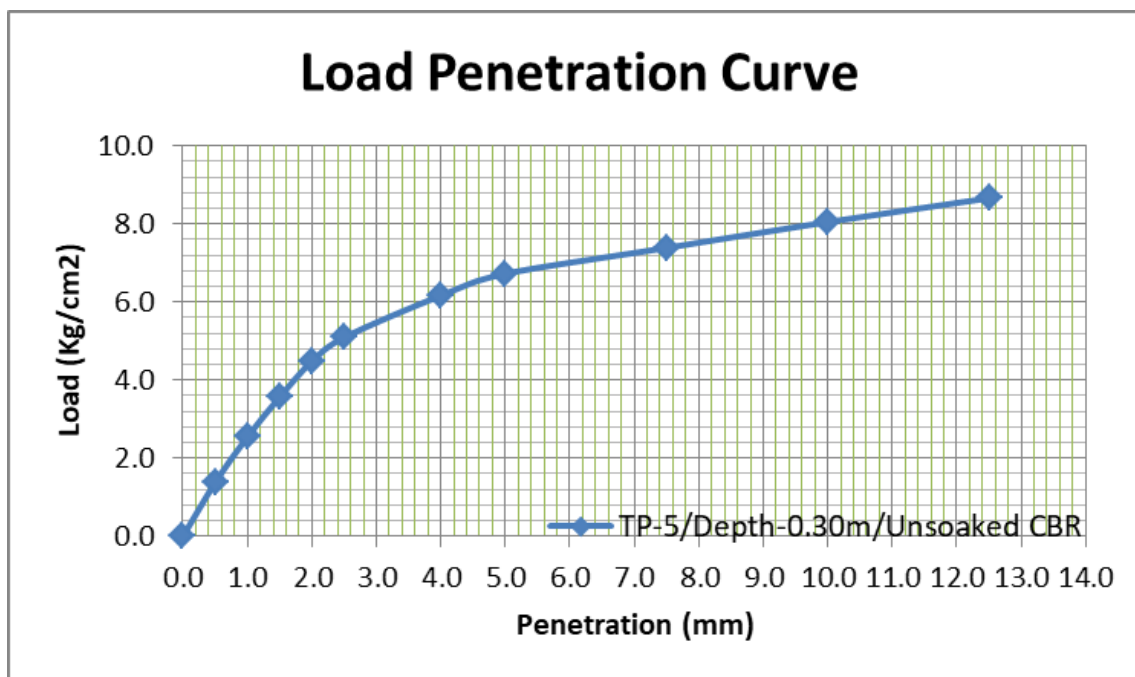
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.27	27	1.38	0.5
0.50	50	2.55	1.0
0.70	70	3.57	1.5
0.88	88	4.48	2.0
1.00	100	5.09	2.5
1.21	121	6.16	4.0
1.32	132	6.72	5.0
1.45	145	7.39	7.5
1.58	158	8.05	10.0
1.70	170	8.66	12.5

CBR at 2.5mm= 7.28

CBR at 5.0mm= 6.40


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

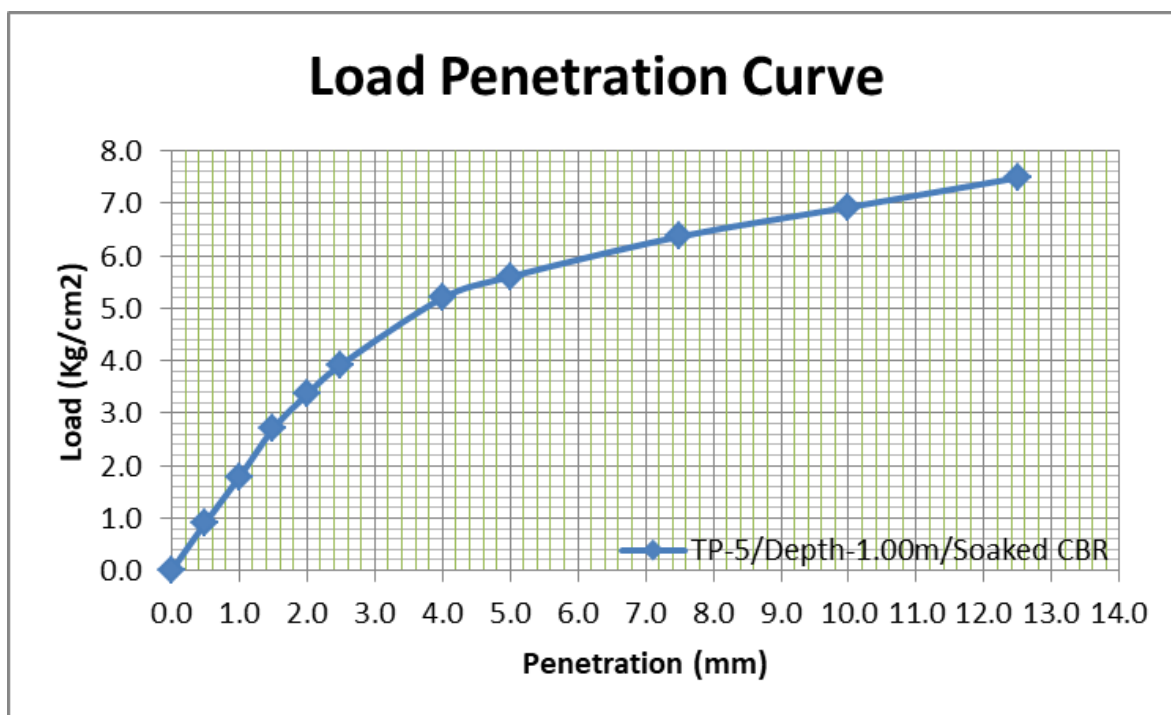
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.18	18	0.92	0.5
0.35	35	1.78	1.0
0.53	53	2.70	1.5
0.66	66	3.36	2.0
0.77	77	3.92	2.5
1.02	102	5.20	4.0
1.10	110	5.60	5.0
1.25	125	6.37	7.5
1.36	136	6.93	10.0
1.47	147	7.49	12.5

CBR at 2.5mm= 5.60

CBR at 5.0mm= 5.34


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

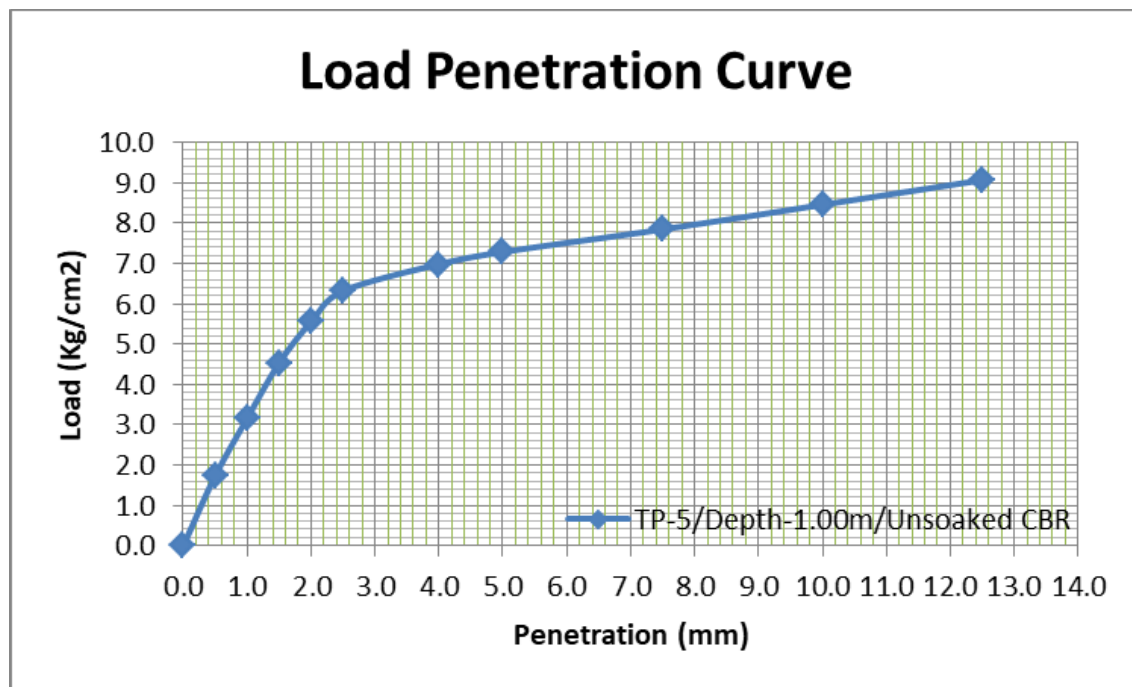
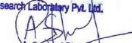
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.34	34	1.73	0.5
0.62	62	3.16	1.0
0.89	89	4.53	1.5
1.09	109	5.55	2.0
1.24	124	6.32	2.5
1.37	137	6.98	4.0
1.43	143	7.28	5.0
1.54	154	7.85	7.5
1.66	166	8.46	10.0
1.78	178	9.07	12.5

CBR at 2.5mm= 9.02

CBR at 5.0mm= 6.94


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

FOR TRIAL PIT-6

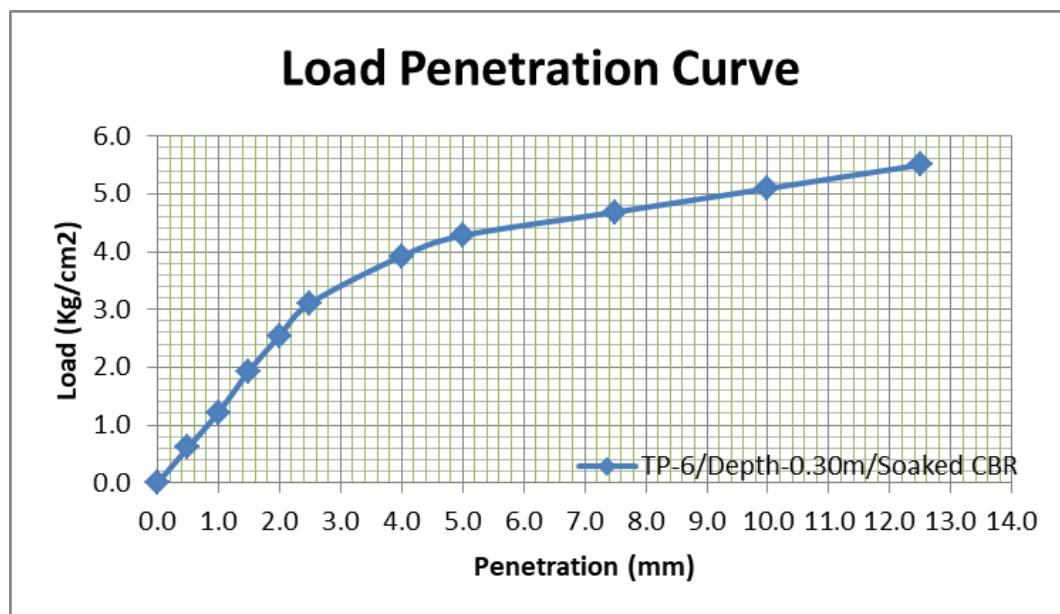
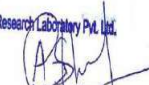
4 days Soaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.12	12	0.61	0.5
0.24	24	1.22	1.0
0.38	38	1.94	1.5
0.50	50	2.55	2.0
0.61	61	3.11	2.5
0.77	77	3.92	4.0
0.84	84	4.28	5.0
0.92	92	4.69	7.5
1.00	100	5.09	10.0
1.08	108	5.50	12.5

CBR at 2.5mm= 4.44

CBR at 5.0mm= 4.08


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

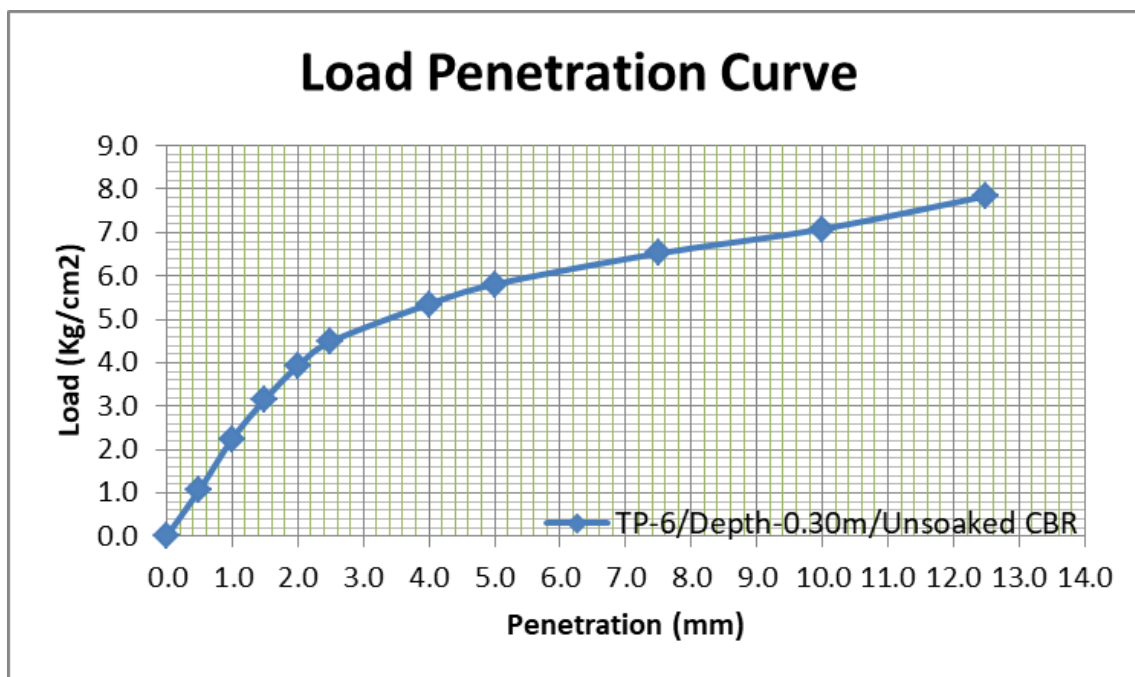
Unsoaked

Location= Trial Pit Depth 0.30m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.21	21	1.07	0.5
0.44	44	2.24	1.0
0.62	62	3.16	1.5
0.77	77	3.92	2.0
0.88	88	4.48	2.5
1.05	105	5.35	4.0
1.14	114	5.81	5.0
1.28	128	6.52	7.5
1.39	139	7.08	10.0
1.54	154	7.85	12.5

CBR at 2.5mm= 6.40

CBR at 5.0mm= 5.53


 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

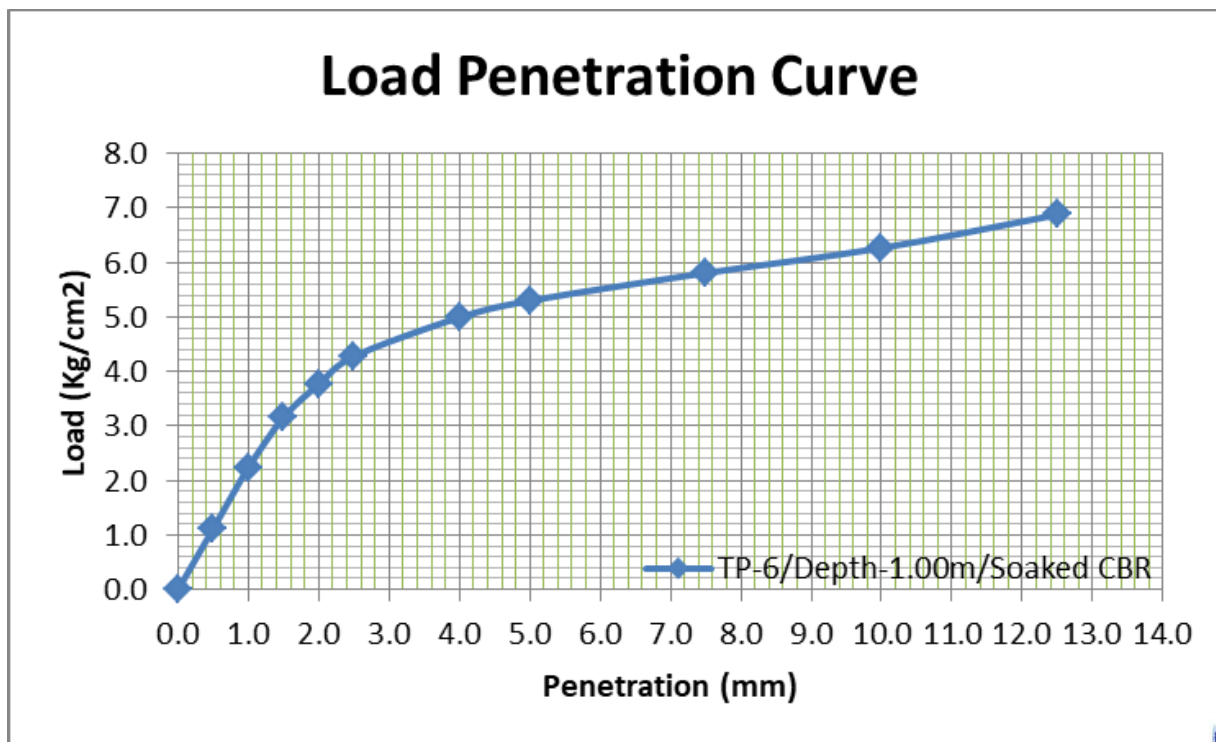
4 days Soaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.22	22	1.12	0.5
0.44	44	2.24	1.0
0.62	62	3.16	1.5
0.74	74	3.77	2.0
0.84	84	4.28	2.5
0.98	98	4.99	4.0
1.04	104	5.30	5.0
1.14	114	5.81	7.5
1.23	123	6.27	10.0
1.35	135	6.88	12.5

CBR at 2.5mm= 6.11

CBR at 5.0mm= 5.05



For Shiva Testing & Research Laboratory Pvt. Ltd.
 Authority Signature

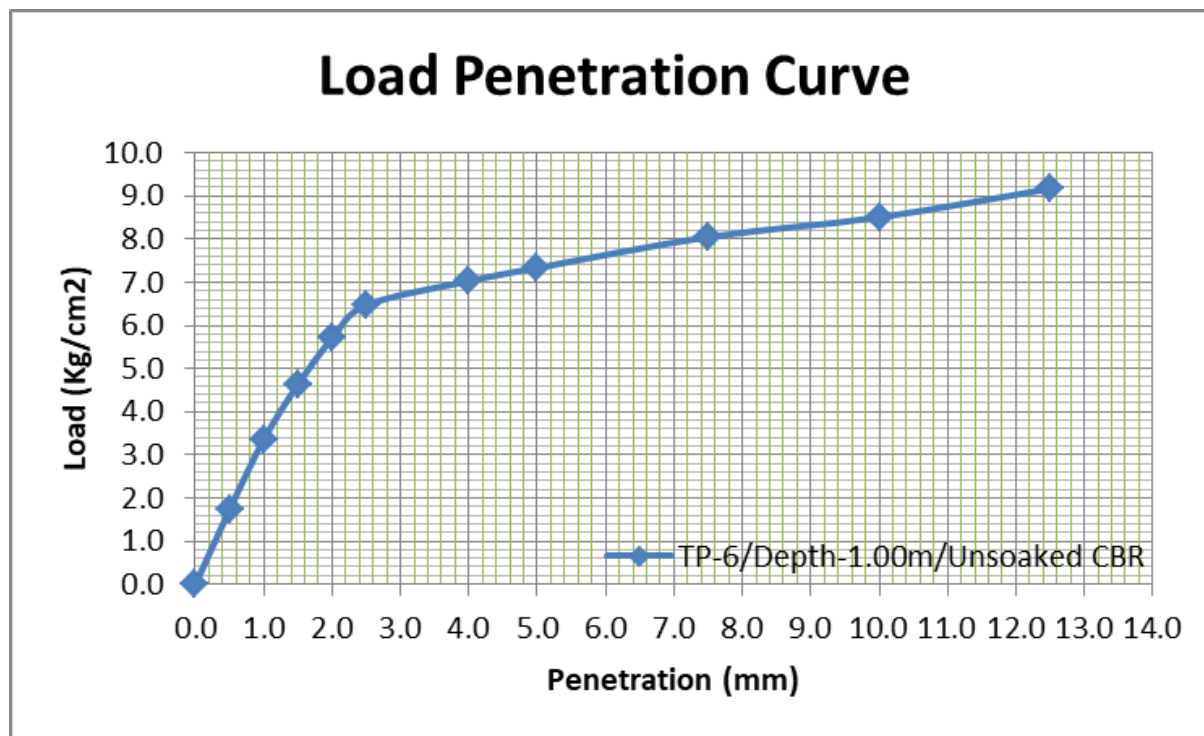
Unsoaked

Location= Trial Pit Depth 1.00m

Load (KN)	Load(Kg)	Load (Kg/cm ²)	Displacement (mm)
0.00	0	0.0	0.0
0.34	34	1.73	0.5
0.66	66	3.36	1.0
0.91	91	4.64	1.5
1.12	112	5.71	2.0
1.27	127	6.47	2.5
1.38	138	7.03	4.0
1.44	144	7.34	5.0
1.58	158	8.05	7.5
1.67	167	8.51	10.0
1.80	180	9.17	12.5

CBR at 2.5mm= 9.24

CBR at 5.0mm= 6.99


 For Shiva Testing & Research Laboratory Pvt. Ltd.
 Authorized Signature

SUMMARY OF FINAL CBR VALUE			
SAMPLE DETAILS		CBR (%)	
TRIAL PIT	DEPTH (M)	4 days Soaked	UNSOAKED
TP-1	0.30	3.78	7.57
	1.00	5.89	9.53
TP-2	0.30	4.29	7.42
	1.00	6.40	9.10
TP-3	0.30	3.06	7.93
	1.00	4.66	9.39
TP-4	0.30	4.00	6.40
	1.00	5.82	8.44
TP-5	0.30	3.42	7.28
	1.00	5.60	9.02
TP-6	0.30	4.44	6.40
	1.00	6.11	9.24

For Shiva Testing & Research Laboratory Pvt. Ltd.



Authorized Signatory

19.0 DESIGN OF SHALLOW FOUNDATION

For: Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

19.1 COMMUNITY CENTRE (BH-4)

SIZE OF FOUNDATION 2M X 2M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-4					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	4	4	4
SPT N Value at end of compressible zone		N	18	18	18
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	2	2	2
Width of Foundation	m	B	2	2	2
Saturated density of soil	kg/cm ³	γ_{sat}	0.00163	0.00163	0.00174
Submerged density of soil	kg/cm ³	γ'	0.00063	0.00063	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.2445	0.489	0.783
	kg/cm ²	$B \times \gamma$	0.126	0.126	0.148
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.64	0.64	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	30.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	21.51
Initial void ratio		e ₀	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	15.00	30.00	58.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
N ϕ			1.000	1.000	3.059
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	16.608
		Nq'	1.000	1.000	7.686
		N γ'	0.000	0.000	7.048
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		S γ	0.8	0.8	0.8
Depth Factors		Dc	1.150	1.300	1.787
		Dq	1.000	1.000	1.394
		D γ	1.000	1.000	1.394
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	3.295	3.725	9.045
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd'/FOS)	13.180	14.900	36.180
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index Cc=0.3*(e ₀ -0.27)		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	1.890	2.363	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.224	3.967	14.500
Consolidation Settlement	mm	sc	68.081	67.162	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔP_c	0.6	2.45	14.5
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.017	0.017	0.017
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	Su'=Su/W'	0.034	0.034	0.034
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	Sg=Su'* ΔP_c /10	2.04	8.33	49.3
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.8	0.69	0.62
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	Sc=sc* λ *Def*Rf	48.335	47.721	49.300
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	3.75	7.5	14.5
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	3.750	7.500	14.500

SIZE OF FOUNDATION 3M X 3M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-4

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	4	4	4
SPT N Value at end of compressible zone		N	18	18	18
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	3	3	3
Width of Foundation	m	B	3	3	3
Saturated density of soil	kg/cm ³	γ_{sat}	0.00163	0.00163	0.00174
Submerged density of soil	kg/cm ³	γ'	0.00063	0.00063	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.2445	0.489	0.783
	kg/cm ²	$B \times \gamma$	0.189	0.189	0.222
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.64	0.64	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	30.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	21.51
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	23.00	53.00	125.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
N ϕ			1.000	1.000	3.059
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	16.608
		Nq'	1.000	1.000	7.686
		Ny'	0.000	0.000	7.048
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.100	1.200	1.525
		Dq	1.000	1.000	1.262
		Dy	1.000	1.000	1.262
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times Ny' \times Sy \times Dy \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	3.152	3.438	8.325
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	12.610	13.750	33.300

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P0	1.890	2.363	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.136	3.769	13.889
Consolidation Settlement	mm	sc	64.143	64.996	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔPc	0.639	2.62	13.889
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.0174	0.0174	0.0174
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	$Su'=Su/W'$	0.0348	0.0348	0.0348
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	$Sg=Su' \times \Delta Pc / 10$	2.223	9.118	48.333
Corrected Settlement Total					
If clay layer is resting on cohesionless soil layer then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.85	0.73	0.63
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	48.567	49.447	48.333
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	2.56	5.89	13.89
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	2.560	5.890	13.890

SIZE OF FOUNDATION 4M X 4M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-4					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	4	4	4
SPT N Value at end of compressible zone		N	18	18	18
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	4	4	4
Width of Foundation	m	B	4	4	4
Saturated density of soil	kg/cm ³	γ_{sat}	0.00163	0.00163	0.00174
Submerged density of soil	kg/cm ³	γ'	0.00063	0.00063	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.245	0.489	0.783
	kg/cm ²	$B \times \gamma$	0.252	0.252	0.296
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.64	0.64	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	30.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	21.51
Initial void ratio		e ₀	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	33.00	78.00	210.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
N ϕ			1.000	1.000	3.059
Bearing Capacity factors for local shear failure		N _{c'}	5.140	5.140	16.608
		N _{q'}	1.000	1.000	7.686
		N _{y'}	0.000	0.000	7.048
Shape Factors		S _c	1.3	1.3	1.3
		S _q	1.2	1.2	1.2
		S _y	0.8	0.8	0.8
Depth Factors		D _c	1.075	1.150	1.394
		D _q	1.000	1.000	1.197
		D _y	1.000	1.000	1.197
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times N_{c'} \times S_c \times D_c + q(N_{q'} - 1) \times S_q \times D_q + 0.5 \times B \times N_{y'} \times S_y \times D_y \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Q _{d'}	3.080	3.295	8.018
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Q _d /FOS)	12.320	13.180	32.070
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index C _c =0.3*(e ₀ -0.27)		C _c	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	1.890	2.363	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.091	3.457	13.125
Consolidation Settlement	mm	s _c	62.106	61.438	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔP_c	0.673	2.58	13.125
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	S _u	0.019	0.019	0.019
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	S _{u'} =S _u /W'	0.038	0.038	0.038
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	S _g =S _{u'} * ΔP_c /10	2.532	9.701	49.35
Corrected Settlement Total					
If clay layer is resting on cohesionless soil layer then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.89	0.77	0.74
Correction factor for rigidity		R _f	1	1	1
Consolidation Settlement after correction	mm	S _c =s _c * λ *Def*R _f	49.515	49.912	49.350
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	S _a	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	2.06	4.88	13.13
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	2.060	4.880	13.130

19.2 GUEST HOUSE (BH-8)

SIZE OF FOUNDATION 2M X 2M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	8	8	8
SPT N Value		N	8	20	30
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	2	2	2
Width of Foundation	m	B	2	2	2
Saturated density of soil	kg/cm ³	γ_{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.243	0.486	0.729
	kg/cm ²	$B \times \gamma$	0.124	0.124	0.124
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	0.00
Initial void ratio		e ₀	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	14.00	25.00	40.00
N ϕ			1.000	1.000	1.000
Bearing Capacity factors for local shear failure		N _{c'}	5.140	5.140	5.140
		N _{q'}	1.000	1.000	1.000
		N _{y'}	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.150	1.300	1.450
		Dq	1.000	1.000	1.000
		Dy	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C' \cdot N_{c'} \cdot Sc \cdot Dc + q(N_q \cdot 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N_{y'} \cdot Sy \cdot Dy \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	3.141	3.550	3.960
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd'/FOS)	12.560	14.200	15.840
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Safe Bearing Capacity Based on Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	2xB	4	4	4
Compression Index Cc=0.3*(e ₀ -0.27)		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	2.170	3.100	4.030
Additional pressure intensity 2:1 method	T/m ²	ΔP	0.875	1.563	2.500
Consolidation Settlement	mm	sc	61.567	74.175	87.713
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	1	1	1
Correction factor for depth		Def	0.78	0.65	0.55
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \cdot \lambda \cdot Def \cdot Rf$	48.022	48.214	48.242
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/(B*L)	3.5	6.25	10
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	3.500	6.250	10.000

SIZE OF FOUNDATION 3M X 3M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation1	Calculation2	Calculation3
Bore Hole Number		BH	8	8	8
SPT N Value		N	8	20	30
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	3	3	3
Width of Foundation	m	B	3	3	3
Saturated density of soil	kg/cm ³	γ_{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.243	0.486	0.729
	kg/cm ²	$B \times \gamma$	0.186	0.186	0.186
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	0.00
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	23.00	34.00	54.00
N ϕ			1.000	1.000	1.000
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140
		Nq'	1.000	1.000	1.000
		N γ'	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		S γ	0.8	0.8	0.8
Depth Factors		Dc	1.100	1.200	1.300
		Dq	1.000	1.000	1.000
		D γ	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	3.004	3.277
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd'/FOS)	12.020	13.110

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

Safe Bearing Capacity Based on Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	2xB	6	6	6
Compression Index Cc=0.3*(e0-0.27)		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	2.790	3.720	4.650
Additional pressure intensity 2:1 method	T/m ²	ΔP	0.639	0.944	1.500
Consolidation Settlement	mm	sc	56.209	61.675	76.216
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85					
		λ	1	1	1
Correction factor for depth		Def	0.86	0.8	0.65
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	48.340	49.340	49.540
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/(B*L)	2.556	3.778	6.000
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	2.556	3.778	6.000

SIZE OF FOUNDATION 4M X 4M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation1	Calculation2	Calculation3
Bore Hole Number		BH	8	8	8
SPT N Value		N	8	20	30
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	4	4	4
Width of Foundation	m	B	4	4	4
Saturated density of soil	kg/cm ³	γ_{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.243	0.486	0.729
	kg/cm ²	$B \times \gamma$	0.248	0.248	0.248
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	0.00
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	35.00	52.00	65.00
N ϕ			1.000	1.000	1.000
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140
		Nq'	1.000	1.000	1.000
		Ny'	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.075	1.150	1.225
		Dq	1.000	1.000	1.000
		Dy	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times Ny' \times Sy \times Dy \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.936	3.141	3.345
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd'/FOS)	11.740	12.560	13.380

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

Safe Bearing Capacity Based on Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	2xB	8	8	8
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	3.410	4.340	5.270
Additional pressure intensity 2:1 method	T/m ²	ΔP	0.547	0.813	1.016
Consolidation Settlement	mm	sc	54.064	62.374	64.056
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	1	1	1
Correction factor for depth		Def	0.89	0.77	0.74
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	48.117	48.028	47.402
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/(B*L)	2.1875	3.25	4.0625
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	2.188	3.250	4.063

19.3 SERVICE INFRASTRUCTURE (BH-11)

SIZE OF FOUNDATION 2M X 2M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-11					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotation	1	2	3
Bore Hole Number		BH	11	11	11
SPT N Value at end of compressible zone		N	25	25	25
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	2	2	2
Width of Foundation	m	B	2	2	2
Saturated density of soil	kg/cm ³	γ_{sat}	0.0016	0.0016	0.00174
Submerged density of soil	kg/cm ³	γ'	0.0006	0.0006	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.24	0.48	0.783
	kg/cm ²	$B \times \gamma$	0.12	0.12	0.148
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.62	0.62	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.84
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	23.37
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	15.00	34.00	98.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
Bearing Capacity factors for local shear failure		N ϕ	1.000	1.000	3.370
		Nc'	5.140	5.140	18.803
		Nq'	1.000	1.000	9.274
		N γ'	0.000	0.000	9.093
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		S γ	0.8	0.8	0.8
		Dc	1.150	1.300	1.826
Depth Factors		Dq	1.000	1.000	1.413
		D γ	1.000	1.000	1.413
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	3.192	3.608	11.365
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	12.770	14.430	45.460
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P0	1.800	2.250	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.224	4.496	24.500
Consolidation Settlement	mm	sc	70.735	74.830	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔPc	0.6	2.78	24.5
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.01	0.01	0.01
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	$Su'=Su/W'$	0.02	0.02	0.02
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	$Sg=Su' \times \Delta Pc/10$	1.2	5.56	49
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.77	0.63	0.61
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	47.496	45.631	49.000
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	3.75	8.5	24.5
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	3.750	8.500	24.500

SIZE OF FOUNDATION 3M X 3M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-11					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	11	11	11
SPT N Value at end of compressible zone		N	25	25	25
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	3	3	3
Width of Foundation	m	B	3	3	3
Saturated density of soil	kg/cm ³	γ_{sat}	0.0016	0.0016	0.00174
Submerged density of soil	kg/cm ³	γ'	0.0006	0.0006	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.24	0.48	0.783
	kg/cm ²	$B \times \gamma$	0.18	0.18	0.222
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.62	0.62	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.84
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	23.37
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	24.00	60.00	185.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
N ϕ			1.000	1.000	3.370
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.803
		Nq'	1.000	1.000	9.274
		Ny'	0.000	0.000	9.093
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.100	1.200	1.551
		Dq	1.000	1.000	1.275
		Dy	1.000	1.000	1.275
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times Ny' \times Sy \times Dy \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	3.053	3.331	10.429
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	12.210	13.320	41.720
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P0	1.800	2.250	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.185	4.267	20.556
Consolidation Settlement	mm	sc	68.952	72.474	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔPc	0.667	2.96	20.556
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.012	0.012	0.012
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	$Su'=Su/W'$	0.024	0.024	0.024
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	$Sg=Su' \times \Delta Pc / 10$	1.6	7.104	49.333
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.85	0.73	0.63
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	51.418	52.074	49.333
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			Not OK	Not OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m2	SBC=Load/BL	2.67	6.67	20.56
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m2	SBP	2.670	6.670	20.560

SIZE OF FOUNDATION 4M X 4M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-11

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	11	11	11
SPT N Value at end of compressible zone		N	25	25	25
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	4	4	4
Width of Foundation	m	B	4	4	4
Saturated density of soil	kg/cm ³	γ_{sat}	0.0016	0.0016	0.00174
Submerged density of soil	kg/cm ³	γ'	0.0006	0.0006	0.00074
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.24	0.48	0.783
	kg/cm ²	$B \times \gamma$	0.24	0.24	0.296
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.62	0.62	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.84
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	23.37
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	36.00	90.00	280.00
Depth of compressible from base of footing	m	H	3.00	1.50	0.00
N ϕ			1.000	1.000	3.370
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.803
		Nq'	1.000	1.000	9.274
		Ny'	0.000	0.000	9.093
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.075	1.150	1.413
		Dq	1.000	1.000	1.207
		Dy	1.000	1.000	1.207
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times Ny' \times Sy \times Dy \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.984	3.192	10.029
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.940	12.770	40.110

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from influence zone	m	H	3.00	1.50	0.00
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P0	1.800	2.250	3.330
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	1.190	3.989	17.500
Consolidation Settlement	mm	sc	69.175	69.505	0.000
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔPc	0.735	2.98	17.5
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.014	0.014	0.014
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	$Su'=Su/W'$	0.028	0.028	0.028
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	$Sg=Su' \times \Delta Pc / 10$	2.057	8.344	49
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.89	0.77	0.74
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	54.388	53.835	49.000
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	50	50	50
Check whether observed settlement is within permissible limits			Not OK	Not OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m2	SBC=Load/BL	2.25	5.63	17.5
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m2	SBP	2.250	5.630	17.500

20.0 DESIGN OF RAFT FOUNDATION

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

20.1 RAFT FOUNDATION FOR GUEST HOUSE (BH-8)

SIZE OF RAFT FOUNDATION-5M X 5M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	8	8	8
SPT N Value at end of compressible zone		N	50	50	50
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	5	5	5
Width of Foundation	m	B	5	5	5
Saturated density of soil	kg/cm ³	γ_{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.243	0.486	0.729
	kg/cm ²	$B \times \gamma$	0.31	0.31	0.31
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	0.00
Initial void ratio		e0	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	85.00	115.00	155.00
Depth of compressible from base of footing	m	H	9.00	7.50	6.00
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140
		Nq'	1.000	1.000	1.000
		Ny'	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.060	1.120	1.180
		Dq	1.000	1.000	1.000
		Dy	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B' \times Ny' \times Sy \times Dy \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.895	3.059	3.223
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.580	12.230	12.890
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from base	m	H	9.00	7.50	6.00
Compression Index $Cc=0.3 \times (e0-0.27)$		Cc	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P0	3.720	4.185	4.650
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	0.942	1.502	2.422
Consolidation Settlement	mm	sc	92.283	104.504	114.291
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔPc	0.434	0.74	1.281
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	Su	0.0065	0.0065	0.0065
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	$Su'=Su/W'$	0.013	0.013	0.013
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	$Sg=Su' \times \Delta Pc/10$	0.564	0.962	1.665
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.92	0.82	0.74
Correction factor for rigidity		Rf	1	1	1
Consolidation Settlement after correction	mm	$Sc=sc \times \lambda \times Def \times Rf$	72.729	73.801	73.554
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	3.40	4.6	6.2
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	3.400	4.600	6.200

SIZE OF RAFT FOUNDATION-10M X 10M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	8	8	8
SPT N Value at end of compressible zone		N	50	50	50
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ _{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	γ x Df	0.243	0.486	0.729
	kg/cm ²	B x γ	0.62	0.62	0.62
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	φ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	0.00
Initial void ratio		e ₀	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	190.00	250.00	340.00
Depth of compressible from base of footing	m	H	9.00	7.50	6.00
Bearing Capacity factors for local shear failure		N _c '	1.000	1.000	1.000
		N _q '	5.140	5.140	5.140
		N _γ '	1.000	1.000	1.000
		S _c	0.000	0.000	0.000
Shape Factors		S _q	1.3	1.3	1.3
		S _γ	1.2	1.2	1.2
		S _γ	0.8	0.8	0.8
Depth Factors		D _c	1.030	1.060	1.090
		D _q	1.000	1.000	1.000
		D _γ	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot N_c' \cdot S_c \cdot D_c + q(N_q' - 1) \cdot S_q \cdot D_q + 0.5 \cdot B \cdot N_{\gamma}' \cdot S_{\gamma} \cdot D_{\gamma} \cdot W'$ [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Q _d '	2.813	2.895
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Q _d /FOS)	11.250	11.580
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from base	m	H	9.00	7.50	6.00
Compression Index C _c =0.3*(e ₀ -0.27)		C _c	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	3.720	4.185	4.650
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	0.904	1.322	2.012
Consolidation Settlement	mm	sc	88.924	93.561	98.008
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔP _c	0.526	0.82	1.328
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	S _u	0.0065	0.0065	0.0065
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	S _u '=S _u /W'	0.013	0.013	0.013
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	S _g =S _u '*ΔP _c /10	0.684	1.066	1.727
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.96	0.92	0.87
Correction factor for rigidity		R _f	1	1	1
Consolidation Settlement after correction	mm	S _c =s _c *λ*Def*R _f	73.246	74.231	74.204
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	S _a	75	75	75
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	1.90	2.5	3.4
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	1.900	2.500	3.400

SIZE OF RAFT FOUNDATION-15M X 15M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-8					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
Parameters	Unit	Denotion	1	2	3
Bore Hole Number		BH	8	8	8
SPT N Value at end of compressible zone		N	50	50	50
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	15	15	15
Width of Foundation	m	B	15	15	15
Saturated density of soil	kg/cm ³	γ _{sat}	0.00162	0.00162	0.00162
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.00062
Overburden Pressure	kg/cm ²	γ x Df	0.243	0.486	0.729
	kg/cm ²	B x γ	0.93	0.93	0.93
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	φ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	0.00
Initial void ratio		e ₀	0.95	0.95	0.95
Column load coming on to footing (Unfactored Load)	Tons	Load	340.00	450.00	600.00
Depth of compressible from base of footing	m	H	9.00	7.50	6.00
Bearing Capacity factors for local shear failure		N _c '	1.000	1.000	1.000
		N _q '	5.140	5.140	5.140
		N _γ '	1.000	1.000	1.000
		S _c	1.3	1.3	1.3
Shape Factors		S _q	1.2	1.2	1.2
		S _γ	0.8	0.8	0.8
		D _c	1.020	1.040	1.060
Depth Factors		D _q	1.000	1.000	1.000
		D _γ	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot N_c' \cdot S_c \cdot D_c + q(N_q' - 1) \cdot S_q \cdot D_q + 0.5 \cdot B \cdot N_{\gamma}' \cdot S_{\gamma} \cdot D_{\gamma} \cdot W'$ [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Q _d '	2.786	2.840
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Q _d /FOS)	11.140	11.360
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
Consolidation Settlement for Clayey Soil Layers					
Height of compressible zone from base	m	H	9.00	7.50	6.00
Compression Index C _c =0.3*(e ₀ -0.27)		C _c	0.204	0.204	0.204
Initial Pressure at the centre of influence zone	T/m ²	P ₀	3.720	4.185	4.650
Additional pressure intensity 2:1 method at mid of clay layer	T/m ²	ΔP	0.894	1.280	1.852
Consolidation Settlement	mm	sc	88.080	90.933	91.382
Settlement criteria for for Sandy Soil Layers					
Pressure intensity 2:1 method at end of clay layer	T/m ²	ΔP _c	0.590	0.89	1.361
Settlement from Fig 9 of IS8009, for 1Kg/cm ² or 10 T/m ²	m	S _u	0.0065	0.0065	0.0065
Corrected Settlement due to Water Table for 1Kg/cm ² or 10 T/m ²	m	S _u '=S _u /W'	0.013	0.013	0.013
Settlement for the granular layer due to pressure intensity at end of clay layer	mm	S _g =S _u '*ΔP _c /10	0.767	1.157	1.769
Corrected Settlement Total					
If clay layer is resting on cohesionless soil laue then value of λ should be taken as 0.85		λ	0.85	0.85	0.85
Correction factor for depth		Def	0.98	0.95	0.92
Correction factor for rigidity		R _f	1	1	1
Consolidation Settlement after correction	mm	S _c =s _c *λ*Def*R _f	74.138	74.585	73.229
Permissible settlement = 50mm (For RC structures and Water tanks IS1904:1986)	mm	S _a	75	75	75
Check whether observed settlement is within permissible limits			OK	OK	OK
Net Safe Bearing Capacity (Settlement Criteria)	T/m ²	SBC=Load/BL	1.51	2	2.67
Safe Bearing Pressure (Without risk of Shear failure or excessive settlement)	T/m ²	SBP	1.510	2.000	2.670

20.2 RAFT FOUNDATION FOR RESIDENTIAL TOWERS

SIZE OF RAFT FOUNDATION-10M X 10M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-1					
STEP 1 : SBC BASED ON SHEAR CRITERIA					
			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	1	1	1
SPT N Value (Average of individual values less than 50% of overall average)		N	17	19	20
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ_{sat}	0.0016	0.00171	0.00171
Submerged density of soil	kg/cm ³	γ'	0.0006	0.00071	0.00071
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.24	0.513	0.770
	kg/cm ²	$B \times \gamma$	0.6	0.71	0.71
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.6	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	31.29	31.29
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	22.14	22.14
$N\phi$			1.000	3.161	3.161
Bearing Capacity factors for local shear failure		Nc'	5.140	17.356	17.356
		Nq'	1.000	8.227	8.227
		Ny'	0.000	7.744	7.744
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.107	1.160
		Dq	1.000	1.053	1.080
		Dy	1.000	1.053	1.080
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot Nc' \cdot Sc \cdot Dc + q(Nq'-1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot Ny' \cdot Sy \cdot Dy \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.767	5.844	8.395
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	$NSBC=(Qd'/FOS)$	11.070	23.380	33.580
STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA					
SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	19.8	17.6	16
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su'=Su/W'$	39.6	35.2	32
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	1.894	2.131	2.344
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	18.939	21.307	23.438
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.070	21.310	23.440

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-2

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3	
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3	
Bore Hole Number		BH	2	2	2	
SPT N Value (Average of individual values less than 50% of overall average)		N	19	24	26	
Depth of embedment	m	Df	1.5	3	4.5	
Factor of Safety		FOS	2.5	2.5	2.5	
Length of foundation	m	L	10	10	10	
Width of Foundation	m	B	10	10	10	
Saturated density of soil	kg/cm ³	γ _{sat}	0.00162	0.00162	0.0017	
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.0007	
Overburden Pressure	kg/cm ²	γ x Df	0.243	0.486	0.765	
	kg/cm ²	B x γ	0.62	0.62	0.7	
Shape of foundation base			Square	Square	Square	
Effect of Water Table		Worst case	0.5	0.5	0.5	
Cohesion	kg/cm ²	C	0.62	0.62	0	
Angle of Internal Friction for General Shear	Degrees	φ	0	0	31.05	
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	21.96	
Nφ			1.000	1.000	3.130	
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	17.134	
		Nq'	1.000	1.000	8.066	
		Nγ'	0.000	0.000	7.537	
Shape Factors		Sc	1.3	1.3	1.3	
		Sq	1.2	1.2	1.2	
		Sy	0.8	0.8	0.8	
Depth Factors		Dc	1.030	1.060	1.159	
		Dq	1.000	1.000	1.080	
		Dγ	1.000	1.000	1.080	
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.859	2.942	8.142
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	11.440	11.770	32.570

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	17.7	14.4	13.6
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	35.4	28.8	27.2
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.119	2.604	2.757
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	21.186	26.042	27.574
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.440	11.770	27.570

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-3

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	3	3	3
SPT N Value (Average of individual values less than 50% of overall average)		N	19	23	31
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ _{sat}	0.00159	0.00159	0.00173
Submerged density of soil	kg/cm ³	γ'	0.00059	0.00059	0.00073
Overburden Pressure	kg/cm ²	γ x Df	0.2385	0.477	0.7785
	kg/cm ²	B x γ	0.59	0.59	0.73
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	φ	0	0	32.01
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	22.71
Nφ			1.000	1.000	3.256
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.025
		Nq'	1.000	1.000	8.711
		Ny'	0.000	0.000	8.368
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.060	1.162
		Dq	1.000	1.000	1.081
		Dy	1.000	1.000	1.081
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot Nc' \cdot Sc \cdot Dc + q(Nq' - 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot Ny' \cdot Sy \cdot Dy \cdot W'$ [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.997	3.085
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd'/FOS)	11.990	12.340

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	17.4	15	9.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	34.8	30	19.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.155	2.500	3.827
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	21.552	25.000	38.265
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.990	12.340	36.440

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-5

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	5	5	5
SPT N Value (Average of individual values less than 50% of overall average)		N	20	21	21
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ_{sat}	0.00167	0.00175	0.00175
Submerged density of soil	kg/cm ³	γ'	0.00067	0.00075	0.00075
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.2505	0.525	0.7875
	kg/cm ²	$B \times \gamma$	0.67	0.75	0.75
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.59	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	33.48	33.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	23.89	23.89
$N\phi$			1.000	3.461	3.461
Bearing Capacity factors for local shear failure		Nc'	5.140	19.408	19.408
		Nq'	1.000	9.711	9.711
		Ny'	0.000	9.657	9.657
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.112	1.167
		Dq	1.000	1.056	1.084
		Dy	1.000	1.056	1.084
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot Nc' \cdot Sc \cdot Dc + q(Nq' - 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot Ny' \cdot Sy \cdot Dy \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.721	7.323
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	$NSBC = (Qd'/FOS)$	10.880	29.290
				10.880	41.960

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	16	15.4	15.4
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u/W'$	32	30.8	30.8
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a/S_u'	2.344	2.435	2.435
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a/S_u') \cdot 10$	23.438	24.351	24.351
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	10.880	24.350	24.350

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-6

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	6	6	6
SPT N Value (Average of individual values less than 50% of overall average)		N	22	25	27
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ _{sat}	0.00168	0.00168	0.00175
Submerged density of soil	kg/cm ³	γ'	0.00068	0.00068	0.00075
Overburden Pressure	kg/cm ²	γ x Df	0.252	0.504	0.788
	kg/cm ²	B x γ	0.68	0.68	0.75
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.63	0.63	0
Angle of Internal Friction for General Shear	Degrees	φ	0	0	33.74
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	24.10
Nφ			1.000	1.000	3.499
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.654
		Nq'	1.000	1.000	9.889
		Nγ'	0.000	0.000	9.887
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.060	1.168
		Dq	1.000	1.000	1.084
		Dγ	1.000	1.000	1.084
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.905	2.990	10.715
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.620	11.960	42.860

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	15	14.1	12.3
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	30	28.2	24.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.500	2.660	3.049
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	25.000	26.596	30.488
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.620	11.960	30.490

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-7

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3	
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3	
Bore Hole Number		BH	7	7	7	
SPT N Value (Average of individual values less than 50% of overall average)		N	15	22	26	
Depth of embedment	m	Df	1.5	3	4.5	
Factor of Safety		FOS	2.5	2.5	2.5	
Length of foundation	m	L	10	10	10	
Width of Foundation	m	B	10	10	10	
Saturated density of soil	kg/cm ³	γ _{sat}	0.00166	0.00166	0.00166	
Submerged density of soil	kg/cm ³	γ'	0.00066	0.00066	0.00066	
Overburden Pressure	kg/cm ²	γ x Df	0.249	0.498	0.747	
	kg/cm ²	B x γ	0.66	0.66	0.66	
Shape of foundation base			Square	Square	Square	
Effect of Water Table		Worst case	0.5	0.5	0.5	
Cohesion	kg/cm ²	C	0.61	0.61	0.61	
Angle of Internal Friction for General Shear	Degrees	φ	0	0	0	
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	0.00	
Nφ			1.000	1.000	1.000	
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140	
		Nq'	1.000	1.000	1.000	
		Nγ'	0.000	0.000	0.000	
Shape Factors		Sc	1.3	1.3	1.3	
		Sq	1.2	1.2	1.2	
		Sγ	0.8	0.8	0.8	
Depth Factors		Dc	1.030	1.060	1.090	
		Dq	1.000	1.000	1.000	
		Dγ	1.000	1.000	1.000	
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sγ *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.813	2.895	2.977
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	11.250	11.580	11.910

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	23	15	13.3
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	46	30	26.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	1.630	2.500	2.820
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	16.304	25.000	28.195
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.250	11.580	11.910

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-9

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	9	9	9
SPT N Value (Average of individual values less than 50% of overall average)		N	16	20	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ_{sat}	0.00158	0.00158	0.00166
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00066
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.237	0.474	0.747
	kg/cm ²	$B \times \gamma$	0.58	0.58	0.66
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.66	0.66	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.39
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	23.01
$N\phi$			1.000	1.000	3.307
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.380
		Nq'	1.000	1.000	8.968
		$N\gamma'$	0.000	0.000	8.699
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.060	1.164
		Dq	1.000	1.000	1.082
		$D\gamma$	1.000	1.000	1.082
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot Nc' \cdot Sc \cdot Dc + q(Nq'-1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	3.043	3.132
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	12.170	12.530

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	22.3	16.3	15
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su'=Su/W'$	44.6	32.6	30
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	1.682	2.301	2.500
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	16.816	23.006	25.000
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	12.170	12.530	25.000

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-10

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	10	10	10
SPT N Value (Average of individual values less than 50% of overall average)		N	20	22	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	10	10	10
Width of Foundation	m	B	10	10	10
Saturated density of soil	kg/cm ³	γ_{sat}	0.00158	0.00158	0.00169
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00069
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.237	0.474	0.7605
	kg/cm ²	$B \times \gamma$	0.58	0.58	0.69
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	33.67
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	24.04
$N\phi$			1.000	1.000	3.488
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.588
		Nq'	1.000	1.000	9.841
		$N\gamma'$	0.000	0.000	9.825
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.030	1.060	1.168
		Dq	1.000	1.000	1.084
		$D\gamma$	1.000	1.000	1.084
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C \cdot Nc' \cdot Sc \cdot Dc + q(Nq'-1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.997	3.085
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	11.990	12.340

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	16.2	15	14.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su'=Su/W'$	32.4	30	29.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.315	2.500	2.534
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	23.148	25.000	25.338
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.990	12.340	25.340

SIZE OF RAFT FOUNDATION - 20M X 20M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-1

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	1	1	1
SPT N Value (Average of individual values less than 50% of overall average)		N	18	20	20
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ _{sat}	0.0016	0.00171	0.00171
Submerged density of soil	kg/cm ³	γ'	0.0006	0.00071	0.00071
Overburden Pressure	kg/cm ²	γ x Df	0.24	0.513	0.7695
	kg/cm ²	B x γ	1.2	1.42	1.42
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.6	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	31.29	31.29
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	22.14	22.14
Nϕ			1.000	3.161	3.161
Bearing Capacity factors for local shear failure		Nc'	5.140	17.356	17.356
		Nq'	1.000	8.227	8.227
		Nγ'	0.000	7.744	7.744
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sγ	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.053	1.080
		Dq	1.000	1.027	1.040
		Dγ	1.000	1.027	1.040
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sγ *Dγ*W' [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.726	6.825	9.227
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	10.910	27.300	36.910

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	18.4	16.4	16
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	36.8	32.8	32
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.038	2.287	2.344
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	20.380	22.866	23.438
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	10.910	22.870	23.440

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-2

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3	
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3	
Bore Hole Number		BH	2	2	2	
SPT N Value (Average of individual values less than 50% of overall average)		N	22	24	26	
Depth of embedment	m	Df	1.5	3	4.5	
Factor of Safety		FOS	2.5	2.5	2.5	
Length of foundation	m	L	20	20	20	
Width of Foundation	m	B	20	20	20	
Saturated density of soil	kg/cm ³	γ _{sat}	0.00162	0.00162	0.0017	
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.0007	
Overburden Pressure	kg/cm ²	γ x Df	0.243	0.486	0.765	
	kg/cm ²	B x γ	1.24	1.24	1.4	
Shape of foundation base			Square	Square	Square	
Effect of Water Table		Worst case	0.5	0.5	0.5	
Cohesion	kg/cm ²	C	0.62	0.62	0	
Angle of Internal Friction for General Shear	Degrees	φ	0	0	31.05	
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	21.96	
Nφ			1.000	1.000	3.130	
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	17.134	
		Nq'	1.000	1.000	8.066	
		Nγ'	0.000	0.000	7.537	
Shape Factors		Sc	1.3	1.3	1.3	
		Sq	1.2	1.2	1.2	
		Sy	0.8	0.8	0.8	
Depth Factors		Dc	1.015	1.030	1.080	
		Dq	1.000	1.000	1.040	
		Dγ	1.000	1.000	1.040	
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.817	2.859	8.939
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	11.270	11.440	35.760

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	15.2	14.4	13.6
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	30.4	28.8	27.2
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a / S_u'	2.467	2.604	2.757
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \cdot 10$	24.671	26.042	27.574
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.270	11.440	27.570

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-3

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	3	3	3
SPT N Value (Average of individual values less than 50% of overall average)		N	19	29	32
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ _{sat}	0.00159	0.00159	0.00173
Submerged density of soil	kg/cm ³	γ'	0.00059	0.00059	0.00073
Overburden Pressure	kg/cm ²	γ x Df	0.2385	0.477	0.7785
	kg/cm ²	B x γ	1.18	1.18	1.46
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	φ	0	0	32.01
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	22.71
Nφ			1.000	1.000	3.256
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.025
		Nq'	1.000	1.000	8.711
		Nγ'	0.000	0.000	8.368
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.030	1.081
		Dq	1.000	1.000	1.041
		Dγ	1.000	1.000	1.041
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.954	2.997	10.039
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.810	11.990	40.150

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	17.4	10.9	9.5
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su' = Su/W'$	34.8	21.8	19
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.155	3.440	3.947
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	21.552	34.404	39.474
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.810	11.990	39.470

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-5

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	5	5	5
SPT N Value (Average of individual values less than 50% of overall average)		N	23	23	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ_{sat}	0.00167	0.00175	0.00175
Submerged density of soil	kg/cm ³	γ'	0.00067	0.00075	0.00075
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.2505	0.525	0.7875
	kg/cm ²	$B \times \gamma$	1.34	1.5	1.5
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.59	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	33.48	33.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	23.89	23.89
N ϕ			1.000	3.461	3.461
Bearing Capacity factors for local shear failure		Nc'	5.140	19.408	19.408
		Nq'	1.000	9.711	9.711
		N γ'	0.000	9.657	9.657
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		S γ	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.056	1.084
		Dq	1.000	1.028	1.042
		D γ	1.000	1.028	1.042
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.681	8.619
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd'/FOS)	10.720	34.470

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers					
Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	14.8	14.6	14.7
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	Su'=Su/W'	29.6	29.2	29.4
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.534	2.568	2.551
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	(Sa/Su')*10	25.338	25.685	25.510
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	10.720	25.680	25.510

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-6

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3	
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3	
Bore Hole Number		BH	6	6	6	
SPT N Value (Average of individual values less than 50% of overall average)		N	24	25	27	
Depth of embedment	m	Df	1.5	3	4.5	
Factor of Safety		FOS	2.5	2.5	2.5	
Length of foundation	m	L	20	20	20	
Width of Foundation	m	B	20	20	20	
Saturated density of soil	kg/cm ³	γ _{sat}	0.00168	0.00168	0.00175	
Submerged density of soil	kg/cm ³	γ'	0.00068	0.00068	0.00075	
Overburden Pressure	kg/cm ²	γ x Df	0.252	0.504	0.7875	
	kg/cm ²	B x γ	1.36	1.36	1.5	
Shape of foundation base			Square	Square	Square	
Effect of Water Table		Worst case	0.5	0.5	0.5	
Cohesion	kg/cm ²	C	0.63	0.63	0	
Angle of Internal Friction for General Shear	Degrees	φ	0	0	33.74	
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	24.10	
Nφ			1.000	1.000	3.499	
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.654	
		Nq'	1.000	1.000	9.889	
		Nγ'	0.000	0.000	9.887	
Shape Factors		Sc	1.3	1.3	1.3	
		Sq	1.2	1.2	1.2	
		Sy	0.8	0.8	0.8	
Depth Factors		Dc	1.015	1.030	1.084	
		Dq	1.000	1.000	1.042	
		Dγ	1.000	1.000	1.042	
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.863	2.905	11.845
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	11.450	11.620	47.380

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	14.4	13.7	12.2
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su'=Su/W'$	28.8	27.4	24.4
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.604	2.737	3.074
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \times 10$	26.042	27.372	30.738
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.450	11.620	30.740

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-7

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	7	7	7
SPT N Value (Average of individual values less than 50% of overall average)		N	21	24	27
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ_{sat}	0.00166	0.00166	0.00166
Submerged density of soil	kg/cm ³	γ'	0.00066	0.00066	0.00066
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.249	0.498	0.747
	kg/cm ²	$B \times \gamma$	1.32	1.32	1.32
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	0.00
$N\phi$			1.000	1.000	1.000
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140
		Nq'	1.000	1.000	1.000
		$N\gamma'$	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.030	1.045
		Dq	1.000	1.000	1.000
		$D\gamma$	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq' - 1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.772	2.813
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd'/FOS)	11.090	11.250

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	15.8	14.5	12.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	31.6	29	25.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a / S_u'	2.373	2.586	2.930
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \times 10$	23.734	25.862	29.297
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.090	11.250	11.420

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-9

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	9	9	9
SPT N Value (Average of individual values less than 50% of overall average)		N	17	21	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ_{sat}	0.00158	0.00158	0.00166
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00066
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.237	0.474	0.747
	kg/cm ²	$B \times \gamma$	1.16	1.16	1.32
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.66	0.66	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.39
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	23.01
$N\phi$			1.000	1.000	3.307
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.380
		Nq'	1.000	1.000	8.968
		$N\gamma'$	0.000	0.000	8.699
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.030	1.082
		Dq	1.000	1.000	1.041
		$D\gamma$	1.000	1.000	1.041
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C' \cdot Nc' \cdot Sc \cdot Dc + q(Nq' - 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.999	3.043
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	12.000	12.170
				39.300	

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm2	mm	S_u	19.6	15.6	15
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	39.2	31.2	30
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a / S_u'	1.913	2.404	2.500
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \cdot 10$	19.133	24.038	25.000
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	12.000	12.170	25.000

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-10

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	10	10	10
SPT N Value (Average of individual values less than 50% of overall average)		N	21	22	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	20	20	20
Width of Foundation	m	B	20	20	20
Saturated density of soil	kg/cm ³	γ _{sat}	0.00158	0.00158	0.00169
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00069
Overburden Pressure	kg/cm ²	γ x Df	0.237	0.474	0.7605
	kg/cm ²	B x γ	1.16	1.16	1.38
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	φ	0	0	33.67
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	24.04
Nφ			1.000	1.000	3.488
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.588
		Nq'	1.000	1.000	9.841
		Nγ'	0.000	0.000	9.825
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sγ	0.8	0.8	0.8
Depth Factors		Dc	1.015	1.030	1.084
		Dq	1.000	1.000	1.042
		Dγ	1.000	1.000	1.042
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sγ *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.954	2.997	11.233
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.810	11.990	44.930

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	15.6	15	14.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u'=S_u/W'$	31.2	30	29.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a/S_u'	2.404	2.500	2.534
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a/S_u') \cdot 10$	24.038	25.000	25.338
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.810	11.990	25.340

SIZE OF RAFT FOUNDATION- 30M X 30M

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-1

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	1	1	1
SPT N Value (Average of individual values less than 50% of overall average)		N	18	20	20
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ_{sat}	0.0016	0.00171	0.00171
Submerged density of soil	kg/cm ³	γ'	0.0006	0.00071	0.00071
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.24	0.513	0.7695
	kg/cm ²	$B \times \gamma$	1.8	2.13	2.13
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.6	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	31.29	31.29
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	22.14	22.14
$N\phi$			1.000	3.161	3.161
Bearing Capacity factors for local shear failure		Nc'	5.140	17.356	17.356
		Nq'	1.000	8.227	8.227
		$N\gamma'$	0.000	7.744	7.744
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.036	1.053
		Dq	1.000	1.018	1.027
		$D\gamma$	1.000	1.018	1.027
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.713	7.886
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	10.850	31.540

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm2	mm	S_u	18.4	16.4	16
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	36.8	32.8	32
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	kg/cm ²	S_a / S_u'	2.038	2.287	2.344
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \times 10$	20.380	22.866	23.438
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	10.850	22.870	23.440

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-2

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	2	2	2
SPT N Value (Average of individual values less than 50% of overall average)		N	22	24	26
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ_{sat}	0.00162	0.00162	0.0017
Submerged density of soil	kg/cm ³	γ'	0.00062	0.00062	0.0007
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.243	0.486	0.765
	kg/cm ²	$B \times \gamma$	1.86	1.86	2.1
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.62	0.62	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	31.05
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	21.96
$N\phi$			1.000	1.000	3.130
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	17.134
		Nq'	1.000	1.000	8.066
		$N\gamma'$	0.000	0.000	7.537
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.053
		Dq	1.000	1.000	1.027
		$D\gamma$	1.000	1.000	1.027
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \times C' \times Nc' \times Sc \times Dc + q(Nq'-1) \times Sq \times Dq + 0.5 \times B \times N\gamma' \times S\gamma \times D\gamma \times W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.803	2.831
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	$NSBC = (Qd'/FOS)$	11.210	11.320
				39.630	

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	15.2	14.4	13.6
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u/W'$	30.4	28.8	27.2
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a/S_u'	2.467	2.604	2.757
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a/S_u') \times 10$	24.671	26.042	27.574
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.210	11.320	27.570

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-3

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	3	3	3
SPT N Value (Average of individual values less than 50% of overall average)		N	19	29	32
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ _{sat}	0.00159	0.00159	0.00173
Submerged density of soil	kg/cm ³	γ'	0.00059	0.00059	0.00073
Overburden Pressure	kg/cm ²	γ x Df	0.2385	0.477	0.7785
	kg/cm ²	B x γ	1.77	1.77	2.19
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	32.01
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	22.71
Nϕ			1.000	1.000	3.256
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.025
		Nq'	1.000	1.000	8.711
		Nγ'	0.000	0.000	8.368
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sγ	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.054
		Dq	1.000	1.000	1.027
		Dγ	1.000	1.000	1.027
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sγ *Dγ*W' [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.939	2.968	11.163
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd'/FOS)	11.760	11.870	44.650

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	17.4	10.9	9.5
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	34.8	21.8	19
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a / S_u'	2.155	3.440	3.947
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \times 10$	21.552	34.404	39.474
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.760	11.870	39.470

 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-5

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	5	5	5
SPT N Value (Average of individual values less than 50% of overall average)		N	23	23	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ_{sat}	0.00167	0.00175	0.00175
Submerged density of soil	kg/cm ³	γ'	0.00067	0.00075	0.00075
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.251	0.525	0.788
	kg/cm ²	$B \times \gamma$	2.01	2.25	2.25
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.59	0	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	33.48	33.48
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	23.89	23.89
$N\phi$			1.000	3.461	3.461
Bearing Capacity factors for local shear failure		Nc'	5.140	19.408	19.408
		Nq'	1.000	9.711	9.711
		$N\gamma'$	0.000	9.657	9.657
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.037	1.056
		Dq	1.000	1.019	1.028
		$D\gamma$	1.000	1.019	1.028
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C' \cdot Nc' \cdot Sc \cdot Dc + q(Nq' - 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.668	10.016
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd/FOS)	10.670	40.060
				25.510	51.710

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	14.8	14.6	14.7
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su' = Su/W'$	29.6	29.2	29.4
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.534	2.568	2.551
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	25.338	25.685	25.510
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	10.670	25.680	25.510

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-6

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	6	6	6
SPT N Value (Average of individual values less than 50% of overall average)		N	24	25	27
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ_{sat}	0.00168	0.00168	0.00175
Submerged density of soil	kg/cm ³	γ'	0.00068	0.00068	0.00075
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.252	0.504	0.788
	kg/cm ²	$B \times \gamma$	2.04	2.04	2.25
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.63	0.63	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	33.74
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	24.10
$N\phi$			1.000	1.000	3.499
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.654
		Nq'	1.000	1.000	9.889
		$N\gamma'$	0.000	0.000	9.887
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.056
		Dq	1.000	1.000	1.028
		$D\gamma$	1.000	1.000	1.028
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C' \cdot Nc' \cdot Sc \cdot Dc + q(Nq'-1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.849	2.877	13.210
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	$NSBC = (Qd'/FOS)$	11.390	11.510	52.840

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	S_u	14.4	13.7	12.2
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u/W'$	28.8	27.4	24.4
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a/S_u'	2.604	2.737	3.074
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a/S_u') \cdot 10$	26.042	27.372	30.738
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.390	11.510	30.740

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-7

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	7	7	7
SPT N Value (Average of individual values less than 50% of overall average)		N	21	24	27
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ _{sat}	0.00166	0.00166	0.00166
Submerged density of soil	kg/cm ³	γ'	0.00066	0.00066	0.00066
Overburden Pressure	kg/cm ²	γ x Df	0.249	0.498	0.747
	kg/cm ²	B x γ	1.98	1.98	1.98
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.61	0.61	0.61
Angle of Internal Friction for General Shear	Degrees	φ	0	0	0
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	0.00
Nφ			1.000	1.000	1.000
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	5.140
		Nq'	1.000	1.000	1.000
		Nγ'	0.000	0.000	0.000
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.030
		Dq	1.000	1.000	1.000
		Dγ	1.000	1.000	1.000
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.758	2.786	2.813
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.030	11.140	11.250

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm2	mm	S_u	15.8	14.5	12.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u / W'$	31.6	29	25.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a / S_u'	2.373	2.586	2.930
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a / S_u') \cdot 10$	23.734	25.862	29.297
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.030	11.140	11.250

CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-9

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	9	9	9
SPT N Value (Average of individual values less than 50% of overall average)		N	17	21	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ _{sat}	0.00158	0.00158	0.00166
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00066
Overburden Pressure	kg/cm ²	γ x Df	0.237	0.474	0.747
	kg/cm ²	B x γ	1.74	1.74	1.98
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.66	0.66	0
Angle of Internal Friction for General Shear	Degrees	φ	0	0	32.39
Angle of Internal Friction for Local Shear	Degrees	φ'	0.00	0.00	23.01
Nφ			1.000	1.000	3.307
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	18.380
		Nq'	1.000	1.000	8.968
		Nγ'	0.000	0.000	8.699
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		Sy	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.055
		Dq	1.000	1.000	1.027
		Dγ	1.000	1.000	1.027
Net Ultimate Bearing Capacity (Local Shear Criteria) = 0.67*C*Nc'*Sc*Dc + q(Nq'-1)*Sq*Dq + 0.5*B*Nγ' * Sy *Dγ*W' [Used for φ less than 29 degrees as per IS 6403-1981]	kg/cm ²	Qd'	2.984	3.014	10.876
Net Safe Bearing Capacity (Local Shear Criteria)	T/m ²	NSBC=(Qd/FOS)	11.940	12.060	43.500

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm2	mm	S_u	19.6	15.6	15
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$S_u' = S_u/W'$	39.2	31.2	30
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	S_a	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	S_a/S_u'	1.913	2.404	2.500
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(S_a/S_u') \times 10$	19.133	24.038	25.000
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.940	12.060	25.000

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATION OF SAFE BEARING CAPACITY OF FOUNDATION FOR DULIAJAN SITE BH-10

STEP 1 : SBC BASED ON SHEAR CRITERIA

			1	2	3
Parameters	Unit	Denotion	Calculation 1	Calculation 2	Calculation 3
Bore Hole Number		BH	10	10	10
SPT N Value (Average of individual values less than 50% of overall average)		N	21	22	23
Depth of embedment	m	Df	1.5	3	4.5
Factor of Safety		FOS	2.5	2.5	2.5
Length of foundation	m	L	30	30	30
Width of Foundation	m	B	30	30	30
Saturated density of soil	kg/cm ³	γ_{sat}	0.00158	0.00158	0.00169
Submerged density of soil	kg/cm ³	γ'	0.00058	0.00058	0.00069
Overburden Pressure	kg/cm ²	$\gamma \times Df$	0.237	0.474	0.7605
	kg/cm ²	$B \times \gamma$	1.74	1.74	2.07
Shape of foundation base			Square	Square	Square
Effect of Water Table		Worst case	0.5	0.5	0.5
Cohesion	kg/cm ²	C	0.65	0.65	0
Angle of Internal Friction for General Shear	Degrees	ϕ	0	0	33.67
Angle of Internal Friction for Local Shear	Degrees	ϕ'	0.00	0.00	24.04
$N\phi$			1.000	1.000	3.488
Bearing Capacity factors for local shear failure		Nc'	5.140	5.140	19.588
		Nq'	1.000	1.000	9.841
		$N\gamma'$	0.000	0.000	9.825
Shape Factors		Sc	1.3	1.3	1.3
		Sq	1.2	1.2	1.2
		$S\gamma$	0.8	0.8	0.8
Depth Factors		Dc	1.010	1.020	1.056
		Dq	1.000	1.000	1.028
		$D\gamma$	1.000	1.000	1.028
Net Ultimate Bearing Capacity (Local Shear Criteria) = $0.67 \cdot C' \cdot Nc' \cdot Sc \cdot Dc + q(Nq' - 1) \cdot Sq \cdot Dq + 0.5 \cdot B \cdot N\gamma' \cdot S\gamma \cdot D\gamma \cdot W'$ [Used for ϕ less than 29 degrees as per IS 6403-1981]		kg/cm ²	Qd'	2.939	2.968
Net Safe Bearing Capacity (Local Shear Criteria)		T/m ²	NSBC=(Qd'/FOS)	11.760	11.870

STEP 2 : CALCULATION OF SBC BASED ON SETTLEMENT CRITERIA

SBC based on settlement criteria for for Sandy Soil Layers

Settlement from Fig 9 of IS8009, for 1Kg/cm ²	mm	Su	15.6	15	14.8
Corrected Settlement due to Water Table for 1Kg/cm ²	mm	$Su' = Su/W'$	31.2	30	29.6
Permissible settlement = 75 mm (For RC structures and Water tanks IS1904:1986)	mm	Sa	75	75	75
Net Safe Bearing Capacity (Settlement Criteria)	Kg/cm ²	Sa/Su'	2.404	2.500	2.534
Safe Bearing Pressure for 75mm Settlement (Settlement Criteria)	T/m ²	$(Sa/Su') \cdot 10$	24.038	25.000	25.338
Safe Bearing Pressure (Minimum of shear criteria and settlement criteria)	T/m ²	SBP	11.760	11.870	25.340

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signature



21.0 DESIGN OF PILE FOUNDATION (BORED CAST IN-SITU PILES)

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

21.1 PILE FOUNDATION FOR GUEST HOUSE

LENGTH OF PILE- 10 M

DIA OF PILE- 0.5 M

BH-8 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	15.876	6.066	Cl	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	15.876	6.066	Cl	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	1.571	0	41.84	65.731	6.573	
0.5	3	4.5	0	15.876	6.066	Cl	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	2.356	0	41.84	98.575	9.858	
0.5	4.5	6	0	16.366	6.556	Cl	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	2.356	0	43.31	102.038	10.204	
0.5	6	7.5	0	16.366	6.556	Cl	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	2.356	0	43.31	102.038	10.204	
0.5	7.5	9	0	16.856	7.046	Cl	0.63	61.74	0	57.53	52.245	42.045	0	0	0.69	2.356	0	42.6	100.366	10.037	
0.5	9	10	0	16.856	7.046	Cl	0.63	61.74	0	64.58	61.055	42.045	0	0	0.69	1.571	0	42.6	66.925	6.692	
																Total Skin Friction		535.673		kN	
																		53.760		Tons	

Total Skin Friction	Capacity of Pile	=	53.760	Tons
Total End Bearing	Capacity of Pile	=	0.000	Tons
Total	Capacity of Pile	=	53.760	Tons
Self Weight of the pile		=	2.403	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile				
		=	15.517	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weigt of Pile				
		=	14.947	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.


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LENGTH OF PILE- 10 M DIA OF PILE- 0.6 M

BH-8 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.876	6.066	Cl	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.876	6.066	Cl	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	1.885	0	41.84	78.868	7.887
0.6	3	4.5	0	15.876	6.066	Cl	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	2.827	0	41.84	118.282	11.828
0.6	4.5	6	0	16.366	6.556	Cl	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	2.827	0	43.31	122.437	12.244
0.6	6	7.5	0	16.366	6.556	Cl	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	2.827	0	43.31	122.437	12.244
0.6	7.5	9	0	16.856	7.046	Cl	0.63	61.74	0	57.53	52.245	52.245	0	0	0.69	2.827	0	42.6	120.430	12.043
0.6	9	10	0	16.856	7.046	Cl	0.63	61.74	0	64.58	61.055	52.245	0	0	0.69	1.885	0	42.6	80.301	8.030
																Total Skin Friction		642.756	kN	
																		64.507	Tons	

Total Skin FrictionCapacity of Pile	=	64.507	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	64.507	Tons
Self Weight of the pile	=	3.461	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	18.042	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	18.512	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 10 M DIA OF PILE- 0.75 M

BH-8 and Pile Diameter =0.75m

BH-8 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K_{P_{Di}} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.75	0	2	0	15.876	6.066	Cl	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.876	6.066	Cl	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	2.356	0	41.84	98.575	9.858
0.75	3	4.5	0	15.876	6.066	Cl	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	3.534	0	41.84	147.863	14.786
0.75	4.5	6	0	16.366	6.556	Cl	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	3.534	0	43.31	153.058	15.306
0.75	6	7.5	0	16.366	6.556	Cl	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	3.534	0	43.31	153.058	15.306
0.75	7.5	9	0	16.856	7.046	Cl	0.63	61.74	0	57.53	52.245	52.245	0	0	0.69	3.534	0	42.6	150.548	15.055
0.75	9	10	0	16.856	7.046	Cl	0.63	61.74	0	64.58	61.055	61.055	0	0	0.69	2.356	0	42.6	100.366	10.037
																Total Skin Friction		803.467 80.636	kN Tons	

Total Skin FrictionCapacity of Pile	=	80.636	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	80.636	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	21.471	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	24.223	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M DIA OF PILE- 0.5 M

BH-8 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.5	0	2	0	15.876	6.066	Cl	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.876	6.066	Cl	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	1.571	0	41.84	65.731	6.573
0.5	3	4.5	0	15.876	6.066	Cl	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	2.356	0	41.84	98.575	9.858
0.5	4.5	6	0	16.366	6.556	Cl	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	2.356	0	43.31	102.038	10.204
0.5	6	7.5	0	16.366	6.556	Cl	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	2.356	0	43.31	102.038	10.204
0.5	7.5	9	0	16.856	7.046	Cl	0.63	61.74	0	57.53	52.245	42.045	0	0	0.69	2.356	0	42.6	100.366	10.037
0.5	9	10.5	0	16.856	7.046	Cl	0.63	61.74	0	68.1	62.815	42.045	0	0	0.69	2.356	0	42.6	100.366	10.037
0.5	10.5	12	0	18.13	8.32	SM	0	0	35.08	80.58	74.34	42.045	0.61	1	0	2.356	29.4	0	69.219	6.922
0.5	12	13.5	0	18.13	8.32	SM	0	0	35.08	93.06	86.82	42.045	0.61	1	0	2.356	29.4	0	69.219	6.922
0.5	13.5	15	0	18.13	8.32	SM	0	0	35.08	105.54	99.3	42.045	0.61	1	0	2.356	29.4	0	69.219	6.922
																Total Skin Friction		776.771 77.957	kN Tons	

Total Skin FrictionCapacity of Pile	=	77.957	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	77.957	Tons
Self Weight of the pile	=	3.905	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	22.080	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	22.095	Tons

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LENGTH OF PILE- 15 M DIA OF PILE- 0.6 M

BH-8 and Pile Diameter =0.6m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K_{P_{Di}} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.6	0	2	0	15.876	6.066	CI	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	3.77	0	CUTOFF	CUTOFF	CUTOFF	
0.6	2	3	0	15.876	6.066	CI	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	1.885	0	41.84	78.868	7.887	
0.6	3	4.5	0	15.876	6.066	CI	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	2.827	0	41.84	118.282	11.828	
0.6	4.5	6	0	16.366	6.556	CI	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	2.827	0	43.31	122.437	12.244	
0.6	6	7.5	0	16.366	6.556	CI	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	2.827	0	43.31	122.437	12.244	
0.6	7.5	9	0	16.856	7.046	CI	0.63	61.74	0	57.53	52.245	52.245	0	0	0.69	2.827	0	42.6	120.430	12.043	
0.6	9	10.5	0	16.856	7.046	CI	0.63	61.74	0	68.1	62.815	52.245	0	0	0.69	2.827	0	42.6	120.430	12.043	
0.6	10.5	12	0	18.13	8.32	SM	0	0	35.08	80.58	74.34	52.245	0.61	1	0	2.827	36.5	0	103.214	10.321	
0.6	12	13.5	0	18.13	8.32	SM	0	0	35.08	93.06	86.82	52.245	0.61	1	0	2.827	36.5	0	103.214	10.321	
0.6	13.5	15	0	18.13	8.32	SM	0	0	35.08	105.54	99.3	52.245	0.61	1	0	2.827	36.5	0	103.214	10.321	
																Total Skin Friction		992.527 99.610		kN Tons	

Total Skin Friction	Capacity of Pile	=	99.610	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	99.610	Tons
Self Weight of the pile		=	5.624	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	27.580	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	28.866	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M DIA OF PILE- 0.75 M

BH-8 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K_{P_{Di}} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.876	6.066	CI	0.61	59.78	0	12.13	6.065	6.065	0	0	0.7	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.876	6.066	CI	0.61	59.78	0	18.2	15.165	15.165	0	0	0.7	2.356	0	41.84	98.575	9.858
0.75	3	4.5	0	15.876	6.066	CI	0.61	59.78	0	27.3	22.75	22.75	0	0	0.7	3.534	0	41.84	147.863	14.786
0.75	4.5	6	0	16.366	6.556	CI	0.65	63.7	0	37.13	32.215	32.215	0	0	0.68	3.534	0	43.31	153.058	15.306
0.75	6	7.5	0	16.366	6.556	CI	0.65	63.7	0	46.96	42.045	42.045	0	0	0.68	3.534	0	43.31	153.058	15.306
0.75	7.5	9	0	16.856	7.046	CI	0.63	61.74	0	57.53	52.245	52.245	0	0	0.69	3.534	0	42.6	150.548	15.055
0.75	9	10.5	0	16.856	7.046	CI	0.63	61.74	0	68.1	62.815	62.815	0	0	0.69	3.534	0	42.6	150.548	15.055
0.75	10.5	12	0	18.13	8.32	SM	0	0	35.08	80.58	74.34	62.815	0.61	1	0	3.534	43.9	0	155.143	15.514
0.75	12	13.5	0	18.13	8.32	SM	0	0	35.08	93.06	86.82	62.815	0.61	1	0	3.534	43.9	0	155.143	15.514
0.75	13.5	15	0	18.13	8.32	SM	0	0	35.08	105.54	99.3	62.815	0.61	1	0	3.534	43.9	0	155.143	15.514
Total Skin Friction																	1319.077 132.383		kN Tons	

Total Skin FrictionCapacity of Pile	=	132.383	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	132.383	Tons
Self Weight of the pile	=	8.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	35.340	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	39.676	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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21.2 PILE FOUNDATION FOR RESIDENTIAL TOWERS LENGTH OF PILE- 15 M DIA OF PILE- 0.5 M

BH-1 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.5	0	2	0	15.68	5.87	Cl	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.142	0	CUTOFF	CUTOFF	CUTOFF	Liquefiable
0.5	2	3	0	15.68	5.87	Cl	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.571	0	42.33	0	0	Liquefiable
0.5	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.356	13.7	0	0	0	Liquefiable
0.5	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.356	19.9	0	46.932	4.693	
0.5	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.356	26.2	0	61.657	6.166	
0.5	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	43.66	0.59	1	0	2.356	29.2	0	68.866	6.887	
0.5	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	13.5	15	0	17.248	7.438	SM	0	0	34.05	104.67	99.09	43.66	0.59	1	0	2.356	29.2	0	68.866	6.887	
																Total Skin Friction		246.320 24.721	kN Tons		

Total Skin FrictionCapacity of Pile	=	24.721	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	24.721	Tons
Self Weight of the pile	=	3.905	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	4.335	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	9.674	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-2 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.142	2.28	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.571	5.7	0	0	0
0.5	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.356	8.56	0	0	0
0.5	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.356	19.4	0	45.801	4.580
0.5	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031
0.5	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031
0.5	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	13.5	15	0	17.15	7.34	SP-SM	0	0	33.18	102.18	96.675	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
																Total Skin Friction		424.457	kN	
																		42.599	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	42.599	Tons
Total End Bearing Capacity of Pile	=	31.460	Tons
Total Capacity of Pile	=	74.059	Tons
Self Weight of the pile	=	3.905	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	20.781	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	13.845	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.5	m
Depth of pile	=	15	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.196	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	42.72	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	31.460	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	31.46	Tons

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

 For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-3 and Pile Diameter =0.5m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.571	0	43.31	0	0
0.5	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.356	0	43.31	0	0
0.5	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.356	19.2	0	45.235	4.524
0.5	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074
0.5	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074
0.5	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	13.5	15	0	17.542	7.732	SP	0	0	34.66	104.53	98.73	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
Total Skin Friction																			437.745	kN
																			43.932	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	43.932	Tons
Total End Bearing Capacity of Pile	=	40.446	Tons
Total Capacity of Pile	=	84.378	Tons
Self Weight of the pile	=	3.905	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	24.221	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	14.156	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer =	34.66	degrees	
Pile dia =	0.5	m	
Depth of pile =	15	m	
Effective unit weight of the soil at pile tip =	7.732	kN/m ³	
Cross-sectional area of pile tip A_p =	0.196	m ²	
Effective depth =	7.5	m	
Effective overburden pressure at pile tip =	42.05	kPa	
N_q =	46.894		
N_γ =	45.525		
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$ =	40.446	Tons	
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p =	0.196	m ²	
N_c =	9		
Average cohesion at pile tip C =	0	kPa	
$A_p \cdot C \cdot N_c$ =	0	Tons	
Total End bearing capacity of pile =	40.446	Tons	

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

 For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-5 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.571	0	42.2	0	0	
0.5	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.356	16.5	0	38.850	3.885	
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.356	23.7	0	55.837	5.584	
0.5	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	47.195	0.63	1	0	2.356	34.4	0	0	0	
0.5	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	47.195	0.63	1	0	2.356	34.4	0	0	0	
																Total Skin Friction		418.967 42.048		kN Tons	

Total Skin FrictionCapacity of Pile	=	42.048	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	42.048	Tons
Self Weight of the pile	=	3.905	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]- Self weight of pile	=	10.110	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	13.716	Tons

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LENGTH OF PILE- 15 M

BH-6 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.571	0	42.6	66.925	6.692
0.5	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.356	0	42.6	100.366	10.037
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.356	23.2	0	54.706	5.471
0.5	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.356	30.4	0	71.693	7.169
0.5	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	13.5	15	0	17.444	7.634	SM	0	0	35.62	109.21	103.485	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
																Total Skin Friction		684.314 68.678	kN Tons	

Total Skin FrictionCapacity of Pile	=	68.678	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	68.678	Tons
Self Weight of the pile	=	3.905	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	18.987	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	19.930	Tons

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LENGTH OF PILE- 15 M

BH-7 and Pile Diameter =0.5m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.5	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.571	0	41.84	0	0
0.5	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.356	0	41.84	0	0
0.5	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.356	22.7	0	53.481	5.348
0.5	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.356	30	0	70.704	7.070
0.5	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	69.45	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	81.49	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	93.53	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	13.5	15	0	17.836	8.026	SM	0	0	35.65	111.59	105.57	105.57	0.62	1	0	2.356	32.7	0	77.041	7.704
Total Skin Friction																			509.391	kN
																			51.122	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	51.122	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	51.122	Tons
Self Weight of the pile	=	3.905	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	13.135	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	15.834	Tons

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LENGTH OF PILE - 15 M

BH-9 and Pile Diameter =0.5m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.5	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.571	0	43.33	0	0
0.5	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.356	0	43.33	0	0
0.5	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.356	19	0	0	0
0.5	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.356	25.1	0	59.159	5.916
0.5	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	13.5	15	0	17.15	7.34	SM	0	0	34.68	99.96	94.455	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
																Total Skin Friction		381.931		
																		38.331		
																		kN		
																		Tons		

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	38.331	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	38.331	Tons
Self Weight of the pile		=	3.905	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	8.871	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	12.849	Tons

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LENGTH OF PILE- 15 M

BH-10 and Pile Diameter =0.5m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.5	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.571	0	43.31	0	0
0.5	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.356	0	43.31	0	0
0.5	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.356	20	0	47.214	4.721
0.5	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.356	26.7	0	62.858	6.286
0.5	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	13.5	15	0	17.15	7.34	SM	0	0	35.14	100.84	95.335	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
Total Skin Friction																			445.331	kN
																			44.693	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	44.693	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	44.693	Tons
Self Weight of the pile	=	3.905	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	10.992	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wght of Pile	=	14.334	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.
Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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DIA OF PILE – 0.6 M

LENGTH OF PILE- 15 M

BH-1 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.68	5.87	CI	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.68	5.87	CI	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.885	0	42.33	0	0
0.6	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.827	13.7	0	0	0
0.6	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.827	19.9	0	56.314	5.631
0.6	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.827	26.2	0	73.983	7.398
0.6	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	2.827	36.5	0	103.044	10.304
0.6	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	13.5	15	0	17.248	7.438	SM	0	0	34.05	104.67	99.09	54.45	0.59	1	0	2.827	36.5	0	103.04415	10.304415
Total Skin Friction																			336.385	kN
																			33.760	Tons

Liquefiable
Liquefiable
Liquefiable

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	33.760	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	33.760	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	5.629	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	13.501	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-2 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.77	2.28	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.885	5.7	0	0	0
0.6	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.827	8.56	0	0	0
0.6	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.827	19.4	0	54.957	5.496
0.6	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.827	25.6	0	72.371	7.237
0.6	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	2.827	31.8	0	89.786	8.979
0.6	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	13.5	15	0	17.15	7.34	SP-SM	0	0	33.18	102.18	96.675	53	0.57	1	0	2.827	34	0	96.033	9.603
Total Skin Friction																			601.246 60.341	kN Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	60.341	Tons
Total End Bearing Capacity of Pile	=	56.130	Tons
Total Capacity of Pile	=	116.471	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	33.200	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	19.703	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.6	m
Depth of pile	=	15	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	53	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	56.127	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	56.13	Tons

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

 For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

Report No. SR-2507170001

LENGTH OF PILE- 15 M

BH-3 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.885	0	43.31	0	0
0.6	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.827	0	43.31	0	0
0.6	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.827	19.2	0	54.278	5.428
0.6	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.827	25.8	0	72.880	7.288
0.6	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	2.827	32.4	0	91.453	9.145
0.6	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	13.5	15	0	17.542	7.732	SP	0	0	34.66	104.53	98.73	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
																Total Skin Friction		626.831		kN
																		62.909		Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	62.909	Tons
Total End Bearing Capacity of Pile	=	72.953	Tons
Total Capacity of Pile	=	135.862	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	39.664	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	20.302	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.6	m
Depth of pile	=	15	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	52.77	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p(0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	72.953	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	72.953	Tons

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.


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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-5 and Pile Diameter =0.6m

BH-5 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	16.366	6.556	Cl	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.366	6.556	Cl	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.885	0	42.2	0	0
0.6	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.827	16.5	0	46.617	4.662
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.827	23.7	0	67.000	6.700
0.6	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.827	34.4	0	97.277	9.728
0.6	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	58.72	0.63	1	0	2.827	42.8	0	0	0
0.6	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	58.72	0.63	1	0	2.827	42.8	0	0	0
																Total Skin Friction		573.966 57.603	kN Tons	

Total Skin FrictionCapacity of Pile	=	57.603	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	57.603	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	13.577	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	19.065	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-6 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	16.464	6.654	Cl	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.464	6.654	Cl	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.885	0	42.6	80.301	8.030
0.6	3	4.5	0	16.464	6.654	Cl	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.827	0	42.6	120.430	12.043
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.827	23.2	0	65.643	6.564
0.6	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.827	30.4	0	86.026	8.603
0.6	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	13.5	15	0	17.444	7.634	SM	0	0	35.62	109.21	103.485	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
																Total Skin Friction		934.479 93.784	kN Tons	

Total Skin FrictionCapacity of Pile	=	93.784	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	93.784	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	25.638	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weigt of Pile	=	27.507	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-7 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.6	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.885	0	41.84	0	0
0.6	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.827	0	41.84	0	0
0.6	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.827	22.7	0	64.173	6.417
0.6	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.827	30	0	84.838	8.484
0.6	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	69.45	0.62	1	0	2.827	41	0	115.850	11.585
0.6	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	81.49	0.62	1	0	2.827	41	0	115.850	11.585
0.6	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	93.53	0.62	1	0	2.827	41	0	115.850	11.585
0.6	13.5	15	0	17.836	8.026	SM	0	0	35.65	111.59	105.57	105.57	0.62	1	0	2.827	41	0	115.850	11.585
Total Skin Friction																			728.263	kN
																			73.089	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	73.089	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	73.089	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapcity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	18.739	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	22.678	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-9 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.885	0	43.33	0	0
0.6	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.827	0	43.33	0	0
0.6	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.827	19	0	0	0
0.6	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.827	25.1	0	70.986	7.099
0.6	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750
0.6	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750
0.6	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750
0.6	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750
0.6	13.5	15	0	17.15	7.34	SM	0	0	34.68	99.96	94.455	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750
Total Skin Friction																			558.502	kN
																			56.051	Tons

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	56.051	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	56.051	Tons
Self Weight of the pile	=	5.624	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	13.060	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	18.702	Tons

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LENGTH OF PILE- 15 M

BH-10 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.6	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.885	0	43.31	0	0
0.6	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.827	0	43.31	0	0
0.6	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.827	20	0	56.653	5.665
0.6	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.827	26.7	0	75.424	7.542
0.6	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	2.827	35.9	0	101.348	10.135
0.6	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	51.295	0.61	1	0	2.827	35.9	0	101.348	10.135
0.6	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	51.295	0.61	1	0	2.827	35.9	0	101.348	10.135
0.6	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	51.295	0.61	1	0	2.827	35.9	0	101.348	10.135
0.6	13.5	15	0	17.15	7.34	SM	0	0	35.14	100.84	95.335	51.295	0.61	1	0	2.827	35.9	0	101.348	10.135
Total Skin Friction																			638.817	kN
																			64.112	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	64.112	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	64.112	Tons
Self Weight of the pile		=	5.624	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]- Self weight of pile		=	15.747	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile		=	20.583	Tons

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DIA OF PILE- 0.75 M

LENGTH OF PILE- 15 M

BH-1 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.75	0	2	0	15.68	5.87	CI	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	4.712	0	CUTOFF	CUTOFF	Liquefiable	
0.75	2	3	0	15.68	5.87	CI	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	2.356	0	42.33	0	0	Liquefiable
0.75	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	3.534	13.7	0	0	0	Liquefiable
0.75	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	3.534	19.9	0	70.397	7.040	
0.75	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	3.534	26.2	0	92.485	9.248	
0.75	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	3.534	36.5	0	128.814	12.881	
0.75	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	13.5	15	0	17.248	7.438	SM	0	0	34.05	104.67	99.09	65.61	0.59	1	0	3.534	43.9	0	155.213	15.521	
																Total Skin Friction		446.910 44.852	kN Tons		

Total Skin FrictionCapacity of Pile	=	44.852	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	44.852	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	6.163	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	19.253	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-2 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	4.712	2.28	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	2.356	5.7	0	0	0
0.75	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	3.534	8.56	0	0	0
0.75	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	3.534	19.4	0	68.701	6.870
0.75	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	3.534	25.6	0	90.470	9.047
0.75	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	3.534	31.8	0	112.240	11.224
0.75	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	13.5	15	0	17.15	7.34	SP-SM	0	0	33.18	102.18	96.675	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
																Total Σ Skin Friction		848.019	kN	
																		85.107	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	85.107	Tons
Total End Bearing Capacity of Pile	=	105.490	Tons
Total Capacity of Pile	=	190.597	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	54.745	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	28.645	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARINGS BH-2, DIA OF PILE-0.75 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.75	m
Depth of pile	=	15	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.645	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	105.485	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	105.49	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-3 and Pile Diameter =0.75m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	2.356	0	43.31	0	0
0.75	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	3.534	0	43.31	0	0
0.75	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	3.534	19.2	0	67.853	6.785
0.75	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	3.534	25.8	0	91.107	9.111
0.75	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	3.534	32.4	0	114.325	11.432
0.75	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	13.5	15	0	17.542	7.732	SP	0	0	34.66	104.53	98.73	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
																	Total Skin Friction		891.452 89.466	kN Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction Capacity of Pile	=	89.466	Tons
Total End Bearing Capacity of Pile	=	138.276	Tons
Total Capacity of Pile	=	227.742	Tons
Self Weight of the pile	=	8.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	67.127	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	29.663	Tons

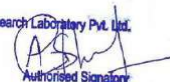
For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARINGS BH-3, DIA OF PILE-0.75 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.75	m
Depth of pile	=	15	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.93	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	138.276	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	138.276	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 15 M

BH-5 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.75	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	4.712	0	CUTOFF	CUTOFF	CUTOFF	
0.75	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	2.356	0	42.2	0	0	
0.75	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	3.534	16.5	0	58.276	5.828	
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	3.534	23.7	0	83.756	8.376	
0.75	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	3.534	34.4	0	121.605	12.160	
0.75	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	3.534	42.8	0	151.291	15.129	
0.75	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232	
0.75	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232	
0.75	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	70.76	0.63	1	0	3.534	51.6	0	0	0	
0.75	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	70.76	0.63	1	0	3.534	51.6	0	0	0	
																Total Skin Friction		779.56506 78.23714942		kN Tons	

Total Skin FrictionCapacity of Pile	=	78.237	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	78.237	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	17.292	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	27.042	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.
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LENGTH OF PILE- 15 M

BH-6 and Pile Diameter =0.75m																			
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN
0.75	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	4.712	0	CUTOFF	CUTOFF
0.75	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	2.356	0	42.6	100.366
0.75	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	3.534	0	42.6	150.548
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	3.534	23.2	0	82.059
0.75	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	3.534	30.4	0	107.540
0.75	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	3.534	41.2	0	145.530
0.75	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	69.135	0.62	1	0	3.534	49.4	0	174.403
0.75	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	69.135	0.62	1	0	3.534	49.4	0	174.403
0.75	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	69.135	0.62	1	0	3.534	49.4	0	174.403
0.75	13.5	15	0	17.444	7.634	SM	0	0	35.62	109.21	103.485	69.135	0.62	1	0	3.534	49.4	0	174.403
Total Skin Friction																			1283.655
																			128.828
																			kN
																			Tons

Total Skin Friction	Capacity of Pile	=	128.828	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	128.828	Tons
Self Weight of the pile		=	8.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	34.155	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile		=	38.847	Tons

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LENGTH OF PILE- 15 M

BH-7 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	2.356	0	41.84	0	0
0.75	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	3.534	0	41.84	0	0
0.75	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	3.534	22.7	0	80.222	8.022
0.75	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	3.534	30	0	106.055	10.606
0.75	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	3.534	41	0	144.823	14.482
0.75	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	13.5	15	0	17.836	8.026	SM	0	0	35.65	111.59	105.57	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
																Total Skin Friction		1031.963	kN	
																		103.568	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	103.568	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	103.568	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	25.735	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	32.953	Tons

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LENGTH OF PILE- 15 M

BH-9 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.75	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	2.356	0	43.33	0	0
0.75	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	3.534	0	43.33	0	0
0.75	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	3.534	19	0	0	0
0.75	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	3.534	25.1	0	88.739	8.874
0.75	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	3.534	34.5	0	121.888	12.189
0.75	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	13.5	15	0	17.15	7.34	SM	0	0	34.68	99.96	94.455	61.425	0.6	1	0	3.534	42	0	148.499	14.850
																Total Skin Friction		804.621	kN	
																		80.752	Tons	

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	80.752	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	80.752	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	18.130	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	27.629	Tons

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LENGTH OF PILE- 15 M

BH-10 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.75	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	2.356	0	43.31	0	0
0.75	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	3.534	0	43.31	0	0
0.75	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	3.534	20	0	70.821	7.082
0.75	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	3.534	26.7	0	94.287	9.429
0.75	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	3.534	35.9	0	126.694	12.669
0.75	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	13.5	15	0	17.15	7.34	SM	0	0	35.14	100.84	95.335	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
																	Total Skin Friction		907.284 91.055	kN Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	91.055	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	91.055	Tons
Self Weight of the pile	=	8.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	21.565	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	30.033	Tons

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DIA OF PILE- 0.5 M

LENGTH OF PILE- 20 M

BH-1 and Pile Diameter =0.5m

BH-1 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	15.68	5.87	Cl	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.142	0	CUTOFF	CUTOFF	CUTOFF	Liquefiable
0.5	2	3	0	15.68	5.87	Cl	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.571	0	42.33	0	0	Liquefiable
0.5	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.356	13.7	0	0	0	Liquefiable
0.5	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.356	19.9	0	46.932	4.693	
0.5	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.356	26.2	0	61.657	6.166	
0.5	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	43.66	0.59	1	0	2.356	29.2	0	68.866	6.887	
0.5	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	43.66	0.59	1	0	2.356	29.2	0	0	0	Liquefiable
0.5	13.5	20	0	17.248	7.438	SM	0	0	34.05	141.86	117.685	43.66	0.59	1	0	10.21	29.2	0	298.438	29.844	
																Total Skin Friction		475.892		kN	
																		47.761		Tons	

Total Skin FrictionCapacity of Pile	=	47.761	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	47.761	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	10.513	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	16.552	Tons

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LENGTH OF PILE- 20 M

BH-2 and Pile Diameter =0.5m

BH-2 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.142	2.28	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.571	5.7	0	0	0
0.5	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.356	8.56	0	0	0
0.5	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.356	19.4	0	45.801	4.580
0.5	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031
0.5	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031
0.5	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451
0.5	13.5	20	0	17.15	7.34	SP-SM	0	0	33.18	138.88	115.025	42.72	0.57	1	0	10.21	27.4	0	279.550	27.955
																Total Skin Friction		639.499 64.180	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	64.180	Tons
Total End Bearing	Capacity of Pile	=	31.460	Tons
Total	Capacity of Pile	=	95.640	Tons
Self Weight of the pile		=	5.407	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	26.473	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	20.383	Tons

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CALCULATIONS OF END BEARING BH-2, Dia of Pile-0.5 m

END BEARING CAPCITY FOR SANDY SOILS		
ϕ at the base layer =	33.18	degrees
Pile dia =	0.5	m
Depth of pile =	20	m
Effective unit weight of the soil at pile tip =	7.34	kN/m ³
Cross-sectional area of pile tip A_p =	0.196	m ²
Effective depth =	7.5	m
Effective overburden pressure at pile tip =	42.72	kPa
N_q =	35.951	
N_γ =	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$ =	31.460	Tons
END BEARING CAPCITY FOR CLAYEY SOILS		
Cross-sectional area of pile tip A_p =	0.196	m ²
N_c =	9	
Average cohesion at pile tip C =	0	kPa
$A_p \cdot C \cdot N_c$ =	0	Tons
Total End bearing capacity of pile =	31.46	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

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LENGTH OF PILE- 20 M

BH-3 and Pile Diameter =0.5m

BH-3 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.571	0	43.31	0	0
0.5	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.356	0	43.31	0	0
0.5	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.356	19.2	0	45.235	4.524
0.5	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074
0.5	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074
0.5	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776
0.5	13.5	20	0	17.542	7.732	SP	0	0	34.66	143.19	118.06	42.05	0.6	1	0	10.21	28.8	0	293.640	29.364
																Total Skin Friction		663.626	kN	
																		66.601	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	66.601	Tons
Total End Bearing Capacity of Pile	=	40.446	Tons
Total Capacity of Pile	=	107.047	Tons
Self Weight of the pile	=	5.407	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	30.275	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile	=	20.948	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING BH-3, Dia of Pile-0.5 m

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.5	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.19635	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	42.05	kPa
N_q	=	46.89	
N_γ	=	45.53	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	40.45	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.1963	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	40.446	Tons

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For. Shiva Testing & Research Laboratory Pvt. Ltd.


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LENGTH OF PILE- 20 M

BH-5 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.571	0	42.2	0	0
0.5	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.356	16.5	0	38.850	3.885
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.356	23.7	0	55.837	5.584
0.5	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107
0.5	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107
0.5	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107
0.5	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107
0.5	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	47.195	0.63	1	0	2.356	34.4	0	0.000	0.000
0.5	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	47.195	0.63	1	0	2.356	34.4	0	0.000	0.000
0.5	15	20	0	17.836	8.026	SM	0	0	36.15	153.03	132.965	47.195	0.63	1	0	7.854	34.4	0	270.256	27.026
																Total Skin Friction		689.224 69.170	kN Tons	

Total Skin FrictionCapacity of Pile	=	69.170	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	69.170	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	17.649	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	21.547	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-6 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.571	0	42.6	66.925	6.692	
0.5	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.356	0	42.6	100.366	10.037	
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.356	23.2	0	54.706	5.471	
0.5	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.356	30.4	0	71.693	7.169	
0.5	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812	
0.5	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812	
0.5	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812	
0.5	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812	
0.5	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	46.455	0.62	1	0	10.21	33.2	0	338.564	33.856	
																Total Skin Friction		944.753 94.815		kN Tons	

Total Skin Friction	Capacity of Pile	=	94.815	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	94.815	Tons
Self Weight of the pile		=	5.407	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	26.198	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile		=	27.531	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-7 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.571	0	41.84	0	0
0.5	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.356	0	41.84	0	0
0.5	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.356	22.7	0	53.481	5.348
0.5	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.356	30	0	70.704	7.070
0.5	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704
0.5	13.5	20	0	17.836	8.026	SM	0	0	35.65	151.72	125.635	45.81	0.62	1	0	10.21	32.7	0	333.867	33.387
																Total Skin Friction		766.217	kN	
																		76.897	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	76.897	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	76.897	Tons
Self Weight of the pile		=	5.407	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	20.225	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	23.350	Tons

For Shiva Testing & Research Laboratory Pvt Ltd.
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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-9 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.571	0	43.33	0	0	
0.5	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.356	0	43.33	0	0	
0.5	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.356	19	0	0	0	
0.5	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.356	25.1	0	59.159	5.916	
0.5	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455	
0.5	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455	
0.5	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455	
0.5	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455	
0.5	13.5	20	0	17.15	7.34	SM	0	0	34.68	136.66	112.805	40.065	0.6	1	0	10.21	27.4	0	279.754	27.975	
																Total Skin Friction		597.131 59.928		kN Tons	

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	59.928	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	59.928	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	14.569	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	19.391	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-10 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.5	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.571	0	43.31	0	0
0.5	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.356	0	43.31	0	0
0.5	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.356	20	0	47.214	4.721
0.5	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.356	26.7	0	62.858	6.286
0.5	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	13.5	20	0	17.15	7.34	SM	0	0	35.14	137.54	113.685	40.725	0.61	1	0	10.21	28.5	0	290.577	29.058
																	Total Skin Friction		668.856	kN
																			67.126	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	67.126	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	67.126	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	16.968	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	21.070	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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DIA OF PILE- 0.6 M LENGTH OF PILE- 20 M

BH-1 and Pile Diameter =0.6m

BH-1 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.68	5.87	CI	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.68	5.87	CI	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.885	0	42.33	0	0
0.6	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.827	13.7	0	0	0
0.6	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.827	19.9	0	56.314	5.631
0.6	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.827	26.2	0	73.983	7.398
0.6	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	2.827	36.5	0	103.044	10.304
0.6	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	54.45	0.59	1	0	2.827	36.5	0	0	0
0.6	13.5	20	0	17.248	7.438	SM	0	0	34.05	141.86	117.685	54.45	0.59	1	0	12.252	36.5	0	446.585	44.659
																Total Skin Friction		679.926 68.237	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	68.237	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	68.237	Tons
Self Weight of the pile		=	7.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile				
		=	14.959	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weigt of Pile				
		=	23.709	Tons

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LENGTH OF PILE- 20 M

BH-2 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	2	kN/m ²	kN	Tons
0.6	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.77	2.28	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.885	5.7	0	0	0
0.6	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.827	8.56	0	0	0
0.6	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.827	19.4	0	54.957	5.496
0.6	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.827	25.6	0	72.371	7.237
0.6	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	2.827	31.8	0	89.786	8.979
0.6	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	13.5	20	0	17.15	7.34	SP-SM	0	0	33.18	138.88	115.025	53	0.57	1	0	12.252	34	0	416.200	41.620
Total Skin Friction																			921.414 92.473	kN Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	92.473	Tons
Total End Bearing Capacity of Pile	=	56.130	Tons
Total Capacity of Pile	=	148.603	Tons
Self Weight of the pile	=	7.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	41.748	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	29.364	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARIG CAPACITY

GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



Shiva Testing & Research Laboratory Pvt. Ltd.

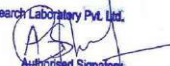
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BH-2, DIA OF PILE-0.6 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.6	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	53	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	56.127	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	56.13	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-3 and Pile Diameter =0.6m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.6	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.77	0	CUTOFF	CUTOFF	CUTOFF	
0.6	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.885	0	43.31	0	0	
0.6	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.827	0	43.31	0	0	
0.6	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.827	19.2	0	54.278	5.428	
0.6	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.827	25.8	0	72.880	7.288	
0.6	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	2.827	32.4	0	91.453	9.145	
0.6	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205	
0.6	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205	
0.6	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205	
0.6	13.5	20	0	17.542	7.732	SP	0	0	34.66	143.19	118.06	52.77	0.6	1	0	12.252	36.1	0	442.297	44.230	
																Total Skin Friction			967.073	kN	
																			97.055	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	97.055	Tons
Total End Bearing Capacity of Pile	=	72.953	Tons
Total Capacity of Pile	=	170.009	Tons
Self Weight of the pile	=	7.787	Tons
Net VerticalCapcity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	48.883	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	30.433	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-3, DIA OF PILE-0.6 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.6	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	52.77	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p(0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	72.953	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	72.953	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For. Shiva Testing & Research Laboratory Pvt. Ltd.

(Signature)
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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-5 and Pile Diameter =0.6m

BH-5 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.6	0	2	0	16.366	6.556	Cl	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.366	6.556	Cl	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.885	0	42.2	0	0
0.6	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.827	16.5	0	46.617	4.662
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.827	23.7	0	67.000	6.700
0.6	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.827	34.4	0	97.277	9.728
0.6	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102
0.6	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	58.72	0.63	1	0	2.827	42.8	0	0	0
0.6	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	58.72	0.63	1	0	2.827	42.8	0	0	0
0.6	15	20	0	17.836	8.026	SM	0	0	36.15	153.03	132.965	58.72	0.63	1	0	9.425	42.8	0	403.484	40.348
																	Total Skin Friction		977.450 98.097	kN Tons

Liquefiable

Liquefiable

Liquefiable

Total Skin Friction	Capacity of Pile	=	98.097	Tons
Total End Bearing	Capacity of Pile	=	0.000	Tons
Total	Capacity of Pile	=	98.097	Tons
Self Weight of the pile		=	7.787	Tons
Net Vertical	Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	24.912	Tons
Uplift	Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weigt of Pile	=	30.676	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-6 and Pile Diameter =0.6m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.6	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.885	0	42.6	80.301	8.0301
0.6	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.827	0	42.6	120.430	12.043
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.827	23.2	0	65.643	6.564
0.6	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.827	30.4	0	86.026	8.603
0.6	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	57.685	0.62	1	0	12.252	41.2	0	504.537	50.454
Total Skin Friction																			1322.601	kN
																			132.736	Tons

Total Skin Friction	Capacity of Pile	=	132.736	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	132.736	Tons
Self Weight of the pile		=	7.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile				
		=	36.459	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile				
		=	38.759	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-7 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.885	0	41.84	0	0
0.6	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.827	0	41.84	0	0
0.6	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.827	22.7	0	64.173	6.417
0.6	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.827	30	0	84.838	8.484
0.6	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	13.5	20	0	17.836	8.026	SM	0	0	35.65	151.72	125.635	57.41	0.62	1	0	12.252	41	0	502.087	50.209
																Total Skin Friction		1114.500 111.851	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	111.851	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	111.851	Tons
Self Weight of the pile	=	7.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	29.497	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	33.885	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-9 and Pile Diameter =0.6m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.6	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.77	0	CUTOFF	CUTOFF	CUTOFF	
0.6	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.885	0	43.33	0	0	
0.6	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.827	0	43.33	0	0	
0.6	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.827	19	0	0	0	
0.6	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.827	25.1	0	70.986	7.099	
0.6	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	13.5	20	0	17.15	7.34	SM	0	0	34.68	136.66	112.805	50.415	0.6	1	0	12.252	34.5	0	422.571	42.257	
																Total Skin Friction		883.570		kN	
																		88.675		Tons	

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	88.675	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	88.675	Tons
Self Weight of the pile	=	7.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	21.772	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	28.478	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-10 and Pile Diameter =0.6m																	
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²
0.6	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.77	0
0.6	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.885	0
0.6	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.827	0
0.6	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.827	20
0.6	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.827	26.7
0.6	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	2.827	35.9
0.6	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	51.295	0.61	1	0	2.827	35.9
0.6	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	51.295	0.61	1	0	2.827	35.9
0.6	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	51.295	0.61	1	0	2.827	35.9
0.6	13.5	20	0	17.15	7.34	SM	0	0	35.14	137.54	113.685	51.295	0.61	1	0	12.252	35.9
																Total Skin Friction	
																976.703	
																kN	
																98.022	
																Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	98.022	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	98.022	Tons
Self Weight of the pile	=	7.787	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	24.887	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	30.659	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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DIA OF PILE- 0.75 M LENGTH OF PILE- 20 M

BH-1 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.75	0	2	0	15.68	5.87	Cl	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	4.712	0	CUTOFF	CUTOFF	CUTOFF	Liquefiable
0.75	2	3	0	15.68	5.87	Cl	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	2.356	0	42.33	0	0	Liquefiable
0.75	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	3.534	13.7	0	0	0	Liquefiable
0.75	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	3.534	19.9	0	70.397	7.040	
0.75	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	3.534	26.2	0	92.485	9.248	
0.75	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	3.534	36.5	0	128.814	12.881	
0.75	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	65.61	0.59	1	0	3.534	43.9	0	0	0	Liquefiable
0.75	13.5	20	0	17.248	7.438	SM	0	0	34.05	141.86	117.685	65.61	0.59	1	0	15.315	43.9	0	672.635	67.263	
																Total Skin Friction		964.331		kN	
																		96.780		Tons	

Total Skin FrictionCapacity of Pile	=	96.780	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	96.780	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	20.093	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	34.749	Tons

For Shiva Testing & Research Laboratory Pvt Ltd.

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LENGTH OF PILE- 20 M

BH-2 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m³	kN/m³		Kg/cm²	kN/m²	Degree	kN/m²	kN/m²	kN/m²				m²	²	kN/m²	kN	Tons
0.75	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	4.712	2.28	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	2.356	5.7	0	0	0
0.75	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	3.534	8.56	0	0	0
0.75	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	3.534	19.4	0	68.701	6.870
0.75	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	3.534	25.6	0	90.470	9.047
0.75	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	3.534	31.8	0	112.240	11.224
0.75	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	13.5	20	0	17.15	7.34	SP-SM	0	0	33.18	138.88	115.025	63.645	0.57	1	0	15.315	40.8	0	624.699	62.470
																	Total Skin Friction		1328.566	kN
																			133.335	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	133.335	Tons
Total End Bearing Capacity of Pile	=	105.490	Tons
Total Capacity of Pile	=	238.825	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	67.441	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	43.278	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY (BH-2, DIA OF PILE-0.75 M)

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.75	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.645	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P D \cdot N_q)$	=	105.485	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	105.49	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-3 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	2.356	0	43.31	0	0
0.75	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	3.534	0	43.31	0	0
0.75	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	3.534	19.2	0	67.853	6.785
0.75	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	3.534	25.8	0	91.107	9.111
0.75	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	3.534	32.4	0	114.325	11.432
0.75	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454
0.75	13.5	20	0	17.542	7.732	SP	0	0	34.66	143.19	118.06	63.93	0.6	1	0	15.315	43.7	0	669.725	66.972
																Total Skin Friction		1406.635	kN	
																		141.170	Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	141.170	Tons
Total End Bearing Capacity of Pile	=	138.276	Tons
Total Capacity of Pile	=	279.446	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	80.982	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	45.106	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY (BH-3, DIA OF PILE-0.75 M)

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.75	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.93	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	138.276	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	138.276	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-5 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	16.366	6.556	Cl	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	16.366	6.556	Cl	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	2.356	0	42.2	0	0
0.75	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	3.534	16.5	0	58.276	5.828
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	3.534	23.7	0	83.756	8.376
0.75	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	3.534	34.4	0	121.605	12.160
0.75	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	3.534	42.8	0	151.291	15.129
0.75	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232
0.75	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232
0.75	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	70.76	0.63	1	0	3.534	51.6	0	0	0
0.75	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	70.76	0.63	1	0	3.534	51.6	0	0	0
0.75	15	20	0	17.836	8.026	SM	0	0	36.15	153.03	132.965	70.76	0.63	1	0	11.781	51.6	0	607.782	60.778
																Total Skin Friction		1387.347 139.234	kN Tons	

Total Skin FrictionCapacity of Pile	=	139.234	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	139.234	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	34.245	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	44.655	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-6 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.75	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	2.356	0	42.6	100.366	10.037
0.75	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	3.534	0	42.6	150.548	15.055
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	3.534	23.2	0	82.059	8.206
0.75	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	3.534	30.4	0	107.540	10.754
0.75	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	3.534	41.2	0	145.530	14.553
0.75	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	69.135	0.62	1	0	15.315	49.4	0	755.795	75.580
																Total Skin Friction		1865.047 187.176	kN Tons	

Total Skin FrictionCapacity of Pile	=	187.176	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	187.176	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	50.225	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	55.841	Tons

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LENGTH OF PILE- 20 M

BH-7 and Pile Diameter =0.75m

BH-7 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K_{P_{Di}} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.75	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	4.712	0	CUTOFF	CUTOFF	CUTOFF	
0.75	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	2.356	0	41.84	0	0	
0.75	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	3.534	0	41.84	0	0	
0.75	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	3.534	22.7	0	80.222	8.022	
0.75	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	3.534	30	0	106.055	10.606	
0.75	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	3.534	41	0	144.823	14.482	
0.75	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522	
0.75	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522	
0.75	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522	
0.75	13.5	20	0	17.836	8.026	SM	0	0	35.65	151.72	125.635	69.45	0.62	1	0	15.315	49.6	0	759.318	75.932	
																Total Skin Friction		1616.065 162.188		kN Tons	

Total Skin Friction	Capacity of Pile	=	162.188	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	162.188	Tons
Self Weight of the pile		=	12.167	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	41.896	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile		=	50.011	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-9 and Pile Diameter =0.75m

Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.75	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	4.712	0	CUTOFF	CUTOFF	CUTOFF	Liquefiable
0.75	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	2.356	0	43.33	0	0	Liquefiable
0.75	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	3.534	0	43.33	0	0	Liquefiable
0.75	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	3.534	19	0	0	0	Liquefiable
0.75	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	3.534	25.1	0	88.739	8.874	
0.75	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	3.534	34.5	0	121.888	12.189	
0.75	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	61.425	0.6	1	0	3.534	42	0	148.499	14.850	
0.75	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	72.435	0.6	1	0	3.534	42	0	148.499	14.850	
0.75	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	83.445	0.6	1	0	3.534	42	0	148.499	14.850	
0.75	13.5	20	0	17.15	7.34	SM	0	0	34.68	136.66	112.805	112.805	0.6	1	0	15.315	42	0	643.536	64.354	
Total Skin Friction																			1299.659	kN	
																			130.434	Tons	

Total Skin FrictionCapacity of Pile	=	130.434	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	130.434	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	31.311	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	42.601	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE- 20 M

BH-10 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.75	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	2.356	0	43.31	0	0
0.75	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	3.534	0	43.31	0	0
0.75	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	3.534	20	0	70.821	7.082
0.75	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	3.534	26.7	0	94.287	9.429
0.75	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	3.534	35.9	0	126.694	12.669
0.75	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387
0.75	13.5	20	0	17.15	7.34	SM	0	0	35.14	137.54	113.685	62.305	0.61	1	0	15.315	43.5	0	666.815	66.682
Total Skin Friction																	1420.229		142.534	kN
																				Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	142.534	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	142.534	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	35.345	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	45.425	Tons

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LENGTH OF PILE-25 M DIA OF PILE-0.5 M

BH-1 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.68	5.87	CI	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.68	5.87	CI	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.571	0	42.33	0	0
0.5	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.356	13.7	0	0	0
0.5	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.356	19.9	0	46.932	4.693
0.5	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.356	26.2	0	61.657	6.166
0.5	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	43.66	0.59	1	0	2.356	29.2	0	68.866	6.887
0.5	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	43.66	0.59	1	0	2.356	29.2	0	0	0
0.5	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	43.66	0.59	1	0	2.356	29.2	0	0	0
0.5	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	43.66	0.59	1	0	2.356	29.2	0	0	0
0.5	13.5	25	0	17.248	7.438	SM	0	0	34.05	179.05	136.28	43.66	0.59	1	0	18.064	29.2	0	528.011	52.801
																Total Skin Friction		705.465	kN	
																		70.800	Tons	

Liquefiable
Liquefiable
Liquefiable

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	70.800	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	70.800	Tons
Self Weight of the pile	=	6.910	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	16.691	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	23.430	Tons

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LENGTH OF PILE-25 M

BH-2 and Pile Diameter =0.5m																						
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance		
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons		
0.5	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.142	2.28	CUTOFF	CUTOFF	CUTOFF		
0.5	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.571	5.7	0	0	0		
0.5	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.356	8.56	0	0	0		
0.5	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.356	19.4	0	45.801	4.580		
0.5	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031		
0.5	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	42.72	0.54	1	0	2.356	25.6	0	60.314	6.031		
0.5	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451		
0.5	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451		
0.5	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	42.72	0.57	1	0	2.356	27.4	0	64.507	6.451		
0.5	13.5	25	0	17.15	7.34	SP-SM	0	0	33.18	175.58	133.375	42.72	0.57	1	0	18.064	27.4	0	494.592	49.459		
																	Total Skin Friction		854.542		kN	
																			85.762		Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	85.762	Tons
Total End Bearing Capacity of Pile	=	31.460	Tons
Total Capacity of Pile	=	117.222	Tons
Self Weight of the pile	=	6.910	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	32.164	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	26.921	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY BH-2, DIA OF PILE-0.5 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.5	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.196	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	42.72	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	31.460	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	31.46	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-3 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.571	0	43.31	0	0	
0.5	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.356	0	43.31	0	0	
0.5	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.356	19.2	0	45.235	4.524	
0.5	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074	
0.5	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	42.05	0.55	1	0	2.356	25.8	0	60.738	6.074	
0.5	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776	
0.5	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776	
0.5	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	42.05	0.6	1	0	2.356	28.8	0	67.759	6.776	
0.5	13.5	25	0	17.542	7.732	SP	0	0	34.66	181.85	137.39	42.05	0.6	1	0	18.064	28.8	0	519.521	51.952	
																Total Skin Friction		889.507 89.271		kN Tons	

 Liquefiable
 Liquefiable
 Liquefiable

Total Skin FrictionCapacity of Pile	=	89.271	Ton
Total End Bearing Capacity of Pile	=	40.446	Ton
Total Capacity of Pile	=	129.717	Ton
Self Weight of the pile	=	6.910	Ton
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	36.329	Ton
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weigt of Pile	=	27.739	Ton

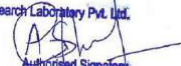
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CALCULATIONS OF END BEARING CAPACITY BH-3, DIA OF PILE-0.5 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.5	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.196	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	42.05	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	40.446	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	40.446	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-5 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.5	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.571	0	42.2	0	0	
0.5	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.356	16.5	0	38.850	3.885	
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.356	23.7	0	55.837	5.584	
0.5	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	47.195	0.63	1	0	2.356	34.4	0	81.070	8.107	
0.5	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	47.195	0.63	1	0	2.356	34.4	0	0	0	
0.5	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	47.195	0.63	1	0	2.356	34.4	0	0	0	
0.5	15	25	0	17.836	8.026	SM	0	0	37.15	193.16	153.03	47.195	0.64	1	0	15.708	35.1	0	551.822	55.182	
																Total Skin Friction		970.790		kN	
																		97.428		Tons	

Total Skin FrictionCapacity of Pile	=	97.428	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	97.428	Tons
Self Weight of the pile	=	6.910	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	25.567	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	29.643	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.
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LENGTH OF PILE-25 M

BH-6 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.571	0	42.6	66.925	6.692
0.5	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.356	0	42.6	100.366	10.037
0.5	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.356	23.2	0	54.706	5.471
0.5	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.356	30.4	0	71.693	7.169
0.5	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	46.455	0.62	1	0	2.356	33.2	0	78.125	7.812
0.5	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	46.455	0.62	1	0	10.21	33.2	0	338.564	33.856
																Total Skin Friction		944.753 94.815	kN Tons	

Total Skin FrictionCapacity of Pile	=	94.815	Tons
Total End Bearing Capacity of Pile	=	52.920	Tons
Total Capacity of Pile	=	147.735	Tons
Self Weight of the pile	=	5.407	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	43.838	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	27.531	Tons

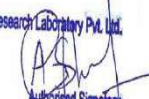
For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY BH-6, DIA OF PILE-0.5 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	35.62	degrees
Pile dia	=	0.5	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.634	kN/m ³
Cross-sectional area of pile tip A_p	=	0.196	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	46.455	kPa
N_q	=	55.840	
N_γ	=	52.992	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	52.920	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	52.920	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-7 and Pile Diameter =0.5m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.5	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.142	0	CUTOFF	CUTOFF	CUTOFF	
0.5	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.571	0	41.84	0	0	
0.5	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.356	0	41.84	0	0	
0.5	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.356	22.7	0	53.481	5.348	
0.5	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.356	30	0	70.704	7.070	
0.5	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704	
0.5	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704	
0.5	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704	
0.5	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	45.81	0.62	1	0	2.356	32.7	0	77.041	7.704	
0.5	13.5	25	0	17.836	8.026	SM	0	0	35.65	191.85	145.7	45.81	0.62	1	0	18.064	32.7	0	590.693	59.069	
																Total Skin Friction		1023.042		kN	
																		102.673		Tons	

 Liquefiable
 Liquefiable
 Liquefiable

Total Skin Friction	Capacity of Pile	=	102.673	Tons
Total End Bearing Capacity of Pile		=	52.600	Tons
Total Capacity of Pile		=	155.273	Tons
Self Weight of the pile		=	6.910	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	44.848	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	30.866	Tons

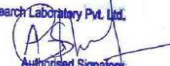
For Shiva Testing & Research Laboratory Pvt. Ltd.


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CALCULATIONS OF END BEARING CAPACITY BH-7, DIA OF PILE-0.5 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	35.65	degrees
Pile dia	=	0.5	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	8.026	kN/m ³
Cross-sectional area of pile tip A_p	=	0.196	m ²
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	45.81	kPa
N_q	=	56.147	
N_γ	=	53.246	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	52.600	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	52.600	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-9 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.5	0	2	0	15.484	5.674	Cl	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.484	5.674	Cl	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.571	0	43.33	0	0
0.5	3	4.5	0	15.484	5.674	Cl	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.356	0	43.33	0	0
0.5	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.356	19	0	0	0
0.5	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.356	25.1	0	59.159	5.916
0.5	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	40.065	0.6	1	0	2.356	27.4	0	64.554	6.455
0.5	13.5	25	0	17.15	7.34	SM	0	0	34.68	173.36	131.155	40.065	0.6	1	0	18.064	27.4	0	494.954	49.495
																Total Skin Friction		812.330	kN	
																		81.525	Tons	

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	81.525	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	81.525	Tons
Self Weight of the pile	=	6.910	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	20.266	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	25.932	Tons

For: Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-10 and Pile Diameter =0.5m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons
0.5	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.142	0	CUTOFF	CUTOFF	CUTOFF
0.5	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.571	0	43.31	0	0
0.5	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.356	0	43.31	0	0
0.5	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.356	20	0	47.214	4.721
0.5	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.356	26.7	0	62.858	6.286
0.5	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	40.725	0.61	1	0	2.356	28.5	0	67.052	6.705
0.5	13.5	25	0	17.15	7.34	SM	0	0	35.14	174.24	132.035	40.725	0.61	1	0	18.064	28.5	0	514.101	51.410
																Total Skin Friction		892.381 89.559	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	89.559	Tons
Total End Bearing Capacity of Pile		=	42.680	Tons
Total Capacity of Pile		=	132.239	Tons
Self Weight of the pile		=	6.910	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile				
		=	37.170	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile				
		=	27.807	Tons

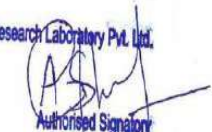
For. Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-10, DIA OF PILE-0.5 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	35.14	degrees
Pile dia	=	0.5	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m3
Cross-sectional area of pile tip A_p	=	0.196	m2
Effective depth	=	7.5	m
Effective overburden pressure at pile tip	=	40.725	kPa
N_q	=	51.162	
N_y	=	49.103	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_y + P_D \cdot N_q)$	=	42.680	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.196	m2
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	42.680	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorised Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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DIA OF PILE- 0.6 M LENGTH OF PILE-25 M

BH-1 and Pile Diameter =0.6m																			
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN
0.6	0	2	0	15.68	5.87	Cl	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	3.77	0	CUTOFF	CUTOFF
0.6	2	3	0	15.68	5.87	Cl	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	1.885	0	42.33	0
0.6	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	2.827	13.7	0	0
0.6	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	2.827	19.9	0	56.314
0.6	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	2.827	26.2	0	73.983
0.6	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	2.827	36.5	0	103.044
0.6	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	65.61	0.59	1	0	2.827	36.5	0	0
0.6	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	76.77	0.59	1	0	2.827	36.5	0	0
0.6	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	87.93	0.59	1	0	2.827	36.5	0	0
0.6	13.5	25	0	17.248	7.438	SM	0	0	34.05	179.05	136.28	136.28	0.59	1	0	21.677	36.5	0	790.127
Total Skin Friction																			1023.467
																			102.715
																			kN
																			Tons

Liquefiable
Liquefiable
Liquefiable

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	102.715	Tons
Total End Bearing Capacity of Pile		=	0.000	Tons
Total Capacity of Pile		=	102.715	Tons
Self Weight of the pile		=	9.950	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	24.289	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	33.917	Tons

For: Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-2 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.876	6.066	Cl	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	3.77	2.28	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.876	6.066	Cl	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	1.885	5.7	0	0	0
0.6	3	4.5	0	15.876	6.066	Cl	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	2.827	8.56	0	0	0
0.6	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	2.827	19.4	0	54.957	5.496
0.6	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	2.827	25.6	0	72.371	7.237
0.6	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	2.827	31.8	0	89.786	8.979
0.6	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	53	0.57	1	0	2.827	34	0	96.033	9.603
0.6	13.5	25	0	17.15	7.34	SP-SM	0	0	33.18	175.58	133.375	53	0.57	1	0	21.677	34	0	736.368	73.637
																Total Skin Friction		1241.581 124.605	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	124.605	Tons
Total End Bearing Capacity of Pile	=	56.130	Tons
Total Capacity of Pile	=	180.735	Tons
Self Weight of the pile	=	9.950	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	50.295	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	39.024	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-2, DIA OF PILE-0.6 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.6	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	53	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	56.127	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	56.13	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

LENGTH OF PILE-25 M

BH-3 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	1.885	0	43.31	0	0
0.6	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	2.827	0	43.31	0	0
0.6	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	2.827	19.2	0	54.278	5.428
0.6	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	2.827	25.8	0	72.880	7.288
0.6	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	2.827	32.4	0	91.453	9.145
0.6	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	52.77	0.6	1	0	2.827	36.1	0	102.055	10.205
0.6	13.5	25	0	17.542	7.732	SP	0	0	34.66	181.85	137.39	52.77	0.6	1	0	21.677	36.1	0	782.540	78.254
																Total Skin Friction		1307.316 131.202	kN Tons	

 Liquefiable
 Liquefiable
 Liquefiable

Total Skin FrictionCapacity of Pile	=	131.202	Tons
Total End Bearing Capacity of Pile	=	72.953	Tons
Total Capacity of Pile	=	204.155	Tons
Self Weight of the pile	=	9.950	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	58.102	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	40.564	Tons

 For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-3, DIA OF PILE-0.6 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.6	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	52.77	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	72.953	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	72.953	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory



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LENGTH OF PILE-25 M

BH-5 and Pile Diameter =0.6m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.6	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	3.77	0	CUTOFF	CUTOFF	CUTOFF	
0.6	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	1.885	0	42.2	0	0	
0.6	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	2.827	16.5	0	46.617	4.662	
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	2.827	23.7	0	67.000	6.700	
0.6	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	2.827	34.4	0	97.277	9.728	
0.6	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102	
0.6	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102	
0.6	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	58.72	0.63	1	0	2.827	42.8	0	121.024	12.102	
0.6	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	58.72	0.63	1	0	2.827	42.8	0	0	0	
0.6	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	58.72	0.63	1	0	2.827	42.8	0	0	0	
0.6	15	25	0	17.836	8.026	SM	0	0	37.15	193.16	153.03	58.72	0.64	1	0	18.85	43.7	0	823.934	82.393	
																Total Skin Friction		1397.899 140.293		kN Tons	

Total Skin FrictionCapacity of Pile	=	140.293	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	140.293	Tons
Self Weight of the pile	=	9.950	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	36.815	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	42.685	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-6 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons
0.6	0	2	0	16.464	6.654	Cl	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.464	6.654	Cl	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	1.885	0	42.6	80.301	8.030
0.6	3	4.5	0	16.464	6.654	Cl	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	2.827	0	42.6	120.430	12.043
0.6	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	2.827	23.2	0	65.643	6.564
0.6	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	2.827	30.4	0	86.026	8.603
0.6	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	57.685	0.62	1	0	2.827	41.2	0	116.416	11.642
0.6	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	57.685	0.62	1	0	12.252	41.2	0	504.537	50.454
Total Skin Friction																			1322.601	kN
																			132.736	Tons

Total Skin Friction	Capacity of Pile	=	132.736	Tons
Total End Bearing Capacity of Pile		=	94.506	Tons
Total Capacity of Pile		=	227.243	Tons
Self Weight of the pile		=	7.787	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	67.961	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	38.759	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-6, DIA OF PILE-0.6 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	35.62	degrees
Pile dia	=	0.6	m
Depth of pile	=	20	m
Effective unit weight of the soil at pile tip	=	7.634	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	57.685	kPa
N_q	=	55.840	
N_γ	=	52.992	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	94.506	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	94.506	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For. Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorised Signatory

LENGTH OF PILE-25 M

BH-7 and Pile Diameter =0.6m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.6	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	3.77	0	CUTOFF	CUTOFF	CUTOFF
0.6	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	1.885	0	41.84	0	0
0.6	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	2.827	0	41.84	0	0
0.6	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	2.827	22.7	0	64.173	6.417
0.6	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	2.827	30	0	84.838	8.484
0.6	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	57.41	0.62	1	0	2.827	41	0	115.850	11.585
0.6	13.5	25	0	17.836	8.026	SM	0	0	35.65	191.85	145.7	57.41	0.62	1	0	21.677	41	0	888.323	88.832
																Total Skin Friction		1500.736 150.614	kN Tons	

 Liquefiable
 Liquefiable
 Liquefiable

Total Skin FrictionCapacity of Pile	=	150.614	Tons
Total End Bearing Capacity of Pile	=	94.764	Tons
Total Capacity of Pile	=	245.378	Tons
Self Weight of the pile	=	9.950	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	71.843	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	45.093	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.


 Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-7, DIA OF PILE-0.6 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	35.65	degrees
Pile dia	=	0.6	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	8.026	kN/m3
Cross-sectional area of pile tip A_p	=	0.283	m2
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	57.410	kPa
N_q	=	56.147	
N_γ	=	53.246	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	94.764	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m2
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	94.764	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-9 and Pile Diameter =0.6m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons	
0.6	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	3.77	0	CUTOFF	CUTOFF	CUTOFF	
0.6	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	1.885	0	43.33	0	0	
0.6	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	2.827	0	43.33	0	0	
0.6	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	2.827	19	0	0	0	
0.6	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	2.827	25.1	0	70.986	7.099	
0.6	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	50.415	0.6	1	0	2.827	34.5	0	97.503	9.750	
0.6	13.5	25	0	17.15	7.34	SM	0	0	34.68	173.36	131.155	50.415	0.6	1	0	21.677	34.5	0	747.640	74.764	
																Total Skin Friction		1208.639		kN	
																		121.299		Tons	

Liquefiable
Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	121.299	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	121.299	Tons
Self Weight of the pile	=	9.950	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	30.483	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	38.253	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.
Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-10 and Pile Diameter =0.6m																			
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	Tons
0.6	0	2	0	15.484	5.674	CI	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	3.77	0	CUTOFF	CUTOFF
0.6	2	3	0	15.484	5.674	CI	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	1.885	0	43.31	0
0.6	3	4.5	0	15.484	5.674	CI	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	2.827	0	43.31	0
0.6	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	2.827	20	0	56.653
0.6	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	2.827	26.7	0	75.424
0.6	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	2.827	35.9	0	101.348
0.6	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	51.295	0.61	1	0	2.827	35.9	0	101.348
0.6	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	51.295	0.61	1	0	2.827	35.9	0	101.348
0.6	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	51.295	0.61	1	0	2.827	35.9	0	101.348
0.6	13.5	25	0	17.15	7.34	SM	0	0	35.14	174.24	132.035	51.295	0.61	1	0	21.677	35.9	0	777.120
Total Skin Friction																		1314.590	kN
																		131.932	Tons

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	131.932	Tons
Total End Bearing Capacity of Pile		=	77.259	Tons
Total Capacity of Pile		=	209.191	Tons
Self Weight of the pile		=	9.950	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	59.781	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	40.734	Tons

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-10, DIA OF PILE-0.6 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	35.14	degrees
Pile dia	=	0.6	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.283	m ²
Effective depth	=	9	m
Effective overburden pressure at pile tip	=	51.295	kPa
N_q	=	51.162	
N_γ	=	49.103	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	77.259	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.283	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	77.259	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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DIA OF PILE- 0.75 M LENGTH OF PILE-25 M

BH-1 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.68	5.87	CI	0.6	58.8	0	11.74	5.87	5.87	0	0	0.72	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.68	5.87	CI	0.6	58.8	0	17.61	14.675	14.675	0	0	0.72	2.356	0	42.33	0	0
0.75	3	4.5	0	16.758	6.948	SM	0	0	31.29	28.03	22.82	22.82	0.54	1	0	3.534	13.7	0	0	0
0.75	4.5	6	0	16.758	6.948	SM	0	0	31.29	38.45	33.24	33.24	0.54	1	0	3.534	19.9	0	70.397	7.040
0.75	6	7.5	0	16.758	6.948	SM	0	0	31.29	48.87	43.66	43.66	0.54	1	0	3.534	26.2	0	92.485	9.248
0.75	7.5	9	0	17.248	7.438	SM	0	0	34.05	60.03	54.45	54.45	0.59	1	0	3.534	36.5	0	128.814	12.881
0.75	9	10.5	0	17.248	7.438	SM	0	0	34.05	71.19	65.61	65.61	0.59	1	0	3.534	43.9	0	0	0
0.75	10.5	12	0	17.248	7.438	SM	0	0	34.05	82.35	76.77	65.61	0.59	1	0	3.534	43.9	0	0	0
0.75	12	13.5	0	17.248	7.438	SM	0	0	34.05	93.51	87.93	65.61	0.59	1	0	3.534	43.9	0	0	0
0.75	13.5	25	0	17.248	7.438	SM	0	0	34.05	179.05	136.28	65.61	0.59	1	0	27.096	43.9	0	1190.056	119.006
																Total Skin Friction		1481.753 148.709	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Liquefiable
Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	148.709	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	148.709	Tons
Self Weight of the pile	=	15.546	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	34.023	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	50.245	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-2 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	2	kN/m ²	kN	Tons
0.75	0	2	0	15.876	6.066	CI	0.62	0	21.01	12.13	6.065	6.065	0.36	1	0	4.712	2.28	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.876	6.066	CI	0.62	0	21.01	18.2	15.165	15.165	0.36	1	0	2.356	5.7	0	0	0
0.75	3	4.5	0	15.876	6.066	CI	0.62	0	21.01	27.3	22.75	22.75	0.36	1	0	3.534	8.56	0	0	0
0.75	4.5	6	0	16.66	6.85	SM	0	0	31.05	37.58	32.44	32.44	0.54	1	0	3.534	19.4	0	68.701	6.870
0.75	6	7.5	0	16.66	6.85	SM	0	0	31.05	47.86	42.72	42.72	0.54	1	0	3.534	25.6	0	90.470	9.047
0.75	7.5	9	0	16.66	6.85	SM	0	0	31.05	58.14	53	53	0.54	1	0	3.534	31.8	0	112.240	11.224
0.75	9	10.5	0	17.15	7.34	SP-SM	0	0	33.18	69.15	63.645	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	10.5	12	0	17.15	7.34	SP-SM	0	0	33.18	80.16	74.655	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	12	13.5	0	17.15	7.34	SP-SM	0	0	33.18	91.17	85.665	63.645	0.57	1	0	3.534	40.8	0	144.152	14.415
0.75	13.5	25	0	17.15	7.34	SP-SM	0	0	33.18	175.58	133.375	63.645	0.57	1	0	27.096	40.8	0	1105.246	110.525
																Total Skin Friction		1809.113 181.563	kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	181.563	Tons
Total End Bearing Capacity of Pile		=	105.490	Tons
Total Capacity of Pile		=	287.053	Tons
Self Weight of the pile		=	15.546	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile		=	80.138	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile		=	57.911	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-2, DIA OF PILE-0.75 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	33.18	degrees
Pile dia	=	0.75	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.645	kPa
N_q	=	35.951	
N_γ	=	36.174	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	105.485	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	105.49	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

 Authorised Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-3 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	2	kN/m ²	kN	Tons	
0.75	0	2	0	15.582	5.772	CL	0.65	63.7	0	11.54	5.77	5.77	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF	
0.75	2	3	0	15.582	5.772	CL	0.65	63.7	0	17.31	14.425	14.425	0	0	0.68	2.356	0	43.31	0	0	
0.75	3	4.5	0	15.582	5.772	CL	0.65	63.7	0	25.97	21.64	21.64	0	0	0.68	3.534	0	43.31	0	0	
0.75	4.5	6	0	16.954	7.144	SP-SM	0	0	32.01	36.69	31.33	31.33	0.55	1	0	3.534	19.2	0	67.853	6.785	
0.75	6	7.5	0	16.954	7.144	SP-SM	0	0	32.01	47.41	42.05	42.05	0.55	1	0	3.534	25.8	0	91.107	9.111	
0.75	7.5	9	0	16.954	7.144	SP-SM	0	0	32.01	58.13	52.77	52.77	0.55	1	0	3.534	32.4	0	114.325	11.432	
0.75	9	10.5	0	17.542	7.732	SP	0	0	34.66	69.73	63.93	63.93	0.6	1	0	3.534	43.7	0	154.542	15.454	
0.75	10.5	12	0	17.542	7.732	SP	0	0	34.66	81.33	75.53	75.53	0.6	1	0	3.534	43.7	0	154.542	15.454	
0.75	12	13.5	0	17.542	7.732	SP	0	0	34.66	92.93	87.13	87.13	0.6	1	0	3.534	43.7	0	154.542	15.454	
0.75	13.5	25	0	17.542	7.732	SP	0	0	34.66	181.85	137.39	137.39	0.6	1	0	27.096	43.7	0	1184.908	118.491	
																Total Skin Friction		1921.818 192.874		kN Tons	

Liquefiable
Liquefiable
Liquefiable

Total Skin Friction	Capacity of Pile	=	192.874	Tons
Total End Bearing Capacity of Pile		=	138.276	Tons
Total Capacity of Pile		=	331.149	Tons
Self Weight of the pile		=	15.546	Tons
Net Vertical Capacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile				
		=	94.837	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weight of Pile				
		=	60.550	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory

CALCULATIONS OF END BEARING CAPACITY BH-3, DIA OF PILE-0.75 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	34.66	degrees
Pile dia	=	0.75	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.732	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	63.93	kPa
N_q	=	46.894	
N_γ	=	45.525	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$	=	138.276	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	138.276	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For. Shiva Testing & Research Laboratory Pvt. Ltd.


 Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-5 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σα C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	°	kN/m ²	kN	Tons	
0.75	0	2	0	16.366	6.556	CI	0.59	57.82	0	13.11	6.555	6.555	0	0	0.73	4.712	0	CUTOFF	CUTOFF	CUTOFF	
0.75	2	3	0	16.366	6.556	CI	0.59	57.82	0	19.67	16.39	16.39	0	0	0.73	2.356	0	42.2	0	0	
0.75	3	4.5	0	17.15	7.34	SM	0	0	33.48	30.68	25.175	25.175	0.58	1	0	3.534	16.5	0	58.276	5.828	
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.48	41.69	36.185	36.185	0.58	1	0	3.534	23.7	0	83.756	8.376	
0.75	6	7.5	0	17.15	7.34	SM	0	0	36.15	52.7	47.195	47.195	0.63	1	0	3.534	34.4	0	121.605	12.160	
0.75	7.5	9	0	17.836	8.026	SM	0	0	36.15	64.74	58.72	58.72	0.63	1	0	3.534	42.8	0	151.291	15.129	
0.75	9	10.5	0	17.836	8.026	SM	0	0	36.15	76.78	70.76	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232	
0.75	10.5	12	0	17.836	8.026	SM	0	0	36.15	88.82	82.8	70.76	0.63	1	0	3.534	51.6	0	182.319	18.232	
0.75	12	13.5	0	17.836	8.026	SM	0	0	36.15	100.86	94.84	70.76	0.63	1	0	3.534	51.6	0	0	0	
0.75	13.5	15	0	17.836	8.026	SM	0	0	36.15	112.9	106.88	70.76	0.63	1	0	3.534	51.6	0	0	0	
0.75	15	25	0	17.836	8.026	SM	0	0	37.15	193.16	153.03	70.76	0.64	1	0	23.562	52.7	0	1241.246	124.125	
																Total Skin Friction		2020.811 202.809		kN Tons	

Liquefiable

Liquefiable
Liquefiable

Total Skin FrictionCapacity of Pile	=	202.809	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	202.809	Tons
Self Weight of the pile	=	15.546	Tons
Net VerticalCapcity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	52.056	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	62.868	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-6 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	16.464	6.654	CI	0.63	61.74	0	13.31	6.655	6.655	0	0	0.69	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	16.464	6.654	CI	0.63	61.74	0	19.96	16.635	16.635	0	0	0.69	2.356	0	42.6	100.366	10.037
0.75	3	4.5	0	16.464	6.654	CI	0.63	61.74	0	29.94	24.95	24.95	0	0	0.69	3.534	0	42.6	150.548	15.055
0.75	4.5	6	0	17.15	7.34	SM	0	0	33.74	40.95	35.445	35.445	0.58	1	0	3.534	23.2	0	82.059	8.206
0.75	6	7.5	0	17.15	7.34	SM	0	0	33.74	51.96	46.455	46.455	0.58	1	0	3.534	30.4	0	107.540	10.754
0.75	7.5	9	0	17.444	7.634	SM	0	0	35.62	63.41	57.685	57.685	0.62	1	0	3.534	41.2	0	145.530	14.553
0.75	9	10.5	0	17.444	7.634	SM	0	0	35.62	74.86	69.135	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	10.5	12	0	17.444	7.634	SM	0	0	35.62	86.31	80.585	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	12	13.5	0	17.444	7.634	SM	0	0	35.62	97.76	92.035	69.135	0.62	1	0	3.534	49.4	0	174.403	17.440
0.75	13.5	20	0	17.444	7.634	SM	0	0	35.62	147.38	122.57	69.135	0.62	1	0	15.315	49.4	0	755.795	75.580
																Total Skin Friction		1865.047	kN	
																		187.176	Tons	

Total Skin FrictionCapacity of Pile	=	187.176	Tons
Total End Bearing Capacity of Pile	=	177.253	Tons
Total Capacity of Pile	=	364.429	Tons
Self Weight of the pile	=	12.167	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	109.310	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Weght of Pile	=	55.841	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY BH-6, DIA OF PILE-0.75 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer =	35.62	degrees	
Pile dia =	0.75	m	
Depth of pile =	20	m	
Effective unit weight of the soil at pile tip =	7.634	kN/m ³	
Cross-sectional area of pile tip A_p =	0.442	m ²	
Effective depth =	11.5	m	
Effective overburden pressure at pile tip =	69.135	kPa	
N_q =	55.840		
N_γ =	52.992		
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + P_D \cdot N_q)$ =	177.253	Tons	
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p =	0.442	m ²	
N_c =	9		
Average cohesion at pile tip C =	0	kPa	
$A_p \cdot C \cdot N_c$ =	0	Tons	
Total End bearing capacity of pile =	177.253	Tons	

 Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

 For Shiva Testing & Research Laboratory Pvt. Ltd.

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Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-7 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	$\sum K P_{Di} \tan \delta$	$\sum \alpha C$	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	16.268	6.458	CL	0.61	59.78	0	12.92	6.46	6.46	0	0	0.7	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	16.268	6.458	CL	0.61	59.78	0	19.38	16.15	16.15	0	0	0.7	2.356	0	41.84	0	0
0.75	3	4.5	0	16.268	6.458	CL	0.61	59.78	0	29.07	24.225	24.225	0	0	0.7	3.534	0	41.84	0	0
0.75	4.5	6	0	17.248	7.438	SM	0	0	33.56	40.23	34.65	34.65	0.58	1	0	3.534	22.7	0	80.222	8.022
0.75	6	7.5	0	17.248	7.438	SM	0	0	33.56	51.39	45.81	45.81	0.58	1	0	3.534	30	0	106.055	10.606
0.75	7.5	9	0	17.836	8.026	SM	0	0	35.65	63.43	57.41	57.41	0.62	1	0	3.534	41	0	144.823	14.482
0.75	9	10.5	0	17.836	8.026	SM	0	0	35.65	75.47	69.45	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	10.5	12	0	17.836	8.026	SM	0	0	35.65	87.51	81.49	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	12	13.5	0	17.836	8.026	SM	0	0	35.65	99.55	93.53	69.45	0.62	1	0	3.534	49.6	0	175.216	17.522
0.75	13.5	25	0	17.836	8.026	SM	0	0	35.65	191.85	145.7	69.45	0.62	1	0	27.096	49.6	0	1343.420	134.342
																	Total Skin Friction		2200.167	kN
																			220.809	Tons

Liquefiable
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Liquefiable

Total Skin FrictionCapacity of Pile	=	220.809	Tons
Total End Bearing Capacity of Pile	=	179.349	Tons
Total Capacity of Pile	=	400.158	Tons
Self Weight of the pile	=	15.546	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	117.840	Tons
Uplift Capacity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	67.069	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY BH-7, DIA OF PILE-0.75 M

END BEARING CAPCITY FOR SANDY SOILS			
ϕ at the base layer	=	35.65	degrees
Pile dia	=	0.75	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	8.026	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	69.45	kPa
N_q	=	56.147	
N_γ	=	53.246	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	179.349	Tons
END BEARING CAPCITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	179.349	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-9 and Pile Diameter =0.75m																				
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	²	kN/m ²	kN	Tons
0.75	0	2	0	15.484	5.674	CI	0.66	64.68	0	11.35	5.675	5.675	0	0	0.67	4.712	0	CUTOFF	CUTOFF	CUTOFF
0.75	2	3	0	15.484	5.674	CI	0.66	64.68	0	17.02	14.185	14.185	0	0	0.67	2.356	0	43.33	0	0
0.75	3	4.5	0	15.484	5.674	CI	0.66	64.68	0	25.53	21.275	21.275	0	0	0.67	3.534	0	43.33	0	0
0.75	4.5	6	0	16.268	6.458	SM	0	0	32.39	35.22	30.375	30.375	0.56	1	0	3.534	19	0	0	0
0.75	6	7.5	0	16.268	6.458	SM	0	0	32.39	44.91	40.065	40.065	0.56	1	0	3.534	25.1	0	88.739	8.874
0.75	7.5	9	0	17.15	7.34	SM	0	0	34.68	55.92	50.415	50.415	0.6	1	0	3.534	34.5	0	121.888	12.189
0.75	9	10.5	0	17.15	7.34	SM	0	0	34.68	66.93	61.425	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	10.5	12	0	17.15	7.34	SM	0	0	34.68	77.94	72.435	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	12	13.5	0	17.15	7.34	SM	0	0	34.68	88.95	83.445	61.425	0.6	1	0	3.534	42	0	148.499	14.850
0.75	13.5	25	0	17.15	7.34	SM	0	0	34.68	173.36	131.155	61.425	0.6	1	0	27.096	42	0	1138.574	113.857
																Total Skin Friction		1794.696 180.116	kN Tons	

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Total Skin FrictionCapacity of Pile	=	180.116	Tons
Total End Bearing Capacity of Pile	=	0.000	Tons
Total Capacity of Pile	=	180.116	Tons
Self Weight of the pile	=	15.546	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	44.492	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	57.573	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LENGTH OF PILE-25 M

BH-10 and Pile Diameter =0.75m																					
Diameter of pile	Depth From	Depth To	Depth of water table	Bulk unit weight or saturated unit weight	Effective Unit Weight	Soil Type	Cohesion (C)		phi	Eff. stress at btm of the present layer	Eff. stress at center of the layer	Effective stress used for calculation based on critical depth 15 D	Delta = phi in rad	K	α	As	Σ K P _{Di} tanδ	Σ α C	Skin Friction resistance	Skin Friction resistance	
m	m	m	m	kN/m ³	kN/m ³		Kg/cm ²	kN/m ²	Degree	kN/m ²	kN/m ²	kN/m ²				m ²	kN/m ²	kN/m ²	kN	Tons	
0.75	0	2	0	15.484	5.674	Cl	0.65	63.7	0	11.35	5.675	5.675	0	0	0.68	4.712	0	CUTOFF	CUTOFF	CUTOFF	
0.75	2	3	0	15.484	5.674	Cl	0.65	63.7	0	17.02	14.185	14.185	0	0	0.68	2.356	0	43.31	0	0	
0.75	3	4.5	0	15.484	5.674	Cl	0.65	63.7	0	25.53	21.275	21.275	0	0	0.68	3.534	0	43.31	0	0	
0.75	4.5	6	0	16.562	6.752	SM	0	0	33.67	35.66	30.595	30.595	0.58	1	0	3.534	20	0	70.821	7.082	
0.75	6	7.5	0	16.562	6.752	SM	0	0	33.67	45.79	40.725	40.725	0.58	1	0	3.534	26.7	0	94.287	9.429	
0.75	7.5	9	0	17.15	7.34	SM	0	0	35.14	56.8	51.295	51.295	0.61	1	0	3.534	35.9	0	126.694	12.669	
0.75	9	10.5	0	17.15	7.34	SM	0	0	35.14	67.81	62.305	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387	
0.75	10.5	12	0	17.15	7.34	SM	0	0	35.14	78.82	73.315	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387	
0.75	12	13.5	0	17.15	7.34	SM	0	0	35.14	89.83	84.325	62.305	0.61	1	0	3.534	43.5	0	153.870	15.387	
0.75	13.5	25	0	17.15	7.34	SM	0	0	35.14	174.24	132.035	62.305	0.61	1	0	27.096	43.5	0	1179.760	117.976	
																	Total Skin Friction		1933.173	kN	
																			194.013	Tons	

Total Skin FrictionCapacity of Pile	=	194.013	Tons
Total End Bearing Capacity of Pile	=	146.797	Tons
Total Capacity of Pile	=	340.811	Tons
Self Weight of the pile	=	15.546	Tons
Net VerticalCapacity of Pile = [Total capacity of pile /3.0(FOS)]-Self weight of pile	=	98.057	Tons
Uplift Capcity of Pile =[Reduced Skin Friction of Pile by 0.7/3.0(FOS)] +Self Wegt of Pile	=	60.816	Tons

For Shiva Testing & Research Laboratory Pvt. Ltd.

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CALCULATIONS OF END BEARING CAPACITY BH-10, DIA OF PILE-0.75 M

END BEARING CAPACITY FOR SANDY SOILS			
ϕ at the base layer	=	35.14	degrees
Pile dia	=	0.75	m
Depth of pile	=	25	m
Effective unit weight of the soil at pile tip	=	7.34	kN/m ³
Cross-sectional area of pile tip A_p	=	0.442	m ²
Effective depth	=	11.5	m
Effective overburden pressure at pile tip	=	62.305	kPa
N_q	=	51.162	
N_γ	=	49.103	
$A_p \cdot (0.5 \cdot D \cdot \gamma \cdot N_\gamma + PD \cdot N_q)$	=	146.797	Tons
END BEARING CAPACITY FOR CLAYEY SOILS			
Cross-sectional area of pile tip A_p	=	0.442	m ²
N_c	=	9	
Average cohesion at pile tip C	=	0	kPa
$A_p \cdot C \cdot N_c$	=	0	Tons
Total End bearing capacity of pile	=	146.797	Tons

Critical Depth 15D for $\phi < 30$ and 20D for $\phi > 40$

For Shiva Testing & Research Laboratory Pvt. Ltd.

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22.0 LATERAL CAPACITY OF PILE

For. Shiva Testing & Research Laboratory Pvt. Ltd.

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22.1 LATERAL CAPACITY OF PILE FOR GUEST HOUSE LENGTH OF PILE- 10 M

LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	10	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.003066	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14	
Modulus of subgrade reaction	1976	kN/m ³
Stiffness factor T	2.117	
Embedment length L	6	m
L/T	2.835	
Category of pile	Long	
L1/T for depth of fixity	0.945	
Lf/T	2.023	
Cantilever length Lf	4.282	m
Lateral capacity	2.033	T
Reduction factor	0.9	
Maximum moment t-m	5.746	T-m

LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	10	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14	
Modulus of subgrade reaction	1976	kN/m ³
Stiffness factor T	2.449	
Embedment length L	6	m
L/T	2.450	
Category of pile	Long	
L1/T for depth of fixity	0.817	
Lf/T	2.042	
Cantilever length Lf	5.000	m
Lateral capacity	3.046	T
Reduction factor	0.9	
Maximum moment t-m	9.594	T-m

LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	10	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14	
Modulus of subgrade reaction	1976	kN/m ³
Stiffness factor T	2.928	
Embedment length L	6	m
L/T	2.049	
Category of pile	Long	
L1/T for depth of fixity	0.683	
Lf/T	2.064	
Cantilever length Lf	6.042	m
Lateral capacity	4.904	T
Reduction factor	0.9	
Maximum moment t-m	17.747	T-m

For Shiva Testing & Research Laboratory Pvt. Ltd.



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LENGTH OF PILE- 15 M

LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.333	
Modulus of subgrade reaction	2744	kN/m ³
Stiffness factor T	1.982	
Embedment length L	11	m
L/T	5.549	
Category of pile	Long	
L1/T for depth of fixity	1.009	
Lf/T	2.014	
Cantilever length Lf	3.992	m
Lateral capacity	2.342	T
Reduction factor	0.9	
Maximum moment t-m	6.314	T-m

LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.333	
Modulus of subgrade reaction	2744	kN/m ³
Stiffness factor T	2.294	
Embedment length L	11	m
L/T	4.796	
Category of pile	Long	
L1/T for depth of fixity	0.872	
Lf/T	2.033	
Cantilever length Lf	4.663	m
Lateral capacity	3.532	T
Reduction factor	0.9	
Maximum moment t-m	10.590	T-m

For Shiva Testing & Research Laboratory Pvt. Ltd.
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LATERAL CAPACITY FOR BH-8 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.333	
Modulus of subgrade reaction	2744	kN/m ³
Stiffness factor T	2.742	
Embedment length L	11	m
L/T	4.012	
Category of pile	Long	
L1/T for depth of fixity	0.729	
Lf/T	2.056	
Cantilever length Lf	5.636	m
Lateral capacity	5.728	T
Reduction factor	0.9	
Maximum moment t-m	19.684	T-m

For Shiva Testing & Research Laboratory Pvt. Ltd.

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22.2 LATERAL CAPACITY OF PILE FOR RESIDENTIAL TOWERS

LENGTH OF PILE- 15 M

DIA OF PILE-0.5 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.5m			LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.5m		
Pile Dia	0.5	m	Pile Dia	0.5	m
Terminal Depth	15	m	Terminal Depth	15	m
Cutoff	2	m	Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND			LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.01	m	Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000.00	Kpa	Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386127.88	Kpa	Youngs modulus E	27386127.9	Kpa
Moment of Inertia I	0.00		Moment of Inertia I	0.00307	
Free length (L1) in	2.00	m	Free length (L1) in	2	m
FIXED HEAD			FIXED HEAD		
Average N values corrected	9.667		Average N values corrected	8.333	
Modulus of subgrade reaction	1333.333	kN/m ³	Modulus of subgrade reaction	1066.667	kN/m ³
Stiffness factor T	2.290		Stiffness factor T	2.395	
Embedment length L	11.000	m	Embedment length L	11.000	m
L/T	4.803		L/T	4.594	
Category of pile	Long		Category of pile	Long	
L1/T for depth of fixity	0.873		L1/T for depth of fixity	0.835	
Lf/T	2.033		Lf/T	2.039	
Cantilever length Lf	4.656	m	Cantilever length Lf	4.882	m
Lateral capacity	1.709	T	Lateral capacity	1.546	T
Reduction factor	0.900		Reduction factor	0.900	
Maximum moment t-m	5.118	T-m	Maximum moment t-m	4.787	T-m

For Shiva Testing & Research Laboratory Pvt. Ltd.

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LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.667	
Modulus of subgrade reaction	1784	kN/m ³
Stiffness factor T	2.160	
Embedment length L	11	m
L/T	5.091	
Category of pile	Long	
L1/T for depth of fixity	0.926	
Lf/T	2.025	
Cantilever length Lf	4.376	m
Lateral capacity	1.944	T
Reduction factor	0.9	
Maximum moment t-m	5.578	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16	
Modulus of subgrade reaction	2264	kN/m ³
Stiffness factor T	2.060	
Embedment length L	11	m
L/T	5.340	
Category of pile	Long	
L1/T for depth of fixity	0.971	
Lf/T	2.019	
Cantilever length Lf	4.159	m
Lateral capacity	2.156	T
Reduction factor	0.9	
Maximum moment t-m	5.977	T-m

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LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15	
Modulus of subgrade reaction	2120	kN/m ³
Stiffness factor T	2.087	
Embedment length L	11	m
L/T	5.270	
Category of pile	Long	
L1/T for depth of fixity	0.958	
Lf/T	2.021	
Cantilever length Lf	4.218	m
Lateral capacity	2.096	T
Reduction factor	0.9	
Maximum moment t-m	5.864	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	2.339	
Embedment length L	11	m
L/T	4.703	
Category of pile	Long	
L1/T for depth of fixity	0.855	
Lf/T	2.036	
Cantilever length Lf	4.761	m
Lateral capacity	1.630	T
Reduction factor	0.9	
Maximum moment t-m	4.960	T-m

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LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	8	
Modulus of subgrade reaction	1000	kN/m ³
Stiffness factor T	2.426	
Embedment length L	11	m
L/T	4.535	
Category of pile	Long	
L1/T for depth of fixity	0.825	
Lf/T	2.040	
Cantilever length Lf	4.949	m
Lateral capacity	1.501	T
Reduction factor	0.9	
Maximum moment t-m	4.695	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	10.333	
Modulus of subgrade reaction	1448	kN/m ³
Stiffness factor T	2.253	
Embedment length L	11	m
L/T	4.883	
Category of pile	Long	
L1/T for depth of fixity	0.888	
Lf/T	2.031	
Cantilever length Lf	4.575	m
Lateral capacity	1.773	T
Reduction factor	0.9	
Maximum moment t-m	5.245	T-m

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LENGTH OF PILE- 15 M

DIA OF PILE 0.6 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9.667	
Modulus of subgrade reaction	1333.333	kN/m ³
Stiffness factor T	2.650	
Embedment length L	11	m
L/T	4.151	
Category of pile	Long	
L1/T for depth of fixity	0.755	
Lf/T	2.052	
Cantilever length Lf	5.436	m
Lateral capacity	2.541	T
Reduction factor	0.9	
Maximum moment t-m	8.503	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	8.333	
Modulus of subgrade reaction	1066.667	kN/m ³
Stiffness factor T	2.771	
Embedment length L	11	m
L/T	3.970	
Category of pile	Long	
L1/T for depth of fixity	0.722	
Lf/T	2.057	
Cantilever length Lf	5.699	m
Lateral capacity	2.289	T
Reduction factor	0.9	
Maximum moment t-m	7.931	T-m

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LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.667	
Modulus of subgrade reaction	1784	kN/m ³
Stiffness factor T	2.500	
Embedment length L	11	m
L/T	4.400	
Category of pile	Long	
L1/T for depth of fixity	0.800	
Lf/T	2.044	
Cantilever length Lf	5.110	m
Lateral capacity	2.907	T
Reduction factor	0.9	
Maximum moment t-m	9.300	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16	
Modulus of subgrade reaction	2264	kN/m ³
Stiffness factor T	2.383	
Embedment length L	11	m
L/T	4.615	
Category of pile	Long	
L1/T for depth of fixity	0.839	
Lf/T	2.038	
Cantilever length Lf	4.858	m
Lateral capacity	3.240	T
Reduction factor	0.9	
Maximum moment t-m	9.997	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15	
Modulus of subgrade reaction	2120	kN/m ³
Stiffness factor T	2.415	
Embedment length L	11	m
L/T	4.555	
Category of pile	Long	
L1/T for depth of fixity	0.828	
Lf/T	2.040	
Cantilever length Lf	4.926	m
Lateral capacity	3.145	T
Reduction factor	0.9	
Maximum moment t-m	9.801	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	2.706	
Embedment length L	11	m
L/T	4.065	
Category of pile	Long	
L1/T for depth of fixity	0.739	
Lf/T	2.054	
Cantilever length Lf	5.559	m
Lateral capacity	2.419	T
Reduction factor	0.9	
Maximum moment t-m	8.229	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	8	
Modulus of subgrade reaction	1000	kN/m ³
Stiffness factor T	2.807	
Embedment length L	11	m
L/T	3.919	
Category of pile	Long	
L1/T for depth of fixity	0.713	
Lf/T	2.059	
Cantilever length Lf	5.778	m
Lateral capacity	2.221	T
Reduction factor	0.9	
Maximum moment t-m	7.772	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	10.333	
Modulus of subgrade reaction	1448	kN/m ³
Stiffness factor T	2.606	
Embedment length L	11	m
L/T	4.221	
Category of pile	Long	
L1/T for depth of fixity	0.767	
Lf/T	2.049	
Cantilever length Lf	5.342	m
Lateral capacity	2.640	T
Reduction factor	0.9	
Maximum moment t-m	8.723	T-m

LENGTH OF PILE- 15 M DIA OF PILE-0.75 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.016	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9.667	
Modulus of subgrade reaction	1333.333	kN/m ³
Stiffness factor T	3.168	
Embedment length L	11	m
L/T	3.473	
Category of pile	Long	
L1/T for depth of fixity	0.631	
Lf/T	2.073	
Cantilever length Lf	6.566	m
Lateral capacity	4.058	T
Reduction factor	0.9	
Maximum moment t-m	15.642	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	8.333	
Modulus of subgrade reaction	1066.667	kN/m ³
Stiffness factor T	3.312	
Embedment length L	11	m
L/T	3.321	
Category of pile	Long	
L1/T for depth of fixity	0.604	
Lf/T	2.078	
Cantilever length Lf	6.883	m
Lateral capacity	3.639	T
Reduction factor	0.9	
Maximum moment t-m	14.547	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.667	
Modulus of subgrade reaction	1784	kN/m ³
Stiffness factor T	2.988	
Embedment length L	11	m
L/T	3.681	
Category of pile	Long	
L1/T for depth of fixity	0.669	
Lf/T	2.066	
Cantilever length Lf	6.174	m
Lateral capacity	4.670	T
Reduction factor	0.9	
Maximum moment t-m	17.178	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16	
Modulus of subgrade reaction	2264	kN/m ³
Stiffness factor T	2.849	
Embedment length L	11	m
L/T	3.861	
Category of pile	Long	
L1/T for depth of fixity	0.702	
Lf/T	2.060	
Cantilever length Lf	5.871	m
Lateral capacity	5.231	T
Reduction factor	0.9	
Maximum moment t-m	18.529	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15	
Modulus of subgrade reaction	2120	kN/m ³
Stiffness factor T	2.887	
Embedment length L	11	m
L/T	3.810	
Category of pile	Long	
L1/T for depth of fixity	0.693	
Lf/T	2.062	
Cantilever length Lf	5.953	m
Lateral capacity	5.071	T
Reduction factor	0.9	
Maximum moment t-m	18.148	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	3.235	
Embedment length L	11	m
L/T	3.400	
Category of pile	Long	
L1/T for depth of fixity	0.618	
Lf/T	2.075	
Cantilever length Lf	6.714	m
Lateral capacity	3.855	T
Reduction factor	0.9	
Maximum moment t-m	15.117	T-m

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LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.016	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	8	
Modulus of subgrade reaction	1000	kN/m ³
Stiffness factor T	3.355	
Embedment length L	11	m
L/T	3.279	
Category of pile	Long	
L1/T for depth of fixity	0.596	
Lf/T	2.080	
Cantilever length Lf	6.977	m
Lateral capacity	3.526	T
Reduction factor	0.9	
Maximum moment t-m	14.243	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	15	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	10.333	
Modulus of subgrade reaction	1448	kN/m ³
Stiffness factor T	3.116	
Embedment length L	11	m
L/T	3.531	
Category of pile	Long	
L1/T for depth of fixity	0.642	
Lf/T	2.071	
Cantilever length Lf	6.453	m
Lateral capacity	4.223	T
Reduction factor	0.9	
Maximum moment t-m	16.065	T-m

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LENGTH OF PILE -20 M DIA OF PILE-0.5 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.25	
Modulus of subgrade reaction	2300	kN/m ³
Stiffness factor T	2.053	
Embedment length L	16	m
L/T	7.792	
Category of pile	Long	
L1/T for depth of fixity	0.974	
Lf/T	2.019	
Cantilever length Lf	4.145	m
Lateral capacity	2.171	T
Reduction factor	0.9	
Maximum moment t-m	6.004	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.75	
Modulus of subgrade reaction	1796	kN/m ³
Stiffness factor T	2.158	
Embedment length L	16	m
L/T	7.416	
Category of pile	Long	
L1/T for depth of fixity	0.927	
Lf/T	2.025	
Cantilever length Lf	4.370	m
Lateral capacity	1.950	T
Reduction factor	0.9	
Maximum moment t-m	5.588	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.25	
Modulus of subgrade reaction	2444	kN/m ³
Stiffness factor T	2.029	
Embedment length L	16	m
L/T	7.887	
Category of pile	Long	
L1/T for depth of fixity	0.986	
Lf/T	2.017	
Cantilever length Lf	4.092	m
Lateral capacity	2.229	T
Reduction factor	0.9	
Maximum moment t-m	6.109	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.75	
Modulus of subgrade reaction	2804	kN/m ³
Stiffness factor T	1.974	
Embedment length L	16	m
L/T	8.107	
Category of pile	Long	
L1/T for depth of fixity	1.013	
Lf/T	2.013	
Cantilever length Lf	3.974	m
Lateral capacity	2.363	T
Reduction factor	0.9	
Maximum moment t-m	6.353	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.25	
Modulus of subgrade reaction	2732	kN/m ³
Stiffness factor T	1.984	
Embedment length L	16	m
L/T	8.065	
Category of pile	Long	
L1/T for depth of fixity	1.008	
Lf/T	2.014	
Cantilever length Lf	3.996	m
Lateral capacity	2.337	T
Reduction factor	0.9	
Maximum moment t-m	6.307	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14.5	
Modulus of subgrade reaction	2048	kN/m ³
Stiffness factor T	2.102	
Embedment length L	16	m
L/T	7.613	
Category of pile	Long	
L1/T for depth of fixity	0.952	
Lf/T	2.022	
Cantilever length Lf	4.249	m
Lateral capacity	2.065	T
Reduction factor	0.9	
Maximum moment t-m	5.806	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00307	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	2.339	
Embedment length L	16	m
L/T	6.841	
Category of pile	Long	
L1/T for depth of fixity	0.855	
Lf/T	2.036	
Cantilever length Lf	4.761	m
Lateral capacity	1.630	T
Reduction factor	0.9	
Maximum moment t-m	4.960	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	13.75	
Modulus of subgrade reaction	1940	kN/m ³
Stiffness factor T	2.125	
Embedment length L	16	m
L/T	7.531	
Category of pile	Long	
L1/T for depth of fixity	0.941	
Lf/T	2.023	
Cantilever length Lf	4.298	m
Lateral capacity	2.017	T
Reduction factor	0.9	
Maximum moment t-m	5.716	T-m

LENGTH OF PILE -20 M DIA OF PILE- 0.6 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.25	
Modulus of subgrade reaction	2300	kN/m ³
Stiffness factor T	2.376	
Embedment length L	16	m
L/T	6.734	
Category of pile	Long	
L1/T for depth of fixity	0.842	
Lf/T	2.038	
Cantilever length Lf	4.842	m
Lateral capacity	3.263	T
Reduction factor	0.9	
Maximum moment t-m	10.045	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.75	
Modulus of subgrade reaction	1796	kN/m ³
Stiffness factor T	2.496	
Embedment length L	16	m
L/T	6.409	
Category of pile	Long	
L1/T for depth of fixity	0.801	
Lf/T	2.044	
Cantilever length Lf	5.103	m
Lateral capacity	2.916	T
Reduction factor	0.9	
Maximum moment t-m	9.319	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.25	
Modulus of subgrade reaction	2444	kN/m ³
Stiffness factor T	2.347	
Embedment length L	16	m
L/T	6.817	
Category of pile	Long	
L1/T for depth of fixity	0.852	
Lf/T	2.036	
Cantilever length Lf	4.779	m
Lateral capacity	3.353	T
Reduction factor	0.9	
Maximum moment t-m	10.230	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.00636	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.75	
Modulus of subgrade reaction	2804	kN/m ³
Stiffness factor T	2.284	
Embedment length L	16	m
L/T	7.006	
Category of pile	Long	
L1/T for depth of fixity	0.876	
Lf/T	2.033	
Cantilever length Lf	4.642	m
Lateral capacity	3.566	T
Reduction factor	0.9	
Maximum moment t-m	10.658	T-m

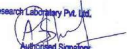
LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.25	
Modulus of subgrade reaction	2732	kN/m ³
Stiffness factor T	2.296	
Embedment length L	16	m
L/T	6.970	
Category of pile	Long	
L1/T for depth of fixity	0.871	
Lf/T	2.033	
Cantilever length Lf	4.668	m
Lateral capacity	3.525	T
Reduction factor	0.9	
Maximum moment t-m	10.576	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14.5	
Modulus of subgrade reaction	2048	kN/m ³
Stiffness factor T	2.432	
Embedment length L	16	m
L/T	6.580	
Category of pile	Long	
L1/T for depth of fixity	0.822	
Lf/T	2.041	
Cantilever length Lf	4.962	m
Lateral capacity	3.096	T
Reduction factor	0.9	
Maximum moment t-m	9.699	T-m

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LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	2.706	
Embedment length L	16	m
L/T	5.913	
Category of pile	Long	
L1/T for depth of fixity	0.739	
Lf/T	2.054	
Cantilever length Lf	5.559	m
Lateral capacity	2.419	T
Reduction factor	0.9	
Maximum moment t-m	8.229	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	13.75	
Modulus of subgrade reaction	1940	kN/m ³
Stiffness factor T	2.458	
Embedment length L	16	m
L/T	6.509	
Category of pile	Long	
L1/T for depth of fixity	0.814	
Lf/T	2.042	
Cantilever length Lf	5.020	m
Lateral capacity	3.020	T
Reduction factor	0.9	
Maximum moment t-m	9.541	T-m

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LENGTH OF PILE -20 M DIA OF PILE 0.75 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.25	
Modulus of subgrade reaction	2300	kN/m ³
Stiffness factor T	2.840	
Embedment length L	16	m
L/T	5.633	
Category of pile	Long	
L1/T for depth of fixity	0.704	
Lf/T	2.060	
Cantilever length Lf	5.851	m
Lateral capacity	5.271	T
Reduction factor	0.9	
Maximum moment t-m	18.622	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.75	
Modulus of subgrade reaction	1796	kN/m ³
Stiffness factor T	2.984	
Embedment length L	16	m
L/T	5.361	
Category of pile	Long	
L1/T for depth of fixity	0.670	
Lf/T	2.066	
Cantilever length Lf	6.166	m
Lateral capacity	4.685	T
Reduction factor	0.9	
Maximum moment t-m	17.215	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.25	
Modulus of subgrade reaction	2444	kN/m ³
Stiffness factor T	2.806	
Embedment length L	16	m
L/T	5.702	
Category of pile	Long	
L1/T for depth of fixity	0.713	
Lf/T	2.059	
Cantilever length Lf	5.776	m
Lateral capacity	5.424	T
Reduction factor	0.9	
Maximum moment t-m	18.982	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.75	
Modulus of subgrade reaction	2804	kN/m ³
Stiffness factor T	2.730	
Embedment length L	16	m
L/T	5.861	
Category of pile	Long	
L1/T for depth of fixity	0.733	
Lf/T	2.055	
Cantilever length Lf	5.611	m
Lateral capacity	5.786	T
Reduction factor	0.9	
Maximum moment t-m	19.817	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.25	
Modulus of subgrade reaction	2732	kN/m ³
Stiffness factor T	2.744	
Embedment length L	16	m
L/T	5.831	
Category of pile	Long	
L1/T for depth of fixity	0.729	
Lf/T	2.056	
Cantilever length Lf	5.642	m
Lateral capacity	5.716	T
Reduction factor	0.9	
Maximum moment t-m	19.657	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	14.5	
Modulus of subgrade reaction	2048	kN/m ³
Stiffness factor T	2.907	
Embedment length L	16	m
L/T	5.504	
Category of pile	Long	
L1/T for depth of fixity	0.688	
Lf/T	2.063	
Cantilever length Lf	5.997	m
Lateral capacity	4.988	T
Reduction factor	0.9	
Maximum moment t-m	17.950	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	9	
Modulus of subgrade reaction	1200	kN/m ³
Stiffness factor T	3.235	
Embedment length L	16	m
L/T	4.946	
Category of pile	Long	
L1/T for depth of fixity	0.618	
Lf/T	2.075	
Cantilever length Lf	6.714	m
Lateral capacity	3.855	T
Reduction factor	0.9	
Maximum moment t-m	15.117	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	13.75	
Modulus of subgrade reaction	1940	kN/m ³
Stiffness factor T	2.939	
Embedment length L	16	m
L/T	5.445	
Category of pile	Long	
L1/T for depth of fixity	0.681	
Lf/T	2.064	
Cantilever length Lf	6.066	m
Lateral capacity	4.861	T
Reduction factor	0.9	
Maximum moment t-m	17.644	T-m

LENGTH OF PILE-25 M DIA OF PILE-0.5 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	18.6	
Modulus of subgrade reaction	2638.4	kN/m ³
Stiffness factor T	1.998	
Embedment length L	21	m
L/T	10.511	
Category of pile	Long	
L1/T for depth of fixity	1.001	
Lf/T	2.015	
Cantilever length Lf	4.026	m
Lateral capacity	2.303	T
Reduction factor	0.9	
Maximum moment t-m	6.244	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386127.9	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15.8	
Modulus of subgrade reaction	2235.2	kN/m ³
Stiffness factor T	2.065	
Embedment length L	21	m
L/T	10.168	
Category of pile	Long	
L1/T for depth of fixity	0.968	
Lf/T	2.019	
Cantilever length Lf	4.171	m
Lateral capacity	2.144	T
Reduction factor	0.9	
Maximum moment t-m	5.955	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.4	
Modulus of subgrade reaction	2753.6	kN/m ³
Stiffness factor T	1.981	
Embedment length L	21	m
L/T	10.602	
Category of pile	Long	
L1/T for depth of fixity	1.010	
Lf/T	2.014	
Cantilever length Lf	3.989	m
Lateral capacity	2.345	T
Reduction factor	0.9	
Maximum moment t-m	6.321	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21.4	
Modulus of subgrade reaction	3041.6	kN/m ³
Stiffness factor T	1.942	
Embedment length L	21	m
L/T	10.815	
Category of pile	Long	
L1/T for depth of fixity	1.030	
Lf/T	2.011	
Cantilever length Lf	3.906	m
Lateral capacity	2.446	T
Reduction factor	0.9	
Maximum moment t-m	6.501	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21	
Modulus of subgrade reaction	2984	kN/m ³
Stiffness factor T	1.949	
Embedment length L	16	m
L/T	8.208	
Category of pile	Long	
L1/T for depth of fixity	1.026	
Lf/T	2.012	
Cantilever length Lf	3.922	m
Lateral capacity	2.427	T
Reduction factor	0.9	
Maximum moment t-m	6.466	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.2	
Modulus of subgrade reaction	2436.8	kN/m ³
Stiffness factor T	2.030	
Embedment length L	21	m
L/T	10.345	
Category of pile	Long	
L1/T for depth of fixity	0.985	
Lf/T	2.017	
Cantilever length Lf	4.095	m
Lateral capacity	2.226	T
Reduction factor	0.9	
Maximum moment t-m	6.104	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.8	
Modulus of subgrade reaction	1803.2	kN/m ³
Stiffness factor T	2.156	
Embedment length L	21	m
L/T	9.741	
Category of pile	Long	
L1/T for depth of fixity	0.928	
Lf/T	2.025	
Cantilever length Lf	4.366	m
Lateral capacity	1.953	T
Reduction factor	0.9	
Maximum moment t-m	5.595	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.5m		
Pile Dia	0.5	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0031	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.6	
Modulus of subgrade reaction	2350.4	kN/m ³
Stiffness factor T	2.045	
Embedment length L	21	m
L/T	10.271	
Category of pile	Long	
L1/T for depth of fixity	0.978	
Lf/T	2.018	
Cantilever length Lf	4.126	m
Lateral capacity	2.191	T
Reduction factor	0.9	
Maximum moment t-m	6.041	T-m

LENGTH OF PILE-25 M DIA OF PILE- 0.6 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	18.6	
Modulus of subgrade reaction	2638.4	kN/m ³
Stiffness factor T	2.312	
Embedment length L	21	m
L/T	9.085	
Category of pile	Long	
L1/T for depth of fixity	0.865	
Lf/T	2.034	
Cantilever length Lf	4.702	m
Lateral capacity	3.470	T
Reduction factor	0.9	
Maximum moment t-m	10.467	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15.8	
Modulus of subgrade reaction	2235.2	kN/m ³
Stiffness factor T	2.390	
Embedment length L	21	m
L/T	8.788	
Category of pile	Long	
L1/T for depth of fixity	0.837	
Lf/T	2.038	
Cantilever length Lf	4.871	m
Lateral capacity	3.221	T
Reduction factor	0.9	
Maximum moment t-m	9.959	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.4	
Modulus of subgrade reaction	2753.6	kN/m ³
Stiffness factor T	2.292	
Embedment length L	21	m
L/T	9.163	
Category of pile	Long	
L1/T for depth of fixity	0.873	
Lf/T	2.033	
Cantilever length Lf	4.660	m
Lateral capacity	3.537	T
Reduction factor	0.9	
Maximum moment t-m	10.601	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21.4	
Modulus of subgrade reaction	3041.6	kN/m ³
Stiffness factor T	2.247	
Embedment length L	21	m
L/T	9.347	
Category of pile	Long	
L1/T for depth of fixity	0.890	
Lf/T	2.031	
Cantilever length Lf	4.562	m
Lateral capacity	3.697	T
Reduction factor	0.9	
Maximum moment t-m	10.919	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21	
Modulus of subgrade reaction	2984	kN/m ³
Stiffness factor T	2.255	
Embedment length L	16	m
L/T	7.094	
Category of pile	Long	
L1/T for depth of fixity	0.887	
Lf/T	2.031	
Cantilever length Lf	4.581	m
Lateral capacity	3.666	T
Reduction factor	0.9	
Maximum moment t-m	10.857	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.2	
Modulus of subgrade reaction	2436.8	kN/m ³
Stiffness factor T	2.349	
Embedment length L	21	m
L/T	8.941	
Category of pile	Long	
L1/T for depth of fixity	0.852	
Lf/T	2.036	
Cantilever length Lf	4.782	m
Lateral capacity	3.349	T
Reduction factor	0.9	
Maximum moment t-m	10.221	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.8	
Modulus of subgrade reaction	1803.2	kN/m ³
Stiffness factor T	2.494	
Embedment length L	21	m
L/T	8.419	
Category of pile	Long	
L1/T for depth of fixity	0.802	
Lf/T	2.044	
Cantilever length Lf	5.098	m
Lateral capacity	2.921	T
Reduction factor	0.9	
Maximum moment t-m	9.331	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.6m		
Pile Dia	0.6	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0064	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.6	
Modulus of subgrade reaction	2350.4	kN/m ³
Stiffness factor T	2.366	
Embedment length L	21	m
L/T	8.877	
Category of pile	Long	
L1/T for depth of fixity	0.845	
Lf/T	2.037	
Cantilever length Lf	4.819	m
Lateral capacity	3.295	T
Reduction factor	0.9	
Maximum moment t-m	10.111	T-m

LENGTH OF PILE-25 M DIA OF PILE-0.75 M

LATERAL CAPACITY FOR BH-1 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	18.6	
Modulus of subgrade reaction	2638.4	kN/m ³
Stiffness factor T	2.763	
Embedment length L	21	m
L/T	7.599	
Category of pile	Long	
L1/T for depth of fixity	0.724	
Lf/T	2.057	
Cantilever length Lf	5.683	m
Lateral capacity	5.623	T
Reduction factor	0.9	
Maximum moment t-m	19.443	T-m

LATERAL CAPACITY FOR BH-2 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	15.8	
Modulus of subgrade reaction	2235.2	kN/m ³
Stiffness factor T	2.857	
Embedment length L	21	m
L/T	7.352	
Category of pile	Long	
L1/T for depth of fixity	0.700	
Lf/T	2.061	
Cantilever length Lf	5.887	m
Lateral capacity	5.200	T
Reduction factor	0.9	
Maximum moment t-m	18.454	T-m

LATERAL CAPACITY FOR BH-3 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	19.4	
Modulus of subgrade reaction	2753.6	kN/m ³
Stiffness factor T	2.740	
Embedment length L	21	m
L/T	7.665	
Category of pile	Long	
L1/T for depth of fixity	0.730	
Lf/T	2.056	
Cantilever length Lf	5.632	m
Lateral capacity	5.737	T
Reduction factor	0.9	
Maximum moment t-m	19.705	T-m

LATERAL CAPACITY FOR BH-5 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21.4	
Modulus of subgrade reaction	3041.6	kN/m ³
Stiffness factor T	2.686	
Embedment length L	21	m
L/T	7.819	
Category of pile	Long	
L1/T for depth of fixity	0.745	
Lf/T	2.053	
Cantilever length Lf	5.515	m
Lateral capacity	6.011	T
Reduction factor	0.9	
Maximum moment t-m	20.327	T-m

LATERAL CAPACITY FOR BH-6 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	20	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	21	
Modulus of subgrade reaction	2984	kN/m ³
Stiffness factor T	2.696	
Embedment length L	16	m
L/T	5.934	
Category of pile	Long	
L1/T for depth of fixity	0.742	
Lf/T	2.054	
Cantilever length Lf	5.537	m
Lateral capacity	5.957	T
Reduction factor	0.9	
Maximum moment t-m	20.206	T-m

LATERAL CAPACITY FOR BH-7 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	17.2	
Modulus of subgrade reaction	2436.8	kN/m ³
Stiffness factor T	2.808	
Embedment length L	21	m
L/T	7.480	
Category of pile	Long	
L1/T for depth of fixity	0.712	
Lf/T	2.059	
Cantilever length Lf	5.780	m
Lateral capacity	5.417	T
Reduction factor	0.9	
Maximum moment t-m	18.964	T-m

LATERAL CAPACITY FOR BH-9 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	12.8	
Modulus of subgrade reaction	1803.2	kN/m ³
Stiffness factor T	2.982	
Embedment length L	21	m
L/T	7.042	
Category of pile	Long	
L1/T for depth of fixity	0.671	
Lf/T	2.066	
Cantilever length Lf	6.160	m
Lateral capacity	4.694	T
Reduction factor	0.9	
Maximum moment t-m	17.237	T-m

LATERAL CAPACITY FOR BH-10 and Pile Diameter =0.75m		
Pile Dia	0.75	m
Terminal Depth	25	m
Cutoff	2	m
LATERAL LOAD CAPACITY OF PILE ON NORMALLY CONSOLIDATED CLAYS AND SAND		
Permissible top deflection mm	0.005	m
Concrete grade (M-30)	30000	Kpa
Youngs modulus E	27386128	Kpa
Moment of Inertia I	0.0155	
Free length (L1) in	2	m
FIXED HEAD		
Average N values corrected	16.6	
Modulus of subgrade reaction	2350.4	kN/m ³
Stiffness factor T	2.828	
Embedment length L	21	m
L/T	7.426	
Category of pile	Long	
L1/T for depth of fixity	0.707	
Lf/T	2.060	
Cantilever length Lf	5.824	m
Lateral capacity	5.325	T
Reduction factor	0.9	
Maximum moment t-m	18.749	T-m



23.0 LIQUEFACTION ANALYSIS

For Shiva Testing & Research Laboratory Pvt. Ltd.

Authorized Signatory



Shiva Testing & Research Laboratory Pvt. Ltd.

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DULIAJAN ASSAM BH-1

DULIAJAN ASSAM BH-1																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	Cl	6	1.600	0.600	80	V	0.36	7.50	0.99	2.40	0.90	0.62	1.70	1.000	1.05	0.75	1.00	8.03	5.00	1.20	14.6	0.16	0.25	FOS<1, Liquefiable
3.00	SM	8	1.710	0.710	22	V	0.36	7.50	0.98	4.97	1.97	0.58	1.70	1.000	1.05	0.80	1.00	11.42	3.93	1.09	16.4	0.17	0.30	FOS<1, Liquefiable
4.50	SM	15	1.710	0.710	13	V	0.36	7.50	0.97	7.53	3.03	0.56	1.70	1.000	1.05	0.85	1.00	22.76	1.89	1.04	25.5	0.30	0.54	FOS<1, Liquefiable
6.00	SM	36	1.710	0.710	16	V	0.36	7.50	0.95	10.10	4.10	0.55	1.56	1.000	1.05	0.95	1.00	56.12	2.77	1.05	61.9	2.24	4.08	FOS>1, Non Liquefiable
7.50	SM	20	1.760	0.760	15	V	0.36	7.50	0.94	12.74	5.24	0.54	1.38	1.000	1.05	0.95	1.00	27.57	2.50	1.05	31.4	0.61	1.14	FOS>1, Non Liquefiable
9.00	SM	18	1.760	0.760	16	V	0.36	7.50	0.93	15.38	6.38	0.53	1.25	1.000	1.05	0.95	1.00	22.49	2.77	1.05	26.5	0.32	0.62	FOS<1, Liquefiable
10.50	SM	21	1.760	0.760	17	V	0.36	7.50	0.89	18.02	7.52	0.50	1.15	1.000	1.05	1	1.00	25.44	3.01	1.06	30.0	0.47	0.93	FOS<1, Liquefiable
12.00	SM	24	1.760	0.760	13	V	0.36	7.50	0.85	20.66	8.66	0.48	1.07	1.000	1.05	1	1.00	27.09	1.89	1.04	30.0	0.47	0.98	FOS<1, Liquefiable
13.50	SM	26	1.760	0.760	15	V	0.36	7.50	0.81	23.30	9.80	0.45	1.01	1.000	1.05	1	1.00	27.58	2.50	1.05	31.4	0.61	1.36	FOS>1, Non Liquefiable
15.00	SM	69	1.760	0.760	13	V	0.36	7.50	0.77	25.94	10.94	0.43	0.96	1.000	1.05	1	1.00	69.28	1.89	1.04	73.7	2.24	5.23	FOS>1, Non Liquefiable
16.50	SM	70	1.760	0.760	14	V	0.36	7.50	0.73	28.58	12.08	0.41	0.91	1.000	1.05	1	1.00	66.89	2.20	1.04	71.9	2.24	5.52	FOS>1, Non Liquefiable
18.00	SM	62	1.760	0.760	13	V	0.36	7.50	0.69	31.22	13.22	0.38	0.87	1.000	1.05	1	1.00	56.63	1.72	1.03	60.3	2.24	5.85	FOS>1, Non Liquefiable
19.50	SM	49	1.760	0.760	14	V	0.36	7.50	0.65	33.86	14.36	0.36	0.83	1.000	1.05	1	1.00	42.94	2.20	1.04	47.0	2.24	6.22	FOS>1, Non Liquefiable
21.00	SM	56	1.760	0.760	13	V	0.36	7.50	0.61	36.50	15.50	0.34	0.80	1.000	1.05	1	1.00	47.24	1.89	1.04	50.9	2.24	6.64	FOS>1, Non Liquefiable
22.50	SM	60	1.760	0.760	13	V	0.36	7.50	0.57	39.14	16.64	0.32	0.78	1.000	1.05	1	1.00	48.85	1.89	1.04	52.5	2.24	7.11	FOS>1, Non Liquefiable
24.00	SM	44	1.760	0.760	15	V	0.36	7.50	0.53	41.78	17.78	0.29	0.75	1.000	1.05	1	1.00	34.65	2.50	1.05	38.8	2.24	7.65	FOS>1, Non Liquefiable
25.50	SM	62	1.760	0.760	14	V	0.36	7.50	0.49	44.42	18.92	0.27	0.73	1.000	1.05	1	1.00	47.33	2.20	1.04	51.5	2.24	8.28	FOS>1, Non Liquefiable
27.00	SM	66	1.760	0.760	13	V	0.36	7.50	0.45	47.06	20.06	0.25	0.71	1.000	1.05	1	1.00	48.94	1.72	1.03	52.3	2.24	9.02	FOS>1, Non Liquefiable
28.50	SM	72	1.760	0.760	13	V	0.36	7.50	0.41	49.70	21.20	0.23	0.69	1.000	1.05	1	1.00	51.93	1.89	1.04	55.7	2.24	9.90	FOS>1, Non Liquefiable
30.00	SM	82	1.760	0.760	15	V	0.36	7.50	0.37	52.34	22.34	0.20	0.67	1.000	1.05	1	1.00	57.61	2.50	1.05	62.9	2.24	10.97	FOS>1, Non Liquefiable

For Shiva Testing & Research Laboratory Pvt. Ltd.
Authorized Signature

DULIAJAN ASSAM BH-2																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m^3)	Submerged Density (t/m^3)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration a_{max}/g	Earthquake Magnitude M_w	Stress reduction coefficient (rd)	Total overburden pressure (s_o), t/m^2	Effective overburden (s_o), t/m^2	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N_1) ₆₀	α	β	$(N_1)_{60cs}$	$CRR_{M=7.5}$	FOS	Conclusion
													C_N	C_E	C_B	C_R	C_S							
1.50	CL	6	1.620	0.620	90	V	0.36	7.50	0.99	2.43	0.93	0.60	1.70	1.000	1.05	0.75	1.00	8.03	5.00	1.20	14.6	0.16	0.26	FOS<1, Liquefiable
3.00	CI	8	1.700	0.700	80	V	0.36	7.50	0.98	4.98	1.98	0.58	1.70	1.000	1.05	0.80	1.00	11.42	5.00	1.20	18.7	0.20	0.35	FOS<1, Liquefiable
4.50	SM	11	1.700	0.700	23	V	0.36	7.50	0.97	7.53	3.03	0.56	1.70	1.000	1.05	0.85	1.00	16.69	4.06	1.10	22.4	0.25	0.44	FOS<1, Liquefiable
6.00	SM	26	1.700	0.700	22	V	0.36	7.50	0.95	10.08	4.08	0.55	1.57	1.000	1.05	0.95	1.00	40.60	3.93	1.09	48.3	2.24	4.07	FOS>1, Non Liquefiable
7.50	SM	28	1.750	0.750	20	V	0.36	7.50	0.94	12.71	5.21	0.54	1.39	1.000	1.05	0.95	1.00	38.71	3.61	1.08	45.4	2.24	4.17	FOS>1, Non Liquefiable
9.00	SP-SM	30	1.750	0.750	7	V	0.36	7.50	0.93	15.33	6.33	0.53	1.26	1.000	1.05	0.95	1.00	37.61	0.12	1.01	38.1	2.24	4.25	FOS>1, Non Liquefiable
10.50	SP-SM	34	1.750	0.750	8	V	0.36	7.50	0.89	17.96	7.46	0.50	1.16	1.000	1.05	1	1.00	41.35	0.20	1.01	42.0	2.24	4.45	FOS>1, Non Liquefiable
12.00	SP-SM	40	1.750	0.750	8	V	0.36	7.50	0.85	20.58	8.58	0.48	1.08	1.000	1.05	1	1.00	45.34	0.30	1.01	46.2	2.24	4.68	FOS>1, Non Liquefiable
13.50	SP-SM	42	1.750	0.750	9	V	0.36	7.50	0.81	23.21	9.71	0.46	1.02	1.000	1.05	1	1.00	44.77	0.42	1.01	45.8	2.24	4.93	FOS>1, Non Liquefiable
15.00	SP-SM	50	1.750	0.750	5	V	0.36	7.50	0.77	25.83	10.83	0.43	0.96	1.000	1.05	1	1.00	50.45	0.00	1.00	50.4	2.24	5.20	FOS>1, Non Liquefiable
16.50	SP-SM	54	1.750	0.750	8	V	0.36	7.50	0.73	28.46	11.96	0.41	0.91	1.000	1.05	1	1.00	51.86	0.30	1.01	52.8	2.24	5.49	FOS>1, Non Liquefiable
18.00	SP-SM	58	1.750	0.750	6	V	0.36	7.50	0.69	31.08	13.08	0.39	0.87	1.000	1.05	1	1.00	53.25	0.01	1.00	53.4	2.24	5.82	FOS>1, Non Liquefiable
19.50	SP-SM	62	1.750	0.750	6	V	0.36	7.50	0.65	33.71	14.21	0.36	0.84	1.000	1.05	1	1.00	54.62	0.03	1.00	54.9	2.24	6.18	FOS>1, Non Liquefiable
21.00	SP-SM	65	1.750	0.750	7	V	0.36	7.50	0.61	36.33	15.33	0.34	0.81	1.000	1.05	1	1.00	55.12	0.06	1.01	55.5	2.24	6.60	FOS>1, Non Liquefiable
22.50	SP-SM	68	1.750	0.750	7	V	0.36	7.50	0.57	38.96	16.46	0.32	0.78	1.000	1.05	1	1.00	55.66	0.12	1.01	56.3	2.24	7.06	FOS>1, Non Liquefiable
24.00	SP-SM	67	1.750	0.750	8	V	0.36	7.50	0.53	41.58	17.58	0.30	0.75	1.000	1.05	1	1.00	53.06	0.20	1.01	53.8	2.24	7.60	FOS>1, Non Liquefiable
25.50	SP-SM	67	1.750	0.750	5	V	0.36	7.50	0.49	44.21	18.71	0.27	0.73	1.000	1.05	1	1.00	51.44	0.00	1.00	51.4	2.24	8.23	FOS>1, Non Liquefiable
27.00	SP-SM	72	1.750	0.750	6	V	0.36	7.50	0.45	46.83	19.83	0.25	0.71	1.000	1.05	1	1.00	53.69	0.03	1.00	54.0	2.24	8.96	FOS>1, Non Liquefiable
28.50	SP-SM	79	1.750	0.750	8	V	0.36	7.50	0.41	49.46	20.96	0.23	0.69	1.000	1.05	1	1.00	57.30	0.30	1.01	58.3	2.24	9.84	FOS>1, Non Liquefiable
30.00	SP-SM	84	1.750	0.750	7	V	0.36	7.50	0.37	52.08	22.08	0.21	0.67	1.000	1.05	1	1.00	59.36	0.06	1.01	59.8	2.24	10.90	FOS>1, Non Liquefiable

 For Shiva Testing & Research Laboratory Pvt. Ltd.

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DULIAJAN ASSAM BH-3

Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s ₀), t/m ²	Effective overburden (s ₀), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion
													overburden	Equipment	BH Dia	Rod Length	std sampler							
													C _N	C _E	C _B	C _R	C _S							
1.50	CL	9	1.620	0.620	80	V	0.36	7.50	0.99	2.43	0.93	0.60	1.70	1.000	1.05	0.75	1.00	12.05	5.00	1.20	19.5	0.21	0.35	FOS<1, Liquefiable
3.00	CL	12	1.700	0.700	80	V	0.36	7.50	0.98	4.98	1.98	0.58	1.70	1.000	1.05	0.80	1.00	17.14	5.00	1.20	25.6	0.30	0.53	FOS<1, Liquefiable
4.50	SP-SM	17	1.700	0.700	9	V	0.36	7.50	0.97	7.53	3.03	0.56	1.70	1.000	1.05	0.85	1.00	25.79	0.56	1.02	26.8	0.33	0.59	FOS<1, Liquefiable
6.00	SP-SM	31	1.700	0.700	7	V	0.36	7.50	0.95	10.08	4.08	0.55	1.57	1.000	1.05	0.95	1.00	48.41	0.06	1.01	48.8	2.24	4.07	FOS>1, Non Liquefiable
7.50	SP-SM	44	1.750	0.750	7	V	0.36	7.50	0.94	12.71	5.21	0.54	1.39	1.000	1.05	0.95	1.00	60.84	0.12	1.01	61.5	2.24	4.17	FOS>1, Non Liquefiable
9.00	SP	45	1.750	0.750	4	V	0.36	7.50	0.93	15.33	6.33	0.53	1.26	1.000	1.05	0.95	1.00	56.42	0.00	1.00	56.4	2.24	4.25	FOS>1, Non Liquefiable
10.50	SP	47	1.750	0.750	4	V	0.36	7.50	0.89	17.96	7.46	0.50	1.16	1.000	1.05	1	1.00	57.16	0.00	1.00	57.2	2.24	4.45	FOS>1, Non Liquefiable
12.00	SP	49	1.750	0.750	5	V	0.36	7.50	0.85	20.58	8.58	0.48	1.08	1.000	1.05	1	1.00	55.54	0.00	1.00	55.5	2.24	4.68	FOS>1, Non Liquefiable
13.50	SP	52	1.750	0.750	3	V	0.36	7.50	0.81	23.21	9.71	0.46	1.02	1.000	1.05	1	1.00	55.42	0.00	1.00	55.4	2.24	4.93	FOS>1, Non Liquefiable
15.00	SP	51	1.750	0.750	2	V	0.36	7.50	0.77	25.83	10.83	0.43	0.96	1.000	1.05	1	1.00	51.46	0.00	1.00	51.5	2.24	5.20	FOS>1, Non Liquefiable
16.50	SP	54	1.750	0.750	3	V	0.36	7.50	0.73	28.46	11.96	0.41	0.91	1.000	1.05	1	1.00	51.86	0.00	1.00	51.9	2.24	5.49	FOS>1, Non Liquefiable
18.00	SP	57	1.750	0.750	4	V	0.36	7.50	0.69	31.08	13.08	0.39	0.87	1.000	1.05	1	1.00	52.33	0.00	1.00	52.3	2.24	5.82	FOS>1, Non Liquefiable
19.50	SP	60	1.750	0.750	3	V	0.36	7.50	0.65	33.71	14.21	0.36	0.84	1.000	1.05	1	1.00	52.86	0.00	1.00	52.9	2.24	6.18	FOS>1, Non Liquefiable
21.00	SP	62	1.750	0.750	2	V	0.36	7.50	0.61	36.33	15.33	0.34	0.81	1.000	1.05	1	1.00	52.58	0.00	1.00	52.6	2.24	6.60	FOS>1, Non Liquefiable
22.50	SP	66	1.750	0.750	3	V	0.36	7.50	0.57	38.96	16.46	0.32	0.78	1.000	1.05	1	1.00	54.02	0.00	1.00	54.0	2.24	7.06	FOS>1, Non Liquefiable
24.00	SP	69	1.750	0.750	2	V	0.36	7.50	0.53	41.58	17.58	0.30	0.75	1.000	1.05	1	1.00	54.64	0.00	1.00	54.6	2.24	7.60	FOS>1, Non Liquefiable
25.50	SP	70	1.750	0.750	1	V	0.36	7.50	0.49	44.21	18.71	0.27	0.73	1.000	1.05	1	1.00	53.74	0.00	1.00	53.7	2.24	8.23	FOS>1, Non Liquefiable
27.00	SP	76	1.750	0.750	2	V	0.36	7.50	0.45	46.83	19.83	0.25	0.71	1.000	1.05	1	1.00	56.67	0.00	1.00	56.7	2.24	8.96	FOS>1, Non Liquefiable
28.50	SP	80	1.750	0.750	4	V	0.36	7.50	0.41	49.46	20.96	0.23	0.69	1.000	1.05	1	1.00	58.03	0.00	1.00	58.0	2.24	9.84	FOS>1, Non Liquefiable
30.00	SP	86	1.750	0.750	3	V	0.36	7.50	0.37	52.08	22.08	0.21	0.67	1.000	1.05	1	1.00	60.77	0.00	1.00	60.8	2.24	10.90	FOS>1, Non Liquefiable

For Shiva Testing & Research Laboratory Pvt. Ltd.

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DULIAJAN ASSAM BH-4																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Accleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _M = 7.5	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	CI	7	1.630	0.630	88	V	0.36	7.50	0.99	2.45	0.95	0.60	1.70	1.000	1.05	0.75	1.00	9.37	5.00	1.20	16.2	0.17	0.29	FOS<1, Liquefiable
3.00	CI	12	1.630	0.630	82	V	0.36	7.50	0.98	4.89	1.89	0.59	1.70	1.000	1.05	0.80	1.00	17.14	5.00	1.20	25.6	0.30	0.51	FOS<1, Liquefiable
4.50	SM	15	1.740	0.740	13	V	0.36	7.50	0.97	7.50	3.00	0.56	1.70	1.000	1.05	0.85	1.00	22.76	1.95	1.04	25.6	0.30	0.54	FOS<1, Liquefiable
6.00	SM	25	1.740	0.740	14	V	0.36	7.50	0.95	10.11	4.11	0.55	1.56	1.000	1.05	0.95	1.00	38.90	2.20	1.04	42.8	2.24	4.09	FOS>1, Non Liquefiable
7.50	SM	37	1.800	0.800	15	V	0.36	7.50	0.94	12.81	5.31	0.53	1.37	1.000	1.05	0.95	1.00	50.65	2.50	1.05	55.6	2.24	4.22	FOS>1, Non Liquefiable
9.00	SM	48	1.800	0.800	14	V	0.36	7.50	0.93	15.51	6.51	0.52	1.24	1.000	1.05	0.95	1.00	59.34	2.20	1.04	64.1	2.24	4.32	FOS>1, Non Liquefiable
10.50	SM	53	1.800	0.800	15	V	0.36	7.50	0.89	18.21	7.71	0.49	1.14	1.000	1.05	1	1.00	63.38	2.50	1.05	68.9	2.24	4.54	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-5																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Accleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s ₀), t/m ²	Effective overburden (s ₀), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	CI	10	1.670	0.670	84	V	0.36	7.50	0.99	2.51	1.01	0.58	1.70	1.000	1.05	0.75	1.00	13.39	5.00	1.20	21.1	0.23	0.40	FOS<1, Liquefiable
3.00	SM	14	1.750	0.750	40	V	0.36	7.50	0.98	5.13	2.13	0.55	1.70	1.000	1.05	0.80	1.00	19.99	5.00	1.20	29.0	0.41	0.74	FOS<1, Liquefiable
4.50	SM	24	1.750	0.750	25	V	0.36	7.50	0.97	7.76	3.26	0.54	1.70	1.000	1.05	0.85	1.00	36.41	4.29	1.12	44.9	2.24	4.17	FOS>1, Non Liquefiable
6.00	SM	31	1.750	0.750	28	V	0.36	7.50	0.95	10.38	4.38	0.53	1.51	1.000	1.05	0.95	1.00	46.72	4.56	1.14	57.7	2.24	4.24	FOS>1, Non Liquefiable
7.50	SM	29	1.820	0.820	18	V	0.36	7.50	0.94	13.11	5.61	0.52	1.34	1.000	1.05	0.95	1.00	38.62	3.23	1.07	44.4	2.24	4.35	FOS>1, Non Liquefiable
9.00	SM	33	1.820	0.820	10	V	0.36	7.50	0.93	15.84	6.84	0.50	1.21	1.000	1.05	0.95	1.00	39.80	0.87	1.02	41.5	2.24	4.45	FOS>1, Non Liquefiable
10.50	SM	35	1.820	0.820	15	V	0.36	7.50	0.89	18.57	8.07	0.48	1.11	1.000	1.05	1	1.00	40.91	2.50	1.05	45.4	2.24	4.66	FOS>1, Non Liquefiable
12.00	SM	44	1.820	0.820	14	V	0.36	7.50	0.85	21.30	9.30	0.46	1.04	1.000	1.05	1	1.00	47.91	2.20	1.04	52.1	2.24	4.90	FOS>1, Non Liquefiable
13.50	SM	22	1.820	0.820	13	V	0.36	7.50	0.81	24.03	10.53	0.43	0.97	1.000	1.05	1	1.00	22.51	1.89	1.04	25.2	0.30	0.68	FOS<1, Liquefiable
15.00	SM	24	1.820	0.820	14	V	0.36	7.50	0.77	26.76	11.76	0.41	0.92	1.000	1.05	1	1.00	23.24	2.20	1.04	26.4	0.32	0.79	FOS<1, Liquefiable
16.50	SM	26	1.820	0.820	19	V	0.36	7.50	0.73	29.49	12.99	0.39	0.88	1.000	1.05	1	1.00	23.95	3.43	1.07	29.1	0.42	1.07	FOS>1, Non Liquefiable
18.00	SM	35	1.820	0.820	18	V	0.36	7.50	0.69	32.22	14.22	0.37	0.84	1.000	1.05	1	1.00	30.82	3.23	1.07	36.1	2.24	6.10	FOS>1, Non Liquefiable
19.50	SM	44	1.820	0.820	13	V	0.36	7.50	0.65	34.95	15.45	0.35	0.80	1.000	1.05	1	1.00	37.17	1.89	1.04	40.4	2.24	6.49	FOS>1, Non Liquefiable
21.00	SM	58	1.820	0.820	17	V	0.36	7.50	0.61	37.68	16.68	0.32	0.77	1.000	1.05	1	1.00	47.15	3.01	1.06	53.0	2.24	6.92	FOS>1, Non Liquefiable
22.50	SM	55	1.820	0.820	16	V	0.36	7.50	0.57	40.41	17.91	0.30	0.75	1.000	1.05	1	1.00	43.15	2.77	1.05	48.2	2.24	7.41	FOS>1, Non Liquefiable
24.00	SM	44	1.820	0.820	15	V	0.36	7.50	0.53	43.14	19.14	0.28	0.72	1.000	1.05	1	1.00	33.39	2.50	1.05	37.5	2.24	7.98	FOS>1, Non Liquefiable
25.50	SM	46	1.820	0.820	14	V	0.36	7.50	0.49	45.87	20.37	0.26	0.70	1.000	1.05	1	1.00	33.84	2.05	1.04	37.2	2.24	8.63	FOS>1, Non Liquefiable
27.00	SM	51	1.820	0.820	14	V	0.36	7.50	0.45	48.60	21.60	0.24	0.68	1.000	1.05	1	1.00	36.44	2.20	1.04	40.2	2.24	9.40	FOS>1, Non Liquefiable
28.50	SM	60	1.820	0.820	13	V	0.36	7.50	0.41	51.33	22.83	0.22	0.66	1.000	1.05	1	1.00	41.70	1.89	1.04	45.1	2.24	10.32	FOS>1, Non Liquefiable
30.00	SM	70	1.820	0.820	15	V	0.36	7.50	0.37	54.06	24.06	0.20	0.64	1.000	1.05	1	1.00	47.38	2.50	1.05	52.2	2.24	11.44	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-6																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	CI	7	1.680	0.680	90	V	0.36	7.50	0.99	2.52	1.02	0.57	1.70	1.000	1.05	0.75	1.00	9.37	5.00	1.20	16.2	0.17	0.30	FOS<1, Liquefiable
3.00	CI	16	1.680	0.680	86	V	0.36	7.50	0.98	5.04	2.04	0.56	1.70	1.000	1.05	0.80	1.00	22.85	5.00	1.20	32.4	0.87	1.54	FOS>1, Non Liquefiable
4.50	SM	22	1.750	0.750	14	V	0.36	7.50	0.97	7.67	3.17	0.55	1.70	1.000	1.05	0.85	1.00	33.38	2.20	1.04	37.0	2.24	4.10	FOS>1, Non Liquefiable
6.00	SM	32	1.750	0.750	18	V	0.36	7.50	0.95	10.29	4.29	0.54	1.53	1.000	1.05	0.95	1.00	48.73	3.23	1.07	55.2	2.24	4.19	FOS>1, Non Liquefiable
7.50	SM	24	1.750	0.750	17	V	0.36	7.50	0.94	12.92	5.42	0.53	1.36	1.000	1.05	0.95	1.00	32.53	3.04	1.06	37.5	2.24	4.26	FOS>1, Non Liquefiable
9.00	SM	37	1.780	0.780	14	V	0.36	7.50	0.93	15.59	6.59	0.52	1.23	1.000	1.05	0.95	1.00	45.48	2.07	1.04	49.4	2.24	4.35	FOS>1, Non Liquefiable
10.50	SM	42	1.780	0.780	14	V	0.36	7.50	0.89	18.26	7.76	0.49	1.14	1.000	1.05	1	1.00	50.08	2.20	1.04	54.4	2.24	4.56	FOS>1, Non Liquefiable
12.00	SM	48	1.780	0.780	15	V	0.36	7.50	0.85	20.93	8.93	0.47	1.06	1.000	1.05	1	1.00	53.35	2.51	1.05	58.4	2.24	4.79	FOS>1, Non Liquefiable
13.50	SM	51	1.780	0.780	17	V	0.36	7.50	0.81	23.60	10.10	0.44	1.00	1.000	1.05	1	1.00	53.30	2.99	1.06	59.5	2.24	5.04	FOS>1, Non Liquefiable
15.00	SM	55	1.780	0.780	15	V	0.36	7.50	0.77	26.27	11.27	0.42	0.94	1.000	1.05	1	1.00	54.41	2.47	1.05	59.5	2.24	5.32	FOS>1, Non Liquefiable
16.50	SM	36	1.780	0.780	18	V	0.36	7.50	0.73	28.94	12.44	0.40	0.90	1.000	1.05	1	1.00	33.90	3.18	1.06	39.3	2.24	5.62	FOS>1, Non Liquefiable
18.00	SM	40	1.780	0.780	19	V	0.36	7.50	0.69	31.61	13.61	0.38	0.86	1.000	1.05	1	1.00	36.01	3.41	1.07	42.0	2.24	5.95	FOS>1, Non Liquefiable
19.50	SM	48	1.780	0.780	20	V	0.36	7.50	0.65	34.28	14.78	0.35	0.82	1.000	1.05	1	1.00	41.46	3.56	1.08	48.2	2.24	6.33	FOS>1, Non Liquefiable
21.00	SM	46	1.780	0.780	14	V	0.36	7.50	0.61	36.95	15.95	0.33	0.79	1.000	1.05	1	1.00	38.25	2.14	1.04	42.0	2.24	6.75	FOS>1, Non Liquefiable
22.50	SM	50	1.780	0.780	19	V	0.36	7.50	0.57	39.62	17.12	0.31	0.76	1.000	1.05	1	1.00	40.13	3.42	1.07	46.5	2.24	7.23	FOS>1, Non Liquefiable
24.00	SM	52	1.780	0.780	19	V	0.36	7.50	0.53	42.29	18.29	0.29	0.74	1.000	1.05	1	1.00	40.38	3.46	1.07	46.8	2.24	7.78	FOS>1, Non Liquefiable
25.50	SM	57	1.780	0.780	18	V	0.36	7.50	0.49	44.96	19.46	0.27	0.72	1.000	1.05	1	1.00	42.91	3.17	1.06	48.8	2.24	8.41	FOS>1, Non Liquefiable
27.00	SM	54	1.780	0.780	17	V	0.36	7.50	0.45	47.63	20.63	0.24	0.70	1.000	1.05	1	1.00	39.48	2.94	1.06	44.7	2.24	9.16	FOS>1, Non Liquefiable
28.50	SM	61	1.780	0.780	15	V	0.36	7.50	0.41	50.30	21.80	0.22	0.68	1.000	1.05	1	1.00	43.39	2.40	1.05	47.8	2.24	10.06	FOS>1, Non Liquefiable
30.00	SM	67	1.780	0.780	16	V	0.36	7.50	0.37	52.97	22.97	0.20	0.66	1.000	1.05	1	1.00	46.42	2.68	1.05	51.5	2.24	11.14	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-7																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Accelration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _M = 7.5	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	CL	5	1.660	0.660	80	V	0.36	7.50	0.99	2.49	0.99	0.58	1.70	1.000	1.05	0.75	1.00	6.69	5.00	1.20	13.0	0.14	0.24	FOS<1, Liquefiable
3.00	CL	11	1.660	0.660	87	V	0.36	7.50	0.98	4.98	1.98	0.58	1.70	1.000	1.05	0.80	1.00	15.71	5.00	1.20	23.8	0.27	0.47	FOS<1, Liquefiable
4.50	CL	11	1.660	0.660	92	V	0.36	7.50	0.97	7.47	2.97	0.57	1.70	1.000	1.05	0.85	1.00	16.69	5.00	1.20	25.0	0.29	0.51	FOS<1, Liquefiable
6.00	SM	31	1.760	0.760	19	V	0.36	7.50	0.95	10.11	4.11	0.55	1.56	1.000	1.05	0.95	1.00	48.23	3.47	1.07	55.3	2.24	4.09	FOS>1, Non Liquefiable
7.50	SM	25	1.760	0.760	18	V	0.36	7.50	0.94	12.75	5.25	0.54	1.38	1.000	1.05	0.95	1.00	34.42	3.25	1.07	40.0	2.24	4.19	FOS>1, Non Liquefiable
9.00	SM	32	1.820	0.820	14	V	0.36	7.50	0.93	15.48	6.48	0.52	1.24	1.000	1.05	0.95	1.00	39.65	2.19	1.04	43.5	2.24	4.31	FOS>1, Non Liquefiable
10.50	SM	34	1.820	0.820	18	V	0.36	7.50	0.89	18.21	7.71	0.49	1.14	1.000	1.05	1	1.00	40.66	3.23	1.07	46.6	2.24	4.54	FOS>1, Non Liquefiable
12.00	SM	41	1.820	0.820	14	V	0.36	7.50	0.85	20.94	8.94	0.47	1.06	1.000	1.05	1	1.00	45.53	2.17	1.04	49.6	2.24	4.80	FOS>1, Non Liquefiable
13.50	SM	39	1.820	0.820	19	V	0.36	7.50	0.81	23.67	10.17	0.44	0.99	1.000	1.05	1	1.00	40.61	3.41	1.07	46.9	2.24	5.06	FOS>1, Non Liquefiable
15.00	SM	43	1.820	0.820	20	V	0.36	7.50	0.77	26.40	11.40	0.42	0.94	1.000	1.05	1	1.00	42.29	3.59	1.08	49.2	2.24	5.35	FOS>1, Non Liquefiable
16.50	SM	47	1.820	0.820	18	V	0.36	7.50	0.73	29.13	12.63	0.40	0.89	1.000	1.05	1	1.00	43.91	3.22	1.07	50.0	2.24	5.67	FOS>1, Non Liquefiable
18.00	SM	41	1.820	0.820	17	V	0.36	7.50	0.69	31.86	13.86	0.37	0.85	1.000	1.05	1	1.00	36.57	2.96	1.06	41.7	2.24	6.02	FOS>1, Non Liquefiable
19.50	SM	44	1.820	0.820	16	V	0.36	7.50	0.65	34.59	15.09	0.35	0.81	1.000	1.05	1	1.00	37.61	2.73	1.05	42.3	2.24	6.40	FOS>1, Non Liquefiable
21.00	SM	50	1.820	0.820	14	V	0.36	7.50	0.61	37.32	16.32	0.33	0.78	1.000	1.05	1	1.00	41.10	2.14	1.04	44.9	2.24	6.84	FOS>1, Non Liquefiable
22.50	SM	47	1.820	0.820	15	V	0.36	7.50	0.57	40.05	17.55	0.31	0.75	1.000	1.05	1	1.00	37.25	2.48	1.05	41.5	2.24	7.33	FOS>1, Non Liquefiable
24.00	SM	56	1.820	0.820	16	V	0.36	7.50	0.53	42.78	18.78	0.28	0.73	1.000	1.05	1	1.00	42.91	2.74	1.05	47.9	2.24	7.89	FOS>1, Non Liquefiable
25.50	SM	53	1.820	0.820	16	V	0.36	7.50	0.49	45.51	20.01	0.26	0.71	1.000	1.05	1	1.00	39.34	2.88	1.06	44.5	2.24	8.55	FOS>1, Non Liquefiable
27.00	SM	61	1.820	0.820	18	V	0.36	7.50	0.45	48.24	21.24	0.24	0.69	1.000	1.05	1	1.00	43.95	3.14	1.06	49.9	2.24	9.32	FOS>1, Non Liquefiable
28.50	SM	66	1.820	0.820	15	V	0.36	7.50	0.41	50.97	22.47	0.22	0.67	1.000	1.05	1	1.00	46.23	2.35	1.05	50.7	2.24	10.23	FOS>1, Non Liquefiable
30.00	SM	71	1.820	0.820	14	V	0.36	7.50	0.37	53.70	23.70	0.20	0.65	1.000	1.05	1	1.00	48.43	2.10	1.04	52.5	2.24	11.34	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-8																													
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion					
													C _N	overburden	C _E	Equipment	C _B								BH Dia	C _R	Rod Length	C _S	std sampler
1.50	CI	8	1.620	0.620	86	V	0.36	7.50	0.99	2.43	0.93	0.60	1.70	1.000	1.05	0.75	1.00	10.71	5.00	1.20	17.9	0.19	0.31	FOS<1, Liquefiable					
3.00	CI	20	1.620	0.620	77	V	0.36	7.50	0.98	4.86	1.86	0.60	1.70	1.000	1.05	0.80	1.00	28.56	5.00	1.20	39.3	2.24	3.76	FOS>1, Non Liquefiable					
4.50	CI	30	1.620	0.620	80	V	0.36	7.50	0.97	7.29	2.79	0.59	1.70	1.000	1.05	0.85	1.00	45.52	5.00	1.20	59.6	2.24	3.80	FOS>1, Non Liquefiable					
6.00	CL	37	1.670	0.670	80	V	0.36	7.50	0.95	9.80	3.80	0.58	1.62	1.000	1.05	0.95	1.00	59.91	5.00	1.20	76.9	2.24	3.89	FOS>1, Non Liquefiable					
7.50	CL	41	1.670	0.670	78	V	0.36	7.50	0.94	12.30	4.80	0.57	1.44	1.000	1.05	0.95	1.00	59.03	5.00	1.20	75.8	2.24	3.97	FOS>1, Non Liquefiable					
9.00	CL	46	1.720	0.720	85	V	0.36	7.50	0.93	14.88	5.88	0.55	1.30	1.000	1.05	0.95	1.00	59.84	5.00	1.20	76.8	2.24	4.07	FOS>1, Non Liquefiable					
10.50	SM	51	1.720	0.720	12	V	0.36	7.50	0.89	17.46	6.96	0.52	1.20	1.000	1.05	1	1.00	64.19	1.55	1.03	67.8	2.24	4.28	FOS>1, Non Liquefiable					
12.00	SM	51	1.850	0.850	13	V	0.36	7.50	0.85	20.24	8.24	0.49	1.10	1.000	1.05	1	1.00	59.01	1.89	1.04	63.1	2.24	4.57	FOS>1, Non Liquefiable					
13.50	SM	51	1.850	0.850	15	V	0.36	7.50	0.81	23.01	9.51	0.46	1.03	1.000	1.05	1	1.00	54.91	2.50	1.05	60.1	2.24	4.87	FOS>1, Non Liquefiable					
15.00	SM	46	1.850	0.850	14	V	0.36	7.50	0.77	25.79	10.79	0.43	0.96	1.000	1.05	1	1.00	46.51	2.20	1.04	50.7	2.24	5.18	FOS>1, Non Liquefiable					
16.50	SM	54	1.850	0.850	17	V	0.36	7.50	0.73	28.56	12.06	0.41	0.91	1.000	1.05	1	1.00	51.63	3.01	1.06	57.7	2.24	5.52	FOS>1, Non Liquefiable					
18.00	SM	58	1.850	0.850	18	V	0.36	7.50	0.69	31.34	13.34	0.38	0.87	1.000	1.05	1	1.00	52.74	3.23	1.07	59.5	2.24	5.88	FOS>1, Non Liquefiable					
20.00	SM	61	1.850	0.850	13	V	0.36	7.50	0.64	35.04	15.04	0.35	0.82	1.000	1.05	1	1.00	52.24	1.89	1.04	56.1	2.24	6.43	FOS>1, Non Liquefiable					

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DULIAJAN ASSAM BH-9																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Accelration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _{M = 7.5}	FOS	Conclusion
													overburden	Equipment	BH Dia	Rod Length	std sampler							
1.50	CI	5	1.580	0.580	92	V	0.36	7.50	0.99	2.37	0.87	0.63	1.70	1.000	1.05	0.75	1.00	6.69	5.00	1.20	13.0	0.14	0.22	FOS<1, Liquefiable
3.00	CI	8	1.580	0.580	98	V	0.36	7.50	0.98	4.74	1.74	0.62	1.70	1.000	1.05	0.80	1.00	11.42	5.00	1.20	18.7	0.20	0.32	FOS<1, Liquefiable
4.50	SM	11	1.580	0.580	36	V	0.36	7.50	0.97	7.11	2.61	0.62	1.70	1.000	1.05	0.85	1.00	16.69	5.00	1.20	25.0	0.29	0.48	FOS<1, Liquefiable
6.00	SM	12	1.660	0.660	31	V	0.36	7.50	0.95	9.60	3.60	0.60	1.67	1.000	1.05	0.95	1.00	19.95	4.77	1.16	28.0	0.37	0.62	FOS<1, Liquefiable
7.50	SM	26	1.660	0.660	21	V	0.36	7.50	0.94	12.09	4.59	0.58	1.48	1.000	1.05	0.95	1.00	38.28	3.78	1.09	45.4	2.24	3.86	FOS>1, Non Liquefiable
9.00	SM	30	1.750	0.750	20	V	0.36	7.50	0.93	14.72	5.72	0.56	1.32	1.000	1.05	0.95	1.00	39.58	3.61	1.08	46.3	2.24	4.00	FOS>1, Non Liquefiable
10.50	SM	33	1.750	0.750	23	V	0.36	7.50	0.89	17.34	6.84	0.53	1.21	1.000	1.05	1	1.00	41.90	4.06	1.10	50.2	2.24	4.23	FOS>1, Non Liquefiable
12.00	SM	41	1.750	0.750	19	V	0.36	7.50	0.85	19.97	7.97	0.50	1.12	1.000	1.05	1	1.00	48.24	3.43	1.07	55.2	2.24	4.48	FOS>1, Non Liquefiable
13.50	SM	39	1.750	0.750	17	V	0.36	7.50	0.81	22.59	9.09	0.47	1.05	1.000	1.05	1	1.00	42.95	3.01	1.06	48.5	2.24	4.74	FOS>1, Non Liquefiable
15.00	SM	43	1.750	0.750	13	V	0.36	7.50	0.77	25.22	10.22	0.45	0.99	1.000	1.05	1	1.00	44.67	1.89	1.04	48.2	2.24	5.02	FOS>1, Non Liquefiable
16.50	SM	41	1.750	0.750	14	V	0.36	7.50	0.73	27.84	11.34	0.42	0.94	1.000	1.05	1	1.00	40.43	2.20	1.04	44.3	2.24	5.32	FOS>1, Non Liquefiable
18.00	SM	44	1.750	0.750	15	V	0.36	7.50	0.69	30.47	12.47	0.40	0.90	1.000	1.05	1	1.00	41.38	2.50	1.05	45.9	2.24	5.66	FOS>1, Non Liquefiable
19.50	SM	46	1.750	0.750	20	V	0.36	7.50	0.65	33.09	13.59	0.37	0.86	1.000	1.05	1	1.00	41.43	3.61	1.08	48.3	2.24	6.03	FOS>1, Non Liquefiable
21.00	SM	43	1.750	0.750	17	V	0.36	7.50	0.61	35.72	14.72	0.35	0.82	1.000	1.05	1	1.00	37.22	2.89	1.06	42.2	2.24	6.44	FOS>1, Non Liquefiable
22.50	SM	45	1.750	0.750	15	V	0.36	7.50	0.57	38.34	15.84	0.32	0.79	1.000	1.05	1	1.00	37.54	2.50	1.05	41.8	2.24	6.91	FOS>1, Non Liquefiable
24.00	SM	48	1.750	0.750	13	V	0.36	7.50	0.53	40.97	16.97	0.30	0.77	1.000	1.05	1	1.00	38.69	1.89	1.04	42.0	2.24	7.45	FOS>1, Non Liquefiable
25.50	SM	53	1.750	0.750	14	V	0.36	7.50	0.49	43.59	18.09	0.28	0.74	1.000	1.05	1	1.00	41.38	2.20	1.04	45.3	2.24	8.07	FOS>1, Non Liquefiable
27.00	SM	55	1.750	0.750	15	V	0.36	7.50	0.45	46.22	19.22	0.26	0.72	1.000	1.05	1	1.00	41.66	2.50	1.05	46.2	2.24	8.80	FOS>1, Non Liquefiable
28.50	SM	61	1.750	0.750	14	V	0.36	7.50	0.41	48.84	20.34	0.23	0.70	1.000	1.05	1	1.00	44.91	2.20	1.04	49.0	2.24	9.67	FOS>1, Non Liquefiable
30.00	SM	64	1.750	0.750	17	V	0.36	7.50	0.37	51.47	21.47	0.21	0.68	1.000	1.05	1	1.00	45.87	3.01	1.06	51.6	2.24	10.72	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-10																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m^3)	Submerged Density (t/m^3)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration a_{max}/g	Earthquake Magnitude M_w	Stress reduction coefficient (rd)	Total overburden pressure (s_o), t/m^2	Effective overburden (s_e), t/m^2	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N_1) ₆₀	α	β	$(N_1)_{60cs}$	$CRR_M = 7.5$	FOS	Conclusion
													C_N	C_E	C_B	C_R	C_S							
1.50	CI	6	1.580	0.580	93	V	0.36	7.50	0.99	2.37	0.87	0.63	1.70	1.000	1.05	0.75	1.00	8.03	5.00	1.20	14.6	0.16	0.25	FOS<1, Liquefiable
3.00	CI	9	1.580	0.580	88	V	0.36	7.50	0.98	4.74	1.74	0.62	1.70	1.000	1.05	0.80	1.00	12.85	5.00	1.20	20.4	0.22	0.35	FOS<1, Liquefiable
4.50	SM	16	1.580	0.580	24	V	0.36	7.50	0.97	7.11	2.61	0.62	1.70	1.000	1.05	0.85	1.00	24.28	4.18	1.11	31.1	0.57	0.92	FOS<1, Liquefiable
6.00	SM	24	1.690	0.690	28	V	0.36	7.50	0.95	9.65	3.65	0.59	1.66	1.000	1.05	0.95	1.00	39.65	4.56	1.14	49.7	2.24	3.80	FOS>1, Non Liquefiable
7.50	SM	26	1.690	0.690	13	V	0.36	7.50	0.94	12.18	4.68	0.57	1.46	1.000	1.05	0.95	1.00	37.91	1.89	1.04	41.2	2.24	3.91	FOS>1, Non Liquefiable
9.00	SM	22	1.750	0.750	16	V	0.36	7.50	0.93	14.81	5.81	0.56	1.31	1.000	1.05	0.95	1.00	28.80	2.77	1.05	33.1	1.38	2.49	FOS>1, Non Liquefiable
10.50	SM	28	1.750	0.750	14	V	0.36	7.50	0.89	17.43	6.93	0.53	1.20	1.000	1.05	1	1.00	35.32	2.20	1.04	39.0	2.24	4.27	FOS>1, Non Liquefiable
12.00	SM	30	1.750	0.750	17	V	0.36	7.50	0.85	20.06	8.06	0.50	1.11	1.000	1.05	1	1.00	35.10	3.01	1.06	40.2	2.24	4.51	FOS>1, Non Liquefiable
13.50	SM	31	1.750	0.750	20	V	0.36	7.50	0.81	22.68	9.18	0.47	1.04	1.000	1.05	1	1.00	33.97	3.53	1.08	40.1	2.24	4.77	FOS>1, Non Liquefiable
15.00	SM	35	1.750	0.750	21	V	0.36	7.50	0.77	25.31	10.31	0.44	0.99	1.000	1.05	1	1.00	36.20	3.78	1.09	43.1	2.24	5.05	FOS>1, Non Liquefiable
16.50	SM	44	1.750	0.750	13	V	0.36	7.50	0.73	27.93	11.43	0.42	0.94	1.000	1.05	1	1.00	43.21	1.89	1.04	46.7	2.24	5.35	FOS>1, Non Liquefiable
18.00	SM	50	1.750	0.750	22	V	0.36	7.50	0.69	30.56	12.56	0.39	0.89	1.000	1.05	1	1.00	46.85	3.93	1.09	55.1	2.24	5.68	FOS>1, Non Liquefiable
19.50	SM	48	1.750	0.750	23	V	0.36	7.50	0.65	33.18	13.68	0.37	0.85	1.000	1.05	1	1.00	43.09	4.06	1.10	51.5	2.24	6.05	FOS>1, Non Liquefiable
21.00	SM	56	1.750	0.750	14	V	0.36	7.50	0.61	35.81	14.81	0.35	0.82	1.000	1.05	1	1.00	48.33	2.05	1.04	52.3	2.24	6.46	FOS>1, Non Liquefiable
22.50	SM	50	1.750	0.750	15	V	0.36	7.50	0.57	38.43	15.93	0.32	0.79	1.000	1.05	1	1.00	41.60	2.50	1.05	46.1	2.24	6.93	FOS>1, Non Liquefiable
24.00	SM	54	1.750	0.750	14	V	0.36	7.50	0.53	41.06	17.06	0.30	0.77	1.000	1.05	1	1.00	43.42	2.20	1.04	47.5	2.24	7.47	FOS>1, Non Liquefiable
25.50	SM	58	1.750	0.750	13	V	0.36	7.50	0.49	43.68	18.18	0.28	0.74	1.000	1.05	1	1.00	45.17	1.89	1.04	48.7	2.24	8.09	FOS>1, Non Liquefiable
27.00	SM	61	1.750	0.750	18	V	0.36	7.50	0.45	46.31	19.31	0.25	0.72	1.000	1.05	1	1.00	46.10	3.23	1.07	52.4	2.24	8.82	FOS>1, Non Liquefiable
28.50	SM	63	1.750	0.750	14	V	0.36	7.50	0.41	48.93	20.43	0.23	0.70	1.000	1.05	1	1.00	46.28	2.20	1.04	50.4	2.24	9.69	FOS>1, Non Liquefiable
30.00	SM	69	1.750	0.750	17	V	0.36	7.50	0.37	51.56	21.56	0.21	0.68	1.000	1.05	1	1.00	49.35	3.01	1.06	55.3	2.24	10.75	FOS>1, Non Liquefiable

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DULIAJAN ASSAM BH-11																								
Depth below EGL, m	Type of Strata	Observed SPT Value	Saturated density (t/m ³)	Submerged Density (t/m ³)	Fine Content (%)	Earthquake Zone	Peak Ground Acceleration amax/g	Earthquake Magnitude Mw	Stress reduction coefficient (rd)	Total overburden pressure (s _o), t/m ²	Effective overburden (s _o), t/m ²	Cyclic Stress ratio (CSR)	Correction Factor for N value					SPT Corrected (N ₁) ₆₀	α	β	(N ₁) _{60cs}	CRR _M = 7.5	FOS	Conclusion
													C _N	C _E	C _B	C _R	C _S							
1.50	CI	9	1.600	0.600	84	V	0.36	7.50	0.99	2.40	0.90	0.62	1.70	1.000	1.05	0.75	1.00	12.05	5.00	1.20	19.5	0.21	0.34	FOS<1, Liquefiable
3.00	CI	13	1.600	0.600	90	V	0.36	7.50	0.98	4.80	1.80	0.61	1.70	1.000	1.05	0.80	1.00	18.56	5.00	1.20	27.3	0.35	0.57	FOS<1, Liquefiable
4.50	SM	25	1.740	0.740	14	V	0.36	7.50	0.97	7.41	2.91	0.58	1.70	1.000	1.05	0.85	1.00	37.93	2.20	1.04	41.7	2.24	3.90	FOS>1, Non Liquefiable
6.00	SM	35	1.740	0.740	15	V	0.36	7.50	0.95	10.02	4.02	0.56	1.58	1.000	1.05	0.95	1.00	55.06	2.50	1.05	60.2	2.24	4.03	FOS>1, Non Liquefiable
7.50	SM	44	1.740	0.740	17	V	0.36	7.50	0.94	12.63	5.13	0.54	1.40	1.000	1.05	0.95	1.00	61.28	3.01	1.06	68.0	2.24	4.13	FOS>1, Non Liquefiable
9.00	SM	42	1.800	0.800	19	V	0.36	7.50	0.93	15.33	6.33	0.53	1.26	1.000	1.05	0.95	1.00	52.66	3.43	1.07	59.9	2.24	4.25	FOS>1, Non Liquefiable
10.50	SM	49	1.800	0.800	13	V	0.36	7.50	0.89	18.03	7.53	0.50	1.15	1.000	1.05	1	1.00	59.29	1.89	1.04	63.4	2.24	4.48	FOS>1, Non Liquefiable

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SUMMARY OF LIQUEFACTION ANALYSIS

Depth below GL	BH-1	BH-2	BH-3	BH-4	BH-5	BH-6	BH-7	BH-8	BH-9	BH-10	BH-11
1.50	L	L	L	L	L	L	L	L	L	L	L
3.00	L	L	L	L	L	NL	L	NL	L	L	L
4.50	L	L	L	L	NL	NL	L	NL	L	L	NL
6.00	NL	NL	NL	NL	NL	NL	NL	NL	L	NL	NL
7.50	NL	NL	NL	NL	NL	NL	NL	NL	NL	NL	NL
9.00	L	NL	NL	NL	NL	NL	NL	NL	NL	NL	NL
10.50	L	NL	NL	NL	NL	NL	NL	NL	NL	NL	NL
12.00	L	NL	NL	-	NL	NL	NL	NL	NL	NL	-
13.50	NL	NL	NL	-	L	NL	NL	NL	NL	NL	-
15.00	NL	NL	NL	-	L	NL	NL	NL	NL	NL	-
16.50	NL	NL	NL	-	NL	NL	NL	NL	NL	NL	-
18.00	NL	NL	NL	-	NL	NL	NL	NL	NL	NL	-
19.50	NL	NL	NL	-	NL	NL	NL	NL	NL	NL	-
21.00	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
22.50	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
24.00	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
25.50	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
27.00	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
28.50	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-
30.00	NL	NL	NL	-	NL	NL	NL	-	NL	NL	-

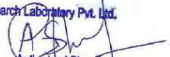
L=Liquefiable and NL= Non Liquefiable

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24.0 CHEMICAL ANALYSIS OF SOIL AND WATER

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24.1 CHEMICAL ANALYSIS OF SOIL

Sr.No	Bore Hole No.	Depth (M)	pH	Organic Matter (mg/l)	Chloride (as Cl) (mg/l)	Sulphate (as SO ₃) (mg/l)	Sulphate (as SO ₄) (mg/l)	Nitrate Content (as NO ₃) (mg/l)	Carbonate content (CO ₃ ²⁻) (Mg/l)
1	BH-1	2.25	6.50	56	28	39	44	18	5
2	BH-2	2.25	6.58	58	23	41	48	20	6
3	BH-3	2.25	6.66	62	26	35	40	23	9
4	BH-4	2.25	6.47	63	22	32	38	19	8
5	BH-5	2.25	6.62	66	19	29	35	21	7
6	BH-6	2.25	6.54	67	21	33	41	24	6
7	BH-7	2.25	6.51	57	17	28	34	22	7
8	BH-8	2.25	6.63	59	24	34	45	18	9
9	BH-9	2.25	6.61	64	20	30	37	20	5
10	BH-10	2.25	6.68	67	27	37	43	21	7
11	BH-11	2.25	6.57	63	25	33	40	23	8

24.2 CHEMICAL ANALYSIS OF WATER

Sr. No	Bore Hole No.	Dept h (M)	Colour	Ordour	Turbidity (NTU)	Specific Conductivity ($\mu\text{S}/\text{cm}$)	pH	Organic Matter (mg/l)	Chloride (as Cl) (mg/l)	Sulphate (as SO ₃) (mg/l)	Sulphate (as SO ₄) (mg/l)	Carbonate content (CO ₃ ²⁻) (Mg/l)	Nitrate Content (as NO ₃) (mg/l)
1	BH-1	2.25	Clear	Unobject i-onable	1.5	55	6.52	36	37	43	48	5	27
2	BH-2	2.25	Clear	Unobject i-onable	1.2	57	6.63	42	28	46	52	6	25
3	BH-3	2.25	Clear	Unobject i-onable	1.4	60	6.70	46	32	42	48	9	31
4	BH-4	2.25	Clear	Unobject i-onable	1.3	54	6.55	48	30	38	46	8	26
5	BH-5	2.25	Clear	Unobject i-onable	1.2	56	6.68	52	25	32	42	7	28
6	BH-6	2.25	Clear	Unobject i-onable	1.4	62	6.62	57	27	39	55	6	31
7	BH-7	2.25	Clear	Unobject i-onable	1.5	57	6.57	47	24	35	43	7	30
8	BH-8	2.25	Clear	Unobject i-onable	1.7	63	6.70	49	24	40	56	9	27
9	BH-9	2.25	Clear	Unobject i-onable	1.6	67	6.67	53	26	36	48	5	32
10	BH-10	2.25	Clear	Unobject i-onable	1.4	58	6.75	58	35	44	53	7	34
11	BH-11	2.25	Clear	Unobject i-onable	1.5	69	6.64	55	31	43	59	8	33

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25.0 SITE PHOTOGRPHS

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GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



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Upper Assam Division
Assam
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For Shiva Testing & Research Laboratory Pvt. Ltd.

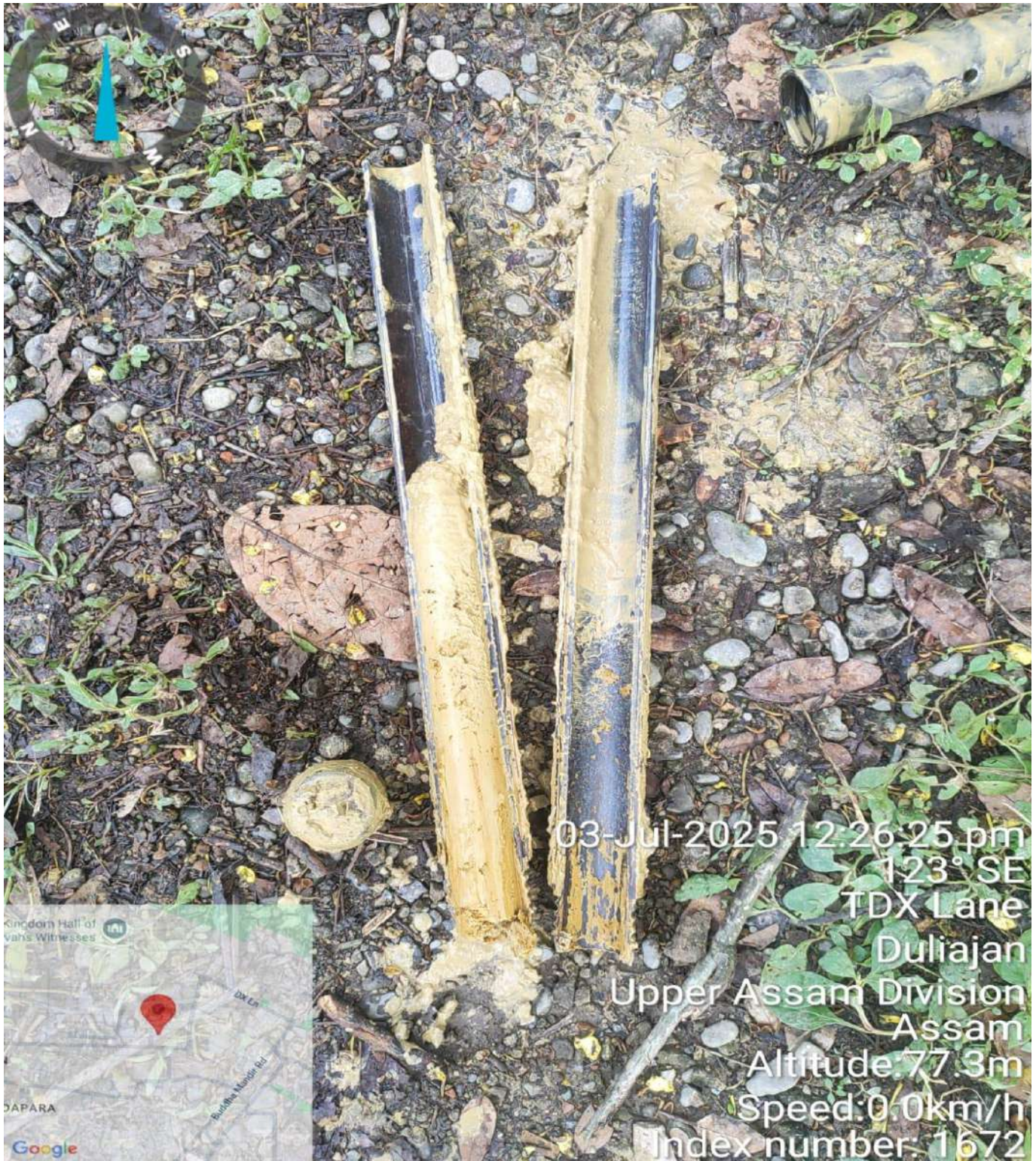
Authorized Signatory





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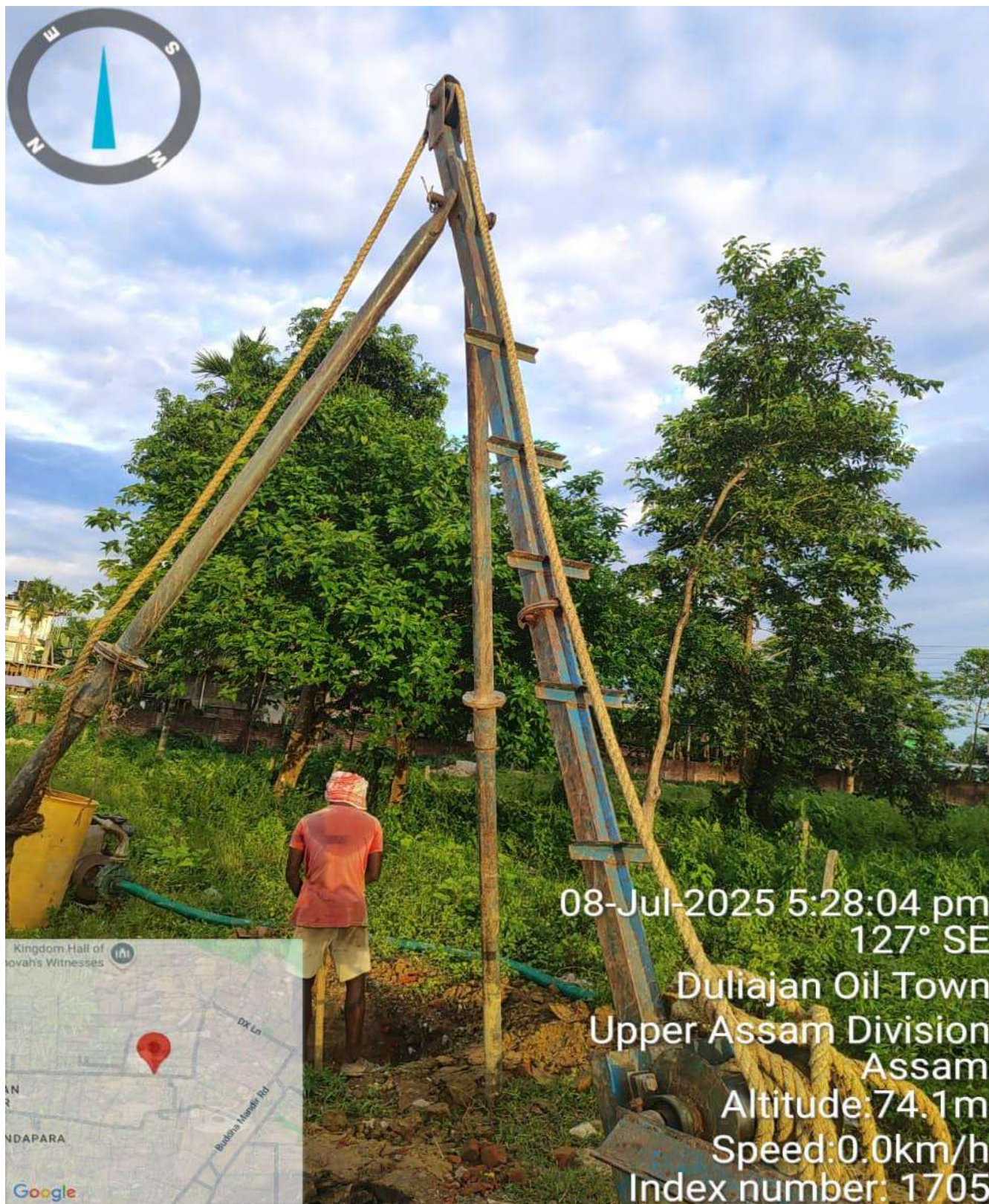
For Shiva Testing & Research Laboratory Pvt. Ltd.

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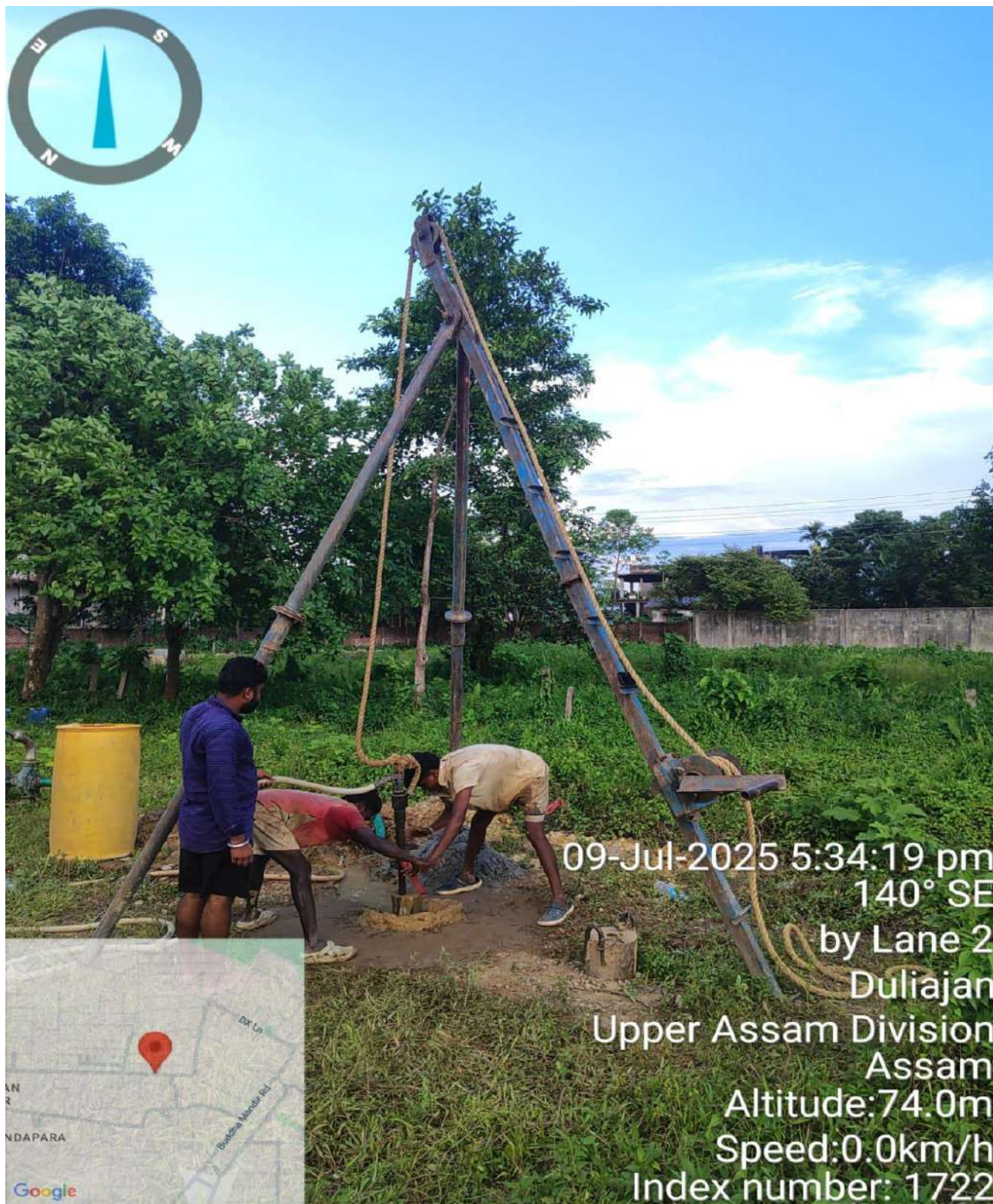


GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM





GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



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GEOTECHNICAL INVESTIGATION FOR THE CONSTRUCTION OF BQ HOUSING COMPLEX AT DULIAJAN, ASSAM



*****END OF REPORT*****

For. Shiva Testing & Research Laboratory Pvt. Ltd.

Akshya Sharma Director

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STR Labs



Geotech. Expert
STR Labs